

RECOVERY OF MATES ON MARTÍN NORMS II: COMPANIONS, CLEAN SUB-WINDOWS AND A DESIGNED OPERATOR

ABSTRACT. This is the second part of a study of the density of norm-attaining operators from Martín's renormings (c_0, p) into ℓ_2^2 . Part I reduced density, for a designed operator T and N blocks, to a single approximation statement (Lemma Z) about first rows with finite base support and signature room. Here we develop tools that work at *nearby* first rows: companions with prescribed data, the threshold equation of a block, the cost of moving the normer, a raise-transfer inequality, a uniform one-sided transfer expansion, and windowed recovery theorems in which the window data live at nearby first rows whose transfer error (for raise companions: the Hilbert footprint of the raise and the change of the normer) is below a fixed multiple of the square of the bottom of the window. We then fix one N -independent admissible operator T_{final} , with a diagonal base whose entries decay doubly exponentially, bounded-gap signature sets, weights chosen after the targets, and windows split into pigeonhole sub-windows at which every rate object of the first row is either tiny or robust. For T_{final} and first rows with finite base support we prove that every first row is recoverable unless, at every clean sub-window of every large level, a near-exact coherent shift resonance occurs (the residual (C^*)); tiny minors of the exact system are made exactly zero at a cheap companion, and the remaining shift obstruction is reduced to a pigeonhole dichotomy. We prove that (C^*) does not occur at constant-sign maximal contact, that some (C^*) rows are recoverable, and that under a lever hypothesis it can be resolved for a modified design. For infinite base support we prove the transfer of peaks to base vectors outside c_{00} , recovery under signature room, under cushion-sparse support swallowing (with the room and peak conditions (W^*) , (H_2) , $(H_3\text{-inf})$) and for box-dominated support (with the block conditions (SC_I) , (W_M)), and, for T_{final} , recovery of first rows whose finitely many bad carriers swallow the support with any profile (under (W^*) , (H_2) , $(H_3\text{-inf})$), and of rows with infinitely many bad carriers under a rate condition. Lemma Z, and the density of $\text{NA}((c_0, p_N), \ell_2^2)$ for any N and of $\text{NA}((c_0, p), \ell_2^2)$ for Martín's norm, remain open; we list the precise remaining cases.

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Date: Research continuation, October 2026.

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1. INTRODUCTION

1.1. The problem and what Part I proved. Martín [8] constructed renormings p of c_0 whose norm-attaining functionals form an algebraically trivial set; the construction starts from the canonical base q of c_0 (a smoothing of the supremum norm in the sense of Debs, Godefroy and Saint-Raymond [2]) and an injective norm-one operator T given by his Lemma B, which rests on [6, Proposition 2.8]. We call such an operator *admissible* [I, Definition 1.3]; p_N denotes the norm built with the blocks $I = \{1, \dots, N\}$ and p the norm built with $I = \mathbb{N}$. The question studied in Part I [I] and here is whether $\text{NA}((c_0, p_N), \ell_2^2)$ and $\text{NA}((c_0, p), \ell_2^2)$ are dense. It is a test case for Lindenstrauss' property B [7] of the two-dimensional Hilbert space (compare [4, 5]): a negative answer for one N and one admissible T would produce a two-dimensional range space without property B [I, Section 9].

An operator into ℓ_2^2 is a pair (f, g) of functionals. For a *first row* $f \in S_{p^*}$ the *mate fibre* is $\mathcal{C}(f) = \{g : (f, g) \text{ is contractive}\}$, and

$$\mathcal{R} := \{f \in S_{p^*} : (f, g) \in \text{cl NA}((c_0, p), \ell_2^2) \text{ for every } g \in \mathcal{C}(f)\}$$

is the recoverable set [I, Definition 1.15]; density holds if and only if $\mathcal{R} = S_{p^*}$ [I, Proposition 1.13]. By [I, Lemma 1.5], density for p_N for infinitely many N implies density for p . Part I proved, among other things:

- (I-1) the transport of finite certificates along every sequence $f_n \rightarrow f$ and an averaging theorem over scales [I, Theorems 3.8 and 4.1];
- (I-2) the resonance obstruction: an admissible T and a first row at which the intrinsically recoverable mates lie on one line while $\mathcal{C}(f)$ is infinite dimensional [I, Theorem 6.6];
- (I-3) engineered recovery: at first rows with finite base support, every mate with d -neutral two-piece data of weighted coefficient at most one is recovered along engineered norm-attaining approximants [I, Theorem 7.18, Corollary 7.21], and every block-tame first row is recoverable [I, Corollary 7.22];
- (I-4) for a *signature-ladder* operator [I, Definition 8.1, Theorem 8.2]: every first row with finite base support and signature room is recoverable [I, Theorem 8.18], and density for p_N is equivalent to *Lemma Z* [I, Theorem 8.20, Problem 8.21]: for every $f \in S_{p_N^*}$, $g \in \mathcal{C}(f)$, $\rho < 1$ and $\varepsilon > 0$ there is f' with finite base support and signature room ($f' \in \mathcal{R}_0$, [I, Definition 8.3]), $p_N^*(f' - f) < \varepsilon$ and $\text{dist}(\rho g, \mathcal{C}(f')) < \varepsilon$;
- (I-5) for that operator, recovery of window-pinned mates, of sign-mixed room and of some exactly swallowed signature sets [I, Theorems 8.31, 8.35 and 8.44].

Lemma Z was left open for every N .

1.2. Lemma Z for the designed operator. In Section 3 we fix one operator T_{final} (Definition 3.5) with base U_{final} (Definition 3.1). It is admissible and satisfies the conclusion of Martín's Lemma B (Theorem 3.12), it does not depend on N , and with any choice of one sub-window of each level it satisfies (T-a)–(T-d) and (P1)–(P3) of [I, Theorem 8.2] verbatim (these are the conclusions of [I, Theorem 8.2], which are all that [I, Section 8] uses; Lemma 3.22(b)). Consequently all results of [I, Section 8] hold for T_{final} ; in particular:

- for $T = T_{\text{final}}$ and every $N \geq 1$, $\text{NA}((c_0, p_N), \ell_2^2)$ is dense if and only if Lemma Z holds for T_{final} and N [I, Theorem 8.20];

- if Lemma Z holds for T_{final} and infinitely many N , then $\text{NA}((c_0, p), \ell_2^2)$ is dense for Martín's norm p built with T_{final} (Remark 3.13, [I, Lemma 1.5]);
- Lemma Z splits into a statement about first rows with finite base support and an approximation statement (LSC-trunc) for infinite base support (Proposition 6.48).

Membership of individual first rows in $\mathcal{R}$ does not transfer from p_N to p : [I, Lemma 1.5] transfers density, not rows, and the hypotheses of the row-wise theorems below depend on N .

1.3. Main results. *Recovery through nearby first rows (Section 2).* The tools of Section 2 hold for every admissible operator. First rows with prescribed data $f[a, z]$ and their companions (Definition 2.1) allow us to move contacts, raise the base vector and change the normer while keeping control of the forced data. The block threshold is the unique root of one explicit equation (Lemma 2.6), with a Lipschitz bound and a quantitative raising property (Lemma 2.7); the cost of moving the normer is bounded by the change of the data, including an unweighted term which cannot be dropped (Lemma 2.12, Remark 2.13); and a same-sign raise of the base vector obeys the raise-transfer inequality (Lemma 2.15), in which the ℓ_1 -mass of the raise cancels against the normalization. The uniform one-sided transfer expansion (Lemma 2.18) holds along every sequence of first rows $f_j \rightarrow f$ with finite supports $F \subseteq F_j$, with constants independent of the quantities that the companions change; for arbitrary supports the corresponding statement is Lemma 6.10, under its flip-sparsity hypothesis. The *Theorem E family* then transfers recovery from nearby first rows to f : if window data for the mate live at first rows f_j whose transfer error (for raise companions: the Hilbert footprint of the raise and the change of the normer; this is not $p^*(f_j - f)$, see Remark 6.35(a)) is smaller than a fixed multiple of the square of the bottom of the j -th window, the mate is recovered (Theorem 2.19 with a transfer error, for arbitrary base support; Theorem 2.20 with exact d -neutral two-piece data; Theorem 5.13 in Section 5 with shifted data). Section 2 also proves an exactness certificate for two-piece data (Lemma 2.22), a shift bound (Lemma 2.23), the rigidity of two-piece data (Lemma 2.24, Corollaries 2.25 and 2.26), a criterion for the failure of lower semicontinuity of the fibre map along block-tame first rows (Theorem 2.28; its hypotheses are realized for T_{final} and $N = 1$ at a self-aligned row in Proposition 5.32 of Section 5), and the violation tolerance of one engineered stage (Lemma 2.30).

The designed operator (Section 3). T_{final} is a signature-ladder operator with four modifications: a diagonal base $U_{\text{final}}k_s = \mu_s^*e_s$ with $\mu_s^* = 2^{-2^{2^s}}$; bounded-gap signature sets $S_l = \{2^l(2i+1) : i \geq 1\}$ with an additional allowedness rule; weights c_l chosen after the target and the signature size of level l ; and windows split into pigeonhole sub-windows governed by a design factor $\text{Des}(l)$ and a finite family of rate objects (Definitions 3.4–3.7). The recursion is well founded and independent of N (Lemma 3.10); T_{final} satisfies Martín's Lemma B (Theorem 3.12); and for every first row and every level there is a *clean* sub-window, at which every rate object of f of that level is either tiny ($\leq b(w)$) or robust ($\geq u(w)$) (Definition 3.18, Theorem 3.19).

Finite base support (Sections 4 and 5). At a clean sub-window the switching amplitudes of two-sided decompositions are pinned diagonally (Lemmas 4.3 and 4.10); far pulls and private banks on the diagonal base tune finitely many carrier values

exactly and two-sidedly (Lemma 4.15); every block has a peak with robust margin whose bank raises the block threshold by a buffer dominating all drifts (Lemmas 4.7, 4.17 and 4.19); and one exactifying companion per clean sub-window carries all exactifications at cost $o(T_{\text{lo}}^2)$ (Definition 4.16, Lemma 4.18, Proposition 4.24). This gives *Master Theorem II* (Theorem 4.25): for T_{final} , a first row with finite base support is in $\mathcal{R}$ if at infinitely many levels some clean sub-window satisfies a shift condition (SH_w) and a ray condition (VR_w). The ray condition is removed by the exactification of all tiny minors at a companion, with a Hoffman constant of the form $C_f^l \text{Des}(l)^2 / u(w)$ (Theorem 5.7, Corollary 5.8), using Hoffman's theorem [3] through minors (Lemma 5.1) and the Łojasiewicz inequality [1] (Lemma 5.3). The shift condition is reduced by the shift dichotomy (Theorem 5.10) to a pigeonhole rate. The result is *Master Theorem III'* (Theorem 5.15) and:

For $T = T_{\text{final}}$, every $N \geq 1$ and every $f \in S_{p_N}^$ with finite base support: if $f \notin \mathcal{R}$, then f is a (C^*) row (Corollary 5.17), i.e. at every clean sub-window of every large level the shift-pinning rate and the shift cost are both tiny (Definition 5.16): a near-exact coherent shift resonance.*

About (C^*) we prove: it does not occur at constant-sign maximal contact (Proposition 5.21); shifted exact data are rigid (Proposition 5.19) and oscillating profiles force them (Proposition 5.23); (C^*) -type resonances are realized by self-aligned first rows, which are block-tame and hence recoverable (Theorem 5.29, Corollary 5.30); near-threshold carriers are Baire generic in every fixed- z fibre (Proposition 5.33) while exact all-negative data are not (Proposition 5.39); exact negative data without dead zones are recovered (Theorem 5.45); and, for a modification $D^{U1'}$ of T_{final} with absorber targets (Definition 5.48), *Master Theorem IV'* (Theorem 5.68) proves $f \in \mathcal{R}$ whenever, at infinitely many main stages, some clean sub-window carries enough scales whose activity class has no active shift, or admits a $\varkappa$ -neutral lever ($\text{KN}_{w,a}$) (in a mixed class, together with the absence of frustrated positive carriers). A lever is needed in this route: a donor raise moves the block scalar $\varkappa$ by the same order as the raise, and the exact shifted system depends on a block only through ratios (Corollary 2.11, Proposition 5.46), so peak pushes alone cannot achieve exactness and threshold protection independently (Remark 5.47; no claim is made that a one-lever argument cannot succeed).

Infinite base support (Section 6). Transfer peaks exist at every base vector, also outside c_{00} (Theorem 6.3); hence signature room, sign-mixed room and window-pinned mates are recovered at arbitrary base support (Theorems 6.20–6.23), which turns [I, Remark 8.46] into a theorem (Theorem 6.21). Engineered recovery at arbitrary base support needs sparsity of flips only for the one-sided parts of the data (Theorems 6.14 and 6.15). Finitely many cushion-sparse swallowed carriers are harmless for every signature-ladder operator (Theorem 6.29, under (W^*) , (H2), (H3-inf)), and box-dominated support is harmless for every admissible operator (Theorem 6.31, under (SC_I) and (W_M)). For T_{final} the raise-transfer inequality removes all profile conditions: support swallowing of any profile by finitely many bad carriers is harmless (Theorems 6.34, 6.38 and 6.40, under (W^*) , (H2), (H3-inf)), and infinitely many bad carriers are harmless under the explicit rate condition (W_{inf}) (Theorem 6.42).

1.4. What remains open. *Lemma Z remains open for T_{final} and every $N \geq 1$, including $N = 1$; so does the density of $\text{NA}((c_0, p_N), \ell_2^2)$ for every N and of*

$\text{NA}((c_0, p), \ell_2^2)$ for Martín's norm p , for every admissible operator, including T_{final} . By the results above, for T_{final} the open part of Lemma Z consists of:

- (i) at finite base support, the (C^*) rows (Problem 5.80) that are not block-tame and do not lie in $\mathcal{R}_0 \cup \mathcal{R}_0^\pm \cup \mathcal{R}_S$; the precise sub-problems are status coherence without a lever, mixed activity classes for $N \geq 2$, exact multi-block coupling, and uniform composition with d -consistent engineered approximants (Problems 7.4–7.7);
- (ii) at infinite base support, (LSC-trunc) for the first rows not covered by Section 6: the transport of Master Theorem III' (only sketched, Remark 6.50), the rate condition (W_{inf}) for infinitely many bad carriers, the tuning regularity modulus, condition (H2) and degenerate swallowing-sign bad peaks, and the scrambling condition along deep raises (Problems 7.9–7.13).

Section 7 states these problems precisely. Nothing proved here or in Part I points to a counterexample; equally, nothing proved here settles any one of the items (i), (ii).

1.5. Conventions. The notation of Part I is used without comment, in particular the forced data $(\xi, q_0, a, w, F, z, \hat{z}, e, \nu)$ of a first row [I, Proposition 1.9], two-piece data and the coefficient Γ_w [I, Definition 7.7], and the signature-ladder notation of [I, Section 8]. References of the form [I, Theorem 8.20] are to Part I [I]. Constants C_f depend on the first row (and on the mate where said) but not on the level, the sub-window or the scale. Statements that are only sketched or heuristic in our sources appear only inside remarks that say so explicitly (“sketch”, “heuristic”, “numerical evidence”); every lemma, proposition, theorem and corollary is given with a complete proof. Where an earlier version of an argument was found to be wrong, the corrected statement is given and the failure is recorded in a remark (for instance Remarks 3.11 and 5.70).

2. COMPANIONS AND TRANSFER

This section collects the tools of Part II that are valid for every admissible operator T ; where a statement uses more (a property of the design, a diagonal base, one block) this is said in the statement. Throughout, T is admissible, the block set I is finite and p is the norm of [I, Section 1] built with I (so $p = p_N$ when $I = \{1, \dots, N\}$). Forced data, the sets F, K, J , the block data $\zeta_m = R_m^{**}\xi$, σ_m , M_m , C_m , P_m , Q_m , α_m , gap_m , $\mu_{k,m}$, two-piece data, Γ_w , κ_w and Δd_m are those of [I, Proposition 1.9, Lemma 1.7, Definition 7.7]. We also write

$$\hat{\zeta}_m := R_m^{**}\hat{z} = \zeta_m/q_0, \quad |\hat{\zeta}_m|_m = \sigma_m/q_0, \quad \psi_m := R_m^*w_m \in \ell_1,$$

and $\phi_z(x) := |x| - zx$ for $|z| \leq 1$, $x \in \mathbb{R}$, as in [I, Lemma 8.23].

The section is organized as follows. Subsection 2.1 introduces first rows with prescribed data and companions. Subsection 2.2 gives the block threshold as the unique root of one explicit equation, with quantitative consequences and the identities behind the block scalar $\varkappa$. Subsection 2.3 bounds the cost of moving the normer and proves the raise-transfer lemma. Subsection 2.4 proves the uniform one-sided transfer expansion along convergent first rows, and Subsection 2.5 the windowed recovery theorems through nearby first rows. Subsection 2.6 studies exact two-piece data: the exactness certificate, the shift bound, the rigidity of two-piece data

and a criterion for the failure of lower semicontinuity of the fibre map along block-tame rows. Subsection 2.7 shows that the engineered approximants of [I, Section 7] tolerate small violations of the sign conditions of two-piece data at a single stage.

2.1. First rows with prescribed data; companions. Let $a' \in \ell_1$ with $q^*(a') = 1$ and $z' \in B_{\ell_\infty}$ with $z'_j = \text{sign } a'_j$ for $j \in \text{supp } a'$. By [I, Remark 8.22(c)] there is a unique $f' \in S_{p^*}$ whose forced data are

$$\begin{aligned} e' &:= U^* a' / \|U^* a'\|_H, & \hat{z}' &:= z' + Ue', & q'_0 &:= \left(1 + \sum_{m \in I} |R_m^{**} \hat{z}'|_m\right)^{-1}, \\ \xi' &:= q'_0 \hat{z}', & w' &:= J_V(L^{**} \xi'), & f' &= a' + L^* w'; \end{aligned}$$

we denote it by $f[a', z']$. Since each J_m is positively homogeneous of degree 0, $w'_m = J_m(R_m^{**} \hat{z}')$ depends on $\hat{z}'$ only. If $a'_n \rightarrow a'$ in ℓ_1 and $z'_n \rightarrow z'$ coordinatewise (with the same constraints), then $f[a'_n, z'_n] \rightarrow f[a', z']$ in norm [I, Remark 8.22(c)]. The row $f[a', z']$ attains its norm if and only if $a' \in c_{00}$ and $z' \in c_0$ [I, Proposition 1.10(c)]; in general it does not.

Definition 2.1 (Companions). Let $f \in S_{p^*}$ have forced data $(a, z, \dots)$.

- (a) A *companion* of f is a row $f^\# := f[a, z^\#]$ with $z^\# \in B_{\ell_\infty}$ and $z^\# = z$ on F . We put $\delta := z^\# - z$; then $e^\# = e$, $\nu^\# = \nu$ and $\hat{z}^\# = \hat{z} + \delta$, and δ vanishes on F .
- (b) A *raise* is a vector $\Delta a \in \ell_1$ with $\text{supp } \Delta a \subseteq F$ and $\text{sign}(a_j) \Delta a_j \geq 0$ for $j \in F$. The *raise companion* is $f^\# := f[a^\#, z]$ with $\lambda := q^*(a + \Delta a)$ and $a^\# := (a + \Delta a)/\lambda$.

Forced data of a companion carry the superscript $\#$; for instance $w_m^\# = J_m(R_m^{**} \hat{z}^\#)$, $C_m^\#, P_m^\#, Q_m^\#, \text{gap}_m^\#, d_m^\#(\omega) = \langle D_m w_m^\#, D_m \omega \rangle / C_m^\#$.

The admissibility of the data in (b) follows from $\text{supp } a^\# = F$ and $\text{sign } a_j^\# = \text{sign } a_j = z_j$ on F . Companions in the sense of (a) are the rows used to *exactify* data (close rooms, flip or raise contacts); raise companions are used to enlarge the base part on its own support. In both cases the contact data off the moved coordinates are those of f .

Lemma 2.2 (d -coefficients through the normer). *Let $f \in S_{p^*}$ and $m \in I$.*

- (a) *For every finitely supported ω with $\text{supp } \omega \subseteq Q_m$,*

$$d_m(\omega) = \frac{(R_m^* \omega)(\hat{z})}{|R_m^{**} \hat{z}|_m} = \frac{\langle \omega, \hat{\zeta}_m \rangle}{|\hat{\zeta}_m|_m}.$$

- (b) *If $f^\#$ is a companion of f (Definition 2.1(a)) and ω is finitely supported in $Q_m \cap Q_m^\#$, then*

$$|R_m^{**} \hat{z}^\#|_m d_m^\#(\omega) - |R_m^{**} \hat{z}|_m d_m(\omega) = (R_m^* \omega)(\delta).$$

Proof. (a) Off P_m , [I, Lemma 1.7] gives $\Phi_m(k)^2 w_m(k)/C_m = \zeta_m(k)/\sigma_m$. Hence $d_m(\omega) = \sum_k \Phi_m(k)^2 w_m(k) \omega(k)/C_m = \langle \omega, \zeta_m \rangle / \sigma_m = \langle \omega, \hat{\zeta}_m \rangle / |\hat{\zeta}_m|_m$, and $\langle \omega, R_m^{**} \hat{z} \rangle = (R_m^* \omega)(\hat{z})$. (b) Apply (a) at f and at $f^\#$ and subtract; $\hat{z}^\# - \hat{z} = \delta$. $\square$

Corollary 2.3 (Moving contacts converts a first-order defect into a d -mismatch). *Let $f^\#$ be a companion of f , let ω be finitely supported in $Q_m \cap Q_m^\#$ and put*

$V := R_m^* \omega \in \ell_1$. Suppose that for every $j \in \text{supp } \delta$ either $V_j = 0$ or $z_j^\# = \text{sign } V_j$. Then

$$|R_m^{**} \hat{z}^\#|_m d_m^\#(\omega) - |R_m^{**} \hat{z}|_m d_m(\omega) = \sum_{j \in \text{supp } \delta} \phi_{z_j}(V_j) \geq 0.$$

Proof. By Lemma 2.2(b) the left side is $V(\delta) = \sum_{j \in \text{supp } \delta} V_j(z_j^\# - z_j)$, and $V_j(z_j^\# - z_j) = |V_j| - z_j V_j = \phi_{z_j}(V_j)$ when $z_j^\# = \text{sign } V_j$; both sides vanish when $V_j = 0$. $\square$

Applied to a switching $\omega = \omega^- - \omega^+$, the corollary says: if the switching is d -neutral at f and a companion makes the coordinates it moves exact contacts with the sign of V , then at the companion the switching has a d -mismatch equal to its first-order defect $\sum_{j \in \text{supp } \delta} \phi_{z_j}(V_j)$, divided by $|R_m^{**} \hat{z}^\#|_m$. Only the defect on $\text{supp } \delta$ is converted; nothing is said about the coordinates that are not moved.

Lemma 2.4 (Approximate resonance and d -neutrality). *Let $u \in \ell_1$, $\varepsilon \in \{\pm 1\}$, and let $f^\#$ be a companion of f such that $\varepsilon u_j z_j^\# = |u_j|$ for every $j \in \text{supp } u \setminus F$. Then*

$$\varepsilon u(\hat{z}^\#) - \varepsilon u(\hat{z}) = \text{Re}(u) := \sum_{j \notin F} \phi_{z_j}(\varepsilon u_j) \geq 0.$$

Consequently, if $u = u_{k,m}$, $k \in Q_m$ with $w_m(k) = 0$ (a d -neutral strict non-peak of f), $\text{Re}(u) > 0$, and k is a strict non-peak of $f^\#$, then

$$\varepsilon \frac{\Phi_m(k) w_m^\#(k)}{m C_m^\#} = \frac{\text{Re}(u)}{|R_m^{**} \hat{z}^\#|_m} > 0;$$

in particular k is not d -neutral at $f^\#$.

Proof. $\hat{z}^\# - \hat{z} = \delta$ vanishes on F , so $\varepsilon u(\hat{z}^\#) - \varepsilon u(\hat{z}) = \sum_{j \notin F} \varepsilon u_j(z_j^\# - z_j)$, and $\varepsilon u_j(z_j^\# - z_j) = |u_j| - \varepsilon u_j z_j = \phi_{z_j}(\varepsilon u_j)$ for $j \in \text{supp } u \setminus F$ (both vanish off $\text{supp } u$). If $w_m(k) = 0$, then $u(\hat{z}) = 0$: off P_m , $w_m(k) = C_m \zeta_m(k) / (\Phi_m(k)^2 \sigma_m)$ by [I, Lemma 1.7]. At the strict non-peak k of $f^\#$ the same formula gives

$$\frac{\Phi_m(k) w_m^\#(k)}{C_m^\#} = \frac{\lambda_{k,m} u(\hat{z}^\#)}{\Phi_m(k) |R_m^{**} \hat{z}^\#|_m} = \frac{m u(\hat{z}^\#)}{|R_m^{**} \hat{z}^\#|_m},$$

and $\varepsilon u(\hat{z}^\#) = \text{Re}(u)$. $\square$

Remark 2.5. (a) If all switching carriers of a block are of the kind described in Lemma 2.4, they all acquire weights of the same sign at $f^\#$, and the cone of exact d -neutral switchings at $f^\#$ meets the nonnegative orthant only in 0. A Hoffman constant of that cone is then at least $|R_m^{**} \hat{z}^\#|_m / \min_l \text{Re}(u_l)$ (test with a unit vector). Since $\text{Re}(u_l)$ is the room of the signature part plus the contribution of the target part, this is of order $1/(\text{room})$ only when the target part is comparable with the room.

(b) By Corollary 2.3 a companion produces nonnegative d -mismatches, which [I, Corollary 7.21] accepts. However, finitely supported two-piece data at a row with nonzero mismatches satisfy, off the supports $\Sigma(\omega)$ of the block parts, $b^+ - b^- = -W_\Delta = -\sum_m \Delta d_m \psi_m$ (Lemma 2.24 below), and ψ_m has infinitely many peak terms; so the mismatch can be carried only where $z_j W_\Delta(j) \leq 0$ at contacts $j \notin \Sigma(\omega)$ and $W_\Delta(j) = 0$ at free coordinates $j \notin \Sigma(\omega)$. This is the content of [I, Remark 8.45(b)].

2.2. The threshold equation. In this subsection we fix one block and drop its index: Φ is a sequence of positive numbers with $\|\Phi\|_1 < 1$, D the diagonal operator with entries $\Phi(k)$, $|\cdot|$ the norm on ℓ_1 with unit ball $B_{\ell_1} + D(B_{\ell_2})$ and $N(w) = \|w\|_\infty + \|Dw\|_2$ its dual norm [I, Remark 1.1]. For $\zeta \in \ell_1 \setminus \{0\}$ let $w = J(\zeta)$, M , C , P , $Q := \mathbb{N} \setminus P$ and α be as in [I, Lemma 1.7], and put

$$\begin{aligned} \bar{\vartheta}(\zeta) &:= \frac{|\zeta|M}{C}, & \mathbf{v}_k(\zeta) &:= \frac{|\zeta(k)|}{\Phi(k)^2}, & \varrho_k(\zeta) &:= \frac{\mathbf{v}_k(\zeta)}{\bar{\vartheta}(\zeta)}, & \Phi_P^2 &:= \sum_{k \in P} \Phi(k)^2, \\ A(x; \zeta) &:= \sum_k (|\zeta(k)| - x\Phi(k)^2)_+, & B(x; \zeta) &:= \sum_k \Phi(k)^2 \min(x, \mathbf{v}_k(\zeta))^2, \\ \Xi(x; \zeta) &:= A(x; \zeta)^2 - B(x; \zeta) \quad (x > 0). \end{aligned}$$

The series converge: $(|\zeta(k)| - x\Phi(k)^2)_+ \leq |\zeta(k)|$ and $\Phi(k)^2 \min(x, \mathbf{v}_k)^2 \leq x\Phi(k)^2 \mathbf{v}_k = x|\zeta(k)|$. All these quantities except ϱ_k are positively homogeneous of degree one in ζ (and ϱ_k , M , C , P of degree zero). For the block vectors of a first row, $\zeta = R_m^{**}\xi$, $\lambda_{k,m} = m\Phi_m(k)$ and $\mathbf{v}_k(\zeta) \geq \bar{\vartheta}(\zeta)$ iff $|\xi(u_{k,m})| \geq \vartheta_m \Phi_m(k)$; thus $\bar{\vartheta}(\zeta_m) = m\vartheta_m$ with the threshold constant ϑ_m of [I, Section 1], and for $\hat{\zeta}_m = R_m^{**}\hat{z}$

$$\begin{aligned} \mu_{k,m} &= \frac{q_0 \Phi_m(k)}{m} (\mathbf{v}_k(\hat{\zeta}_m) - \bar{\vartheta}(\hat{\zeta}_m)) \quad (k \in P_m), \\ \text{gap}_m(k) &= M_m(1 - \varrho_k(\hat{\zeta}_m)) \quad (k \in Q_m) \end{aligned}$$

(the second identity is the clamp formula in part (b) below). We call ϱ_k the *relative position* of k : k is a peak iff $\varrho_k \geq 1$, a degenerate peak iff $\varrho_k = 1$.

Lemma 2.6 (Threshold equation). *Let $\zeta \in \ell_1 \setminus \{0\}$.*

- (a) *For every $\zeta' \in \ell_1 \setminus \{0\}$ the function $\Xi(\cdot; \zeta')$ has exactly one zero in $(0, \infty)$; it is positive to the left and negative to the right of the zero.*
- (b) *The zero of $\Xi(\cdot; \zeta)$ is $\bar{\vartheta}(\zeta)$. Writing $\bar{\vartheta} := \bar{\vartheta}(\zeta)$ and $\mathbf{v}_k := \mathbf{v}_k(\zeta)$:*

$$\begin{aligned} |\zeta| &= A(\bar{\vartheta}; \zeta), \quad |\zeta|^2 = B(\bar{\vartheta}; \zeta), \quad M = \frac{\bar{\vartheta}}{\bar{\vartheta} + |\zeta|}, \quad C = \frac{|\zeta|}{\bar{\vartheta} + |\zeta|}, \\ w(k) &= \text{sign}(\zeta(k)) M \min\left(1, \frac{\mathbf{v}_k}{\bar{\vartheta}}\right), \end{aligned}$$

$P = \{k : \mathbf{v}_k \geq \bar{\vartheta}\}$, and $|\zeta| |\alpha(k)| = |\zeta(k)| - \bar{\vartheta} \Phi(k)^2$ for $k \in P$; in particular $k \in P$ is a degenerate peak iff $\mathbf{v}_k = \bar{\vartheta}$.

- (c) (threshold identities)

$$\sum_{k \in P} \Phi(k)^2 (\mathbf{v}_k - \bar{\vartheta}) = |\zeta|, \quad |\zeta|^2 = \bar{\vartheta}^2 \Phi_P^2 + \sum_{k \in Q} \mathbf{v}_k^2 \Phi(k)^2.$$

- (d) *For $\zeta' \in \ell_1 \setminus \{0\}$: if $\Xi(\bar{\vartheta}(\zeta); \zeta') > 0$ then $\bar{\vartheta}(\zeta') > \bar{\vartheta}(\zeta)$, and if $\Xi(\bar{\vartheta}(\zeta); \zeta') < 0$ then $\bar{\vartheta}(\zeta') < \bar{\vartheta}(\zeta)$.*
- (e) (raising moves) *Let $k' \in \mathbb{N}$ and let ζ' agree with ζ except at k' , where $\text{sign } \zeta'(k') = \text{sign } \zeta(k')$ or $\zeta'(k') = 0$. Then $\bar{\vartheta}(\zeta') > \bar{\vartheta}(\zeta)$ in each of the cases: (i) $k' \in P$ and $|\zeta'(k')| > |\zeta(k')|$; (ii) $k' \notin P$ and $|\zeta'(k')| < |\zeta(k')|$; (iii) k' is a degenerate peak and $|\zeta'(k')| \neq |\zeta(k')|$.*
- (f) (perturbed raising moves) *Let $k' \in \mathbb{N}$, $\zeta' \in \ell_1 \setminus \{0\}$, $\Delta := ||\zeta'(k')| - |\zeta(k')|| > 0$ and $E := \sum_{k \neq k'} |\zeta'(k) - \zeta(k)|$. Then $\bar{\vartheta}(\zeta') > \bar{\vartheta}(\zeta)$ if either (i') $k' \in P$, $|\zeta'(k')| > |\zeta(k')|$ and $E < |\zeta| \Delta / (|\zeta| + \bar{\vartheta})$; or (ii') $\mathbf{v}_{k'} > 0$, $\mathbf{v}_{k'} \leq \bar{\vartheta}$, $|\zeta'(k')| < |\zeta(k')|$ and $E < \mathbf{v}_{k'} \Delta / (2(|\zeta| + \bar{\vartheta}))$.*

Proof. (a) Fix ζ' and write A', B', Ξ', v' . By dominated convergence A' and B' are continuous on $(0, \infty)$; A' is nonincreasing and B' nondecreasing. Let $s^* := \sup_k v'_k \in (0, \infty]$. For $x < s^*$ the set $\{k : v'_k > x\}$ is nonempty; each of its terms of A' is positive and strictly decreasing and each of its terms of B' is strictly increasing near x , so Ξ' is strictly decreasing on $(0, s^*)$. As $x \rightarrow 0^+$, $A'(x) \rightarrow \|\zeta'\|_1 > 0$ and $0 \leq B'(x) \leq x\|\zeta'\|_1 \rightarrow 0$, so $\Xi' > 0$ near 0. For $x \geq s^*$ (if $s^* < \infty$), $A'(x) = 0$ and $\Xi'(x) = -B'(x) < 0$. If $s^* = \infty$, then $A'(x) \rightarrow 0$ as $x \rightarrow \infty$ by dominated convergence, while $B'(x) \geq B'(1) > 0$ for $x \geq 1$, so $\Xi' < 0$ for large x . By continuity and strict monotonicity on $(0, s^*)$, Ξ' has exactly one zero, which lies in $(0, s^*)$, and the sign pattern is as stated.

(b) Let $\bar{\vartheta} > 0$ be the zero of $\Xi(\cdot; \zeta)$, put $a := A(\bar{\vartheta}; \zeta)$ and $s_k := \text{sign } \zeta(k)$. If $a = 0$ then $B(\bar{\vartheta}; \zeta) = 0$, which forces $\zeta = 0$; so $a > 0$, and some k has $v_k > \bar{\vartheta}$. *Primal:* put $x_k := s_k(|\zeta(k)| - \bar{\vartheta}\Phi(k)^2)_+$ and $\beta_k := s_k\Phi(k)\min(v_k, \bar{\vartheta})$. Checking the cases $v_k \geq \bar{\vartheta}$ and $v_k < \bar{\vartheta}$ separately gives $\zeta = x + D\beta$, and $\|x\|_1 = a$, $\|\beta\|_2^2 = B(\bar{\vartheta}; \zeta) = a^2$. Hence $\zeta/a \in B_{\ell_1} + D(B_{\ell_2})$, i.e. $|\zeta| \leq a$. *Dual:* put $v_k := s_k\min(\bar{\vartheta}, v_k)$. Then $\|v\|_\infty = \bar{\vartheta}$, $\|Dv\|_2^2 = B(\bar{\vartheta}; \zeta) = a^2$, so $N(v) = \bar{\vartheta} + a$, and, using $\sum_{v_k \geq \bar{\vartheta}} |\zeta(k)| = a + \bar{\vartheta} \sum_{v_k \geq \bar{\vartheta}} \Phi(k)^2$ and $v_k|\zeta(k)| = \Phi(k)^2 v_k^2$,

$$\langle v, \zeta \rangle = \bar{\vartheta} \sum_{v_k \geq \bar{\vartheta}} |\zeta(k)| + \sum_{v_k < \bar{\vartheta}} \Phi(k)^2 v_k^2 = \bar{\vartheta}a + B(\bar{\vartheta}; \zeta) = a(\bar{\vartheta} + a).$$

So $w' := v/(\bar{\vartheta} + a)$ has $N(w') = 1$ and $\langle w', \zeta \rangle = a \geq |\zeta| \geq \langle w', \zeta \rangle$. Hence $|\zeta| = a$, w' norms ζ , and $w' = J(\zeta) = w$ because $|\cdot|$ is smooth [I, Remark 1.1(d)]. This gives the formulas for $|\zeta|$, $|\zeta|^2$, w , $M = \|w\|_\infty = \bar{\vartheta}/(\bar{\vartheta} + |\zeta|)$, $C = \|Dw\|_2 = |\zeta|/(\bar{\vartheta} + |\zeta|)$, and $|\zeta|M/C = \bar{\vartheta}$; $P = \{|w(k)| = M\} = \{v_k \geq \bar{\vartheta}\}$; finally $\alpha = \zeta/|\zeta| - D^2w/C = x/|\zeta|$, which is the last claim.

(c) The first identity is $A(\bar{\vartheta}; \zeta) = |\zeta|$, the second $B(\bar{\vartheta}; \zeta) = |\zeta|^2$, both read with (b).

(d) By (a) for ζ' : $\Xi(x; \zeta') > 0$ iff $x < \bar{\vartheta}(\zeta')$, and $\Xi(x; \zeta') < 0$ iff $x > \bar{\vartheta}(\zeta')$.

(e) Put $\bar{\vartheta} := \bar{\vartheta}(\zeta)$; $\Xi(\bar{\vartheta}; \zeta) = 0$. In case (i) the k' -term of $A(\bar{\vartheta}; \cdot)$ increases by $|\zeta'(k')| - |\zeta(k')| > 0$ and the k' -term of $B(\bar{\vartheta}; \cdot)$ is $\bar{\vartheta}^2\Phi(k')^2$ before and after; so $\Xi(\bar{\vartheta}; \zeta') > 0$. In case (ii), $v'_{k'} < v_{k'} < \bar{\vartheta}$: the k' -terms of A vanish before and after and the k' -term of B decreases; so $\Xi(\bar{\vartheta}; \zeta') > 0$. Case (iii) is case (i) for an increase and is computed as case (ii) for a decrease. Conclude with (d).

(f) For $k \neq k'$ the k -term of $A(\bar{\vartheta}; \cdot)$ is 1-Lipschitz in $|\zeta(k)|$, and the k -term of $B(\bar{\vartheta}; \cdot)$ is $2\bar{\vartheta}$ -Lipschitz in $|\zeta(k)|$ (since $|\min(x, a)^2 - \min(x, b)^2| \leq 2x|a - b|$ for $a, b \geq 0$). Write $a := |\zeta|$. (i') $A(\bar{\vartheta}; \zeta') \geq a + \Delta - E > 0$ and $B(\bar{\vartheta}; \zeta') \leq a^2 + 2\bar{\vartheta}E$, so $\Xi(\bar{\vartheta}; \zeta') \geq 2a(\Delta - E) - 2\bar{\vartheta}E > 0$ by the hypothesis on E . (ii') The k' -term of $A(\bar{\vartheta}; \cdot)$ vanishes before and after. Since $v'_{k'} < v_{k'} \leq \bar{\vartheta}$, the k' -term of B drops by $\Phi(k')^2(v_{k'}^2 - v'^2_{k'}) \geq \Phi(k')^2 v_{k'}(v_{k'} - v'_{k'}) = v_{k'}\Delta$. Hence, as $A(\bar{\vartheta}; \zeta') \geq (a - E)_+^2 \geq a^2 - 2aE$, $\Xi(\bar{\vartheta}; \zeta') \geq v_{k'}\Delta - 2(a + \bar{\vartheta})E > 0$. Conclude with (d). $\square$

Thus moving block mass toward the peaks (more at a peak, less at a strict non-peak) raises the threshold, and the reverse moves lower it; at a degenerate peak Ξ has a concave kink and both moves raise it. The following quantitative version is used for the threshold buffers of companions.

Lemma 2.7 (Lipschitz bound and quantitative raising). *Let $\zeta \in \ell_1 \setminus \{0\}$, $\bar{\vartheta} := \bar{\vartheta}(\zeta)$, $a := |\zeta|$, $\phi := \|\Phi\|_2^2$; let k_0 be a peak with $v_{k_0} > \bar{\vartheta}$ (it exists, since $\|\alpha\|_1 = 1$), and put $\phi_0 := \Phi(k_0)^2$, $h_0 := \min(1, v_{k_0} - \bar{\vartheta})$, $C_1 := (1 + 2(\bar{\vartheta} + 1)/a)/\phi_0$.*

- (a) If $\zeta' \in \ell_1$ and $E := \|\zeta' - \zeta\|_1$ satisfies $C_1 E < \min(h_0, \bar{\vartheta})$, then $\zeta' \neq 0$ and $|\bar{\vartheta}(\zeta') - \bar{\vartheta}| \leq C_1 E$. If only $C_1 E < h_0$, then $\bar{\vartheta}(\zeta') \leq \bar{\vartheta} + C_1 E$.
- (b) Let c be a peak (degenerate or not), $s > 0$, and let $\zeta' \in \ell_1$ satisfy $|\zeta'(c)| = |\zeta(c)| + s$, $\text{sign } \zeta'(c) = \text{sign } \zeta(c)$ and $E := \sum_{k \neq c} |\zeta'(k) - \zeta(k)| \leq as/(8(a + \bar{\vartheta} + 1))$. Then

$$\bar{\vartheta}(\zeta') - \bar{\vartheta} \geq \min \left\{ 1, \frac{as}{8\phi(a + \bar{\vartheta} + 1)} \right\},$$

and $\bar{\vartheta}(\zeta') - \bar{\vartheta} \leq C_1(s + E)$ if $C_1(s + E) < h_0$.

Proof. For $h \geq 0$ and every ζ'' : $A(x; \cdot)$ changes by at most E'' and $B(x; \cdot)$ by at most $2xE''$ when ζ'' moves by E'' in ℓ_1 (as in the proof of Lemma 2.6(f)); $B(\cdot; \zeta)$ is nondecreasing; and, by Lemma 2.6(b), $A(\bar{\vartheta}; \zeta) = a$, $B(\bar{\vartheta}; \zeta) = a^2$.

(a) $\zeta' \neq 0$ because $E < \phi_0 h_0 \leq |\zeta(k_0)|$. *Upper bound.* Let $h := C_1 E + \varepsilon \leq h_0$ with $\varepsilon > 0$. Since $v_{k_0} \geq \bar{\vartheta} + h$, the terms of $A(\bar{\vartheta} + h; \zeta)$ with $v_k \geq \bar{\vartheta} + h$ are those of $A(\bar{\vartheta}; \zeta)$ minus $h\Phi(k)^2$, and the other terms vanish; so $A(\bar{\vartheta} + h; \zeta) \leq a - h\phi_0$. Put $y := h\phi_0 - E$; then $0 < y \leq a$ (as $C_1\phi_0 > 1$, and $h\phi_0 \leq \Phi(k_0)^2(v_{k_0} - \bar{\vartheta})$, which is one term of $A(\bar{\vartheta}; \zeta)$). Thus $0 \leq A(\bar{\vartheta} + h; \zeta') \leq a - y$ and $B(\bar{\vartheta} + h; \zeta') \geq a^2 - 2(\bar{\vartheta} + 1)E$, whence

$$\Xi(\bar{\vartheta} + h; \zeta') \leq (a - y)^2 - a^2 + 2(\bar{\vartheta} + 1)E \leq -ya + 2(\bar{\vartheta} + 1)E < 0,$$

because $ya > (C_1\phi_0 - 1)Ea = 2(\bar{\vartheta} + 1)E$. So $\bar{\vartheta}(\zeta') < \bar{\vartheta} + h$ by Lemma 2.6(a). *Lower bound* (when $C_1 E < \bar{\vartheta}$). Let $h := C_1 E + \varepsilon < \bar{\vartheta}$. The terms of $A(\bar{\vartheta} - h; \zeta)$ with $v_k \geq \bar{\vartheta}$ exceed those of $A(\bar{\vartheta}; \zeta)$ by $h\Phi(k)^2$, so $A(\bar{\vartheta} - h; \zeta') \geq a + h\phi_0 - E \geq 0$, and $B(\bar{\vartheta} - h; \zeta') \leq a^2 + 2\bar{\vartheta}E$. Hence $\Xi(\bar{\vartheta} - h; \zeta') \geq 2a(h\phi_0 - E) - 2\bar{\vartheta}E > 0$, since $h\phi_0 > E(1 + \bar{\vartheta}/a)$. So $\bar{\vartheta}(\zeta') > \bar{\vartheta} - h$. Let $\varepsilon \rightarrow 0$.

(b) *Lower bound.* Let $h := \min\{1, as/(8\phi(a + \bar{\vartheta} + 1))\}$; then $h \leq s/\phi \leq s/\Phi(c)^2$ and $h\phi \leq s/8$, and $E \leq s/8$. Since $v_c(\zeta') = v_c + s/\Phi(c)^2 \geq \bar{\vartheta} + h$, the c -term of $A(\bar{\vartheta} + h; \zeta')$ is the c -term of $A(\bar{\vartheta}; \zeta)$ plus $s - h\Phi(c)^2$, while every other term of $A(\bar{\vartheta} + h; \zeta)$ is at least the corresponding term of $A(\bar{\vartheta}; \zeta)$ minus $h\Phi(k)^2$; so $A(\bar{\vartheta} + h; \zeta') \geq a + s - h\phi - E$. The c -term of $B(\bar{\vartheta} + h; \zeta')$ is $(\bar{\vartheta} + h)^2\Phi(c)^2$, that of $B(\bar{\vartheta}; \zeta)$ is $\bar{\vartheta}^2\Phi(c)^2$, and $\min(\bar{\vartheta} + h, x)^2 \leq \min(\bar{\vartheta}, x)^2 + 2\bar{\vartheta}h + h^2$; with $h \leq 1$ this gives $B(\bar{\vartheta} + h; \zeta') \leq a^2 + (2\bar{\vartheta} + 1)h\phi + 2(\bar{\vartheta} + 1)E$. As $s - h\phi - E \geq 3s/4 > 0$,

$$\Xi(\bar{\vartheta} + h; \zeta') \geq 2as - 2(a + \bar{\vartheta} + 1)E - 2(a + \bar{\vartheta} + 1)h\phi \geq 2as - \frac{1}{4}as - \frac{1}{4}as > 0,$$

so $\bar{\vartheta}(\zeta') > \bar{\vartheta} + h$. *Upper bound:* part (a) (upper bound) with $\|\zeta' - \zeta\|_1 = s + E$. $\square$

Corollary 2.8 (Common rescaling under a raise). *In the situation of Lemma 2.7(b), for every $k \neq c$,*

$$\varrho_k(\zeta') = \frac{|\zeta'(k)|}{|\zeta(k)|} \varrho_k(\zeta) \frac{\bar{\vartheta}(\zeta)}{\bar{\vartheta}(\zeta')} \quad (\zeta(k) \neq 0);$$

so every coordinate k with $\zeta'(k) = \zeta(k)$ has $\varrho_k(\zeta') = \varrho_k(\zeta)/(1 + \delta)$ with $\delta := \bar{\vartheta}(\zeta')/\bar{\vartheta}(\zeta) - 1 \geq c_{\text{raise}} s$ for $s \leq 8\phi(a + \bar{\vartheta} + 1)/a$, where $c_{\text{raise}} := a/(8\phi\bar{\vartheta}(a + \bar{\vartheta} + 1))$. Moreover, for a first row f , a block m and a companion $f^\#$ of f such that $\hat{\zeta}_m^\# := R_m^{**}\hat{z}^\#$ and $\hat{\zeta}_m$ agree on a set $G \subseteq Q_m \cap Q_m^\#$, every finitely supported ω with $\text{supp } \omega \subseteq G$ satisfies

$$d_m^\#(\omega) = \frac{|\hat{\zeta}_m|_m}{|\hat{\zeta}_m^\#|_m} d_m(\omega) :$$

all d -coefficients of untouched strict non-peaks are multiplied by one common positive factor, and d -neutrality of a switching supported in G is preserved.

Proof. $\varrho_k = |\zeta(k)|/(\Phi(k)^2\bar{\vartheta})$, and Lemma 2.7(b) for the bound on δ . The last claim is Lemma 2.2(a) at f and at $f^\#$: $d_m^\#(\omega) = \langle \omega, \hat{\zeta}_m^\# \rangle / |\hat{\zeta}_m^\#|_m$ and $\langle \omega, \hat{\zeta}_m^\# \rangle = \langle \omega, \hat{\zeta}_m \rangle$. $\square$

The block scalar $\varkappa$. For $\zeta \in \ell_1 \setminus \{0\}$ and $\Omega \subseteq Q(\zeta)$ put

$$\varkappa(\zeta; \Omega) := \frac{|\zeta|(|\zeta| + \bar{\vartheta}) - \sum_{k \in \Omega} \mathbf{v}_k^2 \Phi(k)^2}{\bar{\vartheta}} = |\zeta| + \bar{\vartheta} \Phi_P^2 + \frac{1}{\bar{\vartheta}} \sum_{k \in Q \setminus \Omega} \mathbf{v}_k^2 \Phi(k)^2,$$

with $\bar{\vartheta} = \bar{\vartheta}(\zeta)$, $\mathbf{v}_k = \mathbf{v}_k(\zeta)$; the two expressions agree by Lemma 2.6(c), and $\varkappa \geq |\zeta| > 0$.

Lemma 2.9 (The data identity). *Let $f' \in S_{p^*}$, $m \in I$, $\Omega_m \subseteq Q'_m$ finite, and let $\omega_m^\pm$ be supported in Ω_m . Put $\Delta\omega := \omega_m^- - \omega_m^+$, $\Delta_m := d'_m(\Delta\omega)$ and $\gamma_k := \lambda_{k,m}(\Delta\omega(k) - \Delta_m w'_m(k))$ for $k \in \Omega_m$. Then, with $\hat{\zeta}'_m := R_m^{**} \hat{z}'$,*

$$\Delta_m M'_m \varkappa(\hat{\zeta}'_m; \Omega_m) = \sum_{k \in \Omega_m} u_{k,m}(\hat{z}') \gamma_k.$$

Proof. Write $\zeta := \hat{\zeta}'_m$, $\mathbf{a} := |\zeta|$, $\bar{\vartheta} := \bar{\vartheta}(\zeta)$, and drop m and the primes. For $k \in Q$, [I, Lemma 1.7] gives $\Phi(k)^2 w(k)/C = \zeta(k)/\mathbf{a}$ and hence $\Phi(k)^2 w(k)^2/C = C \mathbf{v}_k^2 \Phi(k)^2/\mathbf{a}^2$. On Ω , $\Delta\omega(k) = \gamma_k/\lambda_k + \Delta w(k)$ and $\zeta(k)/\lambda_k = u_k(\hat{z})$. So

$$\Delta = d(\Delta\omega) = \sum_{k \in \Omega} \frac{\zeta(k)}{\mathbf{a}} \left(\frac{\gamma_k}{\lambda_k} + \Delta w(k) \right) = \frac{1}{\mathbf{a}} \sum_{k \in \Omega} u_k(\hat{z}) \gamma_k + \Delta \frac{C}{\mathbf{a}^2} \sum_{k \in \Omega} \mathbf{v}_k^2 \Phi(k)^2.$$

Multiply by $\mathbf{a}^2/C = \mathbf{a}(\mathbf{a} + \bar{\vartheta})$ (Lemma 2.6(b)); since $\mathbf{a}/C = \mathbf{a} + \bar{\vartheta}$,

$$\Delta \left(\mathbf{a}(\mathbf{a} + \bar{\vartheta}) - \sum_{k \in \Omega} \mathbf{v}_k^2 \Phi(k)^2 \right) = (\mathbf{a} + \bar{\vartheta}) \sum_{k \in \Omega} u_k(\hat{z}) \gamma_k.$$

Divide by $\mathbf{a} + \bar{\vartheta}$ and use $1/(\mathbf{a} + \bar{\vartheta}) = M/\bar{\vartheta}$. $\square$

By [I, (1)], for every carrier (k, m) outside Ω_m the switching contributes the forced coefficient $-\Delta_m \lambda_{k,m} w'_m(k)$ (see Lemma 2.22(a) below), and the lemma shows that the only f' -dependent quantities of the block entering the d -row of an exact switching through Ω_m are the values $u_{k,m}(\hat{z}')$ ($k \in \Omega_m$), M'_m and the single scalar $\varkappa(\hat{\zeta}'_m; \Omega_m)$.

Lemma 2.10 (Continuity and derivatives of $\varkappa$). *Let $\zeta \in \ell_1 \setminus \{0\}$ and $\Omega \subseteq Q(\zeta)$ finite.*

- (a) $\varkappa(\cdot; \Omega)$ is continuous at ζ on the set of $\zeta'' \neq 0$ with $\Omega \subseteq Q(\zeta'')$; in particular it does not jump when a coordinate outside Ω crosses the threshold.
- (b) Let c be a coordinate and ζ_s the vector obtained from ζ by replacing $|\zeta(c)|$ by $|\zeta(c)| + s$ (same sign), for s in an open interval $\mathcal{I} \ni 0$ on which $P(\zeta_s)$ is constant and $\Omega \subseteq Q(\zeta_s)$. Put $X := C \sum_{k \in Q \setminus \Omega} \mathbf{v}_k^2 \Phi(k)^2 / (\bar{\vartheta}^2 \Phi_P^2) \geq 0$. Then $\bar{\vartheta}$, $|\zeta_s|$ and $\varkappa$ are differentiable on $\mathcal{I}$, and at $s = 0$:
 - (i) if $c \in P$: $d\bar{\vartheta}/ds = C/\Phi_P^2$, $d|\zeta_s|/ds = M$, $d\varkappa/ds = 1 - X$;
 - (ii) if $c \in Q \setminus \Omega$: $d\bar{\vartheta}/ds = -\varrho_c M/\Phi_P^2$, $d|\zeta_s|/ds = \varrho_c M$, $d\varkappa/ds = \varrho_c(2 + MX/C)$;
 - (iii) if $c \in \Omega$: $d\varkappa/ds = \varrho_c MX/C$.

Proof. (a) $|\cdot|$ is a norm, $\mathbf{v}_k$ ($k \in \Omega$) are continuous, and $\bar{\vartheta}$ is continuous by Lemma 2.7(a); use the first expression of $\varkappa$, which does not involve P . (In the second expression a coordinate with $\mathbf{v}_k = \bar{\vartheta}$ contributes $\bar{\vartheta}\Phi(k)^2$ both as a peak and as a strict non-peak.) (b) With P fixed, Lemma 2.6(c) reads $\sum_P |\zeta_s(k)| - \bar{\vartheta}\Phi_P^2 = |\zeta_s|$ and $|\zeta_s|^2 = \bar{\vartheta}^2\Phi_P^2 + S_Q(s)$, $S_Q(s) := \sum_{k \in Q} \mathbf{v}_k(\zeta_s)^2\Phi(k)^2$; eliminating $|\zeta_s|$ gives a quadratic equation for $\bar{\vartheta}$ with coefficients affine or quadratic in s and with nonzero derivative in $\bar{\vartheta}$ at the root, so the implicit function theorem gives differentiability. (i) Differentiating, $ds - \Phi_P^2 d\bar{\vartheta} = d|\zeta|$ and $|\zeta|d|\zeta| = \bar{\vartheta}\Phi_P^2 d\bar{\vartheta}$; hence $d\bar{\vartheta} = |\zeta|ds/(\Phi_P^2(|\zeta| + \bar{\vartheta})) = Cds/\Phi_P^2$ and $d|\zeta| = Mds$; with the second expression of $\varkappa$ (the sum over $Q \setminus \Omega$ unchanged), $d\varkappa = d|\zeta| + \Phi_P^2 d\bar{\vartheta} - (S_{Q \setminus \Omega}/\bar{\vartheta}^2) d\bar{\vartheta} = Mds + Cds - XC^{-1}\Phi_P^2 \cdot Cds/\Phi_P^2 = (1 - X)ds$, where $S_{Q \setminus \Omega} := \sum_{k \in Q \setminus \Omega} \mathbf{v}_k^2\Phi(k)^2 = X\bar{\vartheta}^2\Phi_P^2/C$. (ii) Now $-\Phi_P^2 d\bar{\vartheta} = d|\zeta|$ and $|\zeta|d|\zeta| = \bar{\vartheta}\Phi_P^2 d\bar{\vartheta} + \mathbf{v}_c ds$ (as $d(\mathbf{v}_c^2\Phi(c)^2) = 2\mathbf{v}_c ds$); hence $d\bar{\vartheta} = -\mathbf{v}_c ds/(\Phi_P^2(|\zeta| + \bar{\vartheta})) = -\varrho_c Mds/\Phi_P^2$, $d|\zeta| = \varrho_c Mds$, and $d\varkappa = d|\zeta| + \Phi_P^2 d\bar{\vartheta} + 2\mathbf{v}_c ds/\bar{\vartheta} - (S_{Q \setminus \Omega}/\bar{\vartheta}^2) d\bar{\vartheta} = \varrho_c(2 + MX/C)ds$. (iii) As in (ii), but the sum over $Q \setminus \Omega$ does not change. $\square$

Corollary 2.11 (Relative positions as functions of ratios). *Let $\zeta \in \ell_1 \setminus \{0\}$, $\Omega \subseteq Q(\zeta)$ finite, $\varkappa := \varkappa(\zeta; \Omega)$, and put $\mathbf{r}_k := |\zeta(k)|/(\Phi(k)\varkappa)$ for $k \in \Omega$, $R^2 := \sum_{k \in \Omega} \mathbf{r}_k^2$, $a := |\zeta|/\bar{\vartheta}$, $\mathbf{k} := \varkappa/\bar{\vartheta}$ and $s_Q := \bar{\vartheta}^{-2} \sum_{k \in Q \setminus \Omega} \mathbf{v}_k^2\Phi(k)^2$. Then*

$$a^2 = \Phi_P^2 + \mathbf{k}^2 R^2 + s_Q, \quad \mathbf{k} = a + \Phi_P^2 + s_Q, \quad \varrho_k = \mathbf{r}_k \mathbf{k} / \Phi(k) \quad (k \in \Omega),$$

and $R < 1$, $a > \mathbf{k}R$. In particular, for fixed (Φ_P^2, s_Q) the relative positions of the coordinates of Ω are functions of the ratios $(\mathbf{r}_k)_{k \in \Omega}$ alone.

Proof. Divide the second identity of Lemma 2.6(c) by $\bar{\vartheta}^2$, using $\sum_{k \in \Omega} \mathbf{v}_k^2\Phi(k)^2 = \varkappa^2 R^2$; divide the second expression of $\varkappa$ by $\bar{\vartheta}$; and $\varrho_k = \mathbf{v}_k/\bar{\vartheta}$ equals $(|\zeta(k)|/(\Phi(k)\varkappa)) \cdot (\varkappa/\bar{\vartheta})/\Phi(k)$. As $P \neq \emptyset$, $\mathbf{k} > a$; if $R \geq 1$ then $a^2 \geq \mathbf{k}^2 R^2 \geq \mathbf{k}^2 > a^2$, which is absurd; and $a^2 - \mathbf{k}^2 R^2 = \Phi_P^2 + s_Q > 0$.

For the last assertion put $c_0 := \Phi_P^2 + s_Q > 0$, so that $\mathbf{k} = a + c_0$ and a is a root of $G(a) := a^2 - (a + c_0)^2 R^2 - c_0$. The condition $a > \mathbf{k}R = (a + c_0)R$ is equivalent (as $R < 1$) to $a > c_0 R/(1 - R)$. On the half-line $\{a > c_0 R/(1 - R)\}$ we have $G'(a) = 2(a - (a + c_0)R^2) = 2(a - \mathbf{k}R^2) > 0$, because $a > \mathbf{k}R \geq \mathbf{k}R^2$. Hence G is strictly increasing on this half-line and has at most one root there. Since the true value of a lies on it, a is determined by (R, Φ_P^2, s_Q) , and hence so are $\mathbf{k} = a + c_0$ and $\varrho_k = \mathbf{r}_k \mathbf{k} / \Phi(k)$ for $k \in \Omega$ (with $\Phi(k)$ fixed). Compare Proposition 5.46(i) for the smooth dependence. $\square$

For the block vector $\hat{\zeta}_m = R_m^{**} \hat{z}$ one has $|\zeta(k)|/\Phi(k) = m|u_{k,m}(\hat{z})|$, so $\mathbf{r}_k = m|u_{k,m}(\hat{z})|/\varkappa$. A push of a peak (Lemma 2.10(b)(i)) with the values on Ω fixed rescales all ratios of the block by one factor; if the ratios are restored afterwards, all relative positions in Ω return to their old values up to the change of Φ_P^2 and s_Q . Exactness of shifted data (through Lemma 2.9) and the statuses of the switching carriers therefore cannot be adjusted independently by peak pushes alone.

2.3. The cost of moving the normer; raise transfer. For a first row f' write $\delta := \hat{z}' - \hat{z} \in \ell_\infty$ and

$$\Delta_m(\delta) := \sum_k \lambda_{k,m} |u_{k,m}(\delta)| = \|R_m^{**} \delta\|_1,$$

$$c(\delta) := \sum_{m \in I} \left[\Delta_m \log \frac{e}{\Delta_m} + \sum_k \min(\lambda_{k,m}, |u_{k,m}(\delta)|) \right],$$

with $0 \log(e/0) := 0$. We use the elementary counting bound

$$(1) \quad \sum_k \min(2\lambda_{k,m}, x) \leq x (\log_2(2m/x) + 3) \quad (0 < x \leq 2m),$$

which holds because $2\lambda_{k,m} \leq 2m2^{-m-k}$: at most $\log_2(2m/x) + 1$ indices have $2m2^{-m-k} > x$, and the remaining terms form a geometric series bounded by $2x$.

Lemma 2.12 (Cost of moving the normer). *Let $f \in S_{p^*}$. There are $c_f > 0$ and $C_f < \infty$, depending only on f (and on I, T), such that for every $f' \in S_{p^*}$ with $\max_m \Delta_m(\delta) \leq c_f$, $\delta := \hat{z}' - \hat{z}$:*

- (i) $|C'_m - C_m| + |\sigma'_m - \sigma_m| + |q'_0 - q_0| \leq C_f \max_{m'} \Delta_{m'}(\delta)$ for every m ;
- (ii) $\sum_m \|R_m^*(w'_m - w_m)\|_1 \leq C_f c(\delta)$;
- (iii) $\|D_m(w'_m - w_m)\|_2^2 \leq C_f c(\delta)^2$ for every m ;
- (iv) $p^*(L^*(w' - w)) \leq (1 + \|U\|)C_f c(\delta)$ and $p^*(f' - f) \leq (1 + \|U\|)(\|a' - a\|_1 + C_f c(\delta))$.

In particular, for a companion $f^\#$ of f ($a^\# = a$), $p^*(f^\# - f) \leq (1 + \|U\|)C_f c(\delta)$.

Proof. Fix m , drop it, and write $A := |\hat{\zeta}|$, $A' := |\hat{\zeta}'|$ with $\hat{\zeta} = R^{**}\hat{z}$, $\hat{\zeta}' = R^{**}\hat{z}'$; $\Delta := \Delta_m(\delta) = \|\hat{\zeta}' - \hat{\zeta}\|_1$. Recall $|u_k(\hat{z})| \leq q^*(u_k)q^{**}(\hat{z}) = 1$, and $|A' - A| \leq \Delta$ since $|\cdot| \leq \|\cdot\|_1$ [I, Remark 1.1(c)].

Scalars. $1/q_0 = 1 + \sum_m A_m$, so $|1/q'_0 - 1/q_0| \leq \sum_m \Delta_m$ and $|q'_0 - q_0| \leq |I| \max_m \Delta_m$; $\sigma = q_0 A$ and $A' \leq \|\hat{\zeta}'\|_1 \leq \sum_k \lambda_k \leq 1$ give the bound for σ . By Lemma 2.6(b) (applied to $\hat{\zeta}$; M, C are scale invariant), $C = A/(\bar{\vartheta}(\hat{\zeta}) + A)$, and by Lemma 2.7(a) $|\bar{\vartheta}(\hat{\zeta}') - \bar{\vartheta}(\hat{\zeta})| \leq C_1 \Delta$ as soon as $C_1 \Delta < \min(h_0, \bar{\vartheta}(\hat{\zeta}))$ (constants of that lemma for $\hat{\zeta}$). Hence $|C' - C| \leq c_3 \Delta$ with c_3 depending only on f , once Δ is below a constant depending only on f . This is (i).

Clamp formula. By Lemma 2.6(b), $w(k) = \text{sign}(\hat{\zeta}(k)) \min(M, C|\hat{\zeta}(k)|/(A\Phi(k)^2))$ (use $M/\bar{\vartheta} = C/A$), and $|\hat{\zeta}(k)| = m\Phi(k)|u_k(\hat{z})|$; so with $v_k := m|u_k(\hat{z})|/A$,

$$\Phi(k)w(k) = \text{sign}(u_k(\hat{z})) \min(\Phi(k)M, Cv_k),$$

and the same at f' with $v'_k := m|u_k(\hat{z}')|/A'$. For $\Delta \leq A/2$, $|v'_k - v_k| \leq c_1(|u_k(\delta)| + \Delta)$ with $c_1 := 2m(1 + 1/A)/A$. If $u_k(\hat{z})$ and $u_k(\hat{z}')$ have the same sign or one of them vanishes, then $\Phi(k)|w'(k) - w(k)| \leq |C' - C|(\Phi(k) + v_k) + C'|v'_k - v_k|$ (note $M' - M = C - C'$); if they have opposite signs, then $|u_k(\hat{z})|, |u_k(\hat{z}')| \leq |u_k(\delta)|$ and $\Phi(k)(|w'(k)| + |w(k)|) \leq C'v'_k + Cv_k \leq 3m|u_k(\delta)|/A$. Since $v_k \leq m/A$, $|w|, |w'| \leq 1$ and $\lambda_k = m\Phi(k)$, in both cases, with a constant c_5 depending only on f ,

$$\lambda_k |w'(k) - w(k)| \leq \min(2\lambda_k, m\Phi(k)|C' - C| + c_5\Delta + c_5|u_k(\delta)|)$$

$$\leq m\Phi(k)|C' - C| + \min(2\lambda_k, c_5\Delta) + \min(2\lambda_k, c_5|u_k(\delta)|).$$

(ii). Sum over k , using $\sum_k \Phi(k) \leq 1$, (i), (1) with $x = c_5\Delta$ (for $c_5\Delta \leq 2m$), and $\min(2\lambda, c_5y) \leq \max(2, c_5) \min(\lambda, y)$; and $\|R^*x\|_1 \leq \sum_k \lambda_k |x(k)|$ because $\|u_k\|_1 \leq q^*(u_k) = 1$.

(iii). $\|D(w' - w)\|_2^2 \leq \max_k (\Phi(k)|w'(k) - w(k)|) \sum_k \Phi(k)|w'(k) - w(k)|$. The second factor is $\frac{1}{m} \sum_k \lambda_k |w'(k) - w(k)| \leq Cc(\delta)$ by (ii). By the pointwise bounds, the first factor is at most $c_6\Delta + \min(2\Phi(k), c_6|u_k(\delta)|) \leq c_7c(\delta)$, since $\Delta \leq \Delta \log(e/\Delta)$ for $\Delta \leq 1$ and $\min(2\Phi(k), c_6y) \leq \max(2/m, c_6) \min(\lambda_k, y)$.

(iv). $p^* \leq q^* \leq (1 + \|U\|) \cdot \| \cdot \|_1$ on X^* (by [I, Lemma 1.6], $B_{q^*} \subseteq B_{p^*}$), $L^*(w' - w) = \sum_m R_m^*(w'_m - w_m)$, and $f' - f = (a' - a) + L^*(w' - w)$. Finally choose c_f so small that all the smallness conditions used above hold for $\max_m \Delta_m \leq c_f$. $\square$

Remark 2.13. (a) Lemma 2.12 is an upper bound. Its unweighted term, the sum of the numbers $\min(\lambda_{k,m}, |u_{k,m}(\delta)|)$, cannot in general be replaced by a λ -weighted one: at a coordinate k that is a strict non-peak of f and f' , the clamp formula gives

$$\lambda_{k,m}(w'_m(k) - w_m(k)) = m^2(C'_m u_{k,m}(\hat{z}')/A'_m - C_m u_{k,m}(\hat{z})/A_m)$$

($A_m := |R_m^{**}\hat{z}|_m$), which is linear in the unweighted quantity $u_{k,m}(\delta)$ with coefficient $m^2 C_m/A_m$ when the scalars do not move; the changes of the scalars are $O(\Delta_m)$, i.e. of weighted size. That the unweighted term is attained is supported by numerical evidence in our sources (not used). (b) The lemma uses f' only through $\hat{z}'$: w' and q_0' depend on $\hat{z}'$ alone. It applies to companions (where $\delta = z^\# - z$) and to raise companions (where $\delta = U(e^\# - e)$, Lemma 2.14).

Lemma 2.14 (Raise companions). *Let $f \in S_{p^*}$, let Δa be a raise, $\lambda := q^*(a + \Delta a)$, and let $f^\# = f[a^\#, z]$ be the raise companion. Then:*

- (a) $1 + \|\Delta a\|_1 - \|U^* \Delta a\|_H \leq \lambda \leq 1 + \|\Delta a\|_1 + \|U^* \Delta a\|_H$;
- (b) $\|e^\# - e\|_H \leq 2\|U^* \Delta a\|_H / \nu$;
- (c) $\hat{z}^\# - \hat{z} = U(e^\# - e)$; hence $|u(\hat{z}^\#) - u(\hat{z})| \leq \|U^* u\|_H \|e^\# - e\|_H$ for $u \in \ell_1$, and $\Delta_m(\hat{z}^\# - \hat{z}) \leq m2^{-m}\|e^\# - e\|_H$;
- (d) if $\lambda \geq 1$, then $\|a^\# - a\|_1 \leq 2\|\Delta a\|_1 + \|U^* \Delta a\|_H$; and $f^\# \rightarrow f$ in norm as $\|\Delta a\|_1 \rightarrow 0$;
- (e) put $\varepsilon_e := \|e^\# - e\|_H$ and assume $\varepsilon_e \leq 1$ (true when $\|U^* \Delta a\|_H \leq \nu/2$, by (b)); then one has $c(\hat{z}^\# - \hat{z}) \leq C_I \varepsilon_e \log(e/\varepsilon_e)$, C_I depending only on I ; so, if $\max_m \Delta_m(\hat{z}^\# - \hat{z}) \leq c_f$, Lemma 2.12 gives $p^*(L^*(w^\# - w)) \leq C'_f \varepsilon_e \log(e/\varepsilon_e)$.

Proof. (a) On F the signs of a and Δa agree, so $\|a + \Delta a\|_1 = \|a\|_1 + \|\Delta a\|_1$; and $\|U^* a\| - \|U^* \Delta a\| \leq \|U^*(a + \Delta a)\| \leq \|U^* a\| + \|U^* \Delta a\|$; use $q^*(a) = \|a\|_1 + \|U^* a\| = 1$. (b) $e^\#$ is the normalization of $U^*(a + \Delta a)$, and $\|x/\|x\| - y/\|y\|\| \leq 2\|x - y\|/\|y\|$ with $y = U^* a$, $\|y\| = \nu$. (c) z is unchanged, so $\hat{z}^\# - \hat{z} = U(e^\# - e)$ and $u(Uh) = \langle U^* u, h \rangle$; $\|U^* u_{k,m}\| \leq q^*(u_{k,m}) = 1$ and $\sum_k \lambda_{k,m} \leq m2^{-m}$. (d) $a^\# - a = \Delta a/\lambda + (1/\lambda - 1)a$, $\|a\|_1 \leq 1$ and $0 \leq 1 - 1/\lambda \leq \lambda - 1 \leq \|\Delta a\|_1 + \|U^* \Delta a\|$ by (a). If $\|\Delta a\|_1 \rightarrow 0$ then $a^\# \rightarrow a$ in ℓ_1 and z is fixed, so $f^\# \rightarrow f$ [I, Remark 8.22(c)]. (e) By (c), $|u_{k,m}(\delta)| \leq \varepsilon_e$ and $\Delta_m \leq m2^{-m}\varepsilon_e$; since $x \log(e/x)$ is increasing on $(0, 1]$ and by (1), $c(\delta) \leq \sum_m [m2^{-m}\varepsilon_e \log(e2^m/(m\varepsilon_e)) + \varepsilon_e(\log_2(2m/\varepsilon_e) + 3)] \leq C_I \varepsilon_e \log(e/\varepsilon_e)$. $\square$

Lemma 2.15 (Raise transfer). *In the situation of Lemma 2.14 assume $\lambda \geq 1$. Then for every $h \in X^*$ and $r \in \mathbb{R}$*

$$p^*(f^\# + rh) \leq 1 + \frac{p^*(f + \lambda rh) - 1 + 2\|U^* \Delta a\|_H}{\lambda} + p^*(L^*(w^\# - w)).$$

Proof. For $r = 0$ the right side is at least $1 = p^*(f^\#)$. Let $r \neq 0$. By [I, Lemma 1.6], $f + \lambda rh = A + L^*W$ with $\max(q^*(A), \|W\|_{V^*}) = \pi := p^*(f + \lambda rh)$. Since $f = a + L^*w$, $\lambda rh = (A - a) + L^*(W - w)$, and

$$f^\# + rh = \frac{A + \Delta a}{\lambda} + L^*\left(\left(1 - \frac{1}{\lambda}\right)w + \frac{1}{\lambda}W\right) + L^*(w^\# - w).$$

Base. $q^*(A + \Delta a) \leq \pi + \|\Delta a\|_1 + \|U^*\Delta a\| \leq \pi + (\lambda - 1) + 2\|U^*\Delta a\|$ by Lemma 2.14(a), so $q^*((A + \Delta a)/\lambda) \leq 1 + (\pi - 1 + 2\|U^*\Delta a\|)/\lambda$. *Blocks.* As $\lambda \geq 1$, $N_m((1 - 1/\lambda)w_m + W_m/\lambda) \leq (1 - 1/\lambda) + \pi/\lambda = 1 + (\pi - 1)/\lambda$. By [I, Lemma 1.6] the p^* -norm of the first two terms is at most the maximum of these bounds; add $p^*(L^*(w^\# - w))$. $\square$

No smallness of $\|\Delta a\|_1$ is needed: the ℓ_1 -mass of a same-sign raise on F cancels against the normalization $\lambda - 1$, and only the Hilbert footprint $\|U^*\Delta a\|$ and the normer change remain.

Remark 2.16 (Raise with corrections). Suppose that, in addition to the raise Δa , corrections $\Delta a''$ of arbitrary sign are made on a finite set $F_0 \subseteq F \setminus \text{supp } \Delta a$ with $|\Delta a''_j| \leq |a_j|/2$ for $j \in F_0$. Put $D := \Delta a + \Delta a''$, assume $\lambda := q^*(a + D) \geq 1$, and let $f^\# := f[(a + D)/\lambda, z]$ with normer $w^\#$. The condition $|\Delta a''_j| \leq |a_j|/2$ keeps $\text{supp}(a + D) = F$ and the signs of a on F , so $f^\#$ is well defined (Subsection 2.1). Then the proof of Lemma 2.15 gives, for every $h \in X^*$ and $r \in \mathbb{R}$,

$$p^*(f^\# + rh) \leq 1 + \frac{p^*(f + \lambda rh) - 1 + 2\|U^*D\|_H + 2\|\Delta a''\|_1}{\lambda} + p^*(L^*(w^\# - w)).$$

Indeed, $\|a + D\|_1 \geq \|a\|_1 + \|\Delta a\|_1 - \|\Delta a''\|_1$ and $\|U^*(a + D)\|_H \geq \|U^*a\|_H - \|U^*D\|_H$ give $\|\Delta a\|_1 \leq \lambda - 1 + \|\Delta a''\|_1 + \|U^*D\|_H$, hence $q^*(A + D) \leq \pi + \|D\|_1 + \|U^*D\|_H \leq \pi + (\lambda - 1) + 2\|\Delta a''\|_1 + 2\|U^*D\|_H$, and the block part of the proof is unchanged. The error is that of the whole change D ; compared with Lemma 2.15 the extra error is at most $2q^*(\Delta a'')/\lambda$, since $\|U^*D\|_H \leq \|U^*\Delta a\|_H + \|U^*\Delta a''\|_H$.

Corollary 2.17 (Mates under a raise). *In the situation of Lemma 2.15, let $g \in \mathcal{C}(f)$ and $\rho \in (0, 1]$. Then for all $r \in \mathbb{R}$*

$$p^*(f^\# + r\rho g) \leq 1 + \frac{s(\lambda\rho r) - 1}{\lambda} + \varepsilon', \quad \varepsilon' := 2\|U^*\Delta a\|_H + p^*(L^*(w^\# - w)),$$

and $(s(\lambda x) - 1)/\lambda \leq s(x) - 1 + (\lambda - 1) \min(\lambda^2 x^2/2, 1)$ for $x \in \mathbb{R}$.

Proof. Lemma 2.15 with $h = \rho g$ and $p^*(f + \lambda r\rho g) \leq s(\lambda\rho r)$ (as $g \in \mathcal{C}(f)$), and $2\|U^*\Delta a\|/\lambda \leq 2\|U^*\Delta a\|$. For the inequality put $\varphi(\mu) := (s(\mu x) - 1)/\mu$; with $u = \mu x$ and $s'(u) = u/s(u)$ one computes $\varphi'(\mu) = (s(u) - 1)/(\mu^2 s(u))$, and $0 \leq (s(u) - 1)/s(u) \leq \min(u^2/2, 1)$. For $\mu \in [1, \lambda]$ this gives $\varphi'(\mu) \leq \min(\lambda^2 x^2/2, 1)$; integrate over $[1, \lambda]$ and use $\varphi(1) = s(x) - 1$. $\square$

2.4. Uniform one-sided transfer along convergent first rows. For a first row f_j with finite base support $F_j := \text{supp } a_j$ we use the notions of [I, Definition 7.7] at f_j (side- $\pm$ admissible pairs, representation, $d_m^{(j)}, \Gamma_w^{(j)}$), with all data computed at f_j ; a superscript or subscript j refers to f_j . The following coordinate kinds are used for the block parts ω_m of a pair at f_j and a scale $t > 0$, given constants $A_2 \geq 1$ and $\gamma_B \in (0, 1]$:

- [1] $k \in Q_m^{(j)}$ and $|\omega_m(k)| \leq 3 \text{gap}_m^{(j)}(k)/t$;
- [2] $k \in Q_m^{(j)}$, $\text{gap}_m^{(j)}(k) \geq \gamma_B$ and $|\omega_m(k)| \leq A_2/t$;

- [3] *inward coordinates*: k is a degenerate peak or a strict non-peak of $w_{j,m}$, $w_{j,m}(k) \neq 0$, $|\omega_m(k)| \leq A_2/t$, and, with $\varsigma_k := \text{sign } w_{j,m}(k)$, $\varsigma_k \omega_m(k) \leq 0$ for a side+ pair, resp. $\varsigma_k \omega_m(k) \geq 0$ for a side− pair.

For pairs whose block parts live on coordinates of these kinds we put $d_m^{(j)}(\omega_m) := \langle D_m w_{j,m}, D_m \omega_m \rangle / C_{j,m}$, and the pair is said to represent the functional

$$b + \sum_m R_m^*(\omega_m - d_m^{(j)}(\omega_m) w_{j,m}),$$

as in [I, Definition 7.7]; since $\alpha_{j,m}$ vanishes at degenerate peaks and off $P_{j,m}$, [I, Lemma 3.2] applies: the block first-order terms vanish. Kinds [1] and [2] are the coordinates of [I, Lemma 8.42] (with the constant 2 replaced by 3 and without the lower bound $\text{gap} \geq t^2$, which is used there only as a convention of the window certificate); kind [3] allows arbitrarily small gaps, but only in the inward direction.

Lemma 2.18 (Uniform one-sided transfer expansion). *Let $f \in S_{p^*}$ have F finite, and let $f_j \rightarrow f$ in S_{p^*} with $F_j = \text{supp } a_j$ finite and $F \subseteq F_j$. For every $\varepsilon_{\text{tr}} > 0$ and $A_0 \geq 1$ there are $t_1 > 0$, $j_0 \in \mathbb{N}$ and $c_0 \in (0, \frac{1}{8}]$, depending only on f , the sequence (f_j) , ε_{tr} and A_0 , with the following property. Let $A_2 \geq 1$, $\gamma_B \in (0, 1]$, and put $c_b := c_0 \gamma_B / A_2$. Let $j \geq j_0$, $t \in (0, t_1]$, and let (b, ω) , $b \in \ell_1$, $\omega_m \in c_{00}$, satisfy:*

- (OS1) $b(\xi_j) = 0$, $t\|b\|_1 \leq A_0$ and $\Gamma_w^{(j)}(b, \omega) \leq 2$;
- (OS2) $b(i) = 0$ for $i \notin F_j \cup K_j$, $z_j(i)b(i) \geq 0$ for $i \in K_j$, and for every $i \in F_j \setminus F$ either $z_j(i)b(i) \geq 0$ or $t|b(i)| \leq |a_j(i)|$;
- (OS3) every $k \in \text{supp } \omega_m$ is of kind [1], [2] or [3] (side +) at f_j , with the given A_2, γ_B .

Then $G := b + \sum_m R_m^*(\omega_m - d_m^{(j)}(\omega_m) w_{j,m})$ satisfies

$$p^*(f_j + rG) \leq 1 + \frac{r^2}{2} (\Gamma_w^{(j)}(b, \omega) + \varepsilon_{\text{tr}}) \quad (0 < r \leq c_b t).$$

The same holds for side− pairs (in (OS2) and [3] the signs reversed) and $-c_b t \leq r < 0$.

The sequence may be constant: for $f_j = f$ and $F_j = F$ the lemma is [I, Lemma 8.42] with the enlarged coordinate kinds. The essential point is that t_1 and j_0 do not depend on A_2 and γ_B , and that c_b depends on them only through γ_B / A_2 . The coordinates of $F_j \setminus F$ are of two types: *banks*, at which the data have the sign of a_j , and *pulls*, at which the data are small compared with the mass $|a_j(i)|$; nothing is required of f at these coordinates.

Proof. We follow the proofs of [I, Lemmas 8.16 and 8.42] and record where f_j , A_2 and γ_B enter. We treat side +; side − is the same with $-r$, $-b$, $-\omega$.

Step 0: limits and transfer data. We may assume $\varepsilon_{\text{tr}} \leq 1$. Let $a_{\min} := \min_{i \in F} |a_i|$, $C_{\min} := \min_m C_m$, $M_{\min} := \min_m M_m$, and fix $\gamma_m \in (M_m/2, M_m)$ with $|w_m(k)| \neq \gamma_m$ for all k . By [I, Proposition 1.11] the forced data of f_j converge to those of f ; let j'_0 be such that for $j \geq j'_0$: $q_0^{(j)} \geq q_0/2$, $\sigma_{j,m} \geq \sigma_m/2$, $C_{j,m} \geq C_m/2$, $M_{j,m} - \gamma_m \geq (M_m - \gamma_m)/2$, $\nu_j \geq \nu/2$ and $\min_{i \in F} |a_j(i)| \geq a_{\min}/2$. Put $K_3 := 16/\min(q_0, \min_m \sigma_m)$ and $\eta_1 := \min(\frac{1}{4}, \varepsilon_{\text{tr}}/(20|I|K_3))$, and choose transfer data $(\gamma_m, k_{*,m}^s, \Lambda_m^s)$ of f in every block by [I, Lemma 7.4] with this η_1 . By [I, Lemma 7.5] there is $j_0 \geq j'_0$ such that for $j \geq j_0$ the transfer vectors $y_{j,m}^s$ built at f_j (with $L_{j,m} := \{|w_{j,m}| \geq \gamma_m\}$, which contains $P_{j,m}$, and $k_{*,m}^s$ a peak of $w_{j,m}$ of

value $M_{j,m}$) satisfy $e_{y,j,m}^\varsigma \geq \frac{1}{2}$, $\iota_{j,m}^\varsigma \leq 2\eta_1$, $\|D_m y_{j,m}^\varsigma\|_2 \leq K_y$ and $|Y_{j,m}^\varsigma|/q_0^{(j)} \leq K_Y$, where K_y, K_Y depend only on f and the γ_m ([I, Lemma 7.4(b),(d)]). Let $j \geq j_0$, $t \leq t_1$, $0 < r \leq c_b t$, and write $f_j + rG = A + \sum_m R_m^* W_m$ with $A := a_j + rb$ and $W_m := (1 - rd_m)w_{j,m} + r\omega_m$, $d_m := d_m^{(j)}(\omega_m)$. The first-order terms at f_j vanish: $\mathfrak{l}_b(A) = rb(\hat{z}_j) = 0$ by (OS1), and $\mathfrak{l}_m(W_m) = 0$ by [I, Lemma 3.2] at f_j . Put $A_3 := 3 + A_2$; then $t\|\omega_m\|_\infty \leq A_3$ (kind [1]: gap ≤ 1) and $|d_m| \leq \|D_m \omega_m\|_2 \leq \|\Phi_m\|_2 \|\omega_m\|_\infty \leq A_3/(2t)$.

Step 1: base. On F , $|rb(i)| \leq c_b t \|b\|_1 \leq c_b A_0 \leq a_{\min}/2 \leq |a_j(i)|$ if $c_b \leq a_{\min}/(2A_0)$: no flip. On $F_j \setminus F$: if $z_j(i)b(i) \geq 0$ then $a_j(i)$ and $rb(i)$ have the same sign $z_j(i) = \text{sign } a_j(i)$; otherwise $|rb(i)| \leq c_b t |b(i)| \leq c_b |a_j(i)| < |a_j(i)|$; in both cases no flip. Off F_j the summands $|rb(i)| - z_j(i)rb(i)$ vanish by (OS2). By [I, Lemma 7.1(b)] at f_j and the estimate of Ψ in [I, Lemma 3.3],

$$G_b(A) = \nu_j \Psi(rU^*b/\nu_j) \leq \hat{G}_b := \frac{r^2}{2} h^{(j)}(b)(1 + K_A c_b), \quad K_A := 4\|U\|A_0/\nu,$$

provided $c_b \|U\|A_0 \leq \nu/4$ (so that $\|rU^*b\|/\nu_j \leq \frac{1}{2}$).

Step 2: blocks. If $c_b \leq 1/A_3$ then $|rd_m| \leq \frac{1}{2}$. We claim $\|W_m\|_\infty = (1 - rd_m)M_{j,m}$. At a non-degenerate peak $\omega_m = 0$ and $|W_m(k)| = (1 - rd_m)M_{j,m}$; off $\text{supp } \omega_m$, $|W_m(k)| = (1 - rd_m)|w_{j,m}(k)|$. At kinds [1], [2], if $c_b \leq 1/6$ and $c_b \leq \gamma_B/(2A_2)$, then $|r\omega_m(k)| \leq \text{gap}_m^{(j)}(k)/2$ and $|W_m(k)| \leq (1 - rd_m)(M_{j,m} - \text{gap}_m^{(j)}(k)) + \text{gap}_m^{(j)}(k)/2 \leq (1 - rd_m)M_{j,m}$. At kind [3], $\varsigma_k W_m(k) = (1 - rd_m)|w_{j,m}(k)| + r\varsigma_k \omega_m(k) \leq (1 - rd_m)M_{j,m}$, and $\varsigma_k W_m(k) \geq -c_b A_2 \geq -(1 - rd_m)M_{j,m}$ if $c_b A_2 \leq M_{\min}/8$. This proves the claim. The proof of [I, Lemma 3.4(d)] uses its radius condition only to obtain this equality, and the condition $|rd_m| \leq C_{j,m}/(2M_{j,m})$, which holds if $c_b \leq C_{\min}/(2A_3)$. Hence

$$G_m^{(j)}(W_m) = N_m(W_m) - 1 \leq \hat{G}_m := \frac{r^2}{2} H_m^{(j)}(\omega_m)(1 + 2A_3 c_b / C_{\min}).$$

Consequently $\hat{\Gamma} := q_0^{(j)} \hat{G}_b + \sum_m \sigma_{j,m} \hat{G}_m \leq \frac{r^2}{2} \Gamma_w^{(j)}(b, \omega)(1 + (K_A + 2A_3/C_{\min})c_b)$.

Step 3: rebalancing. Since $h^{(j)}(b) \leq 2/q_0^{(j)}$ and $H_m^{(j)}(\omega_m) \leq 2/\sigma_{j,m}$ by (OS1), the numbers $\varepsilon_m := (\hat{G}_m - \hat{\Gamma})/e_{y,j,m}$ (with $y^\downarrow$ or $y^\uparrow$ according to the sign) satisfy $|\varepsilon_m| \leq K_3 r^2$. Now $\|W_m - w_{j,m}\|_\infty \leq r(\|\omega_m\|_\infty + |d_m|) \leq 2A_3 c_b$ and $\|D_m(W_m - w_{j,m})\|_2 \leq A_3 c_b$. If $2A_3 c_b \leq \min_m(M_m - \gamma_m)/16$, $A_3 c_b \leq C_{\min}/4$, $K_3 t_1^2(2 + \Lambda_m^\varsigma) \leq \gamma_m/2$, $K_3 t_1^2 \leq \min_m(M_m - \gamma_m)/8$ and $|I|K_3 K_Y t_1^2 \leq 1$, then [I, Lemma 7.2] at f_j gives [I, (9)] and $|\lambda| \leq 1$, and [I, Proposition 7.3] at f_j gives $p^*(f_j + rG) \leq 1 + \hat{\Gamma} + E$ with

$$E \leq 2|I|K_3 \eta_1 r^2 + 2|I|K_3 K_Y(1 + \|U\|)A_0 c_b r^2 + \frac{8K_3 K_y A_3 c_b r^2 + 2K_3^2 K_y^2 r^4}{C_{\min}},$$

because $|q^*(A) - 1| + q^*(A - a_j) \leq 2q^*(rb) \leq 2(1 + \|U\|)c_b A_0$.

Step 4: constants. Require in addition $2(K_A + 2A_3/C_{\min})c_b \leq \varepsilon_{\text{tr}}/5$, $4|I|K_3 K_Y(1 + \|U\|)A_0 c_b \leq \varepsilon_{\text{tr}}/5$, $16K_3 K_y A_3 c_b / C_{\min} \leq \varepsilon_{\text{tr}}/5$ and $4K_3^2 K_y^2 t_1^2 / C_{\min} \leq \varepsilon_{\text{tr}}/5$; with $4|I|K_3 \eta_1 \leq \varepsilon_{\text{tr}}/5$ and $\Gamma_w^{(j)} \leq 2$ this gives $\hat{\Gamma} + E \leq \frac{r^2}{2}(\Gamma_w^{(j)}(b, \omega) + \varepsilon_{\text{tr}})$. Every condition on c_b above has the form $c_b \leq \kappa_1$, $c_b \leq \kappa_1/A_3$, $c_b \leq \kappa_1/A_2$ or $c_b \leq \gamma_B/(2A_2)$ with $\kappa_1 > 0$ depending only on f , ε_{tr} , A_0 ; since $A_3 \leq 4A_2$ and $\gamma_B \leq 1 \leq A_2$, all of them hold for $c_b = c_0 \gamma_B/A_2$ with $c_0 > 0$ small enough. The conditions on t_1 and j_0 do not involve A_2 or γ_B . $\square$

2.5. Windowed recovery through nearby first rows. The mechanism of [I, Theorem 8.17] (windowed averaging) and of [I, Theorem 8.44] (averaged exact window data, then [I, Corollary 7.21]) does not require the window data to live at f itself: they may live at a nearby first row f_j , provided f_j is closer to f than the square of the *bottom* of the window. We first prove an abstract averaging theorem that also allows a relative factor λ_j in the comparison of f_j with f (this is what raise companions produce, Corollary 2.17), and then the version with exact two-piece data. For a window (T, n) , $T \in (0, 1]$, $n \in \mathbb{N}$, we write $\mathcal{S}(T, n) := \{T2^{1-i} : 1 \leq i \leq n\}$; its least element is $2T2^{-n}$.

Theorem 2.19 (Windowed recovery with a transfer error). *Let $f \in S_{p^*}$ (F arbitrary), $g \in \mathcal{C}(f)$, $\rho \in (0, 1)$ and $\eta_0 \in (0, 2]$ with $\rho^2(1 + \eta_0) \leq (1 + \rho^2)/2$. Suppose that there are first rows $f_j \in S_{p^*}$, numbers $\lambda_j \geq 1$, $\varepsilon'_j \geq 0$, $c_j \in (0, 1]$, and windows (T_j, n_j) with $T_j \in (0, 1]$, $n_j \in \mathbb{N}$, together with numbers $K_j \geq 0$, such that:*

- (a) $p^*(f_j + r\rho g) \leq 1 + (s(\lambda_j \rho r) - 1)/\lambda_j + \varepsilon'_j$ for all $r \in \mathbb{R}$;
- (b) for every $t \in \mathcal{S}_j := \mathcal{S}(T_j, n_j)$ there is $g_{j,t} \in X^*$ with $p^*(g - g_{j,t}) \leq K_j t$ and $p^*(f_j + r\rho g_{j,t}) \leq 1 + \frac{\rho^2 r^2}{2}(1 + \eta_0)$ for $0 < \rho|r| \leq c_j t$;
- (c) $K_j T_j \rightarrow 0$, $\lambda_j \rightarrow 1$, $\varepsilon'_j \rightarrow 0$, $p^*(f_j - f) \rightarrow 0$, and $n_j \geq 48\rho^2 K_j / (c_j(1 - \rho^2))$ for all large j ;
- (d) $\varepsilon'_j \leq \theta_* c_j^2 (T_j 2^{-n_j})^2$ for all large j , where $\theta_* := (1 - \rho^2)/(12\rho^2)$;
- (e) for every j : if $\rho \bar{g}_j \in \mathcal{C}(f_j)$, where $\bar{g}_j := \frac{1}{n_j} \sum_{t \in \mathcal{S}_j} g_{j,t}$, then $(f_j, \rho \bar{g}_j) \in \text{clNA}((c_0, p), \ell_2^2)$.

Then $(f, \rho g) \in \text{clNA}((c_0, p), \ell_2^2)$.

Proof. Put $r_0 := \sqrt{1 - \rho^2} \in (0, 1)$ and let j be so large that (c), (d) are in force and

$$(\lambda_j - 1)(1 + \lambda_j^2) \leq \frac{(1 - \rho^2)r_0^2}{24}, \quad \varepsilon'_j \leq \frac{(1 - \rho^2)r_0^2}{12}, \quad \frac{2\rho K_j T_j}{n_j} \leq \frac{(1 - \rho^2)r_0}{12}.$$

Drop the index j . For $t \in \mathcal{S}$ and $r \in \mathbb{R}$ we have two bounds:

- (A) if $\rho|r| \leq ct$: $p^*(f_j + r\rho g_t) \leq 1 + \frac{\rho^2 r^2}{2}(1 + \eta_0) \leq 1 + \frac{1+\rho^2}{4}r^2$, by (b) and the choice of η_0 ;
- (B) always: $p^*(f_j + r\rho g_t) \leq p^*(f_j + r\rho g) + \rho|r|Kt \leq s(\rho r) + (\lambda - 1)\min(\lambda^2 \rho^2 r^2/2, 1) + \varepsilon' + \rho|r|Kt$, by (a), (b) and Corollary 2.17.

By convexity, $p^*(f_j + r\rho \bar{g}) \leq \frac{1}{n} \sum_{t \in \mathcal{S}} p^*(f_j + r\rho g_t)$.

Case $|r| \leq r_0$. Let $\mathcal{S}_r := \{t \in \mathcal{S} : ct < \rho|r|\}$. If $\mathcal{S}_r = \emptyset$, every term obeys (A), and $1 + \frac{1+\rho^2}{4}r^2 \leq s(r)$ (see below). Otherwise $\rho|r| > c \min \mathcal{S} = 2cT2^{-n}$, so $\varepsilon' \leq \theta_* c^2 (T2^{-n})^2 < \theta_* \rho^2 r^2/4 \leq (1 - \rho^2)r^2/48$; moreover $\sum_{t \in \mathcal{S}_r} t < 2\rho|r|/c$ (the scales are dyadic), so $\frac{1}{n} \sum_{t \in \mathcal{S}_r} \rho|r|Kt \leq 2\rho^2 K r^2/(cn) \leq (1 - \rho^2)r^2/24$; and $(\lambda - 1)\lambda^2 \rho^2 r^2/2 \leq (1 - \rho^2)r^2/48$. Using (A) off $\mathcal{S}_r$ and (B) on $\mathcal{S}_r$,

$$p^*(f_j + r\rho \bar{g}) \leq \max \left\{ 1 + \frac{1 + \rho^2}{4}r^2, s(\rho r) \right\} + \frac{(1 - \rho^2)r^2}{12}.$$

Now $1 + \frac{1+\rho^2}{4}r^2 + \frac{1-\rho^2}{12}r^2 = 1 + \frac{2+\rho^2}{6}r^2 \leq 1 + \frac{r^2}{2} - \frac{r^4}{8} \leq s(r)$, because $r^2 \leq 1 - \rho^2 \leq \frac{4}{3}(1 - \rho^2)$ and $\sqrt{1+x} \geq 1 + \frac{x}{2} - \frac{x^2}{8}$; and $s(\rho r) + \frac{1-\rho^2}{12}r^2 \leq s(r)$ by [I, Lemma 3.6(a)] ($|r| \leq 1$).

Case $|r| > r_0$. Using (B) for every t and $\frac{1}{n} \sum_t t \leq 2T/n$,

$$\begin{aligned} p^*(f_j + r\rho\bar{g}) &\leq s(\rho r) + (\lambda - 1) + \varepsilon' + \frac{2\rho KT}{n}|r| \\ &\leq s(\rho r) + \frac{1 - \rho^2}{12}(r_0^2 + r_0^2 + r_0|r|) \leq s(\rho r) + \frac{(1 - \rho^2)r_0|r|}{4}, \end{aligned}$$

and $s(r) - s(\rho r) \geq (1 - \rho^2) \min(r^2, |r|)/3 \geq (1 - \rho^2)r_0|r|/3$ by [I, Lemma 3.6(a)], since $r_0 \leq 1$.

Hence $\rho\bar{g}_j \in \mathcal{C}(f_j)$ for all large j , and by (e) $(f_j, \rho\bar{g}_j) \in \text{clNA}$. Finally $p^*(\rho\bar{g}_j - \rho g) \leq \frac{\rho}{n_j} \sum_t K_j t \leq 2\rho K_j T_j/n_j \rightarrow 0$ and $p^*(f_j - f) \rightarrow 0$, so $\|(f_j, \rho\bar{g}_j) - (f, \rho g)\| \leq p^*(f_j - f) + p^*(\rho\bar{g}_j - \rho g) \rightarrow 0$; as clNA is closed, $(f, \rho g) \in \text{clNA}$. $\square$

Only pieces of the window that are finer than the probing scale are compared with g at f_j , where g is a mate only up to the error ε'_j ; such pieces exist exactly when $\rho|r| \gtrsim c_j T_j 2^{-n_j}$, which is the reason for (d). With $\lambda_j = 1$ and $\varepsilon'_j := p^*(f_j - f)$, hypothesis (a) holds by the triangle inequality ($p^*(f_j + r\rho g) \leq p^*(f + r\rho g) + \varepsilon'_j$). For raise companions, Corollary 2.17 gives (a) with $\varepsilon'_j = 2\|U^* \Delta a_j\| + p^*(L^*(w_j^\# - w))$, which involves the Hilbert footprint of the raise and the normer change but not $\|\Delta a_j\|_1$.

Theorem 2.20 (Windowed recovery through nearby first rows). *Let $f \in S_{p^*}$ have F finite, $g \in \mathcal{C}(f)$, $\rho \in (0, 1)$, and $\eta_0 \in (0, 2]$ with $\rho^2(1 + \eta_0) \leq (1 + \rho^2)/2$. Let $f_j \rightarrow f$ in S_{p^*} with $F_j = \text{supp } a_j$ finite and $F \subseteq F_j$, and let t_1, j_0, c_0 be given by Lemma 2.18 for $\varepsilon_{\text{tr}} := \eta_0/2$ and a constant $A_0 \geq 1$. Suppose that for every j there are $A_{2,j} \geq 1$, $\gamma_j \in (0, 1]$, a window (T_j, n_j) and $K_j \geq 1$, and for every $t \in \mathcal{S}(T_j, n_j)$ d -neutral two-piece data $(b^\pm, \omega^\pm) = (b_{j,t}^\pm, \omega_{j,t}^\pm)$ at f_j ([I, Definition 7.7] at f_j) for a functional $g_{j,t}$, such that, with $c_j := c_0 \gamma_j / A_{2,j}$:*

- (E-a) $b^\pm(\xi_j) = 0$, $t\|b^\pm\|_1 \leq A_0$ and $\Gamma_w^{(j)}(b^\pm, \omega^\pm) \leq 1 + \eta_0/2$;
- (E-b) every $k \in \text{supp } \omega_m^\pm$ is of kind [1], [2] (with $A_{2,j}, \gamma_j$) or [3] (for the side of the pair) at f_j , and for every $i \in F_j \setminus F$: $z_j(i)b^+(i) \geq 0$ or $t|b^+(i)| \leq |a_j(i)|$, and $z_j(i)b^-(i) \leq 0$ or $t|b^-(i)| \leq |a_j(i)|$;
- (E-c) $p^*(g - g_{j,t}) \leq K_j t$;
- (E-d) $K_j T_j \rightarrow 0$ and $n_j \geq 48\rho^2 K_j / (c_j(1 - \rho^2))$ for all large j ;
- (E-e) $\varepsilon_j := p^*(f_j - f) \leq \theta_* c_j^2 (T_j 2^{-n_j})^2$ for all large j , $\theta_* = (1 - \rho^2)/(12\rho^2)$.

Then $(f, \rho g) \in \text{clNA}((c_0, p), \ell_2^2)$. If the hypotheses hold for every $\rho < 1$, then $(f, g) \in \text{clNA}((c_0, p), \ell_2^2)$.

Proof. We verify the hypotheses of Theorem 2.19 with $\lambda_j := 1$, $\varepsilon'_j := \varepsilon_j$ and c_j as given. (a) holds by the triangle inequality, since $p^*(f + r\rho g) \leq s(\rho r)$. For (b): as $K_j \geq 1$ and $K_j T_j \rightarrow 0$, $T_j \leq t_1$ for large j , and $j \geq j_0$. Both pairs represent $g_{j,t}$ at f_j , are side-admissible at f_j , and satisfy (OS1)–(OS3) of Lemma 2.18 with $(A_{2,j}, \gamma_j)$ (note $\Gamma_w^{(j)} \leq 1 + \eta_0/2 \leq 2$ and, as two-piece data at f_j , they satisfy the sign conditions of (OS2) off F_j). Lemma 2.18 applied to the $+$ pair for $0 < \rho r \leq c_j t$ and to the $-$ pair for $-c_j t \leq \rho r < 0$ gives $p^*(f_j + \rho r g_{j,t}) \leq 1 + \frac{\rho^2 r^2}{2}(1 + \eta_0/2 + \eta_0/2)$. (c), (d) are (E-d), (E-e), together with $\varepsilon_j \rightarrow 0$. For (e): the averaged data $\bar{D}_j := \frac{1}{n_j} \sum_t (b_{j,t}^\pm, \omega_{j,t}^\pm)$ are d -neutral two-piece data at f_j for $\bar{g}_j$: the vanishing off $F_j \cup K_j$, the sign conditions on K_j , the finite supports in $Q_m^{(j)}$ (coordinates of kind [3] in two-piece data are strict non-peaks), the two representations and $\Delta d_m^{(j)} = 0$ are linear

conditions, preserved under averaging. As $\Gamma_w^{(j)}$ is convex, $\kappa_w(\bar{D}_j) \leq 1 + \eta_0/2$, so the data $\rho\bar{D}_j$ of $\rho\bar{g}_j$ have $\kappa_w \leq \rho^2(1 + \eta_0/2) \leq 1$. If $\rho\bar{g}_j \in \mathcal{C}(f_j)$, [I, Corollary 7.21] at f_j (which has finite base support F_j) gives $(f_j, \rho\bar{g}_j) \in \text{clNA}$. Theorem 2.19 applies. The last sentence follows because $(f, g) = \lim_{\rho \rightarrow 1} (f, \rho g)$ and clNA is closed. $\square$

Remark 2.21 (Special cases and comments). (a) With $f_j = f$ (so $\varepsilon_j = 0$) the theorem is the mechanism of the proof of [I, Theorem 8.44]; there (E-d) is provided by the window arithmetic of the design. (b) The case $F_j = F$, $A_{2,j} = A_2$, $\gamma_j = \gamma_B$ fixed and $\varepsilon_j \leq \theta_j(T_j 2^{-n_j})^2$ with $\theta_j \rightarrow 0$ is the original form; window-dependent constants $A_{2,j}$, γ_j (hence $c_j \rightarrow 0$ possibly) are needed when the gaps of the switching carriers at the companions are only controlled by the window; kind [3] allows used degenerate peaks of f that have become strict non-peaks of f_j with arbitrarily small gap, provided the data move them only inward. (c) The coordinates of $F_j \setminus F$ (banks and pulls) need not be contacts of f : in the proof of Lemma 2.18 only the sign of $a_j(i)$ at f_j and the size condition enter. (d) The approximants f_j need not belong to any recovered class, and no lower semicontinuity of $\mathcal{C}$ along $f_j \rightarrow f$ is used: [I, Corollary 7.21] is applied at each f_j to the mate $\rho\bar{g}_j$ of f_j obtained by averaging. (e) Windows are arbitrary: any sub-window (T_j, n_j) of a design window can be used, which improves the exactification thresholds in applications.

2.6. Exact two-piece data: certificate, shift bound and rigidity. Let $f' \in S_{p^*}$ have finite base support F' . A vector $V \in \ell_1$ is z' -admissible off F' if $V(j) = 0$ for $j \notin F' \cup K'$ and $z'_j V(j) \geq 0$ for $j \in K'$. For two-piece data $(b^\pm, \omega^\pm)$ at a row f (with F finite) we use the *data convention* $\Delta d_m = d_m(\omega_m^-) - d_m(\omega_m^+)$ of [I, Definition 7.7], and put

$$W_\Delta := \sum_m \Delta d_m \psi_m, \quad \Sigma(\omega) := F \cup \bigcup_{m \in I} \bigcup_{k \in \text{supp } \omega_m^+ \cup \text{supp } \omega_m^-} \text{supp } u_{k,m}.$$

Lemma 2.22 (Exactness certificate). *Let $f' \in S_{p^*}$ have F' finite, let $\Omega_m \subseteq Q'_m$ be finite, and let $\omega_m^\pm \in c_{00}$ be supported in Ω_m ($m \in I$). Put $\Delta\omega_m := \omega_m^- - \omega_m^+$, $\Delta_m := d'_m(\Delta\omega_m)$ and*

$$\mathcal{V}(\Delta\omega) := \sum_m R_m^*(\Delta\omega_m - \Delta_m w'_m) \in \ell_1.$$

- (a) $\mathcal{V}(\Delta\omega) = \sum_m \sum_k \gamma_{k,m} u_{k,m}$ (absolutely convergent in ℓ_1) with $\gamma_{k,m} := \lambda_{k,m}(\Delta\omega_m(k) - \Delta_m w'_m(k))$; in particular every (k, m) with $k \notin \Omega_m$ enters with the forced coefficient $-\Delta_m \lambda_{k,m} w'_m(k)$, which equals $-\Delta_m \lambda_{k,m} \varsigma_k M'_m$ at a peak k of sign ς_k .
- (b) If $(b^\pm, \omega^\pm)$ are two-piece data at f' , then $b^+ - b^- = \mathcal{V}(\Delta\omega)$; hence $\mathcal{V}(\Delta\omega)$ is z' -admissible off F' .
- (c) Conversely, suppose $\mathcal{V} := \mathcal{V}(\Delta\omega)$ is z' -admissible off F' , let $\beta \in \ell_1$ be supported in F' and $\chi : K' \rightarrow [0, 1]$, and put $b^+ := \beta + \chi \mathcal{V} 1_{K'}$, $b^- := b^+ - \mathcal{V}$. Then (b^+, ω^+) and (b^-, ω^-) are two-piece data at f' for $g := b^+ + \sum_m R_m^*(\omega_m^+ - d'_m(\omega_m^+) w'_m)$, with $\Delta d_m = \Delta_m$.

So the existence of exact two-piece data with prescribed block parts is a property of the single vector $\Delta\omega$: $\mathcal{V}(\Delta\omega)$ must be z' -admissible off F' . Off F' its restriction is a finite sum over the carriers of Ω plus the infinite series

$$- \sum_m \Delta_m \sum_{k \notin \Omega_m} \lambda_{k,m} w'_m(k) u_{k,m}.$$

Proof. (a) $R_m^* x = \sum_k \lambda_{k,m} x(k) u_{k,m}$ [I, (1)], and $\sum_{k,m} \lambda_{k,m} \|u_{k,m}\|_1 < \infty$. (b) Subtracting the two representations of the functional gives $b^+ - b^- = \sum_m R_m^* ((\omega_m^- - \omega_m^+) - (d'_m(\omega_m^-) - d'_m(\omega_m^+))w'_m) = \mathcal{V}(\Delta\omega)$. Off $F' \cup K'$ both $b^\pm$ vanish, and on K' , $z'b^+ \geq 0 \geq z'b^-$, so $z'(b^+ - b^-) \geq 0$. (c) Off F' , $b^+ = \chi \mathcal{V} 1_{K'}$ vanishes off K' and is z' -signed on K' ; $b^- = \chi \mathcal{V} - \mathcal{V} = -(1 - \chi)\mathcal{V}$ on K' and $b^- = 0$ at the free coordinates (where $\mathcal{V} = 0$), so b^- is $(-z')$ -signed. The block parts are finitely supported in Q'_m . By linearity of d'_m , $b^- + \sum_m R_m^* (\omega_m^- - d'_m(\omega_m^-)w'_m) = b^+ - \mathcal{V} + \sum_m R_m^* (\omega_m^+ - d'_m(\omega_m^+)w'_m) + \mathcal{V} = g$. $\square$

Lemma 2.23 (Shift bound). *Let $f' \in S_{p^*}$ have F' finite and let $(b^\pm, \omega^\pm)$ be two-piece data at f' with $\Gamma'_w(b^\pm, \omega^\pm) \leq 2$. Then for every m , with $\Phi_{P'_m}^2 := \sum_{k \in P'_m} \Phi_m(k)^2 > 0$,*

$$|\Delta d_m| \leq \left(\frac{8 C_m'^3}{\sigma'_m M_m'^2 \Phi_{P'_m}^2} \right)^{1/2}.$$

Proof. Drop m and the primes. $\sigma H(\omega^\pm) \leq \Gamma_w \leq 2$, and $\sqrt{H}$ is a seminorm (it is $\|P^\perp D \cdot\|_2 / \sqrt{C}$, $P^\perp$ the projection onto $(Dw)^\perp$), so $H(\Delta\omega) \leq 8/\sigma$ for $\Delta\omega := \omega^- - \omega^+$, which is supported in a finite set $\Omega \subseteq Q$. Put $v := D(w1_\Omega)$ and $\chi := \|v\|_2^2 / C^2$. As $\|Dw\|_2^2 = C^2 \geq \|v\|_2^2 + \|D(w1_P)\|_2^2 = \|v\|_2^2 + M^2 \Phi_P^2$, we have $1 - \chi \geq M^2 \Phi_P^2 / C^2 > 0$. If $v = 0$ then $\Delta d = \langle Dw, D\Delta\omega \rangle / C = \langle v, D\Delta\omega \rangle / C = 0$. Otherwise write $D\Delta\omega = av + y$ with $y \perp v$; then $\Delta d = a\|v\|^2 / C = aC\chi$ and $\|P^\perp D\Delta\omega\|^2 = \|D\Delta\omega\|^2 - \langle Dw, D\Delta\omega \rangle^2 / C^2 \geq a^2 C^2 \chi (1 - \chi)$. Hence $CH(\Delta\omega) \geq a^2 C^2 \chi (1 - \chi)$ and $\Delta d^2 = a^2 C^2 \chi^2 \leq H(\Delta\omega) C \chi / (1 - \chi) \leq (8/\sigma) C^3 / (M^2 \Phi_P^2)$. $\square$

At companions $f^\#$ of f with small cost (Lemma 2.12) the quantities $C_m^\#, \sigma_m^\#, M_m^\#$ are close to those of f , and a fixed non-degenerate peak of f remains a peak, so the shifts of all data with $\Gamma_w^\# \leq 2$ are bounded by a constant depending only on f .

Lemma 2.24 (Sandwich). *Let $f \in S_{p^*}$ have F finite and let $h \in X^*$ carry two-piece data $(b^\pm, \omega^\pm)$. Put $c^\pm := \sum_m d_m(\omega_m^\pm) \psi_m$. Then $b^+ - b^- = \sum_m R_m^* (\omega_m^- - \omega_m^+) - W_\Delta$, and for every $j \notin \Sigma(\omega)$:*

- (a) $h(j) = b^+(j) - c^+(j) = b^-(j) - c^-(j)$;
- (b) if j is a free coordinate ($j \notin F \cup K$), then $b^\pm(j) = 0$, $h(j) = -c^+(j) = -c^-(j)$ and $W_\Delta(j) = 0$;
- (c) if $j \in K$, then $z_j(h(j) + c^+(j)) \geq 0 \geq z_j(h(j) + c^-(j))$; in particular $z_j W_\Delta(j) \leq 0$.

Proof. The identity is the difference of the two representations. For $j \notin \Sigma(\omega)$ every $(R_m^* \omega_m^\pm)(j) = \sum_k \lambda_{k,m} \omega_m^\pm(k) u_{k,m}(j)$ vanishes, and $R_m^*(d w_m) = d \psi_m$; so the representations read $h(j) = b^\pm(j) - c^\pm(j)$. Side admissibility gives $b^\pm(j) = 0$ off $F \cup K$, and $z_j b^+(j) \geq 0 \geq z_j b^-(j)$ on K . Finally $c^+ - c^- = -W_\Delta$. $\square$

No smallness or dominance hypothesis is needed; the lemma holds at every coordinate outside the supports of the carriers used by the data.

Corollary 2.25 (One block: constancy of the profile). *Let $I = \{1\}$, $f \in S_{p^*}$ with F finite, $\psi := R_1^* w_1$, and let h carry two-piece data, $d^\pm := d_1(\omega_1^\pm)$, $\Delta := d^- - d^+$. For $j \notin \Sigma(\omega)$ with $\psi(j) \neq 0$ put $\pi_h(j) := h(j)/\psi(j)$ (the profile); call a contact j aligned if $z_j \psi(j) > 0$ and anti-aligned if $z_j \psi(j) < 0$. For $j \notin \Sigma(\omega)$ with $\psi(j) \neq 0$:*

- (i) if j is free, then $\Delta = 0$ and $\pi_h(j) = -d^+$;
- (ii) if j is an aligned contact, then $-d^+ \leq \pi_h(j) \leq -d^-$ (so $\Delta \leq 0$);

(iii) if j is an anti-aligned contact, then $-d^- \leq \pi_h(j) \leq -d^+$ (so $\Delta \geq 0$).

Consequently the oscillation of π_h over the aligned contacts outside $\Sigma(\omega)$ is at most $|\Delta|$; and if outside $\Sigma(\omega)$ there are an aligned contact and, in addition, a free coordinate or an anti-aligned contact with $\psi \neq 0$, then $\Delta = 0$ and $h = -d^+\psi$ at every coordinate outside $\Sigma(\omega)$.

Proof. Lemma 2.24 with $c^\pm = d^\pm\psi$. (i) is (b) ($W_\Delta(j) = \Delta\psi(j) = 0$). (ii), (iii): divide the inequalities of (c) by $z_j\psi(j)$. For the last claim, the hypotheses give $\Delta \leq 0$ and $\Delta \geq 0$ (or $\Delta = 0$ directly); with $d^+ = d^-$, (b) and (c) give $h(j) = -d^+\psi(j)$ at every free coordinate and every contact outside $\Sigma(\omega)$, and $\Sigma(\omega) \supseteq F$. $\square$

Corollary 2.26 (Finitely many contacts: no switching, no shift). *If F and K are finite (in particular if f attains its norm, [I, Proposition 1.10(c)]), then all two-piece data at f have $b^+ = b^-$, $\omega^+ = \omega^-$ and $\Delta d_m = 0$ for every m .*

Proof. By Lemma 2.24, $v := b^+ - b^- = L^*X$ with $X_m := (\omega_m^- - \omega_m^+) - \Delta d_m w_m$, so $v \in Y$; and v is supported in the finite set $F \cup K$. By (T-c), $v = 0$, and as L^* is injective on V^* [I, (1)], $\omega_m^- - \omega_m^+ = \Delta d_m w_m$ for every m . The left side vanishes on $P_m \neq \emptyset$, where $|w_m| = M_m > 0$; so $\Delta d_m = 0$, $\omega^- = \omega^+$ and $b^+ = b^-$. $\square$

At norm-attaining rows exact data cannot carry any shift or switching; the scale-dependent structure of the mates of such rows lives only in their decompositions at positive scales (compare [I, Remark 7.11(a)]).

Proposition 2.27 (Following a non-constant profile). *Let $I = \{1\}$, $f \in S_{p^*}$ with F finite, $g \in \mathcal{C}(f)$, $\rho \in (0, 1]$, and let $s_1, s_2 \notin F$ with $\psi(s_1)\psi(s_2) \neq 0$ and $\pi_g(s_1) \neq \pi_g(s_2)$, where $\pi_g := g/\psi$; put $\kappa_0 := |\psi(s_1)\psi(s_2)| |\pi_g(s_1) - \pi_g(s_2)| > 0$. Let $f_n \rightarrow f$ be first rows with finite base supports F_n at which every mate carries two-piece data (for instance rows satisfying (BT), [I, Theorem 7.10(c)]); let $\psi_n := R_1^* w_{n,1}$, $\Sigma_n := F_n \cup \bigcup_{k \in Q_{n,1}} \text{supp } u_{k,1}$, and define aligned and anti-aligned contacts of f_n with $z^{(n)}$ and ψ_n . Suppose that for infinitely many n :*

- (1) $s_1, s_2 \notin \Sigma_n$, and
- (2) outside Σ_n , f_n has an aligned contact and, in addition, a free coordinate with $\psi_n \neq 0$ or an anti-aligned contact.

Then, along these n ,

$$\text{dist}_{\ell_1}(\rho g, \mathcal{C}(f_n)) \geq 2\rho \left(\kappa_0 - \|g\|_\infty (|\psi_n - \psi|(s_1) + |\psi_n - \psi|(s_2)) \right) \longrightarrow 2\rho\kappa_0,$$

and $\text{dist}_{p^*}(\rho g, \mathcal{C}(f_n)) \geq \frac{1}{3} \text{dist}_{\ell_1}(\rho g, \mathcal{C}(f_n))$; in particular $\rho g \notin \text{Li}_n \mathcal{C}(f_n)$.

Proof. Let $G \in \mathcal{C}(f_n)$ with two-piece data at f_n ; their block parts are supported in $Q_{n,1}$, so $\Sigma(\omega) \subseteq \Sigma_n$. By (2) and Corollary 2.25 at f_n , $G = -d^+\psi_n$ outside Σ_n , in particular at s_1, s_2 by (1). So the functional $\Lambda_n(h) := h(s_1)\psi_n(s_2) - h(s_2)\psi_n(s_1)$ vanishes at G . Since $\|\psi_n\|_\infty \leq \sum_k \lambda_{k,1} \|u_{k,1}\|_\infty \leq \sum_k \lambda_{k,1} \leq \frac{1}{2}$, we have $|\Lambda_n(h)| \leq \frac{1}{2} \|h\|_1$, so $\|\rho g - G\|_1 \geq 2|\Lambda_n(\rho g)|$. With $\Lambda(g) := g(s_1)\psi(s_2) - g(s_2)\psi(s_1) = \psi(s_1)\psi(s_2)(\pi_g(s_1) - \pi_g(s_2))$, $|\Lambda(g)| = \kappa_0$ and $|\Lambda_n(g) - \Lambda(g)| \leq \|g\|_\infty (|\psi_n - \psi|(s_1) + |\psi_n - \psi|(s_2))$. By [I, Proposition 1.11], $\psi_n \rightarrow \psi$ in ℓ_1 . Finally, if $h = A + L^*W$ with $\max(q^*(A), \|W\|) = p^*(h)$ [I, Lemma 1.6], then $\|h\|_1 \leq q^*(A) + \sum_{k,m} \lambda_{k,m} \|W\| \leq 3p^*(h)$, since $\sum_{k,m} \lambda_{k,m} \leq \sum_m m2^{-m} \leq 2$. $\square$

To recover a mate whose profile is not constant on $\{s_1, s_2\}$, approximants at which all mates have exact data must therefore, for all large n , put s_1 or s_2 into Σ_n

(make it a support coordinate, a *bank*, or cover it by the support of a strict non-peak), or be aligned with maximal contact outside Σ_n (no free coordinate and no anti-aligned contact with $\psi_n \neq 0$). The engineered approximants of [I, Section 7] use the first alternative (window masses are banks on contacts); norm-attaining rows can use only the first alternative, since they have cofinitely many free coordinates.

Theorem 2.28 (A criterion for the failure of lower semicontinuity along block-tame rows). *Let $I = \{1\}$, let $f \in S_{p^*}$ have F finite, $g \in \mathcal{C}(f)$ and $s_1, s_2 \notin F$ with $\psi(s_1)\psi(s_2) \neq 0$ and $\pi_g(s_1) \neq \pi_g(s_2)$. Let $f_n \rightarrow f$ be first rows satisfying (BT) such that, for every n , $s_1, s_2 \notin \Sigma_n$ and, outside Σ_n , f_n has an aligned contact and a free coordinate with $\psi_n \neq 0$ or an anti-aligned contact. Then every f_n belongs to $\mathcal{R}$, and for every $\rho \in (0, 1]$*

$$\liminf_n \text{dist}_{p^*}(\rho g, \mathcal{C}(f_n)) \geq \frac{2}{3}\rho |\psi(s_1)\psi(s_2)| |\pi_g(s_1) - \pi_g(s_2)| > 0.$$

In particular the fibre map $f' \mapsto \mathcal{C}(f')$ is not lower semicontinuous at f along the block-tame rows f_n ; if f itself satisfies (BT), then nevertheless $f \in \mathcal{R}$.

Proof. $f_n \in \mathcal{R}$ and (if (BT) holds at f) $f \in \mathcal{R}$ by [I, Corollary 7.22]; every mate of f_n carries two-piece data by [I, Theorem 7.10(c)]. Apply Proposition 2.27 to all n . $\square$

Remark 2.29 (Realization at self-aligned rows). For T_{final} and $N = 1$ the hypotheses of Theorem 2.28 are realized, with complete proof, in Proposition 5.32 below (with $f = f_{\text{SA}}$ of Theorem 5.29 and the mates of Proposition 5.31). *Sketch, not proved here:* the same construction should work, with f satisfying (BT), for signature-ladder designs that satisfy the dominance condition (SF*) and contain a target supported in two coordinates outside all signature sets (condition (Z0)), with a diagonal base: there f can be taken to be a *self-aligned* row (every carrier except one exceptional strict non-peak k_- is an exactly swallowed peak of its own sign, every coordinate off F is a contact, $Q = \{k_-\}$), f_n is obtained from f by flipping z on the signature set of one far carrier o_n whose target is supported in F (which becomes an exactly swallowed anti-type peak, so its contacts are anti-aligned) and re-assigning the later carriers by a hysteretic owner rule, and g is a coherent-shift mate of f with a coordinate-dependent split of its contact part, whose profile on a signature set of a carrier $l \neq k_-$ is not constant. The construction of these rows and the verification of these properties belong to the design-specific part of our sources and are not reproduced here in this generality. For T_{final} (Proposition 5.32), and for any such design for which the construction is granted, it follows from Theorem 2.28 that membership in $\mathcal{R}$ cannot be propagated along an *arbitrary* dense sequence of block-tame rows, and approximants for mates with oscillating profiles must either bank the oscillation coordinates or keep every peak of a shifted block exactly swallowed and aligned outside the data supports.

2.7. Violation tolerance of the engineered approximants. In the proof of [I, Theorem 7.18] the sign conditions of two-piece data enter only through the claim on the base excess in Step 3. We show that a small violation of these conditions costs a first-order term that is affordable at a single stage of the construction. We use the notation of [I, Definition 7.12] and of the proof of [I, Theorem 7.18]: for pairs $(b^\pm, \omega^\pm)$, $b^\theta := \frac{1}{2}(b^+ + b^-)$, $\omega^\theta := \frac{1}{2}(\omega^+ + \omega^-)$, $v := b^+ - b^-$; the stage parameters (N_w, s_1, N'') ; the row f' with data $a', z', \hat{x}', c'$; the target $g' = \rho(g'' - ca')$; the

vectors $\beta^\theta := b^\theta 1_{[1, N_w]}$, $\beta^\pm := b^\pm 1_{[1, N_w]} \pm \frac{1}{2} v 1_{(N_w, \infty)}$, $B^\diamond := \rho(\beta^\diamond - ca')$; and $V_{>N''} := \sum_{j \in K, j > N''} |v_j|$.

Lemma 2.30 (Violation tolerance at a single stage). *Let I be finite, $f \in S_{p^*}$ with F finite, $\rho \in (0, 1)$, and $g \in X^*$ with $g(\xi) = 0$. Let (b^+, ω^+) and (b^-, ω^-) be pairs that represent g in the sense of [I, Definition 7.7] ($\omega_m^\pm \in c_{00}$, $\text{supp } \omega_m^\pm \subseteq Q_m$), not necessarily side-admissible. For $j \notin F$ put*

$$\text{viol}(j) := (z_j b_j^+)_- + (z_j b_j^-)_+ \quad (j \in K), \quad \text{viol}(j) := |b_j^+| + |b_j^-| \quad (j \notin F \cup K),$$

and $\epsilon := \sum_{j \notin F} \text{viol}(j)$ (two-piece data are exactly the case $\epsilon = 0$). Assume

- (V1) $\rho^2 \kappa_w < 1$, where $\kappa_w := \max_{\pm} \Gamma_w(b^\pm, \omega^\pm)$;
- (V2) $I_- := \{m : \Delta d_m < 0\}$ satisfies (SC) [I, Definition 7.16] (no assumption if $I_- = \emptyset$);
- (V3) $b_j^\theta = 0$ at every free coordinate $j \notin F \cup K$ with $\text{viol}(j) > 0$.

Choose $\delta, T_1, K_\#, \eta_1$, the transfer data and T_0 as in Step 0 of the proof of [I, Theorem 7.18] for $(f, b^\pm, \omega^\pm, \rho)$, with $K_\#$ replaced by $K_\# + 1$. Let (N_w, s_1, N'') be a stage at which all inequalities used in Steps 1–5 of that proof hold ([I, (11)–(13)], the persistence of the transfer data, $|l'_m| \leq K_\# |\tau| s_1$, $\hat{G}_b, \hat{G}_m \leq K_\# \tau^2$ for the exact parts, the bracket bound $\Gamma_\diamond + \delta/8$ of Step 5, and $\frac{1}{2} \rho |\tau| V_{>N''} + |\tau| o(s_1) \leq \delta \tau^2 / 16$ for $|\tau| > s_1$), and at which

$$(VT) \quad 2\rho\epsilon \leq \delta s_1 / 16.$$

Then the engineered approximant f' (norm attaining) and g' satisfy

$$p^*(f' + \tau g') \leq 1 + \frac{\tau^2}{2}(1 - \delta) \quad (0 < |\tau| \leq T_0).$$

Proof. Where side admissibility is used. Step 0 uses the data only through κ_w (and $\Gamma_\theta \leq \kappa_w$ by convexity of Γ_w , which does not need admissibility) and through finitely many constants. Steps 1 and 2 use only the two representations: $g'' \rightarrow g$, $g'(\hat{x}') = 0$, and the identities $\beta^\pm - \beta^\theta = \pm \frac{1}{2} v$ and $v = \sum_m R_m^*((\omega_m^- - \omega_m^+) - \Delta d_m w_m)$, which holds for any two representations of the same functional. Step 3 (Blocks) uses only $\text{supp } \omega_m^\diamond \subseteq Q_m$ and the radius conditions, and [I, Lemma 7.13] uses b^θ only through the masses $m_j = 4\rho s_1 |b_j^\theta|$ and $\|b^\theta\|_1$. The sign conditions enter only the claim of Step 3 (Base), $\text{Exc}'(\tau B^\diamond) \leq \frac{1}{2} \rho |\tau| V_{>N''}$ (and $= 0$ for $\diamond = \theta$). We replace it by

$$(2) \quad \begin{aligned} \text{Exc}'(\tau B^\diamond) &\leq \frac{1}{2} \rho |\tau| V_{>N''} + 2\rho |\tau| \epsilon & (\diamond = \text{sign } \tau, s_1 < |\tau| \leq T_0), \\ \text{Exc}'(\tau B^\theta) &= 0 & (0 < |\tau| \leq s_1). \end{aligned}$$

Proof of (2). By [I, Lemma 7.1(b)] at f' , $\text{Exc}'(B) = \sum_j (|a'_j + B_j| - |a'_j| - z'_j B_j)$. On $\text{supp } a'$ the summand is $2(-\text{sign}(a'_j) B_j - |a'_j|)_+$, and $a'_j + \tau B_j^\diamond = (1 - \tau \rho c) a'_j + \tau \rho \beta_j^\diamond$ with $1 - \tau \rho c \geq \frac{1}{2}$, so the summand is at most $2(x)_-$ with $x := z'_j \tau \rho \beta_j^\diamond$ (recall $\text{sign } a'_j = z'_j$). Off $\text{supp } a'$ the summand is $|\tau \rho \beta_j^\diamond| - z'_j \tau \rho \beta_j^\diamond$. We go through the coordinates.

- $j \in F$: no flip by condition (T_1) ; summand 0.
- Window contacts with mass ($j \in K, j \leq N_w, b_j^\theta \neq 0; z'_j = z_j$): for $\diamond = \theta$, $|\tau \rho b_j^\theta| \leq m_j/4 \leq |a'_j|/2$ [I, Lemma 7.13(a)], summand 0 whatever the sign of b_j^θ ; for $\diamond = +$ ($\tau > 0$), $2(x)_- = 2\rho \tau (z_j b_j^+)_-$; for $\diamond = -$ ($\tau < 0$), $2(x)_- = 2\rho |\tau| (z_j b_j^-)_+$; both $\leq 2\rho |\tau| \text{viol}(j)$.

- Window contacts without mass ($b_j^\theta = 0$, so $\beta_j^\theta = 0$ and $b_j^\pm = -b_j^\mp$): off $\text{supp } a'$; summand 0 for θ , and $2\rho|\tau|(z_j b_j^\pm)_-$, resp. $2\rho|\tau|(z_j b_j^\pm)_+$, for $\diamond = \pm$.
- Free $j \leq N_w$: off $\text{supp } a'$; for $\diamond = \theta$, $\beta_j^\theta = b_j^\theta = 0$ by (V3) if $\text{viol}(j) > 0$, and $b_j^\pm = 0$ if $\text{viol}(j) = 0$: summand 0; for $\diamond = \pm$ the summand is at most $2\rho|\tau||b_j^\diamond| \leq 2\rho|\tau|\text{viol}(j)$.
- Contacts $j \in (N_w, N'']$: $\beta_j^\theta = 0$, and for $\diamond = \text{sign } \tau$, $\tau\beta_j^\diamond = \frac{1}{2}|\tau|v_j$, so the summand is $\rho|\tau|(z_j v_j)_-$; since $z_j v_j \geq -(z_j b_j^\pm)_- - (z_j b_j^\pm)_+$, this is at most $\rho|\tau|\text{viol}(j)$.
- Free $j > N_w$: $\beta_j^\theta = 0$ and $|\beta_j^\pm| = |v_j|/2 \leq \text{viol}(j)/2$; summand at most $\rho|\tau|\text{viol}(j)$.
- Contacts $j > N''$: $z_j' = 0$; summand $\frac{1}{2}\rho|\tau||v_j|$ for $\diamond = \pm$ and 0 for θ (unchanged; this is the term with $V_{>N''}$).

Free coordinates never belong to $\text{supp } a' = F \cup \{j \in K \cap [1, N_w] : b_j^\theta \neq 0\}$. Summing gives (2).

Constants. For $|\tau| > s_1$, (VT) gives $2\rho|\tau|\epsilon \leq \delta|\tau|s_1/16 \leq \delta\tau^2/16 \leq \tau^2$; so the bound $\hat{G}_b \leq (K_\# + 1)\tau^2$ holds, and Step 4 (rebalancing) is unchanged with $K_\# + 1$ in place of $K_\#$ (the choice of η_1 and the conditions (T_0) were made with $K_\# + 1$), giving $E \leq \delta\tau^2/8$. In Step 5, $\hat{\Gamma} = c'\hat{G}_b + \sum_m \sigma'_m \hat{G}_m$ gains at most $c' \cdot 2\rho|\tau|\epsilon \leq \delta\tau^2/16$ in the regimes $\diamond = \pm$ and nothing in the regime $\diamond = \theta$ ($c' \leq 1$). Hence, for $0 < |\tau| \leq T_0$,

$$p^*(f' + \tau g') \leq 1 + \frac{\tau^2}{2} \left(\rho^2 \kappa_w + \frac{\delta}{8} + \frac{\delta}{8} + \frac{\delta}{8} + \frac{\delta}{4} + \frac{\delta}{8} \right) = 1 + \frac{\tau^2}{2} \left(1 - 2\delta + \frac{3\delta}{4} \right) \leq 1 + \frac{\tau^2}{2} (1 - \delta),$$

since $\rho^2 \kappa_w = 1 - 2\delta$. □

Remark 2.31 (Scope of Lemma 2.30). (a) Contact violations inside the window need no further repair when (VT) holds: their cost is the linear term $2\rho|\tau|\text{viol}(j)$, paid only for $|\tau| > s_1$; in the regime $|\tau| \leq s_1$ the window masses protect every window contact, whatever the sign of b_j^θ . Free violations inside the window with $b_j^\theta \neq 0$ are not tolerated: they would cost $2\rho|\tau||b_j^\theta|$ for $|\tau| \leq s_1$, which is not $O(\tau^2)$, and a free coordinate cannot carry a mass (that would change $|z_j'| < 1$ into 1); this is the reason for (V3). (b) The lemma concerns a single stage and does not contain Step 6 of the proof of [I, Theorem 7.18]: a functional with violated data is in general not a mate of f , so [I, Lemma 7.6] has to be applied relative to some pair (f_0, g_0) with $g_0 \in \mathcal{C}(f_0)$, which requires T_0 to be large compared with $\sqrt{p^*(f' - f_0)}$. Moreover (VT) bounds s_1 from *below*, whereas the lateness conditions bound s_1 from above; the lemma is useful only when ϵ is small compared with the lateness threshold of the data. (Numerical evidence; not used.) A finite-model computation indicates that the threshold (VT) is real: below it the bound fails in the predicted window of linear flip cost. (c) A uniform use of Lemma 2.30 across the scales of a window meets a further difficulty: with window data whose switching has size $1/t$, the window masses create a mismatch of order up to s_1/t^2 (an upper bound; that it is attained is generic and supported only by numerical evidence) between the θ -decomposition and the $\pm$ -decompositions. *Sketch, not proved here:* re-tuning at f' the normalized values of the finitely many coarse switching carriers exactly, so that $d'_m = d_m$ on every switching vector, would remove this mismatch.

3. THE DESIGNED OPERATOR T_{final}

Martín's construction works with every admissible operator T [I, Remark 1.2, Definition 1.3], and the recovery results of [I, Section 8] use a designed operator only through a short list of properties [I, Theorem 8.2 and the remark following it]. In this section we fix *one* operator T_{final} , together with its base U_{final} , which carries every design feature used in the rest of this part, and we prove that it is admissible, that it does not depend on the number N of blocks, and that it has the window properties on which the later sections rely. The operator is a signature-ladder design in the sense of [I, Definition 8.1] with four modifications: a diagonal base whose entries decay doubly exponentially; bounded-gap signature sets and an additional allowedness rule; weights c_l chosen *after* the target and the signature size of level l , as the minimum of finitely many explicit upper bounds; and windows split into pigeonhole sub-windows governed by a design factor $\text{Des}(l)$ and a finite family of rate objects.

Throughout, $j : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ is a bijection with $j(k, m) < j(k', m)$ for $k < k'$, and $(k(l), m(l)) := j^{-1}(l)$. Then $j(k, m) \geq k$ for all (k, m) , and $k(1) = 1$. The notation of [I, Sections 1, 7 and 8] is used without comment; in particular, for $f \in S_{p^*}$ we use the forced data $(\xi, q_0, a, w, F, z, \hat{z}, e, \nu)$ of [I, Proposition 1.9], $M_m := \|w_m\|_\infty$, $C_m := \|D_m w_m\|_2$, and the threshold constant $\vartheta_m := |R_m^{**} \xi|_m M_m / (m C_m)$ of [I, Section 1.3].

3.1. The base.

Definition 3.1 (The base U_{final}). Let $H := \ell_2$ with orthonormal basis $(k_s)_{s \geq 1}$, put

$$\mu_s^* := 2^{-2^{2^s}} \quad (s \geq 1),$$

and let $U = U_{\text{final}} : H \rightarrow c_0$ be the diagonal operator $Uk_s := \mu_s^* e_s$.

Lemma 3.2 (Properties of the base). (a) U is compact, its range contains c_{00} and is therefore dense, $U^* e_s^* = \mu_s^* k_s$, $U^* a = \sum_s a_s \mu_s^* k_s$ is injective on ℓ_1 , $\|U\| = \mu_1^* = \frac{1}{16}$, and

$$q^*(a) = \|a\|_1 + \left(\sum_s (\mu_s^* a_s)^2 \right)^{1/2} \quad (a \in \ell_1).$$

Consequently $\text{NA}((c_0, q), \mathbb{R}) = c_{00}$, and every result of [I] that is stated for an arbitrary compact operator U with dense range holds for U_{final} .

(b) (Diagonal identities.) Let $b \in \ell_1$ and $s \notin \text{supp } b$; then $\langle U^* b, k_s \rangle = 0$. Let $a \in \ell_1$, let $J \subseteq \mathbb{N}$ with $J \cap \text{supp } a = \emptyset$ and let $(c_s)_{s \in J}$ be summable; put $A := a + \sum_{s \in J} c_s e_s^*$ and $X := \sum_{s \in J} c_s \mu_s^* k_s$. Then $U^* A = U^* a + X$, $X \perp U^* a$, $\|U^* A\|^2 = \|U^* a\|^2 + \|X\|^2$, and $\langle U^* u, X \rangle = \sum_{s \in J} c_s (\mu_s^*)^2 u(s)$ for every $u \in \ell_1$.

(c) (Decay.) (μ_s^*) is strictly decreasing, $\mu_{s+1}^* = (\mu_s^*)^{2^{2^s}}$, and for every $K > 0$ there is C_K with $\mu_s^* \leq C_K 2^{-Ks}$ for all s (the decay is faster than every exponential).

(d) (Depth function.) For $x \in (0, \frac{1}{2}]$ let $J(x) := \min\{J \geq 1 : \mu_J^* \leq x^2\}$. Then

$$2^{J(x)} \leq 2 \log_2 (2 \log_2 (1/x)).$$

Proof. (a) U is the norm limit of its finite-rank truncations because $\mu_s^* \rightarrow 0$; $Uk_s = \mu_s^* e_s$ gives $c_{00} \subseteq U(H)$. For $a \in \ell_1$ and $h \in H$, $\langle a, Uh \rangle = \sum_s a_s \mu_s^* \langle h, k_s \rangle$, so $U^* a = \sum_s a_s \mu_s^* k_s$; this vanishes only for $a = 0$. Since μ^* is decreasing (part (c)), $\|U\| = \sup_s \mu_s^* = \mu_1^* = 2^{-4}$. The formula for q^* is $q^*(a) = \|a\|_1 + \|U^* a\|_H$ [I,

Section 1.1]. The statement on $\text{NA}((c_0, q), \mathbb{R})$ is [I, Proposition 1.10(c)], whose proof uses only that U is compact with dense range.

(b) $\langle U^*b, k_s \rangle = \mu_s^* b_s = 0$. The vector X is supported (in the basis (k_s)) in J and U^*a in $\text{supp } a$, which is disjoint from J ; this gives orthogonality and the Pythagoras identity, and $\langle U^*u, X \rangle = \sum_{s \in J} u(s) \mu_s^* c_s \mu_s^*$.

(c) $2^{2^{s+1}} = (2^{2^s})^2$, so $\mu_{s+1}^* = 2^{-2^{2^s} \cdot 2^{2^s}} = (\mu_s^*)^{2^{2^s}} < \mu_s^*$. Since $2^{2^s}/s \rightarrow \infty$, $2^{2^s} \geq Ks$ for $s \geq s_K$, and $C_K := \max_{s < s_K} 2^{Ks}$ works.

(d) $\mu_J^* \leq x^2$ is equivalent to $2^{2^J} \geq 2 \log_2(1/x)$, i.e. to $2^J \geq \log_2(2 \log_2(1/x)) =: Y$; note $Y \geq 1$ because $x \leq \frac{1}{2}$. If $J(x) = 1$ the claim reads $2 \leq 2Y$. If $J(x) \geq 2$, minimality gives $2^{J(x)-1} < Y$. $\square$

Remark 3.3 (Why a doubly exponential base). This remark is motivation only; nothing below depends on it. The infinite-support arguments of the later sections raise support coefficients only beyond a depth J and pay a Hilbert footprint of order μ_J^* , while the pinning constant of the coordinates left unraised is of order 2^J . A raise of footprint at most x^2 forces $J \geq J(x)$, and the arguments require $2^{J(x)} = o(\log(1/x))$ as $x \rightarrow 0$, i.e. $\log_2(1/\mu_J)/2^J \rightarrow \infty$ for the base entries μ_J . By Lemma 3.2(d) this holds for μ^* . It fails for $\mu_s = 2^{-s^2-1}$ (then $2^{J(x)}$ is of order $2^{\sqrt{2 \log_2(1/x)}}$) and for $\mu_s = 2^{-2^s}$ (then $2^{J(x)}$ is of order $\log_2(1/x)$). Faster than exponential decay of the base entries is also what makes the raise-room condition of the earlier support swallowing arguments hold for every power profile $|a_s| \approx v_l(s)^{1+b}$; with entries 2^{-s} it fails for every $b \geq 2$.

3.2. The recursion.

Definition 3.4 (Fixed data). (i) *Signature sets*. $j_0 := 1$ and $S_l := \{2^l(2i+1) : i \geq 1\}$ ($l \geq 1$). The sets S_l are pairwise disjoint, infinite, consist of even numbers (so $j_0 \notin S_l$), have *bounded gaps* $G_l := 2^{l+1}$ (consecutive elements differ by G_l), and $\min S_l = 3 \cdot 2^l$. Put $Z_0 := \mathbb{N} \setminus \bigcup_l S_l$, the set of odd numbers together with the powers 2^r , $r \geq 1$.

(ii) *Signatures*. $h_l := \sum_{s \in S_l} 2^{-s} e_s^*$, $H_l := \|h_l\|_1 = 2^{-3 \cdot 2^l} / (1 - 2^{-2^{l+1}})$ and $\delta_l^{\max} := \min\{2^{-l}, (4(1 + \|U\|)H_l)^{-1}\}$.

(iii) *Tolerance sequence*. $u_l := 2^{-l}$.

(iv) *Targets*. A sequence $(y^{(i)})_{i \geq 1}$ in $c_{00} \cap S_{q^*}$ which is norm dense in S_{q^*} , with $y^{(1)} := e_1^*/q^*(e_1^*)$, and which contains $y^* := (e_3^* - e_5^*)/q^*(e_3^* - e_5^*)$ and $y^{**} := (e_7^* - e_9^*)/q^*(e_7^* - e_9^*)$.

(v) *Schedule*. $K_{\text{FD}} := \{2^r : r \geq 1\}$; write $\mathbb{N} \setminus K_{\text{FD}} = \{k_1 < k_2 < \dots\}$; $(i_r)_{r \geq 1}$ is a sequence in which every positive integer occurs infinitely often.

Definition 3.5 (The operator T_{final}). The following quantities are defined by recursion on the stage $l = 1, 2, \dots$; at stage l all quantities of the stages $l' < l$ are known, and the steps are carried out in the order listed.

(1) *Target*. If $k(l) \in K_{\text{FD}}$, let j_l be the least odd integer exceeding $\max(\{9\} \cup \bigcup_{l' < l} \text{supp } y_{l'})$ and $y_l := e_{j_l}^*/q^*(e_{j_l}^*)$ (a *fresh-dominated target*). Otherwise $k(l) = k_r$ for a unique r ; call $y \in c_{00}$ *allowed at l* if

(a) $\text{supp } y \cap S_{l'} = \emptyset$ for every $l' \geq l$, and

(c) $\text{supp } y \cap S_{l'} \subseteq [1, l]$ for every $l' < l$,

and put $y_l := y^{(i_r)}$ if $y^{(i_r)}$ is allowed at l , and $y_l := y^{(1)}$ otherwise. (Every $y \in c_{00}$ supported in Z_0 is allowed at every l .)

(2) Signature size. $\delta_1 := \delta_1^{\max}$. For $l \geq 2$ put $g_l := \delta_l^{\max} H_l / (32 \cdot 3^{|\text{supp } y_l|})$ and let δ_l be the largest element of the set A_l of all $\delta \in [\frac{1}{2}\delta_l^{\max}, \delta_l^{\max}]$ such that

$$(GM) \quad \left| \sum_{j \in \text{supp } y_l} \sigma_j y_l(j) + \epsilon \delta H_l \right| \geq g_l \quad \text{for all } \sigma \in \{-1, 0, 1\}^{\text{supp } y_l}, \epsilon = \pm 1.$$

(3) Normalisation. $n_l := q^*(y_l + \delta_l h_l)$, $u_l := (y_l + \delta_l h_l)/n_l$, $v_l := \delta_l h_l/n_l$, $\delta_l^\circ := \delta_l H_l/n_l$, $\underline{y}_l := \min\{1, \min_{j \in \text{supp } y_l} |y_l(j)|\}$ and $\Lambda^\circ(l) := \prod_{l'' \leq l} (1 + 2/\delta_{l''}^\circ)$.

(4) Weight. $c_1 := 1$. For $l \geq 2$, c_l is the minimum of the following finitely many positive numbers:

- (W1) $c_{l-1}/4$;
- (W2) $b(l-1, M(l-1))^2$ (defined in step (6) of stage $l-1$);
- (W3) $(\delta_l H_l)^2$;
- (W4) $u_{l'} 2^{-2s} c_{l'} \delta_{l'}/2$, for every $l' < l$ and every $s \in \text{supp } y_l \cap S_{l'}$;
- (W5) $c_{l-1} \delta_l 2^{-\min S_{l-1} - l - 10} / (1 + \|U\|)$;
- (W6) $g_l 2^{m(l)+k(l)-l}$;
- (W7) $3 \cdot 2^{-10} u_{l-1} \lambda_{l-1} \delta_{l-1} 2^{-(m(l-1)+k(l-1)+\min S_{l-1})} \underline{y}_{l-1} / (n_{l-1}(1 + \|U\|))$.

Then $\Phi_l := 2^{-m(l)-k(l)} c_l$ and $\lambda_l := m(l) \Phi_l$.

(5) Level data.

$$\begin{aligned} T(l) &:= \bigcup_{l'' \leq l} \text{supp } y_{l''}, & s_{\max}(l) &:= \max(T(l) \cup \{1\}), \\ s_b(l) &:= \max_{l'' \leq l} \min(S_{l''} \cap (s_{\max}(l), \infty)), & s_t(l) &:= s_b(l) + 2^{l+2}, \\ \mathbf{m}_{l''}^{\text{nat}}(l) &:= \|v_{l''} 1_{S_{l''} \setminus \langle [1, l] \cup T(l) \rangle}\|_1 \quad (l'' \leq l), \\ D_{\text{room}}(l) &:= 1 + \sum_{l'' \leq l} \left(\frac{1}{\mathbf{m}_{l''}^{\text{nat}}(l)} + \frac{1}{\Phi_{l''}} \right), \\ \delta_{\min}(l) &:= \min_{l'' \leq l} \delta_{l''}, & G(l) &:= 2^{l+1}, & B^*(l) &:= \frac{2^{s_t(l)}}{\delta_{\min}(l) (\mu_{s_t(l)}^*)^2}. \end{aligned}$$

Further, the design constants of level l of Definition 3.6 and the finite set $\mathcal{O}(l)$ of rate objects of level l of Definition 3.7 are formed, and $\omega(l) := |\mathcal{O}(l)|$. The *design factor* of level l is

$$\begin{aligned} \text{Des}(l) &:= \left[l 2^{l^3} B^*(l) 2^{s_{\max}(l)} 2^{G(l)} \delta_{\min}(l)^{-1} \Lambda^\circ(l) (1 + |T(l)|) \right. \\ &\quad \times D_{\text{room}}(l) H_{\text{conf}}(l) G^{**}(l) H_{\text{tune}}(l) \left. \right]^6 \\ &\quad \times \text{Lip}(l) \left(2 + \frac{1}{\delta_{\text{comb}}(l)} \right) (r(l) c'(l))^l C_*(l) C_L(l) N_L(l) \left(1 + \frac{1}{\delta_{\text{sh}}(l)} \right). \end{aligned}$$

(6) Sub-windows. $M(l) := \omega(l) + 1$. The *sub-windows of level l* are the pairs $w = (l, i)$, $1 \leq i \leq M(l)$; all sub-windows are ordered lexicographically, and w^- (resp. w^+) denotes the predecessor (resp. successor) of w . For $w = (l, 1), \dots, (l, M(l))$ in

this order:

$$\begin{aligned}
u(w) &:= \frac{1}{4} \quad (w = (1, 1)), \quad u(w) := b(w^-) \quad (w \neq (1, 1)), \\
Q(w) &:= \left(\frac{4 \operatorname{Des}(l)}{u(w)} \right)^{\omega(l)+20}, \\
n(w) &:= \lceil l 2^{l^3} Q(w) \rceil, \quad T_{\text{lo}}(w) := 2^{-n(w)} T_{\text{hi}}(w), \\
T_{\text{hi}}(w) &:= \min \left\{ T_{\text{lo}}(w^-), \frac{2^{-l^3}}{l Q(w)} \right\} \quad (T_{\text{lo}}((1, 1)^-) := 1), \\
\eta(w) &:= \frac{T_{\text{lo}}(w)^4}{l \operatorname{Des}(l)}, \quad b(w) := \frac{\min\{\eta(w), (\eta(w)/C_L(l))^{N_L(l)/2}\}}{1 + \operatorname{Lip}(l) C_*(l)}.
\end{aligned}$$

The interval $\mathcal{W}(w) := [T_{\text{lo}}(w), T_{\text{hi}}(w)]$, which contains the $n(w)$ dyadic scales $T_{\text{hi}}(w) 2^{1-i}$ ($1 \leq i \leq n(w)$), is the *sub-window* w , and $\operatorname{Band}(w) := (b(w), u(w))$ is its *band*.

(D2) $T_{\text{final}} e_{k,m} := c_{j(k,m)} u_{j(k,m)}$, extended linearly and continuously to $\ell_1(\mathbb{N} \times \mathbb{N})$.

For $l = j(k, m)$ we write $u_{k,m} = u_l$, $\Phi_m(k) = \Phi_l$ and $\lambda_{k,m} = \lambda_l$, in accordance with [I, Section 1.2] (this is justified in Theorem 3.12). Note that $\Phi_l \leq c_l/4$ and $\lambda_l = m(l) 2^{-m(l)-k(l)} c_l \leq c_l/4$, because $m 2^{-m} \leq \frac{1}{2}$ and $2^{-k} \leq \frac{1}{2}$.

3.3. Design constants and rate objects. In the next two definitions $l \geq 1$ is fixed and all data of the stages $\leq l$ up to step (5) of Definition 3.5 are known ($y_{l''}, \delta_{l''}, n_{l''}, u_{l''}, c_{l''}, \Phi_{l''}$ for $l'' \leq l$, and $T(l)$, $s_{\max}(l)$, $s_t(l)$, $\mathbf{m}_{l''}^{\text{nat}}(l)$, $D_{\text{room}}(l)$, $\Lambda^\circ(l)$). A number is called a *design number of level l* if it is determined by these data and by finitely many combinatorial choices, and does not depend on N .

Definition 3.6 (Patterns and design constants). (a) *Patterns*. A *pattern of level l* is a tuple $\kappa = (E, P, \epsilon, F', \text{type})$ with $E \subseteq [1, l]$, $P \subseteq E$, $\epsilon \in \{\pm 1\}^E$, $F' \subseteq T(l)$ and $\text{type} : T(l) \setminus F' \rightarrow \{1, -1, 0\}$; its *kept set* is $\operatorname{Kp}(\kappa) := E \setminus P$. With $L_j(\tau) := \sum_{l'' \in E} \epsilon_{l''} \tau_{l''} u_{l''}(j)$, its *zero-cost cone* is

$$\begin{aligned}
C(\kappa) &:= \{\tau \in \mathbb{R}^E : \tau_{l''} \geq 0 \ (l'' \in \operatorname{Kp}(\kappa)), \tau_{l''} = 0 \ (l'' \in P), \\
&\quad \text{type}(j) L_j(\tau) \geq 0 \ (\text{type}(j) = \pm 1), L_j(\tau) = 0 \ (\text{type}(j) = 0)\}.
\end{aligned}$$

$C(\kappa) \subseteq [0, \infty)^E$ is a pointed polyhedral cone; $\operatorname{Ext}(\kappa)$ denotes the finite set of its extreme rays normalised by $\|r\|_1 = 1$. A *component* of κ is a pair (r, m) with $r \in \operatorname{Ext}(\kappa)$ and $m \in \{m(l'') : l'' \in \operatorname{supp} r\}$.

(b) *Tuning constant*. For a set Σ of components of a pattern κ let R_Σ be the matrix with one row $(r(l'') 1_{m(l'')=m})_{l'' \leq l}$ per component $(r, m) \in \Sigma$, and R_Σ^+ its Moore–Penrose inverse. $H_{\text{tune}}(l) := \max\{1, \max_{\kappa, \Sigma} (\#\Sigma)^{1/2} \|R_\Sigma^+\|_2\}$.

(c) *Determinantal data*. For a pattern κ and $v \in \mathbb{R}^{[1, l]}$ let $A_\kappa(v)$ be the matrix with columns indexed by E and the following rows: the unit rows $e_{l''}$ ($l'' \in E$); the *coordinate rows* $(\epsilon_{l''} u_{l''}(j))_{l'' \in E}$ for $j \in T(l) \setminus F'$; and, for every block $m \in \{m(l'') : l'' \in E\}$, the *value row* $(v_{l''} 1_{[l'' \in \operatorname{Kp}(\kappa), m(l'') = m]})_{l'' \in E}$. For row and column index sets J, K with $|J| = |K| \geq 1$ put $\pi_{\kappa, J, K}(v) := \det A_\kappa(v)_{J, K}$, a polynomial in v whose coefficients are design numbers. Let $\delta_{\text{comb}}(l)$ be the least nonzero value of $|\pi_{\kappa, J, K}|$ over all (κ, J, K) with J free of value rows ($\delta_{\text{comb}}(l) := 1$ if there is none); let $\operatorname{Lip}(l) \geq 1$ be a common Lipschitz constant, with respect to the maximum norm $\|\cdot\|_\infty$ on $\mathbb{R}^{[1, l]}$, on $[-2, 2]^{[1, l]}$ of all $\pi_{\kappa, J, K}$ whose J contains a value row (polynomials are Lipschitz for every norm on a compact cube, so such a constant exists); and

let $r(l)$, $c'(l)$ be the maximal numbers of rows and of columns of the matrices A_κ . Finally let $N_L(l) \geq 1$ be the least integer for which there is $C \geq 1$ with the following property, and $C_L(l) := 1 + \inf\{C\}$ for this integer: for every subfamily $\mathcal{S}$ of these polynomials and every $v \in [-2, 2]^{[1, l]}$ with $\max_{\pi \in \mathcal{S}} |\pi(v)| \leq \beta \leq 1$, either $\beta \geq 1/C$ or there is v' with $\pi(v') = 0$ for all $\pi \in \mathcal{S}$ and $\|v' - v\|_2 \leq C\beta^{2/N_L(l)}$. Such integers exist by the Łojasiewicz inequality [1, Corollary 2.6.7], applied to the finitely many sums of squares $\sum_{\pi \in \mathcal{S}} \pi^2$ and the distance functions to their zero sets on the compact set $[-2, 2]^{[1, l]}$; and $1 + \inf\{C\}$ is admissible because admissibility is preserved under enlarging C . (If the common zero set $Z_{\mathcal{S}}$ of $\mathcal{S}$ is empty, then $\sum_{\pi \in \mathcal{S}} \pi^2 \geq c_{\mathcal{S}} > 0$ on the cube, and for C large only the alternative $\beta \geq 1/C$ can occur; otherwise the exponent $2/N_L$ comes from $\text{dist}(\cdot, Z_{\mathcal{S}})^N \leq C \sum_{\pi \in \mathcal{S}} \pi^2 \leq C|\mathcal{S}|\beta^2$.)

(d) *Configuration and shift constants.* Let $\mathcal{R}(l)$ be the finite set of pairs $(E, \mathbf{r})$ with $E \subseteq [1, l]$ and $\mathbf{r} \in \mathbb{R}^E$ either a unit vector $\pm e_{l''}$ ($l'' \in E$) or a coordinate row $(\epsilon_{l''} u_{l''}(j))_{l'' \in E}$ ($\epsilon \in \{\pm 1\}^E$, $j \in T(l)$). A *configuration system of level l* is a system of linear conditions $\langle \mathbf{r}, \tau \rangle \leq \beta_{\mathbf{r}}$ or $\langle \mathbf{r}, \tau \rangle = \beta_{\mathbf{r}}$ in $\tau \in \mathbb{R}^E$, each row $\mathbf{r}$ taken from $\{\mathbf{r} : (E, \mathbf{r}) \in \mathcal{R}(l)\}$ at most once; its (ℓ_1) -Hoffman constant is the least H such that $\text{dist}_1(\tau, \mathcal{P}) \leq H \cdot (\ell_1\text{-norm of the violations at } \tau)$ for all τ and all right-hand sides β for which the solution set $\mathcal{P}$ is nonempty; it is finite by Hoffman's theorem [3]. $H_{\text{conf}}(l)$ is the maximum of these constants over the finitely many configuration systems of level l (and 1). For a pattern κ of level l and box radii $\beta \in (0, \infty)^E$ put $Z_\kappa(\beta) := \{\tau \in C(\kappa) : \tau_{l''} \leq \beta_{l''} \text{ } (l'' \in \text{Kp}(\kappa))\}$. The system of linear conditions defining $Z_\kappa(\beta)$ is a configuration system of level l ; let $G^{**}(l)$ be the maximum over all patterns κ of level l of its ℓ_1 -Hoffman constant (and 1). It does not depend on β , and $G^{**}(l) \leq H_{\text{conf}}(l)$ (see also Definition 4.11). Further $C_*(l) := |T(l)| + 2$, and $\delta_{\text{sh}}(l)$ is the number defined in (e) below.

(e) *Combinatorial shift cost.* Let $\kappa^{\text{sh}} = (\kappa, N_0, R_7, I_{\text{sh}}, \text{src}, G_{\text{pk}})$ be an enlarged shift pattern of level l (the index set of (O9) in Definition 3.7 below), $\kappa = (E, P, \epsilon, F', \text{type})$, and let $\text{vs} \in \{\pm 1\}^{G_{\text{pk}}}$. Put $\Pi_m := \sum_{l'' \in G_{\text{pk}}, m(l'')=m} \text{vs}_{l''} \lambda_{l''} u_{l''}$ for blocks $m \in I_{\text{sh}}$, $G_{\text{np}} := E \setminus G_{\text{pk}}$, and, for $\delta \in \mathbb{R}^{I_{\text{sh}}}$ and $x \in [0, \infty)^{G_{\text{np}}}$, $L := \sum_m \delta_m \Pi_m + \sum_{l'' \in G_{\text{np}}} x_{l''} \epsilon_{l''} u_{l''}$. Let

$$c^{\text{comb}}(\delta; \kappa^{\text{sh}}, \text{vs}) := \inf_{x \geq 0} \left[\sum_{j \in T(l) \setminus F', \text{type}(j)=\pm 1} 2(\text{type}(j)L(j)) - \sum_{j \in T(l) \setminus F', \text{type}(j)=0} |L(j)| + \sum_{l'' \in E} 2\mathbf{m}_{l''}^{\text{nat}}(l)(\epsilon_{l''} \text{coef}_{l''})_- \right],$$

where $\text{coef}_{l''}$ is the coefficient of $v_{l''}$ in L on $S_{l''} \setminus ([1, l] \cup T(l))$, namely $\epsilon_{l''} x_{l''}$ for $l'' \in G_{\text{np}}$ and $\text{vs}_{l''} \lambda_{l''} \delta_{m(l'')}$ for $l'' \in G_{\text{pk}}$ (on this set every $u_{l''}$ with $l''' \leq l$, $l''' \neq l''$, vanishes, by the disjointness of the signature sets and Definition 3.5(1)). Let $c_*^{\text{comb}}(\kappa^{\text{sh}}, \text{vs})$ be the minimum of $c^{\text{comb}}(\cdot; \kappa^{\text{sh}}, \text{vs})$ over the compact set $\{\delta : \|\delta\|_1 = 1, \delta_m \leq 0 \text{ if } \text{src}(m) \text{ is } \leq, \delta_m \geq 0 \text{ if it is } \geq\}$ (it is attained: c^{comb} is the minimum of a polyhedral function of (δ, x) with design coefficients, and it is continuous in δ). Then $\delta_{\text{sh}}(l)$ is the least positive value of $c_*^{\text{comb}}(\kappa^{\text{sh}}, \text{vs})$ over all pairs $(\kappa^{\text{sh}}, \text{vs})$ of level l , and $\delta_{\text{sh}}(l) := 1$ if there is none. Since there are finitely many pairs and the coefficients involve only $u_{l''}$ ($l'' \leq l$), $\lambda_{l''}$ and $\mathbf{m}_{l''}^{\text{nat}}(l)$, $\delta_{\text{sh}}(l)$ is a design number of level l .

Only the following properties of the constants of Definition 3.6 are used in this section; their precise role is explained where they are used.

- (K1) Each of $H_{\text{tune}}, \text{Lip}, r, c', C_L, N_L, H_{\text{conf}}, G^{**}, C_*$ is a real number ≥ 1 , and $\delta_{\text{comb}}(l), \delta_{\text{sh}}(l) \in (0, \infty)$.
- (K2) Each is a design number of level l : the patterns, cones, extreme rays and polynomials depend only on the finitely many numbers $u_{l''}(j)$ ($l'' \leq l$, $j \in T(l)$), on the blocks $m(l'')$ ($l'' \leq l$) and on finitely many sign, contact and status choices; the Hoffman constants are finite (Hoffman's theorem [3]), and do not depend on right-hand sides.
- (K3) None of them depends on N : blocks enter only as the blocks $m(l'')$ of carriers $l'' \leq l$.

Definition 3.7 (Rate objects). Let $N \geq 1$ and $f \in S_{p_N^*}$ with forced data $(\xi, q_0, a, w, F, z, \hat{z}, e, \nu)$ (F finite or infinite). For a carrier $l'' = j(k, m)$ with $m \leq N$ put $\varrho_{l''}(f) := |u_{l''}(\xi)|/(\Phi_{l''}\vartheta_m)$, and, given a sign $\epsilon_{l''}$, $\text{val}_{l''}(f) := \epsilon_{l''}u_{l''}(\hat{z})$; put $\gamma_k(f) := \langle U^*u_k, U^*a \rangle$. The *rate objects of level l* are the following indexed families; the *value* of each object at f is a number in $[0, \infty]$. *Convention*: an object that involves a carrier l'' with $m(l'') > N$ has the value 1.

- (O1) *Rooms*, indexed by $l'' \leq l$: with $S^{\text{nat}} := S_{l''} \setminus (F \cup T(l))$, the value $\min_{\sigma=\pm 1} \sum_{s \in S^{\text{nat}}} v_{l''}(s)(1 + \sigma z_s) / \|v_{l''}1_{S^{\text{nat}}}\|_1$ ($:= 1$ if $S^{\text{nat}} = \emptyset$).
- (O2) *Target rooms*, indexed by $j \in T(l)$: $1 - |z_j|$ ($:= 1$ if $j \in F$).
- (O3) *Threshold distances*, indexed by $l'' \leq l$: $|\varrho_{l''}(f) - 1|$.
- (O4) *Relative positions*, indexed by $l'' \leq l$: $\varrho_{l''}(f)$.
- (O5) *Ray components*, indexed by (κ, r, m) , (r, m) a component of a pattern κ : $|D_{r,m}(f)|/\Phi_{\max}(r, m)$, where $D_{r,m}(f) := \sum_{l'' \in \text{supp } r, m(l'')=m} r(l'')\epsilon_{l''}u_{l''}(\hat{z})$ and $\Phi_{\max}(r, m) := \max\{\Phi_{l''} : l'' \in \text{supp } r, m(l'') = m\}$ (the signs ϵ are those of κ).
- (O6) *Shift costs*, indexed by the shift patterns $\pi = (I_{\text{up}}, I_{\text{lo}}, P^*, \text{vs}, B_f, \epsilon')$ of level l : $I_{\text{up}}, I_{\text{lo}}$ disjoint sets of blocks of carriers $\leq l$, $P^* \subseteq [1, l]$ with $m(P^*) \subseteq I_{\text{up}} \cup I_{\text{lo}}$, $\text{vs} \in \{\pm 1\}^{P^*}$, $B_f \subseteq [1, l] \setminus P^*$, $\epsilon' \in \{\pm 1\}^{B_f}$. With $\Pi_m := \sum_{l'' \in P^*, m(l'')=m} \text{vs}_{l''}\lambda_{l''}u_{l''}1_{F^c}$ and $\phi_z(x) = |x| - zx$,

$$c(\delta; \pi) := \inf_{x \in [0, \infty)^{B_f}} \sum_{j \notin F} \phi_{z_j} \left(\sum_m \delta_m \Pi_m(j) + \sum_{l'' \in B_f} x_{l''} \epsilon'_{l''} u_{l''}(j) \right),$$

and the value is $c_\pi(f) := \inf\{c(\delta; \pi) : \|\delta\|_1 = 1, \delta_m \geq 0 \text{ } (m \in I_{\text{up}}), \delta_m \leq 0 \text{ } (m \in I_{\text{lo}})\}$ ($:= \infty$ if $I_{\text{up}} \cup I_{\text{lo}} = \emptyset$).

- (O7) *Joint ray objects*, indexed by $(\kappa, \mathcal{J})$ with $\emptyset \neq \mathcal{J} \subseteq \text{Ext}(\kappa)$: the smallest singular value of the matrix with columns $(D_{r,m}(f)/\Phi_{\max}(r, m))_m$, $r \in \mathcal{J}$ (rows indexed by the blocks of carriers $\leq l$, entries 0 for blocks not meeting $\text{supp } r$; the smallest singular value is 0 if there are more columns than rows).
- (O8) *Minors*, indexed by (κ, J, K) as in Definition 3.6(c) with J containing a value row: $|\pi_{\kappa, J, K}((\text{val}_{l''}(f))_{l'' \in \text{KP}(\kappa)})|$.
- (O9) *Shift-pinning rates*, indexed by the enlarged shift patterns $\kappa^{\text{sh}} = (\kappa, N_0, R_7, I_{\text{sh}}, \text{src}, G_{\text{pk}})$: $\kappa = (E, P, \epsilon, F')$ a pattern, $N_0, R_7, G_{\text{pk}} \subseteq E$, I_{sh} a set of blocks of carriers $\leq l$, $\text{src} : I_{\text{sh}} \rightarrow \{\leq, \geq, \text{free}\}$. Let Σ^{sh} be the system in $(\delta, \tau) \in \mathbb{R}^{I(l)} \times \mathbb{R}^E$, $I(l) := \{m(l'') : l'' \leq l\}$: (S1) $\tau_{l''} \geq 0$ ($l'' \in E$); (S2) $\tau_{l''} = \epsilon_{l''} \text{vs}_{l''} \lambda_{l''} \delta_{m(l'')} \text{ } (l'' \in G_{\text{pk}})$, $\text{vs}_{l''} := \text{sign } w_{m(l'')}(k(l''))$; (S3) $\text{type}(j)V_j(\tau) \geq 0$ if $\text{type}(j) = \pm 1$ and $V_j(\tau) = 0$ if $\text{type}(j) = 0$, $j \in T(l) \setminus F'$,

where $V_j(\tau) := \sum_{l'' \in E} \epsilon_{l''} \tau_{l''} u_{l''}(j)$; (S4) $\delta_m + \sum_{l'' \in E, m(l'')=m} q'_{l''} \tau_{l''} = 0$ ($m \in I(l)$), $q'_{l''} := \epsilon_{l''} \Phi_{l''} w_m(k(l'')) / (m C_m)$ for $l'' \notin N_0$ and $q'_{l''} := 0$ for $l'' \in N_0$; (S5) $\delta_m = 0$ ($m \notin I_{\text{sh}}$); (S6) $\delta_m \leq 0$, resp. $\delta_m \geq 0$, if $\text{src}(m)$ is $\leq$, resp. $\geq$; (S7) $\tau_{l''} + \lambda_{l''}(|w_m(k(l''))|/M_m) \delta_{m(l'')} \leq 0$ ($l'' \in R_7$). With $\text{viol}(\delta, \tau)$ the sum of the positive parts of the violations of the inequalities and of the moduli of the defects of the equations, the value is $\rho^{\text{sh}}(\kappa^{\text{sh}}, f) := \inf\{\text{viol}(\delta, \tau) : \|\delta\|_1 = 1, \tau \in \mathbb{R}^E\}$.

(O10) *Support moduli*, indexed by $s \in [1, s_t(l)]$: $|a_s|$.

(O11) *Spreads*, indexed by (l'', s, s') , $l'' \leq l$, $s < s'$ among the first three elements of $S_{l''} \cap (s_{\max}(l), \infty)$: $|1 - \mathbf{a}_{s'}/\mathbf{a}_s|$ with $\mathbf{a}_s := a_s/v_{l''}(s)$ if $a_s \neq 0 \neq a_{s'}$, and 1 otherwise.

(O12) *Resource minors*, indexed by (L, K) with $\emptyset \neq L \subseteq [1, l]$, $K \subseteq [1, s_t(l)]$, $|K| = |L|$: $|\det(\Gamma(f)_{L,K})|$, where $\Gamma(f)_{k,s} := u_k(s) - \gamma_k(f) a_s/\nu^2$.

$\mathcal{O}(l)$ is the disjoint union of the index sets of (O1)–(O12), and $\omega(l) := |\mathcal{O}(l)|$.

Lemma 3.8 (The rate scheme is a design datum). *For every l , $\mathcal{O}(l)$ is a finite set which is determined by the data of the stages $\leq l$ up to step (5), and does not depend on N or on f ; $\omega(l) \geq 3l + |T(l)|$. For every N and every $f \in S_{p_N^*}$, each object of $\mathcal{O}(l)$ has a well-defined value in $[0, \infty]$.*

Proof. The index sets of (O1)–(O4) are $[1, l]$, $T(l)$, $[1, l]$, $[1, l]$. Patterns, shift patterns and enlarged shift patterns of level l are tuples of subsets of $[1, l]$, of $T(l)$ and of the set of blocks $\{m(l'') : l'' \leq l\}$, together with signs and types: finitely many. $\text{Ext}(\kappa)$ is finite (Minkowski–Weyl), and the row and column index sets of A_κ are finite. The index sets of (O10)–(O12) are subsets of $[1, s_t(l)]$ and of $[1, l]$ and triples of coordinates of the sets $S_{l''}$ determined by $s_{\max}(l)$. None of these involves N or f . The bound on $\omega(l)$ counts (O1)–(O4). The values: $\varrho_{l''}$ is defined because $\Phi_{l''} > 0$ and $\vartheta_m > 0$ ($M_m, C_m > 0$ by [I, Lemma 1.7]); the denominator in (O1) is positive when $S^{\text{nat}} \neq \emptyset$; the series in (O6) converge absolutely because $\sum_l \lambda_l \|u_l\|_1 < \infty$ and $|\phi_z(x)| \leq 2|x|$; $\nu > 0$ by [I, Proposition 1.9(c)]. All other values are finite minima, infima or determinants. $\square$

Remark 3.9 (Interpretation of (O10)–(O12)). The families (O1)–(O9) are the rate objects of the finite-support theory; (O10)–(O12) are needed only when F is infinite. If $s \in S_{l''} \cap F$ lies beyond $s_{\max}(l)$, a change of a_s by Δ (with z fixed) changes val_k at first order by $((\mu_s^*)^2/\nu) \epsilon_k \Gamma(f)_{k,s} \Delta$, and for $s \notin F$ the column $\Gamma(f)_{\cdot, s} = (u_k(s))_k$ is the column of a private bank. So the minors (O12) control the regularity of the first-order tuning resources at depth $\leq s_t(l)$. In the proportional case $a = c_k v_k$ on chosen coordinates $s_k \in S_k$ beyond $s_{\max}(l)$ ($k \in L$) one has $\Gamma_{k', s_k} = v_k(s_k)(1[k' = k] - \gamma_{k'} c_k/\nu^2)$ for $k' \in L$, hence $\det \Gamma_{L, \{s_k\}} = \prod_{k \in L} v_k(s_k) (1 - \sum_{k \in L} c_k \gamma_k/\nu^2)$, the Sherman–Morrison denominator. These statements are used and proved where needed; here only the finiteness of the families matters.

3.4. Well-foundedness and admissibility.

Lemma 3.10 (The recursion is well founded). (a) *Every quantity of stage l in Definition 3.5 is a function of the fixed data of Definition 3.4, of quantities of the stages $l' < l$, and of quantities of earlier steps of stage l .*

(b) *All these quantities are finite; A_l is a nonempty compact set, so δ_l exists, and $\delta_l \in [\frac{1}{2}\delta_l^{\max}, \delta_l^{\max}]$; $n_l \in [\frac{3}{4}, \frac{5}{4}]$; $c_l, \mathbf{m}_{l''}^{\text{nat}}(l), g_l, u(w), Q(w), T_{\text{hi}}(w), T_{\text{lo}}(w), \eta(w), b(w)$ are positive; $D_{\text{room}}(l), B^*(l), \text{Des}(l) \in [1, \infty)$.*

(c) *No quantity depends on N .*

Proof. (a) Step (1) uses $\text{supp } y_{l'}$ ($l' < l$), the fixed sets $S_{l'}$ and the schedule; allowedness consists of the support conditions (a), (c) only. Step (2) uses y_l , H_l and $\delta_l^{\max}$. Step (3) uses y_l and δ_l . Step (4) uses $c_{l'}$, $\delta_{l'}$, $\lambda_{l'}$, $n_{l'}$, $\underline{y}_{l'}$ ($l' < l$), the number $b(l-1, M(l-1))$ from step (6) of stage $l-1$, and δ_l , H_l , g_l , $\text{supp } y_l$ from steps (1)–(2). Step (5) uses data of index $\leq l$, including c_l (through $\Phi_{l''}$ in $D_{\text{room}}(l)$); the design constants and the rate objects are design numbers of level l (properties (K2) and Lemma 3.8). Step (6) uses $\text{Des}(l)$, $\omega(l)$, $C_L(l)$, $N_L(l)$, $\text{Lip}(l)$, $C_*(l)$ and the values $T_{\text{lo}}(w^-)$, $b(w^-)$ of the predecessor. Note that δ_l must be chosen after y_l , since both g_l and the inequalities (GM) involve y_l , and that c_l must be chosen after δ_l ; the order of the steps is the only consistent one.

(b) For fixed (σ, ϵ) the set of δ violating (GM) is an open interval of length $2g_l/H_l$; there are $2 \cdot 3^{|\text{supp } y_l|}$ pairs, so the excluded set is open of total length at most $4 \cdot 3^{|\text{supp } y_l|} g_l/H_l = \delta_l^{\max}/8$. Hence A_l is closed, bounded and of measure at least $\frac{1}{2}\delta_l^{\max} - \frac{1}{8}\delta_l^{\max} > 0$, and its largest element exists. Next, $q^*(y_l) = 1$ and $q^*(\delta_l h_l) \leq (1 + \|U\|)\delta_l H_l \leq \frac{1}{4}$ because $\delta_l \leq \delta_l^{\max}$; so $n_l \in [\frac{3}{4}, \frac{5}{4}]$. The numbers in (W1)–(W7) are positive: (W2) by induction, (W3), (W5) because $\delta_l, H_l > 0$, (W4) is a finite list ($\text{supp } y_l$ is finite), (W6) because $g_l > 0$, (W7) because $\underline{y}_{l-1} > 0$ and $\lambda_{l-1} > 0$; so $c_l > 0$, and $\Phi_l, \lambda_l > 0$. The set $S_{l''} \setminus ([1, l] \cup T(l))$ is infinite and $v_{l''} > 0$ on $S_{l''}$, so $m_{l''}^{\text{nat}}(l) > 0$. All factors of $\text{Des}(l)$ are finite and ≥ 1 by (K1) (note $\delta_{\min}(l) \leq 1$, $\mu_{s_t(l)}^* < 1$). Inductively along the sub-windows, $u(w) > 0$, hence $Q(w)$, $n(w) < \infty$, and $T_{\text{hi}}(w)$, $T_{\text{lo}}(w)$, $\eta(w)$, $b(w) > 0$.

(c) The fixed data and steps (1)–(4) do not involve N ; the design constants and the rate scheme are N -free by (K3) and Lemma 3.8; steps (5) and (6) are built from these. $\square$

Remark 3.11 (Two corrections of the literal union of earlier designs). (a) If the condition (GM) together with $\Phi_l \leq g_l 2^{-l}$ were imposed also at $l = 1$, it would contradict $c_1 = 1$: $S_1 = \{6, 10, 14, \dots\}$ gives $H_1 = 2^{-6}/(1 - 2^{-4}) = \frac{1}{60}$, and $\delta_1^{\max} = \frac{1}{2}$ (because $4(1 + \|U\|)H_1 < 2$); (GM) with $\sigma = 0$ requires $g_1 \leq \delta_1 H_1 \leq \frac{1}{120}$, and then $\Phi_1 = 2^{-m(1)-k(1)}c_1 \leq g_1/2$ forces $c_1 \leq 2^{m(1)+k(1)}/240 = 2^{m(1)+1}/240$ (recall $k(1) = 1$), e.g. $c_1 \leq \frac{1}{60}$ if $j(1, 1) = 1$. Since $c_l \leq c_1/4$ for $l \geq 2$, this gives $\|T\| = c_1 < 1$ whenever $m(1) \leq 6$, in conflict with the equality clause of (T-a) and with “norm one” in Martín’s Lemma B. This is why (GM) and (W6) are imposed only for $l \geq 2$; their only use concerns carriers of large index.

(b) A bank factor calibrated to the base 2^{-j} (a factor $8^{s_b(l)}$ in place of $B^*(l)$) does not dominate the bank efficiency $(\mu_{s'}^*)^2 v_c(s')$ of a fast-decaying base: target supports may contain coordinates of Z_0 far beyond everything fixed at earlier stages, so that $s_b(l)$ is not controlled by the other factors of the design. (For the base 2^{-s^2-1} an explicit choice of pure fresh targets $e_{s_i}^*/q^*(e_{s_i}^*)$ at a sparse set of first schedulings makes the uncalibrated inequality fail at infinitely many levels. This is stated for orientation only and not used.) The factor $B^*(l)$ removes the problem, see Proposition 3.16.

Theorem 3.12 (Admissibility; Martín’s Lemma B for T_{final}). *Let $T = T_{\text{final}}$, $U = U_{\text{final}}$.*

- (a) *T is admissible:* (T-a) $q^*(Te_{k,m}) = c_{j(k,m)} \leq 1$, with equality at $(k,m) = j^{-1}(1)$; (T-b) *T is injective*; (T-c) $Y := T(\ell_1(\mathbb{N} \times \mathbb{N}))$ satisfies $Y \cap c_{00} = \{0\}$; (T-d) *for every m the set $\{u_{j(k,m)} : k \in \mathbb{N}\}$ is norm dense in S_{q^*} .*
- (b) (Conclusion of Martín's Lemma B.) $\|T\| = 1$, *T is injective*, *Y is a separable operator range which is dense in $(\ell_1, q^*) = (c_0, q)^*$* , $Y \cap \text{NA}((c_0, q), \mathbb{R}) = \{0\}$, *and for every m the normalised vectors $Te_{k,m}/q^*(Te_{k,m}) = u_{j(k,m)}$, $k \in \mathbb{N}$, are dense in S_{q^*} , hence in the unit sphere of Y.*
- (c) *For $l = j(k,m)$: $u_{k,m} = u_l$, $\Phi_m(k) = \Phi_l$, $\lambda_{k,m} = \lambda_l$. The pair (y_l, c_l) satisfies allowedness (a) and (b) of [I, Definition 8.1]: $\text{supp } y_l \cap S_{l'} = \emptyset$ for $l' \geq l$, and $2c_l \leq c_{l'} 2^{-2s} \delta_{l'} \leq 2^{-2s} c_{l'} \delta_{l'}$ for $l' < l$ and $s \in \text{supp } y_l \cap S_{l'}$.*
- (d) (P1) $y_l \in c_{00}$, $\|y_l\|_1 \leq 1$, $n_l \in [\frac{3}{4}, \frac{5}{4}]$, $\text{supp } y_l \cap S_{l'} = \emptyset$ ($l' \geq l$); for $s \in S_l$: $u_l(s) = v_l(s) = \delta_l 2^{-s}/n_l$, $u_{l'}(s) = 0$ ($l' < l$) and $u_{l'}(s) = y_{l'}(s)/n_{l'}$ ($l' > l$).
(P2) $c_{l+1} \leq c_l/4$, $\sum_{l' > l} c_{l'} \leq \frac{4}{3} c_{l+1}$, $\sum_{l' > l} \lambda_{l'} \leq \frac{1}{3} c_{l+1} \leq \frac{1}{3} b(l, M(l))^2$.

Proof. (c), (d). $q^*(Te_{k,m}) = c_l q^*(u_l) = c_l$, so $\Phi_m(k) = 2^{-m-k} q^*(Te_{k,m}) = \Phi_l$, $\lambda_{k,m} = m \Phi_m(k) = \lambda_l$, and $u_{k,m} = Te_{k,m}/q^*(Te_{k,m}) = u_l$. The support statement holds for targets chosen by allowedness (a) and for targets supported in Z_0 (fresh-dominated targets and $y^{(1)}$). The weight inequality is (W4), since $c_{l'} \leq 1$. (P1): $\|y_l\|_1 \leq q^*(y_l) = 1$ and $n_l \in [\frac{3}{4}, \frac{5}{4}]$ (Lemma 3.10(b)); for $s \in S_l$ the signatures $h_{l'}$, $l' \neq l$, vanish at s (disjointness), and $y_{l'}(s) = 0$ for $l' \leq l$ by the support statement. (P2): (W1) gives $c_{l+1} \leq c_l/4$, hence $\sum_{l' > l} c_{l'} \leq c_{l+1} \sum_{i \geq 0} 4^{-i} = \frac{4}{3} c_{l+1}$, and $\lambda_{l'} \leq c_{l'}/4$ gives the bound on $\sum \lambda$; finally $c_{l+1} \leq b(l, M(l))^2$ by (W2).

(T-a): $c_l \leq c_1 = 1$ by (W1), and $q^*(Te_{j^{-1}(1)}) = c_1 = 1$.

(T-b), (T-c). Let $x \in \ell_1(\mathbb{N} \times \mathbb{N})$, indexed by l , with $\mathcal{Z} := Tx = \sum_l x_l c_l u_l \in c_{00}$, and suppose $x_{l'} \neq 0$. For $s \in S_{l'}$, by (P1),

$$\mathcal{Z}(s) = \frac{x_{l'} c_{l'} \delta_{l'} 2^{-s}}{n_{l'}} + \sum_{l > l', s \in \text{supp } y_l} \frac{x_l c_l y_l(s)}{n_l}.$$

If the sum is not empty, let $L(s)$ be its least index; by (c) at stage $L(s)$, $2c_{L(s)} \leq 2^{-2s} c_{l'} \delta_{l'}$, and since $|y_l(s)|/n_l \leq \frac{4}{3}$ the sum is at most $\frac{4}{3} \|x\|_\infty \sum_{l \geq L(s)} c_l \leq \frac{16}{9} \|x\|_\infty c_{L(s)} \leq \frac{8}{9} \|x\|_\infty 2^{-2s} c_{l'} \delta_{l'}$. Hence $|\mathcal{Z}(s)| \geq c_{l'} \delta_{l'} 2^{-s} (|x_{l'}/n_{l'} - \frac{8}{9} \|x\|_\infty 2^{-s}) > 0$ for all large $s \in S_{l'}$, which contradicts $\mathcal{Z} \in c_{00}$. So $x = 0$. This proves both (T-b) and (T-c).

(T-d). Fix m and i . The finite set $\text{supp } y^{(i)}$ meets only finitely many $S_{l'}$, so $y^{(i)}$ satisfies (a) and (c) at every $l \geq l_i := 1 + \max(\text{supp } y^{(i)} \cup \{l' : S_{l'} \cap \text{supp } y^{(i)} \neq \emptyset\})$. There are infinitely many r with $i_r = i$, and for them $l = j(k_r, m) \geq k_r \rightarrow \infty$; for all such $l \geq l_i$ we have $k(l) = k_r \notin K_{\text{FD}}$ and $y_l = y^{(i)}$. Along these l , $|n_l - 1| \leq q^*(\delta_l h_l) \leq \frac{5}{4} 2^{-l} H_l \rightarrow 0$ and $q^*(u_l - y^{(i)}) \leq |1/n_l - 1| + q^*(\delta_l h_l)/n_l \rightarrow 0$. Hence every $y^{(i)}$ is a limit of elements of $\{u_{j(k,m)} : k \in \mathbb{N}\}$, and the targets are dense in S_{q^*} .

(b). On $\ell_1(\mathbb{N} \times \mathbb{N})$, $\|T\| = \sup_{(k,m)} q^*(Te_{k,m}) = c_1 = 1$. Y is the range of a bounded operator on a separable Banach space, i.e. a separable operator range. The closure of the linear space Y contains the dense subset $\{u_{j(k,1)}\}$ of S_{q^*} , hence the unit ball, so Y is dense. $\text{NA}((c_0, q), \mathbb{R}) = c_{00}$ (Lemma 3.2(a)), so $Y \cap \text{NA}((c_0, q), \mathbb{R}) = \{0\}$ is (T-c). The last statement is (T-d), since $u_{j(k,m)} \in Y$. $\square$

Remark 3.13 (Blocks and Martín's construction). At the stages $l = j(2^r, m)$, $r \geq 1$, the target is a fresh-dominated target which differs from block to block, so the literal

requirement “every block follows the same target sequence” of [I, Remark 1.2] holds only at the stages with $k(l) \notin K_{\text{FD}}$, where $y_{j(k_r, m)} = y^{(i_r)}$ whenever this target is allowed. This is harmless: Martín’s proof, and everything below, uses only the conclusion of Lemma B, i.e. (T-a)–(T-d) [I, Remark 1.2]. Since T_{final} does not depend on N , [I, Lemma 1.5] applies to it: density of $\text{NA}((c_0, p_N), \mathcal{F})$ for infinitely many N implies density of $\text{NA}((c_0, p), \mathcal{F})$ for Martín’s norm p built with T_{final} .

3.5. Window arithmetic and design features.

Proposition 3.14 (Sub-windows). *For every sub-window $w = (l, i)$ of T_{final} :*

- (a) $u(w) \leq \frac{1}{4}$, $b(w) \leq 2^{-64/u(w)} < u(w)$ and $u(w^+) = b(w)$; hence b is strictly decreasing along the sub-windows and the bands $\text{Band}(w)$ of all sub-windows (of all levels) are pairwise disjoint.
- (b) (P3) on sub-windows. $l 2^{l^3} Q(w) T_{\text{hi}}(w) \leq 1$, $n(w) \geq l 2^{l^3} Q(w) \geq l 2^{l^3} \text{Des}(l) \geq l 2^{l^3} \Lambda^\circ(l)$, $T_{\text{hi}}(w) 2^{l^3} \Lambda^\circ(l) \leq 1/l$, and $T_{\text{hi}}(w^+) \leq T_{\text{lo}}(w)$ (also when w^+ is the first sub-window of level $l+1$).
- (c) (Fine weights.) $\sum_{l' > l} c_{l'} \leq \frac{4}{3} b(w)^2$ and $\sum_{l' > l} \lambda_{l'} \leq \frac{1}{3} b(w)^2 \leq T_{\text{lo}}(w)^8$.
- (d) With $\beta(w) := (1 + \text{Lip}(l) C_*(l)) b(w)$: $\beta(w) \leq \eta(w) < 1/\text{Des}(l) \leq 1/C_L(l)$ and $C_L(l) \beta(w)^{2/N_L(l)} \leq \eta(w)$.
- (e) $\text{Des}(l)^k T_{\text{lo}}(w) \leq u(w)/8$ for every $k \leq 40$.
- (f) $\log_2(1/T_{\text{hi}}(w)) \leq l^3 + l + Q(w)$, $\log_2(1/T_{\text{lo}}(w)) \leq 2n(w)$ and $\log_2(1/b(w)) \leq 11 N_L(l) n(w) \leq 11 \text{Des}(l) n(w)$.

Proof. (a) $u((1, 1)) = \frac{1}{4}$; otherwise $u(w) = b(w^-) \leq \eta(w^-) \leq T_{\text{lo}}(w^-)^4 \leq 2^{-4}$, because $b \leq \eta$ (the denominator of b is ≥ 1 and $\min\{\eta, \cdot\} \leq \eta$) and $T_{\text{lo}} \leq 2^{-n} \leq \frac{1}{2}$. Since $4/u(w) \geq 16$, $n(w) \geq Q(w) \geq (4/u(w))^{20} \geq 16/u(w)$, hence $b(w) \leq T_{\text{lo}}(w)^4 \leq 2^{-4n(w)} \leq 2^{-64/u(w)}$, and $2^{-64/y} < 1/y$ for $y = 1/u(w) \geq 4$. $u(w^+) = b(w)$ is the definition. So the endpoints form one decreasing sequence $u(w_1) > u(w_2) = b(w_1) > u(w_3) = b(w_2) > \dots$ along the lexicographic order, and $\text{Band}(w_j) = (u(w_{j+1}), u(w_j))$ are disjoint.

(b) The first two inequalities are the definitions of $T_{\text{hi}}(w)$ and $n(w)$. $Q(w) \geq \text{Des}(l)$ because $4/u(w) \geq 1$ and $\text{Des}(l) \geq 1$; $\text{Des}(l) \geq \Lambda^\circ(l)$ because $\Lambda^\circ(l)$ is a factor of the bracket and all factors are ≥ 1 . Then $T_{\text{hi}}(w) 2^{l^3} \Lambda^\circ(l) \leq \Lambda^\circ(l)/(lQ(w)) \leq 1/l$. $T_{\text{hi}}(w^+) \leq T_{\text{lo}}(w)$ is part of the definition of $T_{\text{hi}}(w^+)$.

(c) By Theorem 3.12(d), the sums are bounded by $\frac{4}{3} c_{l+1}$ and $\frac{1}{3} c_{l+1}$, and $c_{l+1} \leq b(l, M(l))^2 \leq b(w)^2$ by (W2) and (a). Finally $b(w)^2 \leq T_{\text{lo}}(w)^8$.

(d) $\beta(w) = \min\{\eta, (\eta/C_L)^{N_L/2}\} \leq \eta$, and $\beta^{2/N_L} \leq \eta/C_L$. $\eta(w) = T_{\text{lo}}(w)^4/(l\text{Des}(l)) < 1/\text{Des}(l)$ since $T_{\text{lo}}(w) < 1$, and $C_L(l) \leq \text{Des}(l)$ since $C_L(l)$ is one of the factors of $\text{Des}(l)$, all of which are ≥ 1 .

(e) $Q(w) \geq 16$, so

$$T_{\text{lo}}(w) \leq 2^{-n(w)} \leq 2^{-Q(w)} \leq Q(w)^{-2} = \left(\frac{u(w)}{4\text{Des}(l)} \right)^{2\omega(l)+40} \leq \left(\frac{u(w)}{4\text{Des}(l)} \right)^{40};$$

hence $\text{Des}(l)^k T_{\text{lo}}(w) \leq 4^{-40} u(w)^{40} \leq u(w)/8$ for $k \leq 40$.

(f) $\log_2(1/T_{\text{hi}}(w)) = \max\{\log_2(1/T_{\text{lo}}(w^-)), l^3 + \log_2(lQ(w))\}$. For $w \neq (1, 1)$, $u(w) = b(w^-) \leq T_{\text{lo}}(w^-)^4$, so $\log_2(1/T_{\text{lo}}(w^-)) \leq \frac{1}{4} \log_2(1/u(w)) \leq \log_2 Q(w)$; for $w = (1, 1)$ this term is 0. Hence $\log_2(1/T_{\text{hi}}(w)) \leq l^3 + \log_2 l + \log_2 Q(w) \leq l^3 + l + Q(w)$. Since $Q(w) \geq \text{Des}(l) \geq 2^{6l^3} \geq l^3 + l$ and $l 2^{l^3} \geq 2$, we get $n(w) \geq 2Q(w) \geq Q(w) + l^3 + l$, so $\log_2(1/T_{\text{lo}}(w)) = n(w) + \log_2(1/T_{\text{hi}}(w)) \leq 2n(w)$. Next

$\log_2(1/\eta(w)) = 4\log_2(1/T_{\text{io}}(w)) + \log_2(l\text{Des}(l)) \leq 9n(w)$. With $x := \log_2(1/\eta(w))$ and $c := \log_2 C_L(l) \leq \text{Des}(l) \leq n(w)$,

$$\begin{aligned} \log_2 \frac{1}{b(w)} &= \log_2(1 + \text{Lip}(l)C_*(l)) + \max \left\{ x, \frac{N_L(l)}{2}(x + c) \right\} \\ &\leq n(w) + N_L(l)(9n(w) + n(w)), \end{aligned}$$

because $\log_2(1 + \text{Lip}(l)C_*(l)) \leq \text{Des}(l) \leq n(w)$. So $\log_2(1/b(w)) \leq 11N_L(l)n(w)$, and $N_L(l) \leq \text{Des}(l)$. $\square$

Proposition 3.15 (Design features). *T_{final} and U_{final} have the following properties.*

- (a) (Bounded gaps, rule (c).) *Every S_l has gaps $G_l = 2^{l+1}$; every scheduled target y_l with $k(l) \notin K_{\text{FD}}$ satisfies $\text{supp } y_l \cap S_{l'} \subseteq [1, l]$ for $l' < l$; every $y \in c_{00}$ is allowed at all large l .*
- (b) $c_{l+1} \leq \min\{c_l/4, b(l, M(l))^2, (\delta_{l+1}H_{l+1})^2\}$ for every l .
- (c) $(\text{SF}_l), (\text{SF}^*)$. For every l ,

$$(1 + \|U\|) \sum_{l'' > l} \lambda_{l''} \leq 2^{-10} \iota_l \lambda_l \delta_l 2^{-(m(l)+k(l)+\min S_l)} \underline{y}_l / n_l,$$

with (ι_l) decreasing to 0; and $c_l \leq c_{l-1} \delta_l 2^{-\min S_l - l - 10} / (1 + \|U\|)$ for $l \geq 2$.

- (d) (b'). $2c_l \leq \iota_l 2^{-2s} c_{l'} \delta_{l'}$ for $l' < l$ and $s \in \text{supp } y_l \cap S_{l'}$.
- (e) (Z0). $3, 5, 7, 9$ are distinct elements of Z_0 , and y^*, y^{**} belong to the target family.
- (f) (GM) for $l \geq 2$. $|\sum_{j \in \text{supp } y_l} \sigma_j y_l(j) + \epsilon \delta_l H_l| \geq g_l > 0$ for all $\sigma \in \{-1, 0, 1\}^{\text{supp } y_l}$, $\epsilon = \pm 1$, and $\Phi_l \leq g_l 2^{-l}$.
- (g) (FD). For every block m and every $r \geq 1$ the carrier $l = j(2^r, m)$ has $y_l = e_{j_l}^* / q^*(e_{j_l}^*)$ with j_l odd, $j_l \notin \bigcup_i S_i$ and $j_l \notin \text{supp } y_{l'}$ for $l' < l$; hence $u_{l'}(j_l) = 0$ for $l' < l$, $u_l(j_l) \neq 0$, and

$$|y_l(j_l)| > (1 + \|U\|)(\|y_l\|_1 - |y_l(j_l)| + \delta_l H_l).$$

- (h) The base is diagonal with positive, strictly decreasing entries, and $\|U\| = \frac{1}{16} \leq \frac{1}{2}$.

Proof. (a) Definition 3.4(i), allowedness (c), and the proof of (T-d). (b) (W1), (W2), (W3) at stage $l+1$. (c) By (W7) at stage $l+1$ and (W1) at the later stages, $\sum_{l'' > l} \lambda_{l''} \leq \frac{1}{4} \sum_{l'' \geq l+1} c_{l''} \leq \frac{1}{3} c_{l+1} \leq 2^{-10} \iota_l \lambda_l \delta_l 2^{-(m(l)+k(l)+\min S_l)} \underline{y}_l / (n_l(1 + \|U\|))$. The second inequality is (W5). (d) is (W4). (e) Definition 3.4(i), (iv). (f) The first statement is $\delta_l \in A_l$; $g_l > 0$ trivially; $\Phi_l = 2^{-m(l)-k(l)} c_l \leq g_l 2^{-l}$ by (W6). (g) $k(l) = 2^r \in K_{\text{FD}}$, so step (1) chooses the fresh-dominated target; j_l is odd, hence in Z_0 , and exceeds every element of $\text{supp } y_{l'}$, $l' < l$. For $l' < l$, $u_{l'}(j_l) = (y_{l'}(j_l) + \delta_{l'} h_{l'}(j_l)) / n_{l'} = 0$, and $u_l(j_l) = y_l(j_l) / n_l \neq 0$. Finally $|y_l(j_l)| = 1/(1 + \mu_{j_l}^*) \geq \frac{16}{17}$, $\|y_l\|_1 - |y_l(j_l)| = 0$ and $(1 + \|U\|)\delta_l H_l \leq (1 + \|U\|)\delta_l^{\max} H_l \leq \frac{1}{4}$. (h) Lemma 3.2. $\square$

Proposition 3.16 (Bank efficiency and depths). *For every level l :*

- (a) $s_{\max}(l) < s_b(l) \leq \max\{3 \cdot 2^l, s_{\max}(l) + 2^{l+1}\}$.
- (b) For every $l'' \leq l$ the first three elements of $S_{l''} \cap (s_{\max}(l), \infty)$ lie in $(s_{\max}(l), s_t(l)]$.

(c) For every $c \leq l$ and every $s' \in S_c$ with $s' \leq s_t(l)$:

$$v_c(s') \geq \frac{4}{5} \delta_{\min}(l) 2^{-s_t(l)}, \quad (\mu_{s'}^*)^2 v_c(s') \geq \frac{4}{5 B^*(l)} \geq \text{Des}(l)^{-1/6},$$

$$(\mu_{s'}^* v_c(s'))^{-2} \leq \text{Des}(l)^{1/3}.$$

(d) $2^{s_t(l)+2}/\delta_{\min}(l) \leq \text{Des}(l)^{1/6}$ and $8^{s_b(l)} \leq \text{Des}(l)^{1/6}$.

(e) $\text{Des}(l)^{1/6}$ dominates each factor inside the bracket of $\text{Des}(l)$, and $\text{Des}(l)$ dominates each of $\text{Lip}(l)$, $1/\delta_{\text{comb}}(l)$, $(r(l)c'(l))^l$, $C_*(l)$, $C_L(l)$, $N_L(l)$, $1/\delta_{\text{sh}}(l)$; moreover $\text{Des}(l) \geq 2^{6l^3}$.

Proof. (a) Each $\min(S_{l''} \cap (s_{\max}(l), \infty))$ exceeds $s_{\max}(l)$. If $s_{\max}(l) < \min S_{l''} = 3 \cdot 2^{l''}$, this minimum is $3 \cdot 2^{l''} \leq 3 \cdot 2^l$; otherwise it is at most $s_{\max}(l) + G_{l''} \leq s_{\max}(l) + 2^{l+1}$ (bounded gaps). (The cruder bound $s_b(l) \leq s_{\max}(l) + 2^{l+1}$ fails whenever $s_{\max}(l) + 2^{l+1} < 3 \cdot 2^l$, i.e. whenever $s_{\max}(l) < 2^l$.) (b) The first element is $\leq s_b(l)$, and the next two are at most $2G_{l''} \leq 2^{l+2}$ further. (c) $n_c \leq \frac{5}{4}$ and $\delta_c \geq \delta_{\min}(l)$ give $v_c(s') = \delta_c 2^{-s'}/n_c \geq \frac{4}{5} \delta_{\min}(l) 2^{-s_t(l)}$. Since μ^* is decreasing, $(\mu_{s'}^*)^2 \geq (\mu_{s_t(l)}^*)^2$, so $(\mu_{s'}^*)^2 v_c(s') \geq \frac{4}{5} B^*(l)$, and $(\mu_{s'}^* v_c(s'))^{-2} \leq \frac{25}{16} (\mu_{s_t(l)}^*)^{-2} 4^{s_t(l)} \delta_{\min}(l)^{-2} \leq \frac{25}{16} B^*(l)^2$. Since $l 2^{l^3} \geq 2$ and $2^{G(l)} \geq 4$, $\text{Des}(l)^{1/6} \geq 8B^*(l) \geq \frac{5}{4} B^*(l)$, which gives both remaining inequalities. (d) $B^*(l) \geq 2^{s_t(l)}/\delta_{\min}(l)$ and $B^*(l) \geq (\mu_{s_b(l)}^*)^{-2} = 2^{2 \cdot 2^{s_b(l)}} \geq 8^{s_b(l)}$, and $\text{Des}(l)^{1/6} \geq 8B^*(l)$. (e) All factors are ≥ 1 by (K1) and Lemma 3.10(b); $\text{Des}(l) \geq (l 2^{l^3})^6$. $\square$

Remark 3.17 (The role of $B^*(l)$). In the diagonal-base arguments of the later sections (private banks, far pulls, donor raises, exact two-sided tuning, pair and anchor tuning on support coordinates) the base enters a design constant only through the bank efficiency $(\mu_{s'}^*)^2 v_c(s')$ at coordinates $s' \leq s_t(l)$ and through $\|U\| \leq \frac{1}{2}$; the remaining uses of the base are the identities of Lemma 3.2(b) and constants depending on f . Proposition 3.16(c) turns the bank efficiency into a power of $\text{Des}(l)$, for instance the donor bank mass is at most $\text{Des}(l)\Lambda$ when the required threshold raise is Λ , and the contraction constant of the tuning fixed point, which contains $(\mu_{s'}^* v_c(s'))^{-2}$, is at most $\text{Des}(l)^{1/3}$ times f -dependent constants. This is the sense in which the factor $B^*(l)$ makes the bank and pull machinery independent of the choice of the diagonal base; each such use is checked where it occurs. The depth $s_t(l)$ (rather than $s_b(l)$) is needed because pair tuning on support coordinates uses the first three coordinates of each $S_{l''}$ beyond $s_{\max}(l)$ (Proposition 3.16(b)), and because deep support raises must be taken beyond $s_t(l)$; the pinning constant at that depth, $2^{s_t(l)+2}/\delta_{\min}(l)$, is a design factor by Proposition 3.16(d).

3.6. Clean sub-windows.

Definition 3.18 (Clean sub-windows). Let $N \geq 1$ and $f \in S_{p_N}^*$. A sub-window w of level l is *clean for f* if no object of $\mathcal{O}(l)$ has its value at f in $\text{Band}(w) = (b(w), u(w))$. At a clean sub-window a value is called *tiny* if it is $\leq b(w)$ and *robust* if it is $\geq u(w)$.

Theorem 3.19 (Clean sub-windows). Let $N \geq 1$ and $f \in S_{p_N}^*$, with $F = \text{supp } a$ finite or infinite. For every level l there is $i \in \{1, \dots, M(l)\}$ such that the sub-window (l, i) is clean for f . At such a sub-window the value at f of every rate object of level l is tiny or robust.

Proof. The $M(l) = \omega(l) + 1$ bands $\text{Band}(l, i)$ are pairwise disjoint (Proposition 3.14(a)), so each of the $\omega(l)$ values lies in at most one of them, and at least one band contains none of them. The second statement is the definition, since every value lies in $[0, \infty]$ and $[0, \infty] \setminus (b(w), u(w)) = [0, b(w)] \cup [u(w), \infty]$. $\square$

Corollary 3.20 (Consequences of cleanness). *Let w be a clean sub-window of level l for $f \in S_{p_N}^*$, and write $b = b(w)$, $u = u(w)$.*

- (a) *Every maximum or minimum of finitely many values of objects of $\mathcal{O}(l)$ is tiny or robust.*
- (b) *For every carrier $l'' \leq l$ with $m(l'') \leq N$, $\varrho_{l''}(f) \in [0, b] \cup [u, 1 - u] \cup [1 - b, 1 + b] \cup [1 + u, \infty)$.*
- (c) *$b \leq 2^{-64/u}$; in particular $u/b \geq 2^{64}$, and $\text{Des}(l)^k T_{\text{lo}}(w) \leq u/8$ for $k \leq 40$.*

Proof. (a) The set $[0, b] \cup [u, \infty]$ is closed under finite maxima and minima. (b) By (O3), (O4), $\varrho := \varrho_{l''}(f) \notin (b, u)$ and $|\varrho - 1| \notin (b, u)$. If $\varrho \geq u$, then $|\varrho - 1| \leq b$ or $|\varrho - 1| \geq u$, i.e. $\varrho \in [1 - b, 1 + b]$, $\varrho \leq 1 - u$ or $\varrho \geq 1 + u$. (c) Proposition 3.14(a), (e). $\square$

Remark 3.21 (Quantifiers). The order of choices is: the fixed data and T_{final} (no dependence on N); then N and $f \in S_{p_N}^*$; then, for each level l , a clean sub-window w_l (which depends on f); then the exactifying companion $f_{w_l}^\#$ (Definition 4.16; it depends on f and w_l only); then the mate g and $\rho < 1$; then the data at each scale $t \in \mathcal{W}(w_l)$. The rate objects are fixed before f ; only their values depend on f . An f -dependent quantity used in a window argument must therefore be an f -constant (absorbed by Lemma 3.22(a)(GW2)) or the value of a rate object of the level (tiny or robust at w_l).

3.7. Generic window use.

Lemma 3.22 (Generic window use). (a) *For every sub-window w of level l :*

- (GW1) $\mathcal{W}(w)$ contains the $n(w)$ dyadic scales $T_{\text{hi}}(w)2^{1-i}$, $1 \leq i \leq n(w)$, and $T_{\text{hi}}(w)2^{l^3} \Lambda^\circ(l) \leq 1/l$, $n(w) \geq l2^{l^3} \Lambda^\circ(l)$.
- (GW2) Let $C, K \geq 1$, and for every l let w_l be a sub-window of level l and $R_l \geq 1$ with $R_l \leq Kl2^{l^3} \Lambda^\circ(l)$. Put $X_l := C^{l^2} (C/u(w_l))^{\omega(l)+8} \text{Des}(l)^8 R_l$. Then $T_{\text{hi}}(w_l)X_l \rightarrow 0$ and $n(w_l)/X_l \rightarrow \infty$ as $l \rightarrow \infty$.
- (GW3) $\sum_{l' > l} \lambda_{l'} \leq T_{\text{lo}}(w)^8$; consequently, for every two-sided decomposition at a scale $t \in \mathcal{W}(w)$ (in the sense of [I, Section 8.3]), $\sum_{l' > l} |\Delta\theta_{l'}| \leq 6t^7 \leq 6t^2$.
- (GW4) $T_{\text{hi}}(w') \leq T_{\text{lo}}(w)$ for every sub-window w' following w in the lexicographic order.

(b) *Let $(w_l)_{l \geq 1}$ be any choice of one sub-window w_l of each level l , and use $\mathcal{W}(w_l)$, $T_{\text{hi}}(w_l)$, $T_{\text{lo}}(w_l)$, $n(w_l)$ in place of the l -th window $\mathcal{W}(l)$, $T_{\text{hi}}(l)$, $T_{\text{lo}}(l)$, n_l^w of [I, Definition 8.1]. Then T_{final} satisfies (T-a)–(T-d), (P1), (P2) and (P3) of [I, Theorem 8.2] verbatim; consequently every result of [I, Section 8] holds for T_{final} , every N , and this choice of windows.*

Proof. (GW1) Definition 3.5(6) and Proposition 3.14(b).

(GW2) Since $\text{Des}(l) \geq 2^{6l^3} \rightarrow \infty$, for all large l we have $C \leq 4\text{Des}(l)$, so $(C/u)^{\omega+8} \leq (4\text{Des}/u)^{\omega+8} = Q(w_l)(u/(4\text{Des}))^{12}$ (we write $u = u(w_l)$, $\text{Des} = \text{Des}(l)$, $\omega = \omega(l)$). Using $T_{\text{hi}}(w_l) \leq 2^{-l^3}/(lQ(w_l))$, $u \leq 1$ and $\text{Des} \geq (l2^{l^3} \Lambda^\circ(l))^6 \geq$

$$2^{6l^3} \Lambda^\circ(l),$$

$$T_{\text{hi}}(w_l)X_l \leq \frac{C^{l^2} \text{Des}^8 R_l}{l 2^{l^3} \text{Des}^{12}} \leq \frac{K C^{l^2} \Lambda^\circ(l)}{\text{Des}^4} \leq K C^{l^2} 2^{-24l^3} \rightarrow 0.$$

Similarly, using $n(w_l) \geq l 2^{l^3} Q(w_l)$ and $(4\text{Des}/u)^{12} \geq \text{Des}^{12}$,

$$\frac{n(w_l)}{X_l} \geq \frac{l 2^{l^3} \text{Des}^{12}}{C^{l^2} \text{Des}^8 R_l} \geq \frac{\text{Des}^4}{K C^{l^2} \Lambda^\circ(l)} \geq \frac{2^{24l^3}}{K C^{l^2}} \rightarrow \infty.$$

(GW3) The first inequality is Proposition 3.14(c). By the box bound [I, Lemma 8.10], $|\theta_{l'}^\pm| \leq 3\lambda_{l'}/t$, hence $|\Delta\theta_{l'}| \leq 6\lambda_{l'}/t$, so $\sum_{l' \geq l} |\Delta\theta_{l'}| \leq 6T_{\text{lo}}(w)^8/t \leq 6t^7$ for $t \geq T_{\text{lo}}(w)$, and $t \leq 1$.

(GW4) $T_{\text{hi}}(w^+) \leq T_{\text{lo}}(w)$ by the definition of $T_{\text{hi}}(w^+)$, and $T_{\text{lo}}(w^+) \leq T_{\text{hi}}(w^+)$; hence T_{hi} and T_{lo} are nonincreasing along the order, and induction gives the claim.

(b) (T-a)–(T-d), (P1) and the first part of (P2) are Theorem 3.12. The window part of (P2): $\sum_{l' \geq l} c_{l'} \leq \frac{4}{3}c_{l+1} \leq 2c_{l+1}$ and $c_{l+1} \leq b(w_l)^2 \leq T_{\text{lo}}(w_l)^8 \leq T_{\text{lo}}(w_l)^3$, and $\sum_{l' \geq l} \lambda_{l'} \leq T_{\text{lo}}(w_l)^8 \leq T_{\text{lo}}(w_l)^3$ (Proposition 3.14(c); note that $c_{l+1} \leq b(l, M(l))^2 \leq b(w_l)^2$). (P3): $T_{\text{hi}}(w_l)2^{l^3} \Lambda^\circ(l) \leq 1/l$ and $n(w_l) \geq l 2^{l^3} \Lambda^\circ(l)$ by (GW1), and $T_{\text{hi}}(w_{l+1}) \leq T_{\text{lo}}(w_l)$ by (GW4). The targets are allowed in the sense of [I, Definition 8.1(D1)(a),(b)] (Theorem 3.12(c)), the vectors u_l have the form $(y_l + \delta_l h_l)/n_l$ with $q^*(\delta_l h_l) \leq \frac{1}{4}$ and $\delta_l \rightarrow 0$, and $Te_{k,m} = c_{j(k,m)} u_{j(k,m)}$. As stated after [I, Theorem 8.2], only (T-a)–(T-d), (P1), (P2) and (P3) are used in [I, Section 8]; the value of δ_l enters only through $q^*(\delta_l h_l) \leq \frac{1}{4}$ and $\delta_l \rightarrow 0$ (both hold here although δ_l may differ from $\delta_l^{\max}$ by a factor at most 2). Hence those results hold for T_{final} with the windows $\mathcal{W}(w_l)$. $\square$

Remark 3.23 (How the window properties are used). In every window argument of [I, Section 8] and of the later sections of this part, the window enters only through (i) the dyadic scales of one window, (ii) conditions “ $T_{\text{hi}} X_l \rightarrow 0$ ” and “ $n/X_l \rightarrow \infty$ ” along the levels used, where X_l is an f -constant to the power l^2 , times a product of at most 48 design factors of level l (each $\leq \text{Des}(l)^{1/6}$, so the product is $\leq \text{Des}(l)^8$), times at most $\omega(l) + 8$ factors $C_f/u(w)$ coming from robust rates, times a room product compared with $l 2^{l^3} \Lambda^\circ(l)$; (iii) the box bound (GW3); and (iv) the ordering (GW4). Lemma 3.22(a) provides all four on every sub-window, so these arguments may use any sub-window of level l , in particular a clean one, as “the window of level l ”. All four conditions are monotone in $\text{Des}(l)$: enlarging the design factor only lengthens and deepens the sub-windows. This is a description of the proofs, verified for each window theorem where it is proved; it is the reason why the factors $N_L(l)$ (Proposition 3.14(f)) and $B^*(l)$ could be added to the design factor without affecting any estimate.

Proposition 3.24 (Window factor for the infinite-support arguments). *For a sub-window w of level l put $\Omega_K(l) := s_{\max}(l) 2^{s_{\max}(l)} / \delta_{\min}(l)$ and*

$$\Omega(w) := \Omega_K(l) \left(2 + \log_2 (2 + \log_2 (1/T_{\text{lo}}(w^-))) \right) \left(2 + \log_2 (l 2^{l^3} \Lambda^\circ(l) \Omega_K(l)) \right)$$

$(T_{\text{lo}}((1, 1)^-) = 1)$. Then

- (a) $\Lambda^\circ(l) \Omega(w) \leq Q(w)$; hence $n(w) \geq l 2^{l^3} \Lambda^\circ(l) \Omega(w)$ and $T_{\text{hi}}(w) \leq 2^{-l^3} / (l \Omega(w) \Lambda^\circ(l))$;
- (b) $(\log_2(1/T_{\text{hi}}(w)) + \log_2 n(w)) / n(w) \leq (2l^3 + 2l + 4) / (l 2^{l^3})$;

- (c) for every $x \in [T_{\text{lo}}(w)^2, \frac{1}{2}]$, $2^{J(x)} \leq 2 \log_2(4 \log_2(1/T_{\text{lo}}(w))) \leq 2 \log_2(8n(w))$, with $J(\cdot)$ as in Lemma 3.2(d).

Proof. (a) Write $\text{Des} = \text{Des}(l)$, $u = u(w)$. Since $s \leq 2^s$ and $\delta_{\min}(l) \leq 1$, $\Omega_K(l) \leq (2^{s_{\max}(l)}/\delta_{\min}(l))^2$, so $l2^{l^3} \Lambda^\circ(l) \Omega_K(l) \leq (l2^{l^3} \Lambda^\circ(l) 2^{s_{\max}(l)} \delta_{\min}(l)^{-1})^2 \leq \text{Des}^{1/3}$. The last factor of $\Omega(w)$ is therefore at most $2 + \log_2 \text{Des} \leq \text{Des}^{1/2}$, because $\text{Des} \geq 64$ and $y \mapsto y^{1/2} - \log_2 y - 2$ vanishes at 64 and increases on $[64, \infty)$. For the middle factor: if $w = (1, 1)$ it equals $3 \leq 4/u$; otherwise $T_{\text{lo}}(w^-)^4 \geq b(w^-) = u$, so with $x := \log_2(1/u) \geq 2$ the middle factor is at most $2 + \log_2(2 + x/4) \leq 3 + x/4 \leq 2 + x \leq 2^{x+2} = 4/u$ (we used $\log_2(2 + y) \leq 1 + y$ for $y \geq 0$). Hence $\Lambda^\circ(l) \Omega(w) \leq \text{Des}^{1/3} (4/u) \text{Des}^{1/2} \leq 4 \text{Des}/u \leq Q(w)$, and the two consequences follow from the definitions of $n(w)$ and $T_{\text{hi}}(w)$.

(b) By Proposition 3.14(f), $\log_2(1/T_{\text{hi}}(w)) \leq l^3 + l + Q(w)$; also $\log_2 n(w) \leq \log_2(l2^{l^3} Q(w) + 1) \leq l^3 + l + Q(w) + 1$. Divide by $n(w) \geq l2^{l^3} Q(w)$ and use $Q(w) \geq 1$.

(c) J is nonincreasing in x , so $2^{J(x)} \leq 2^{J(T_{\text{lo}}(w)^2)} \leq 2 \log_2(2 \log_2(T_{\text{lo}}(w)^{-2}))$ by Lemma 3.2(d), and $\log_2(1/T_{\text{lo}}(w)) \leq 2n(w)$ by Proposition 3.14(f). $\square$

Remark 3.25 (Scope of T_{final}). (a) T_{final} contains the features of the earlier signature-ladder designs with sub-windows: the sub-window recursion and the rate objects (O1)–(O4); the bounded gaps and allowedness rule (c); the diagonal base with private bank and pull coordinates (every coordinate of every S_l beyond $s_{\max}(l)$ is available); the design factors of the configuration, tuning and determinantal arguments; the Łojasiewicz-adjusted value of $b(w)$ (Proposition 3.14(d)); the rate objects (O5)–(O9); the weight conditions (SF $_l$), (b'), (GM) for $l \geq 2$, (Z0), (FD) (Proposition 3.15); and, for infinite supports, the doubly exponential base, the depth $s_t(l)$ with the factor $B^*(l)$, the window factor $\Omega(w)$ (Proposition 3.24) and the rate objects (O10)–(O12).

(b) Two features of earlier designs are deliberately *absent*: an explosive window function (replaced by the pigeonhole sub-windows) and lacunary signature sets (incompatible with bounded gaps). Results that depend on these features are not claimed for T_{final} ; no result of this part uses them.

(c) The fresh-dominated targets serve only a negative statement (dead zones forced by positive mismatch) and, below, the fact that $s_{\max}(l) \rightarrow \infty$ (Lemma 4.1(e)) and the stages of the modified design $D^{U1'}$ (Definition 5.48). We do not claim that they can be removed from the schedule.

4. CLEAN SUB-WINDOWS AND THE EXACTIFYING COMPANION

In this section we prove the second master theorem: for the operator T_{final} and a first row f with finite base support, recovery of every mate of f follows from two structural conditions at one clean sub-window of infinitely many levels (Theorem 4.25). The proof combines four ingredients. First, at a clean sub-window the switching amplitudes of a two-sided decomposition are pinned *diagonally*: no product of rooms and no slaving between carriers occurs, because the pinning sets exclude all coarse target coordinates (Subsection 4.1). Second, with the diagonal base U_{final} , masses placed far out on signature sets act on the values of single carriers: a *far pull* lowers one value and a *private bank* raises one value at first order, while all other coarse values move only at second order; together they tune finitely many values exactly and two-sidedly (Subsection 4.2). Third, every block

has, at every clean sub-window of large level, a peak with robust margin, and a bank at a far signature coordinate of that peak raises the block threshold by a prescribed amount; this *threshold buffer* dominates every drift caused by the other moves (Subsection 4.3). Fourth, one companion per clean sub-window carries all exactifications at once, at cost $o(T_{\text{lo}}^2)$, and exact d -neutral two-piece data are transplanted to it (Subsections 4.3 and 4.4). Theorem 2.20 then transfers recovery from the companions to f .

Standing assumptions and notation. Throughout this section $T = T_{\text{final}}$, $U = U_{\text{final}}$ (Section 3), $N \geq 1$, $I = \{1, \dots, N\}$, $p = p_N$, and $f \in S_{p^*}$ has forced data $(\xi, q_0, a, w, F, z, \hat{z}, e, \nu)$ [I, Proposition 1.9] with $F = \text{supp } a$ finite. We use the notation of Sections 2 and 3. In particular, for a block m ,

$$\hat{\zeta}_m := R_m^{**} \hat{z}, \quad A_m := |\hat{\zeta}_m|_m = \sigma_m/q_0, \quad \bar{\vartheta}_m := \bar{\vartheta}(\hat{\zeta}_m) = A_m M_m / C_m,$$

so that $\hat{\zeta}_m(k) = \lambda_{k,m} u_{k,m}(\hat{z})$. The *coarse carriers* of level l are the elements of $\mathcal{C}(l) := \{l'' \leq l : m(l'') \leq N\}$; for a carrier $l'' = j(k, m)$ we write $k(l'') = k$, $m(l'') = m$ and use $u_{l''} = u_{k,m}$, $\Phi_{l''} = \Phi_m(k)$, $\lambda_{l''} = m\Phi_{l''}$ (Theorem 3.12(c)). Carriers $l'' > l$ are *fine*. The relative position $\varrho_{l''} = \varrho_{l''}(f)$ of Definition 3.7(O4) equals $\varrho_k(\hat{\zeta}_m) = m|u_{l''}(\hat{z})|/(\Phi_{l''}\bar{\vartheta}_m)$ (Subsection 2.2); hence $k(l'')$ is a peak iff $\varrho_{l''} \geq 1$, the margin of a peak is $\mu_{k,m} = q_0\Phi_{l''}\bar{\vartheta}_m(\varrho_{l''} - 1)/m$, the gap of a strict non-peak is $\text{gap}_m(k) = M_m(1 - \varrho_{l''})$, and $|w_m(k)| = M_m \min(1, \varrho_{l''})$ (Lemma 2.6(b)). Recall $|u_{l''}(\hat{z})| \leq 1$. Given a sign $\epsilon_{l''}$ we write $\text{val}_{l''} := \epsilon_{l''}u_{l''}(\hat{z})$, and $\text{val}'_{l''} := \epsilon_{l''}u_{l''}(\hat{z}')$ at another first row f' .

For a sub-window w of level l we abbreviate

$$b_w := b(w), \quad u_w := u(w), \quad \Lambda_w := T_{\text{lo}}(w)^3, \quad \beta_w := b_w + \text{Des}(l)^2 \Lambda_w^2, \\ D_r := D_{\text{room}}(l), \quad \text{Des} := \text{Des}(l).$$

For $l'' \leq l$ put $S_{l''}^{\text{nat}}(l) := S_{l''} \setminus (F \cup T(l))$, $\mathbf{m}_{l''} := \|v_{l''} 1_{S_{l''}^{\text{nat}}(l)}\|_1$ and

$$r_{l''}^{\text{nat}} := \min_{\sigma=\pm 1} \sum_{s \in S_{l''}^{\text{nat}}(l)} v_{l''}(s)(1 + \sigma z_s),$$

so that the value of the room object (O1) of l'' is $r_{l''}^{\text{nat}}/\mathbf{m}_{l''}$. An f -constant is a number depending only on f (and on N and the fixed design), not on l , w or the scale; C_f and $c_f > 0$ denote f -constants whose values may change from line to line. l_f denotes a level threshold depending only on f ; it is enlarged finitely many times below, each time by a condition that depends only on f . Statements “for $l \geq l_f$ ” refer to sub-windows of level l .

Lemma 4.1 (Scale relations). *Let w be a sub-window of level l and $t \in \mathcal{W}(w)$.*

- (a) $b_w \leq T_{\text{lo}}(w)^4/(l \text{Des}) \leq t^4/(l \text{Des})$ and $\sum_{l' > l} \lambda_{l'} \leq b_w^2/3$.
- (b) $T_{\text{lo}}(w) \leq (u_w/(4\text{Des}))^{40}$; hence $\text{Des}^k T_{\text{lo}}(w) \leq u_w/8$ for $0 \leq k \leq 40$, and $\text{Des}^k \beta_w \leq u_w \Lambda_w/(4l)$ for $0 \leq k \leq 38$.
- (c) For $l'' \leq l$: $\lambda_{l''} \geq \Phi_{l''} \geq 1/D_r$; if $l \geq \max F$, then $\mathbf{m}_{l''} \geq \mathbf{m}_{l''}^{\text{nat}}(l) \geq 1/D_r$. Each of l , D_r , $2^{G(l)}$, $1 + |T(l)|$, $G^{**}(l)$, $H_{\text{tune}}(l)$, $H_{\text{conf}}(l)$ is at most $\text{Des}^{1/6}$.
- (d) For $l'' \leq l$ let $s'_{l''}(l) := \min(S_{l''} \cap (s_{\max}(l), \infty))$. Then $s'_{l''}(l) \leq s_b(l) \leq s_t(l)$, and for every $c \leq l$ and $s' \in S_c$ with $s' \leq s_t(l)$: $(\mu_{s'}^*)^2 v_c(s') \geq \text{Des}^{-1/6}$ and $(\mu_{s'}^* v_c(s'))^{-2} \leq \text{Des}^{1/3}$.

- (e) $s_{\max}(l) \rightarrow \infty$, $u(w) \rightarrow 0$ and $b(w) \rightarrow 0$ as $l \rightarrow \infty$ (uniformly over the sub-windows of level l).

Proof. (a) $b_w \leq \eta(w)$ (Definition 3.5(6)) and $t \geq T_{\text{lo}}(w)$; the second claim is Proposition 3.14(c). (b) $T_{\text{hi}}(w) \leq 1$, $n(w) \geq Q(w) \geq 16$ and $2^{-Q} \leq Q^{-2}$ give $T_{\text{lo}}(w) \leq 2^{-n(w)} \leq Q(w)^{-2} = (u_w/(4\text{Des}))^{2\omega(l)+40}$. Hence $\text{Des}^k T_{\text{lo}} \leq 4^{-40} u_w^{40} \leq u_w/8$ for $k \leq 40$. By (a), $\text{Des}^k b_w \leq (\Lambda_w/l)\text{Des}^{k-1} T_{\text{lo}} \leq u_w \Lambda_w/(8l)$ for $k \leq 41$, and $\text{Des}^{k+2} \Lambda_w^2 = \Lambda_w T_{\text{lo}}^2 \text{Des}^{k+2} T_{\text{lo}} \leq u_w \Lambda_w/(8l)$ for $k \leq 38$, because $T_{\text{lo}} \leq T_{\text{hi}} \leq 2^{-l^3}/l$ gives $T_{\text{lo}}^2 \leq 1/l$. (c) $\Phi_{l''} = 2^{-m-k} c_{l''} \leq \lambda_{l''}$, and $1/\Phi_{l''}$, $1/\mathbf{m}_{l''}^{\text{nat}}(l)$ are summands of $D_{\text{room}}(l)$. If $l \geq \max F$, then $S_{l''} \setminus ([1, l] \cup T(l)) \subseteq S_{l''}^{\text{nat}}(l)$. The listed numbers are factors of the bracket in $\text{Des}(l)$ or are dominated by such factors ($G^{**}(l) \leq H_{\text{conf}}(l)$ by Definition 4.11 below), and $\text{Des}(l)$ is at least the sixth power of the bracket. (d) Proposition 3.16(a)–(c). (e) At the stages $l' = j(2^r, m')$ the targets are fresh-dominated (Proposition 3.15(g)) with $j_{l'}$ larger than every earlier target coordinate, and $s_{\max}$ is nondecreasing. For w of level $l \geq 2$, $u(w) = b(w^-) \leq T_{\text{lo}}(w^-)^4 \leq 2^{-4(l-1)^3}$, and $b(w) < u(w)$. $\square$

In particular we may and do assume $l_f \geq \max F$ and $s_{\max}(l_f) \geq \max F$, so that for $l \geq l_f$

$$(3) \quad F \cup T(l) \subseteq [1, s_{\max}(l)], \quad S_{l''} \setminus [1, s_{\max}(l)] \subseteq S_{l''}^{\text{nat}}(l) \quad (l'' \leq l).$$

4.1. Classification and pinning at a clean sub-window. Let w be a sub-window of level $l \geq l_f$ which is clean for f (Definition 3.18); by Corollary 3.20(b), every $l'' \in \mathcal{C}(l)$ has $\varrho_{l''} \in [0, b_w] \cup [u_w, 1 - u_w] \cup [1 - b_w, 1 + b_w] \cup [1 + u_w, \infty)$, and every room object and every target room object of level l is tiny or robust.

Definition 4.2 (Classification at a clean sub-window). Let w be clean for f , of level $l \geq l_f$.

- (a) A carrier $l'' \in \mathcal{C}(l)$ is of *class R* if $r_{l''}^{\text{nat}} \geq u_w \mathbf{m}_{l''}$ and of *class G* if $r_{l''}^{\text{nat}} \leq b_w \mathbf{m}_{l''}$. For l'' of class G, $\epsilon_{l''} \in \{\pm 1\}$ is the sign with $r_{l''}^{\text{nat}} = \sum_{s \in S_{l''}^{\text{nat}}(l)} v_{l''}(s)(1 - \epsilon_{l''} z_s)$ (it is unique, because the two signed rooms add up to $2\mathbf{m}_{l''} > 2b_w \mathbf{m}_{l''}$), and the *d-weight* of l'' is

$$q_{l''} := \frac{\epsilon_{l''} \Phi_{l''} w_m(k)}{m C_m} \quad (l'' = j(k, m)).$$

l'' is of *swallowing type* if $\epsilon_{l''} \text{sign } w_m(k) = 1$, of *anti type* if $\epsilon_{l''} \text{sign } w_m(k) = -1$, and *d-neutral* if $w_m(k) = 0$ (then $q_{l''} = 0$). $G(w)$ is the set of class-G carriers.

- (b) For a block m , $\Sigma_m(w)$ is the set of $l'' \in G(w)$ of block m that are swallowing-type peaks, or strict non-peaks with $q_{l''} > b_w$ (hence $q_{l''} \geq u_w$) that are not of anti type with $q_{l''} \geq 1 - b_w$. Block m is *σ -one-signed at w* ($\sigma \in \{\pm 1\}$) if $\text{sign } q_{l''} = \sigma$ for every $l'' \in \Sigma_m(w)$.
- (c) The *dropped set* $P(w) \subseteq G(w)$ consists of (P-i) anti-type peaks; (P-ii) anti-type strict non-peaks with $q \geq 1 - b_w$; (P-iii) swallowing-type peaks with $q \geq 1 + u_w$; (P-iv) the sets $\Sigma_m(w)$ of the blocks m that are one-signed at w . The *kept set* is $\text{Kp}(w) := G(w) \setminus P(w)$; it consists of (K1) swallowing-type strict non-peaks with $q \in [u_w, 1 - u_w]$; (K2) anti-type strict non-peaks with $q \in [u_w, 1 - u_w]$; (K3) carriers with $q \leq b_w$; (K4) swallowing-type carriers with $q \in [1 - b_w, 1 + b_w]$ (strict non-peaks, peaks with small margin, degenerate peaks); in a one-signed block only (K3) carriers are kept.

- (d) A target coordinate $j \in T(l) \setminus F$ is *contact-like* if $1 - |z_j| \leq b_w$ and *free-robust* if $1 - |z_j| \geq u_w$.

Every carrier of $\mathcal{C}(l)$ is of class R or of class G, and every target coordinate off F is contact-like or free-robust, because w is clean. In the remainder of this subsection we fix, in addition, $g \in \mathcal{C}(f)$, $\eta \leq \eta_*$ with $\eta_\Gamma(\eta) \leq 1$ [I, Lemma 8.7], a scale $t \in \mathcal{W}(w)$ with $t \leq \min(t_\eta, 1)$ [I, Lemma 8.6], and a two-sided decomposition $(B_\pm, \Theta_\pm)$ of g at scale t [I, Lemma 8.5]. As in [I, Section 8.3] and [I, (15)] we write $\Delta\theta_{l''} := \lambda_{l''}(\Theta_+ - \Theta_-)_{m(l'')}(k(l''))$ (and $\Delta\theta_{l''} := 0$ if $m(l'') > N$), $\Delta B := B_+ - B_- = -\sum_{l''} \Delta\theta_{l''} u_{l''}$, and, with $d_{\pm, m}$ and $\omega_{\pm, m}$ as in [I, Lemma 8.8], $\Delta d_m := d_{+, m} - d_{-, m}$. For $l'' \in G(w)$ the *switching amplitude* is $\tau_{l''} := -\epsilon_{l''} \Delta\theta_{l''}$. By [I, Lemma 8.10] and Lemma 4.1(a),

$$(4) \quad |\Delta\theta_{l''}| \leq 6\lambda_{l''}/t \quad (\text{all } l''), \quad \sum_{l'' > l} |\Delta\theta_{l''}| \leq \frac{2b_w^2}{t} \leq t^7.$$

Lemma 4.3 (Diagonal pinning). *Put $K_g := (1/q_0 + 1)D_r/u_w$. Then*

- (a) $\sum_{l'' \text{ of class R}} |\Delta\theta_{l''}| \leq K_g t$;
(b) $\sum_{l'' \in G(w)} (\tau_{l''})_- \leq K_g t$, and $e_0 := \Delta B 1_{F^c} - \sum_{l'' \in G(w)} \epsilon_{l''} \tau_{l''} u_{l''} 1_{F^c}$ satisfies $\|e_0\|_1 \leq K_g t + 2b_w^2/t \leq (K_g + 1)t$.

Proof. Let $l'' \in \mathcal{C}(l)$ and $s \in S_{l''}^{\text{nat}}(l)$. For $l' \leq l$, $l' \neq l''$, we have $h_{l'}(s) = 0$ (disjoint signature sets) and $y_{l'}(s) = 0$ ($s \notin T(l)$), so $u_{l'}(s) = 0$; also $u_{l''}(s) = v_{l''}(s)$, and $u_{l'}(s) = y_{l'}(s)/n_{l'}$ for $l' > l$. By [I, (15)],

$$-\Delta B(s) = \Delta\theta_{l''} v_{l''}(s) + r(s), \quad r(s) := \sum_{l' > l} \Delta\theta_{l'} y_{l'}(s)/n_{l'}.$$

The sets $S_{l''}^{\text{nat}}(l)$, $l'' \in \mathcal{C}(l)$, are pairwise disjoint subsets of F^c , so the numbers

$$E_{l''} := \sum_{s \in S_{l''}^{\text{nat}}(l)} \phi_{z_s}(\Delta B(s))$$

satisfy $\sum_{l''} E_{l''} \leq t/q_0$ [I, Lemma 8.24], and $\sum_{l''} \sum_s |r(s)| \leq \frac{4}{3} \sum_{l' > l} |\Delta\theta_{l'}| \leq \frac{8b_w^2}{3t}$ by (4) and $\|y_{l'}\|_1/n_{l'} \leq \frac{4}{3}$. By [I, Lemma 8.23(c),(d)],

$$E_{l''} \geq |\Delta\theta_{l''}| \sum_{s \in S_{l''}^{\text{nat}}(l)} v_{l''}(s) (1 + z_s \text{sign } \Delta\theta_{l''}) - 2 \sum_s |r(s)| \geq r_{l''}^{\text{nat}} |\Delta\theta_{l''}| - 2 \sum_s |r(s)|.$$

(a) For l'' of class R, $r_{l''}^{\text{nat}} \geq u_w \mathbf{m}_{l''} \geq u_w/D_r$ (Lemma 4.1(c)); sum over class R and use $16b_w^2/(3t) \leq t$. (b) For $l'' \in G(w)$ we have $-\Delta\theta_{l''} v_{l''}(s) = \epsilon_{l''} \tau_{l''} v_{l''}(s)$, and by [I, Lemma 8.23(d)], $\phi_{z_s}(\epsilon_{l''} \tau_{l''} v_{l''}(s)) = |\tau_{l''}| v_{l''}(s) (1 - \epsilon_{l''} z_s \text{sign } \tau_{l''})$, which is $(\tau_{l''})_- v_{l''}(s) (1 + \epsilon_{l''} z_s)$ when $\tau_{l''} < 0$. Summing over s , $(\tau_{l''})_- (2\mathbf{m}_{l''} - r_{l''}^{\text{nat}}) \leq E_{l''} + 2 \sum_s |r(s)|$, and $2\mathbf{m}_{l''} - r_{l''}^{\text{nat}} \geq \mathbf{m}_{l''} \geq 1/D_r$. Finally $e_0 = -\sum \Delta\theta_{l''} u_{l''} 1_{F^c}$ over class R and fine carriers, $\|u_{l''}\|_1 \leq q^*(u_{l''}) = 1$; use (a) and (4). $\square$

No triangular unrolling and no product of rooms enter: the pinning sets $S_{l''}^{\text{nat}}(l)$ meet no coarse target, so each coarse amplitude is compared only with its own signature and with fine carriers.

Lemma 4.4 (Peak relations). *Let $l'' \in \mathcal{C}(l)$ with $k := k(l'') \in P_m$, $\varsigma := \text{sign } w_m(k)$, $\mu := \mu_{k, m}$, and $Y := |\omega_{+, m}(k)| + |\omega_{-, m}(k)|$. Then $Y = -\varsigma \Delta\theta_{l''}/\lambda_{l''} - \Delta d_m M_m \geq 0$, and $Y \leq t/(\lambda_{l''} \mu)$ if $\mu > 0$. Consequently:*

- (a) if l'' is of class R, then $\Delta d_m M_m \leq |\Delta \theta_{l''}|/\lambda_{l''}$, and if $\mu > 0$ also $\Delta d_m M_m \geq -|\Delta \theta_{l''}|/\lambda_{l''} - t/(\lambda_{l''}\mu)$;
- (b) if $l'' \in G(w)$ is of anti type, then $\tau_{l''} \leq -\lambda_{l''}\Delta d_m M_m$, hence $\Delta d_m M_m \leq (\tau_{l''})_-/\lambda_{l''}$;
- (c) if $l'' \in G(w)$ is of swallowing type, then $\tau_{l''} \geq \lambda_{l''}\Delta d_m M_m$, and $\tau_{l''} \leq \lambda_{l''}\Delta d_m M_m + t/\mu$ if $\mu > 0$.

If $\varrho_{l''} \geq 1 + u_w$, then $1/\mu \leq mD_r/(q_0\bar{\vartheta}_m u_w)$.

Proof. By [I, (16)] and [I, Lemma 8.27], $Y = -\varsigma\Delta\Theta_m(k) - \Delta d_m M_m \geq 0$ and $Y \leq t/(\sigma_m|\alpha_m(k)|) = t/(\lambda_{l''}\mu)$ by [I, (2)]; and $\Delta\Theta_m(k) = \Delta\theta_{l''}/\lambda_{l''}$. (a) $-\varsigma\Delta\theta_{l''} \leq |\Delta\theta_{l''}|$. (b), (c): $\Delta\theta_{l''} = -\epsilon_{l''}\tau_{l''}$, so $-\varsigma\Delta\theta_{l''}/\lambda_{l''} = \varsigma\epsilon_{l''}\tau_{l''}/\lambda_{l''}$, with $\varsigma\epsilon_{l''} = -1$ in (b) and $= +1$ in (c). Finally $\mu = q_0\Phi_{l''}\bar{\vartheta}_m(\varrho_{l''} - 1)/m \geq q_0\bar{\vartheta}_m u_w/(mD_r)$. $\square$

Definition 4.5 (Shift sources). Let w be clean of level $l \geq l_f$ and $m \leq N$. An *upper source* of block m at w is one of: (U1) a peak $k(l'')$ of block m with l'' of class R; (U2) an anti-type peak in $G(w)$ of block m ; (U3) every $l'' \in G(w)$ of block m with $q_{l''} < 0$ is an anti-type peak, an anti-type strict non-peak with $\varrho_{l''} \geq 1 - b_w$, or has $\varrho_{l''} \leq b_w$. A *lower source* is one of: (L1) a peak of block m of class R with $\varrho \geq 1 + u_w$; (L2) a swallowing-type peak in $G(w)$ of block m with $\varrho \geq 1 + u_w$; (L3) every $l'' \in G(w)$ of block m with $q_{l''} > 0$ has $\varrho_{l''} \leq b_w$.

A block that is $(+1)$ -one-signed at w satisfies (U3), and a block that is (-1) -one-signed at w satisfies (L3): if $\text{sign } q = +1$ on $\Sigma_m(w)$, every $l'' \in G(w)$ of block m with $q_{l''} < 0$ lies outside $\Sigma_m(w)$, i.e. is an anti-type peak, an anti-type strict non-peak with $\varrho \geq 1 - b_w$, or has $\varrho \leq b_w$; if $\text{sign } q = -1$ on $\Sigma_m(w)$, every carrier with $q > 0$ lies outside $\Sigma_m(w)$ and is not a peak (swallowing-type peaks belong to $\Sigma_m(w)$), so it has $\varrho \leq b_w$.

Lemma 4.6 (Sources bound the shift on one side). *There is $K_d = C_f D_r^2/u_w$ such that: if block m has an upper source at w , then $\Delta d_m M_m \leq K_d t$; if it has a lower source, then $\Delta d_m M_m \geq -K_d t$. Each half uses only its own source.*

Proof. We use [I, (17)]: $\Delta d_m M_m = \frac{1}{mC_m} \sum_k \Phi_m(k) w_m(k) \Delta\theta_{j(k,m)} + r_m$ with $|r_m| \leq 2t/\sigma_m$. For $l'' \in G(w)$ the summand equals $-q_{l''}\tau_{l''}$; for class R and fine carriers it is at most $|\Delta\theta_{l''}|/(mC_m)$ in modulus. Recall $|q_{l''}| \leq \Phi_{l''}M_m/(mC_m) \leq 1/C_m$, and $|q_{l''}| = \Phi_{l''}M_m\varrho_{l''}/(mC_m) \leq b_w/C_m$ when $\varrho_{l''} \leq b_w$.

Upper bounds. (U1): Lemma 4.4(a), Lemma 4.3(a) and $\lambda \geq 1/D_r$ give $\Delta d_m M_m \leq D_r K_g t$. (U2): Lemma 4.4(b) and Lemma 4.3(b) give the same bound. (U3): In [I, (17)], the class R and fine terms are at most $(K_g t + t^7)/(mC_m)$ (Lemma 4.3(a), (4)); carriers with $q > 0$ contribute $-q\tau \leq q(\tau)_-$, in total at most $K_g t/C_m$; carriers with $q = 0$ contribute 0. For $q < 0$ the contribution is $|q|\tau$, and by (U3) the carrier is of one of three kinds. At an anti-type peak, $\tau \leq -\lambda\Delta d_m M_m$ (Lemma 4.4(b)). At an anti-type strict non-peak k with $\varrho \geq 1 - b_w$, put $\varsigma := \text{sign } w_m(k)$; by [I, Lemma 8.8(f)] and $|d_{\pm t}| \leq \frac{1}{2}$, $\varsigma(\omega_{-,m} - \omega_{+,m})(k) \geq -3\text{gap}_m(k)/t$, while $\omega_{-,m} - \omega_{+,m} = -\Delta\Theta_m - \Delta d_m w_m$ gives $\varsigma(\omega_{-,m} - \omega_{+,m})(k) = -\tau/\lambda - \Delta d_m |w_m(k)|$; so $\tau \leq \lambda(3\text{gap}_m(k)/t - \Delta d_m |w_m(k)|)$ with $\text{gap}_m(k) = M_m(1 - \varrho) \leq b_w$. At a carrier with $\varrho \leq b_w$, $|q|\tau \leq (b_w/C_m)6\lambda/t$. Collecting, with $\sum \lambda \leq 1$ and $b_w \leq t^4$,

$$\Delta d_m M_m \left(1 + \sum_{\text{a.p.}} |q|\lambda + \sum_{\text{a.n.}} |q|\lambda \frac{|w_m(k)|}{M_m} \right) \leq \frac{K_g t + t^7}{mC_m} + \frac{K_g t}{C_m} + \frac{9t^3}{C_m} + \frac{2t}{\sigma_m},$$

the sums running over the anti-type peaks and the anti-type near-threshold strict non-peaks. If $\Delta d_m \geq 0$ the bracket is ≥ 1 ; otherwise the bound is trivial.

Lower bounds. (L1): Lemma 4.4(a) with $1/\mu \leq mD_r/(q_0\bar{\vartheta}_m u_w)$: $\Delta d_m M_m \geq -(D_r K_g + mD_r^2/(q_0\bar{\vartheta}_m u_w))t$. (L2): Lemma 4.4(c) gives $\Delta d_m M_m \geq \tau/\lambda - t/(\lambda\mu) \geq -(\tau)_-/\lambda - t/(\lambda\mu)$, and the same bound follows. (L3): carriers with $q > 0$ contribute $-q\tau \geq -(b_w/C_m)6\lambda/t$, carriers with $q < 0$ contribute $|q|\tau \geq -|q|(\tau)_-$, in total $\geq -K_g t/C_m$; with the class R and fine terms and r_m , $\Delta d_m M_m \geq -C_f K_g t$. In all cases the constant is at most $C_f D_r^2/u_w$. $\square$

Lemma 4.7 (Robust-margin peaks). *There is l_f such that for every sub-window w of level $l \geq l_f$ which is clean for f and every block $m \leq N$ there is $c \in \mathcal{C}(l)$ of block m such that $k(c)$ is a peak with $\varrho_c \geq 1 + u_w$; in particular its margin is at least $q_0\bar{\vartheta}_m u_w/(mD_r)$. Every such c is of class R, or belongs to $P(w)$ (types (P-i) or (P-iii)).*

Proof. By [I, Lemma 1.7], $\sum_{k \in P_m} |\alpha_m(k)| = 1$, and by [I, (2)], $\sigma_m |\alpha_m(k)| = \lambda_{k,m} \mu_{k,m}$. A peak $k = k(l'')$ with $l'' \in \mathcal{C}(l)$ and $\varrho_{l''} \leq 1 + b_w$ has $\mu_{k,m} \leq q_0 \Phi_{l''} \bar{\vartheta}_m b_w/m$, hence $|\alpha_m(k)| \leq q_0 \bar{\vartheta}_m b_w \Phi_{l''}^2/\sigma_m$, and the total of these is at most $(M_m/C_m)b_w$, since $q_0 \bar{\vartheta}_m/\sigma_m = M_m/C_m$ and $\sum_k \Phi_m(k)^2 \leq 1$. A peak of a fine carrier has $\mu_{k,m} \leq |u_{k,m}(\xi)| \leq q_0$, and these contribute at most $q_0 \sum_{l' > l} \lambda_{l'}/\sigma_m \leq q_0 b_w^2/\sigma_m$. By Lemma 4.1(e) we may take l_f so large that $(M_m/C_m)b_w + q_0 b_w^2/\sigma_m < 1$ for every $m \leq N$ and every sub-window of level $\geq l_f$. Then some coarse peak of block m has $\varrho > 1 + b_w$, hence $\varrho \geq 1 + u_w$, since w is clean. If c is of class G, it is an anti-type peak (P-i) or a swallowing-type peak with $\varrho \geq 1 + u_w$ (P-iii). $\square$

Since $\|\alpha_m\|_1 = 1$, tiny-margin peaks and fine peaks carry only $O(b_w)$ of the α -mass; this is why a robust-margin peak, and hence a *threshold donor* (Lemma 4.17), exists in every block.

Definition 4.8 (Shift pattern of f at w ; condition (SH_w)). Let w be clean of level $l \geq l_f$. Let $I_{\text{up}}(w)$ (resp. $I_{\text{lo}}(w)$) be the set of blocks $m \leq N$ without an upper (resp. lower) source at w ; $P^*(w)$ the set of $l'' \in \mathcal{C}(l)$ of blocks in $I_{\text{up}}(w) \cup I_{\text{lo}}(w)$ such that $k(l'')$ is a peak with $\varrho_{l''} \geq 1 + u_w$; $\text{vs}_{l''} := \text{sign } w_m(l'')(k(l''))$; $B_f(w) := G(w) \setminus P^*(w)$ with the signs $\epsilon_{l''}$ of Definition 4.2. The *shift pattern of f at w* is $\pi(w) := (I_{\text{up}}(w), I_{\text{lo}}(w), P^*(w), \text{vs}, B_f(w), \epsilon)$, and $c_{\pi(w)}(f)$ is the value at f of the corresponding rate object (O6) (Definition 3.7). We say that (SH_w) holds if $I_{\text{up}}(w) \cup I_{\text{lo}}(w) = \emptyset$ or $c_{\pi(w)}(f) \geq u_w$.

By Lemma 4.9(a) below, $I_{\text{up}}(w) \cap I_{\text{lo}}(w) = \emptyset$ and $P^*(w) \subseteq G(w)$, so $\pi(w)$ is a shift pattern of level l in the sense of (O6) (every block $m \leq N$ contains a coarse carrier by Lemma 4.7). As w is clean, (SH_w) fails exactly when $I_{\text{up}}(w) \cup I_{\text{lo}}(w) \neq \emptyset$ and $c_{\pi(w)}(f) \leq b_w$.

Lemma 4.9 (Shift pinning at a clean sub-window). *Let w be clean of level $l \geq l_f$.*

- (a) $I_{\text{up}}(w) \cap I_{\text{lo}}(w) = \emptyset$, and $P^*(w) \subseteq G(w)$.
- (b) If (SH_w) holds, then $|\Delta d_m| M_m \leq K'_d t$ for every block m , with $K'_d := C_f l D_r^2 u_w^{-2}$.

Proof. (a) Let $m \in I_{\text{up}}(w) \cap I_{\text{lo}}(w)$. Then (U1), (U2) and (L2) fail for m , so every coarse peak of block m is a swallowing-type peak in $G(w)$ with $\varrho \leq 1 + b_w$; this contradicts Lemma 4.7. A class-R peak with $\varrho \geq 1 + u_w$ is a source of both kinds ((U1) and (L1)), so blocks in $I_{\text{up}}(w) \cup I_{\text{lo}}(w)$ contain none, and $P^*(w) \subseteq G(w)$.

(b) If $I_{\text{up}}(w) \cup I_{\text{lo}}(w) = \emptyset$, apply Lemma 4.6. Otherwise $c := c_{\pi(w)}(f) \geq u_w$. For $l'' \in P^*(w)$ of block m , Lemma 4.4 gives $-\Delta\theta_{l''} = \text{vs}_{l''}\lambda_{l''}(\Delta d_m M_m + Y_{l''})$ with $0 \leq \lambda_{l''} Y_{l''} \leq t/\mu_{k(l''),m} \leq tmD_r/(q_0\bar{\vartheta}_m u_w)$. For $m \in I_{\text{up}}(w)$ the block has a lower source, so $\Delta d_m M_m \geq -K_d t$ (Lemma 4.6); put $\delta_m := (\Delta d_m)_+ M_m \geq 0$. For $m \in I_{\text{lo}}(w)$ put $\delta_m := -(\Delta d_m)_- M_m \leq 0$; then $(\Delta d_m)_+ M_m \leq K_d t$. In both cases $|\Delta d_m M_m - \delta_m| \leq K_d t$. With $\Pi_m := \sum_{l'' \in P^*(w), m(l'')=m} \text{vs}_{l''}\lambda_{l''} u_{l''} 1_{F^c}$ (so $\|\Pi_m\|_1 \leq \sum \lambda \leq \frac{1}{3}$) and $-\Delta\theta_{l''} = \epsilon_{l''}\tau_{l''} = \epsilon_{l''}(\tau_{l''})_+ - \epsilon_{l''}(\tau_{l''})_-$ on $B_f(w)$, [I, (15)] gives

$$\Delta B 1_{F^c} = \sum_m \delta_m \Pi_m + \sum_{l'' \in B_f(w)} (\tau_{l''})_+ \epsilon_{l''} u_{l''} 1_{F^c} + R,$$

where, by Lemma 4.3 and (4),

$$\|R\|_1 \leq K_d t + \frac{ltmD_r}{q_0\bar{\vartheta}_m u_w} + \sum_{G(w)} (\tau)_- + \sum_{\text{class R}} |\Delta\theta| + \sum_{l' > l} |\Delta\theta_{l'}| \leq C_f l D_r^2 t / u_w.$$

The vector $x := ((\tau_{l''})_+)_{l'' \in B_f(w)}$ is admissible in the infimum defining $c(\delta; \pi(w))$, and δ lies in the sign cone of (O6). By [I, Lemma 8.23(c)] and [I, Lemma 8.24], $c(\delta; \pi(w)) \leq \sum_{j \notin F} \phi_{z_j}(\Delta B(j)) + 2\|R\|_1 \leq t/q_0 + 2\|R\|_1$. As $c(\cdot; \pi(w))$ is positively homogeneous, $c(\delta; \pi(w)) \geq c\|\delta\|_1 \geq u_w\|\delta\|_1$. Hence $\|\delta\|_1 \leq C_f l D_r^2 t / u_w^2$ and $|\Delta d_m| M_m \leq |\delta_m| + K_d t \leq C_f l D_r^2 t / u_w^2$. $\square$

Lemma 4.10 (Pinned carriers). *Let w be clean of level $l \geq l_f$, and suppose $|\Delta d_m| M_m \leq K' t$ for every block m , with some $K' \geq 1$. Put $K_P := K_g + K' + C_f D_r / u_w$ and $K_O := C_f D_r (K' + K_g + l K_P) / u_w$. Then $|\tau_{l''}| \leq K_P t$ for l'' of type (P-i), (P-ii) or (P-iii), $|\tau_{l''}| \leq K_O t$ for l'' of type (P-iv), and $|\Delta\theta_{l''}| \leq K_g t$ for l'' of class R. Under (SH_w) we may take $K' = K'_d$, and then*

$$K_{\text{pin}} := \max(K_g, K_P, K_O, K'_d) \leq C_f l^2 D_r^3 u_w^{-3}.$$

Proof. $(\tau)_- \leq K_g t$ on $G(w)$ by Lemma 4.3(b). (P-i): $\tau \leq -\lambda \Delta d_m M_m \leq K' t$ (Lemma 4.4(b), $\lambda \leq 1$). (P-ii): as in the proof of Lemma 4.6, $\tau \leq \lambda(3\text{gap}/t + |\Delta d_m| M_m) \leq 3b_w/t + K' t \leq (1 + K')t$. (P-iii): $\tau \leq \lambda \Delta d_m M_m + t/\mu \leq K' t + tmD_r/(q_0\bar{\vartheta}_m u_w)$ (Lemma 4.4(c)). Class R: Lemma 4.3(a).

(P-iv): let block m be σ -one-signed. In [I, (17)] the summands of $l'' \in \Sigma_m(w)$ are $-q_{l''}\tau_{l''}$ with $\text{sign } q_{l''} = \sigma$, so

$$\begin{aligned} \sum_{\Sigma_m(w)} |q|(\tau)_+ &\leq \mathcal{A}_m := |\Delta d_m| M_m + \frac{2t}{\sigma_m} \\ &\quad + \left| \sum_{l'' \notin \Sigma_m(w)} \frac{\Phi_{l''} w_m(k(l'')) \Delta\theta_{l''}}{m C_m} \right| + \sum_{\Sigma_m(w)} |q|(\tau)_-. \end{aligned}$$

The carriers of block m outside $\Sigma_m(w)$ are of class R or fine (total $\leq (K_g t + t^7)/(m C_m)$), of types (P-i), (P-ii) (at most l carriers, each $\leq K_P t / C_m$), or in $G(w)$ with $\varrho \leq b_w$ ($|q\tau| \leq (b_w/C_m)6\lambda/t$, in total $\leq t^3/C_m$). So $\mathcal{A}_m \leq C_f(K' + K_g + l K_P)t$. On $\Sigma_m(w)$, $|q| \geq \Phi M_m u_w / (m C_m) \geq M_m u_w / (m C_m D_r)$ (at peaks $|w| = M_m$, at strict non-peaks $|w| = M_m \varrho \geq M_m u_w$). Hence $(\tau)_+ \leq K_O t$ after enlarging C_f .

Under (SH_w), Lemma 4.9(b) gives $K' = K'_d$; then $K_P \leq C_f l D_r^2 u_w^{-2}$ and $K_O \leq C_f l^2 D_r^3 u_w^{-3}$. $\square$

By Lemma 4.7 and Lemma 4.10, every robust-margin peak of Lemma 4.7 is pinned: $|\Delta\theta_c| \leq K_{\text{pin}} t$ under (SH_w). The anti-type near-threshold strict non-peaks

(P-ii) are pinned like anti-type peaks; their tiny gaps are therefore not rates that need exactification. Their swallowing-type counterparts, the (K4) carriers, are not pinned.

Patterns, rays and condition (VR_w) .

Definition 4.11 (The pattern of f at w ; ray components; (VR_w)). Let w be clean of level $l \geq l_f$. The *pattern of f at w* is $\kappa(w) := (G(w), P(w), \epsilon, F \cap T(l), \text{type}_w)$ with $\text{type}_w(j) := \text{sign } z_j$ for contact-like and $\text{type}_w(j) := 0$ for free-robust $j \in T(l) \setminus F$; it is a pattern of level l (Definition 3.6(a)), with kept set $\text{Kp}(w)$. For $r \in \text{Ext}(\kappa(w))$ and a block m meeting $\text{supp } r$, the *d -component* of r in block m at a first row f' is

$$D_{r,m}(f') := \sum_{l'' \in \text{supp } r, m(l'')=m} r(l'') \epsilon_{l''} u_{l''}(\hat{z}'),$$

and the value of the rate object (O5) of $(\kappa(w), r, m)$ at f is $|D_{r,m}(f)|/\Phi_{\max}(r, m)$. The component is *tiny* or *robust* according to this value. Condition (VR_w) holds if every $r \in \text{Ext}(\kappa(w))$ has at most one robust component.

For box radii $\beta \in (0, \infty)^E$ the box $Z_\kappa(\beta) = \{\tau \in C(\kappa) : \tau_{l''} \leq \beta_{l''} \ (l'' \in \text{Kp}(\kappa))\}$ and the constant $G^{**}(l)$ (the maximum over all patterns κ of level l of the ℓ_1 -Hoffman constant of the system defining $Z_\kappa(\beta)$, which does not depend on β) are those of Definition 3.6(d). The system is a configuration system of level l , so $G^{**}(l) \leq H_{\text{conf}}(l)$.

Remark 4.12 (Components in one-signed blocks). (a) If block m is one-signed at w , every kept carrier of block m has $\varrho \leq b_w$, so $|\epsilon_{l''} u_{l''}(\hat{z})| = \varrho \Phi_{l''} \bar{\vartheta}_m/m \leq b_w \Phi_{l''} \bar{\vartheta}_m/m$, and every component in block m has rate at most $b_w \bar{\vartheta}_m/m$ (as $\|r\|_1 = 1$). Since $b_w \leq T_{\text{lo}}(w)^4 \leq (u_w/(4\text{Des}))^{160}$ (Lemma 4.1(a),(b)), $b_w/u_w \leq u_w^{159} \rightarrow 0$, while $\bar{\vartheta}_m/m$ is an f -constant, the rate is $< u_w$ for $l \geq l_f$, hence tiny: one-signed blocks carry no robust components. (b) If $N = 1$, or if every extreme ray of $C(\kappa(w))$ is supported in one block, (VR_w) holds. (c) For a kept carrier that is a strict non-peak, $q_{l''} = \epsilon_{l''} \Phi_{l''} w_m(k)/(mC_m) = \text{val}_{l''}/A_m$ by the clamp formula ($w_m(k) = C_m \hat{c}_m(k)/(\Phi_{l''}^2 A_m)$, [I, Lemma 1.7]); so $D_{r,m}/A_m$ is the d -sum of the switching direction r in block m . For a (K4) carrier that is a peak, $q_{l''} = \text{val}_{l''}/(\varrho_{l''} A_m)$ with $\varrho_{l''} \in [1, 1 + b_w]$. (d) $C(\kappa) \subseteq [0, \infty)^E$, so for $P' \subseteq E$ the set $C(\kappa) \cap \{\tau_{l''} = 0, l'' \in P'\}$ is a face of $C(\kappa)$, and the extreme rays of a face are extreme rays of $C(\kappa)$: dropping more carriers can only remove rays.

4.2. Far pulls and private banks. For $A \in \ell_1 \setminus \{0\}$ and $z' \in B_{\ell_\infty}$ with $z'_j = \text{sign } A_j$ on $\text{supp } A$ we write $f[A, z'] := f[A/q^*(A), z']$ for the first row with prescribed data (Subsection 2.1); its forced data are $e' = U^*A/\|U^*A\|_H$ (independent of the normalization), $\hat{z}' = z' + Ue'$, and $w'_m = J_m(R_m^{**}\hat{z}')$. In this subsection (A^0, z^0) is such a pair, $\nu_0 := \|U^*A^0\|_H$ and $e^0 := U^*A^0/\nu_0$.

Definition 4.13 (Banks and far pulls). Let J be a finite set disjoint from $\text{supp } A^0$ and $(c_j)_{j \in J}$ real, $A := A^0 + \sum_{j \in J} c_j e_j^*$. A coordinate $j \in J$ with $|z_j^0| = 1$ and $\text{sign } c_j = z_j^0$ (the sign vector is not changed at j) is a *bank* of mass $|c_j|$. A coordinate $j \in J \cap S_{l''}$ with $z_j^0 = \epsilon$, at which $c_j = -\epsilon \mathfrak{p}$ with $\mathfrak{p} > 0$ and the sign vector is changed to $-\epsilon$, is a *far pull* (of carrier l'' , with sign ϵ) of mass $\mathfrak{p}$. In both cases the resulting pair (A, z) satisfies $z_j = \text{sign } A_j$ on $\text{supp } A$, so $f[A, z]$ is defined.

Lemma 4.14 (Masses off the support). *In the situation of Definition 4.13 put $X := \sum_{j \in J} c_j \mu_j^* k_j$ and $e^A := U^* A / \|U^* A\|_H$. Then $\|U^* A\|_H^2 = \nu_0^2 + \|X\|^2$ and, for every $u \in \ell_1$,*

$$(5) \quad \langle U^* u, e^A \rangle = \frac{\nu_0 \langle U^* u, e^0 \rangle + \sum_{j \in J} c_j (\mu_j^*)^2 u(j)}{(\nu_0^2 + \|X\|^2)^{1/2}}.$$

Consequently: (i) if u vanishes on J , then

$$|\langle U^* u, e^A \rangle - \langle U^* u, e^0 \rangle| \leq \|U\| \|u\|_1 \|X\|^2 / (2\nu_0^2);$$

(ii) in general $\langle U^* u, e^A \rangle - \langle U^* u, e^0 \rangle = \sum_{j \in J} c_j (\mu_j^*)^2 u(j) / \nu_0 + R_u$ with $|R_u| \leq (\|U\| \|u\|_1 + \sum_{j \in J} |c_j| (\mu_j^*)^2 |u(j)| / \nu_0) \|X\|^2 / (2\nu_0^2)$; (iii) $\|e^A - e^0\|_H \leq 2\|X\| / \nu_0$.

Proof. Lemma 3.2(b) with $a := A^0$ gives $U^* A = U^* A^0 + X$, $X \perp U^* A^0$, the Pythagoras identity and $\langle U^* u, X \rangle = \sum_j c_j (\mu_j^*)^2 u(j)$; divide by $\|U^* A\|$. Put $x := \|X\|^2 / \nu_0^2$, $p := \langle U^* u, e^0 \rangle$ and $s := \sum_j c_j (\mu_j^*)^2 u(j)$; then $\langle U^* u, e^A \rangle = (p + s / \nu_0)(1 + x)^{-1/2}$ and $0 \leq 1 - (1 + x)^{-1/2} \leq x/2$, while $|p| \leq \|U^* u\| \leq \|U\| \|u\|_1$. This gives (i) ($s = 0$) and (ii). (iii): $\|x' / \|x'\| - y / \|y\|\| \leq 2\|x' - y\| / \|y\|$ with $x' = U^* A$, $y = U^* A^0$. $\square$

For a carrier l'' and a coordinate $j \in S_{l''}$ beyond $s_{\max}(l)$, $l \geq l''$, the only carrier $\leq l$ whose vector is nonzero at j is l'' , with $u_{l''}(j) = v_{l''}(j)$ (Theorem 3.12(d)). Hence a pull at j changes $\epsilon u_{l''}(\hat{z})$ by $-2v_{l''}(j)$ through the sign vector, a bank at j changes it by $+b(\mu_j^*)^2 v_{l''}(j) / \nu_0$ at first order through the Hilbert part, and by (5) every other carrier $\leq l$ moves only at second order. This is the diagonal-base mechanism used below.

Lemma 4.15 (Exact two-sided tuning). *Let $l \geq l_f$, let (A^2, z^2) be as above with $F^2 := \text{supp } A^2$ finite, $\nu_2 := \|U^* A^2\|_H \geq \nu$ and $\|A^2\|_1 \leq 2$, let $L_0 \subseteq \mathcal{C}(l)$ and signs $(\epsilon_{l''})_{l'' \in L_0}$, and assume: (T1) $z^2 = \epsilon_{l''}$ on $S_{l''} \setminus [1, s_{\max}(l)]$ for every $l'' \in L_0$; (T2) $F^2 \cap S_{l''} \cap (s_{\max}(l), \infty) = \emptyset$ for $l'' \in L_0$. Let $x \in \mathbb{R}^{L_0}$, $\epsilon_x := \|x\|_\infty$. There are $c_T = c_f \text{Des}^{-1}$ and $C_T = C_f \text{Des}$ such that if $\epsilon_x \leq c_T$ the following holds. For $l'' \in L_0$ let $j_{l''}' := s_{l''}'(l)$ (the bank coordinate), let $j_{l''} \in S_{l''}$ be the largest element with $v_{l''}(j_{l''}) \geq \epsilon_x$ (the pull coordinate; if $x = 0$ there are no moves), and $\mathbf{p}_{l''} := 24\lambda_{l''} v_{l''}(j_{l''})$. Then $j_{l''} > j_{l''}'$, $v_{l''}(j_{l''}) \in [\epsilon_x, 2^{G(l)} \epsilon_x]$, and there is $\mathbf{n} \in (0, \infty)^{L_0}$ such that the row $f^\# := f[A^\#, z^\#]$ with*

$$A^\# := A^2 - \sum_{l'' \in L_0} \epsilon_{l''} \mathbf{p}_{l''} e_{j_{l''}}^* + \sum_{l'' \in L_0} \mathbf{n}_{l''} \epsilon_{l''} e_{j_{l''}}^*, \quad z_{j_{l''}}^\# := -\epsilon_{l''}, \quad z^\# := z^2 \text{ elsewhere,}$$

satisfies:

- (a) $\text{val}_{l''}^\# = \text{val}_{l''}^{(2)} + x_{l''}$ exactly for every $l'' \in L_0$, where $\text{val}^{(2)}$ refers to $f[A^2, z^2]$;
- (b) $|u_k(\hat{z}^\#) - u_k(\hat{z}^{(2)})| \leq C_T \epsilon_x^2$ for every carrier $k \leq l$ with $k \notin L_0$;
- (c) $\|\mathbf{n}\|_\infty \leq C_T \epsilon_x$, $\mathbf{p}_{l''} \leq 6 \cdot 2^{G(l)} \epsilon_x$, and the Hilbert vector

$$X_{\text{tu}} := \sum_{l'' \in L_0} (-\epsilon_{l''} \mathbf{p}_{l''} \mu_{j_{l''}}^* k_{j_{l''}} + \mathbf{n}_{l''} \epsilon_{l''} \mu_{j_{l''}}^* k_{j_{l''}}')$$

has $\|X_{\text{tu}}\|^2 \leq C_T \epsilon_x^2$;

- (d) for $l'' \in L_0$, $|u_{l''}(\hat{z}^\#) - u_{l''}(\hat{z}^{(2)})| \leq \epsilon_x$, and $(A^\#, z^\#)$ is admissible: the pulls carry masses of the new sign $-\epsilon_{l''} = z_{j_{l''}}^\#$, and the banks carry masses of sign $\epsilon_{l''} = z_{j_{l''}}^\#$.

Proof. Coordinates. By (T1), (T2) and (3), all $j_{l''}, j'_{l''}$ lie outside F^2 , are pairwise distinct (distinct signature sets) and carry $z^2 = \epsilon_{l''}$. The values $v_{l''}(s) = \delta_{l''} 2^{-s}/n_{l''}$, $s \in S_{l''}$, decrease by the factor $2^{-G_{l''}} \geq 2^{-G(l)}$ between consecutive elements, so the largest s with $v_{l''}(s) \geq \varepsilon_x$ has $v_{l''}(s) < 2^{G(l)} \varepsilon_x$. By Lemma 4.1(c),(d), $v_{l''}(j'_{l''}) \geq \text{Des}^{-1/6}$ and $2^{G(l)} \leq \text{Des}^{1/6}$; so for $\varepsilon_x < \text{Des}^{-1/3}$ we get $v_{l''}(j'_{l''}) > 2^{G(l)} \varepsilon_x$, hence $j_{l''} > j'_{l''}$. For a carrier $k \leq l$ and $l'' \in L_0$, u_k vanishes at $j_{l''}$ and $j'_{l''}$ unless $k = l''$. The admissibility statement in (d) is clear.

Step 1 (pulls). Let $A^p := A^2 - \sum \epsilon_{l''} \mathbf{p}_{l''} e_{j_{l''}}^*$, $z^p := z^\#$, $\nu_p := \|U^* A^p\| \geq \nu_2 \geq \nu$, and $X_p := \sum -\epsilon_{l''} \mathbf{p}_{l''} \mu_{j_{l''}}^* k_{j_{l''}}$, so $\|X_p\|^2 \leq \sum \mathbf{p}^2 \leq l(24 \cdot 2^{G(l)} \varepsilon_x)^2$. For $l'' \in L_0$, the sign change at $j_{l''}$ changes $\text{val}_{l''}$ by $-2v_{l''}(j_{l''})$, and Lemma 4.14(ii) with base A^2 gives

$$\text{val}_{l''}^p - \text{val}_{l''}^{(2)} = -2v_{l''}(j_{l''}) - \mathbf{p}_{l''}(\mu_{j_{l''}}^*)^2 v_{l''}(j_{l''})/\nu_2 + R_{l''}, \quad |R_{l''}| \leq C\|X_p\|^2/\nu^2.$$

Step 2 (banks; explicit solution). For $\mathbf{n} \in [0, \infty)^{L_0}$ let $A(\mathbf{n}) := A^p + \sum \mathbf{n}_{l''} \epsilon_{l''} e_{j'_{l''}}^*$, $\mathcal{N}(\mathbf{n}) := (\nu_p^2 + \sum_{l''} \mathbf{n}_{l''}^2 (\mu_{j'_{l''}}^*)^2)^{1/2}$, and write $s'_{l''} := \mu_{j'_{l''}}^*$, $v'_{l''} := v_{l''}(j'_{l''})$, $\varphi_{l''} := \epsilon_{l''} \langle U^* u_{l''}, e^p \rangle$ ($|\varphi_{l''}| \leq \|U\|$). By (5) with base A^p (the sign vector is not changed at banks),

$$\text{val}_{l''}(\mathbf{n}) - \text{val}_{l''}^p = \frac{\mathbf{n}_{l''} s_{l''}^{\prime 2} v'_{l''} + \varphi_{l''} \nu_p}{\mathcal{N}(\mathbf{n})} - \varphi_{l''}.$$

The required increments $\Delta_{l''} := x_{l''} + \text{val}_{l''}^{(2)} - \text{val}_{l''}^p$ satisfy $\varepsilon_x/2 \leq \Delta_{l''} \leq 4 \cdot 2^{G(l)} \varepsilon_x$ for $\varepsilon_x \leq c_T$ (Step 1, $v(j) \geq \varepsilon_x \geq |x_{l''}|$, and $\mathbf{p}(\mu_j^*)^2 v(j) \leq 24v(j)^2$). For $y \geq \nu_p$ put

$$\mathbf{n}_{l''}(y) := \frac{\Delta_{l''} y + \varphi_{l''}(y - \nu_p)}{s_{l''}^{\prime 2} v'_{l''}}, \quad \mathcal{M}(y) := \left(\nu_p^2 + \sum_{l''} s_{l''}^{\prime 2} \mathbf{n}_{l''}(y)^2 \right)^{1/2}.$$

If $\mathcal{M}(y) = y$, then $\mathcal{N}(\mathbf{n}(y)) = y$ and $\mathbf{n}(y)$ solves all equations $\text{val}_{l''}(\mathbf{n}) - \text{val}_{l''}^p = \Delta_{l''}$ simultaneously. By Lemma 4.1(c),(d), $(s'_{l''} v'_{l''})^{-2} \leq \text{Des}^{1/3}$ and $(s_{l''}^{\prime 2} v'_{l''})^{-1} \leq \text{Des}^{1/6}$. Put $\kappa_T := 2(\mathcal{M}(\nu_p) - \nu_p)$; since $\mathcal{M}(\nu_p)^2 - \nu_p^2 = \sum \Delta_{l''}^2 \nu_p^2 / (s_{l''}^{\prime 2} v'_{l''})^2$, $\kappa_T \leq C_f \text{Des} \varepsilon_x^2$ (we use $\nu_p \geq \nu$, $|L_0| \leq l$ and $l 4^{G(l)} \leq \text{Des}^{1/2}$). On $\mathcal{I} := [\nu_p, \nu_p + \kappa_T]$,

$$\begin{aligned} |\mathcal{M}'(y)| &= \left| \sum_{l''} \frac{\mathbf{n}_{l''}(y)(\Delta_{l''} + \varphi_{l''})}{s_{l''}^{\prime 2} v'_{l''} \mathcal{M}(y)} \right| \\ &\leq \sum_{l''} \frac{(2\Delta_{l''} + \kappa_T)(\Delta_{l''} + \|U\|)}{(s_{l''}^{\prime 2} v'_{l''})^2 \nu} \leq C_f \text{Des}^{2/3} \varepsilon_x \leq \frac{1}{2} \end{aligned}$$

for $\varepsilon_x \leq c_T$ (we used $|L_0| \leq l \leq \text{Des}^{1/6}$, $2^{G(l)} \leq \text{Des}^{1/6}$, $\nu_p \leq 1$). Hence $\nu_p \leq \mathcal{M}(y) \leq \mathcal{M}(\nu_p) + (y - \nu_p)/2 \leq \nu_p + \kappa_T$ on $\mathcal{I}$, and the continuous self-map $\mathcal{M}$ of $\mathcal{I}$ has a fixed point y^* . Put $\mathbf{n} := \mathbf{n}(y^*)$; then $\mathbf{n}_{l''} \geq (\Delta_{l''} \nu - \|U\| \kappa_T) / (s_{l''}^{\prime 2} v'_{l''}) > 0$ and $\mathbf{n}_{l''} \leq \text{Des}^{1/6} (4 \cdot 2^{G(l)} \varepsilon_x \cdot 2 + \kappa_T) \leq C_f \text{Des}^{1/3} \varepsilon_x$. By construction $\text{val}_{l''}^\# = \text{val}_{l''}^p + \Delta_{l''} = \text{val}_{l''}^{(2)} + x_{l''}$: this is (a), and (d) follows.

(c) $\mathbf{p}_{l''} = 24\lambda_{l''} v_{l''}(j_{l''}) \leq 24 \cdot 2^{G(l)} \varepsilon_x / 4$, and $\|X_{\text{tu}}\|^2 \leq \sum \mathbf{p}^2 + \sum \mathbf{n}^2 \leq C_f \text{Des} \varepsilon_x^2$. (b) For $k \leq l$, $k \notin L_0$, u_k vanishes at all pull and bank coordinates, so $u_k(z^\# - z^2) = 0$, and Lemma 4.14(i) with base A^2 and J the set of all pull and bank coordinates gives $|u_k(\hat{z}^\#) - u_k(\hat{z}^{(2)})| \leq \|U\| \|X_{\text{tu}}\|^2 / (2\nu^2)$. $\square$

The contraction constant of the fixed point contains the bank efficiency $(\mu_{j'}^* v_{l''}(j'))^{-2}$, not only $(\mu_{j'}^*)^{-2} v_{l''}(j')^{-1}$; both are design numbers of level l

(Lemma 4.1(d)), and this is the only place where the base enters the constants of the tuning. The pulls lower and the banks raise the tuned values; since every pull lowers its value by $2v_{l''}(j_{l''}) \geq 2\varepsilon_x$, the required bank increments are positive whatever the signs of the $x_{l''}$: the tuning is two-sided.

4.3. Donor peaks, bank donors and the exactifying companion.

Definition 4.16 (The exactifying companion $f_w^\#$). Let w be clean for f , of level $l \geq l_f$, and $\Lambda := \Lambda_w = T_{\text{lo}}(w)^3$. The following moves are made in the order listed.

- (C1) *Close class-G rooms*: $z_s^1 := \epsilon_{l''}$ for $s \in S_{l''}^{\text{nat}}(l)$, $l'' \in G(w)$.
- (C2) *Close tiny target rooms*: $z_j^1 := \text{sign } z_j$ for $j \in T(l) \setminus F$ with $0 < 1 - |z_j| \leq b_w$; $z^1 := z$ at all other coordinates.
- (C3) *Donor raise*. Let $I_D(w)$ be the set of blocks containing a (K4) carrier. For $m \in I_D(w)$ let c_m be the least carrier given by Lemma 4.7 for block m (the *donor*), $\text{vs}_m := \text{sign } w_m(k(c_m)) = \text{sign } u_{c_m}(\hat{z})$, and $s_m := s'_{c_m}(l) = \min(S_{c_m} \cap (s_{\max}(l), \infty))$.
 - (a) If $\lambda_{c_m} v_{c_m}(s_m)(1 - \text{vs}_m z_{s_m}^1) \geq \Lambda$, put $z_{s_m}^2 := z_{s_m}^1 + \eta_m \text{vs}_m$ with $\eta_m := \Lambda / (\lambda_{c_m} v_{c_m}(s_m))$ (a z -move, no mass).
 - (b) Otherwise put $z_{s_m}^2 := \text{vs}_m$ and place at s_m a bank of mass $\mathfrak{d}_m := 2\nu\Lambda / (\lambda_{c_m}(\mu_{s_m}^*)^2 v_{c_m}(s_m))$ with sign vs_m .
 Put $A^2 := a + \sum_{m \text{ in case (b)}} \mathfrak{d}_m \text{vs}_m e_{s_m}^*$, $z^2 := z^1$ at all other coordinates, and $f^{(2)} := f[A^2, z^2]$.
- (C4) *Exact tuning of tiny ray components*. Let $\Sigma_t(w)$ be the set of tiny components (r, m) of $\kappa(w)$. If $\Sigma_t(w) = \emptyset$, put $f_w^\# := f^{(2)}$. Otherwise let $L_0 := \bigcup_{(r, m) \in \Sigma_t(w)} \{l'' \in \text{supp } r : m(l'') = m\}$, $R := R_{\Sigma_t(w)}$ (Definition 3.6(b)), $V^{(2)} := (D_{r, m}(f^{(2)}))_{(r, m) \in \Sigma_t(w)}$ and $x := -R^+ V^{(2)}$. Then x is supported in L_0 (it lies in the row space of R), and $Rx = -V^{(2)}$, because $V^{(2)} = R \text{val}^{(2)}$ with $\text{val}^{(2)} := (\epsilon_{l''} u_{l''}(\hat{z}^{(2)}))_{l'' \leq l}$ lies in the range of R . Let $f_w^\#$ be the row given by Lemma 4.15 at (A^2, z^2) with L_0 , the signs $\epsilon_{l''}$, and $x|_{L_0}$.

We write $f^\# = f_w^\# = f[A^\#, z^\#]$, $a^\# := A^\# / q^*(A^\#)$, $F^\# := \text{supp } a^\# = F \cup \text{Ba} \cup \text{Pu}$, where Ba is the set of bank coordinates (the s_m of case (b) and the $j_{l''}$, $l'' \in L_0$) and Pu the set of pull coordinates $j_{l''}$, $l'' \in L_0$. Forced data of $f^\#$ carry the superscript $\#$, those of $f^{(2)}$ the superscript (2).

The moves act on pairwise disjoint sets of coordinates, except that (C3) may act after (C1) on the coordinate s_m when $c_m \in G(w)$: (C1) acts on the sets $S_{l''}^{\text{nat}}(l)$, $l'' \in G(w)$, which are disjoint and avoid $F \cup T(l)$; (C2) on $T(l)$; (C3) on $s_m \in S_{c_m}$ beyond $s_{\max}(l)$; (C4) on $S_{l''} \setminus [1, s_{\max}(l)]$ with $l'' \in L_0 \subseteq \text{Kp}(w)$, while $c_m \notin \text{Kp}(w)$ by Lemma 4.7. In case (a) of (C3), $\text{vs}_m z_{s_m}^1 \leq \text{vs}_m z_{s_m}^2 = \text{vs}_m z_{s_m}^1 + \eta_m \leq 1$, so $|z^2| \leq 1$; all moves are off F , and (A^2, z^2) satisfies $z^2 = \text{sign } A^2$ on $\text{supp } A^2$. The companion depends on f and w only, not on a mate or a scale.

Lemma 4.17 (Donor raise). *Let $l \geq l_f$ and $m \in I_D(w)$, $c := c_m$.*

- (i) $\mathfrak{d}_m \leq \text{Des}^{1/3} \Lambda_w / 8$.
- (ii) $\text{vs}_m u_c(\hat{z}^{(2)}) - \text{vs}_m u_c(\hat{z}) \in [\Lambda_w / \lambda_c - 2b_w - C_f \text{Des}^2 \Lambda_w^2, 3\Lambda_w / \lambda_c + 2b_w + C_f \text{Des}^2 \Lambda_w^2]$.
- (iii) *For every carrier $k \leq l$ which is not a donor, (C3) changes $u_k(\hat{z})$ by at most $C_f \text{Des}^2 \Lambda_w^2$, and by 0 if no bank is placed.*

Proof. The only carrier $\leq l$ whose vector meets s_m is c , with $u_c(s_m) = v_c(s_m)$. (i) $s_m \leq s_t(l)$, so $(\mu_{s_m}^*)^2 v_c(s_m) \geq \text{Des}^{-1/6}$ and $1/\lambda_c \leq D_r \leq \text{Des}^{1/6}$ (Lemma 4.1); $\nu \leq \|U\| = \frac{1}{16}$. (iii) In case (a) only z_{s_m} moves and $u_k(s_m) = 0$. Banks: Lemma 4.14(i) with base (a, z^1) ($\nu_0 = \nu$) and J the set of donor bank coordinates, $\|X\|^2 \leq \sum_m \mathfrak{d}_m^2 \leq N \text{Des}^{2/3} \Lambda^2$. (ii) (C1) changes $u_c(\hat{z})$ by at most $r_c^{\text{nat}} \leq b_w$ (only if $c \in G(w)$), and (C2) by at most $b_w \|u_c\|_1 \leq b_w$. In case (a), (C3) raises $\text{vs}_m u_c$ by $v_c(s_m) \eta_m = \Lambda/\lambda_c$ exactly, plus second-order effects of the banks of other blocks (bounded as in (iii)). In case (b) the closing raises $\text{vs}_m u_c$ by $v_c(s_m)(1 - \text{vs}_m z_{s_m}^1) \in [0, \Lambda/\lambda_c]$, and by Lemma 4.14(ii) the banks raise it by $\mathfrak{d}_m(\mu_{s_m}^*)^2 v_c(s_m)/\nu + R = 2\Lambda/\lambda_c + R$ with $|R| \leq C_f \|X\|^2 \leq C_f \text{Des}^2 \Lambda^2$. $\square$

Lemma 4.18 (Cost and data of the companion). *For $l \geq l_f$ and every sub-window w of level l that is clean for f :*

- (a) *in (C4) the hypotheses of Lemma 4.15 hold, and $\varepsilon_x \leq C_f \text{Des}^{1/6} \beta_w \leq c_T$;*
- (b) *for every $l'' \in \mathcal{C}(l)$ that is not a donor,*

$$u_{l''}(\hat{z}^\#) - u_{l''}(\hat{z}) = \Delta_{l''}^{C1} + \Delta_{l''}^{C2} + x_{l''} 1[l'' \in L_0] + \Delta_{l''}^H,$$

where $\Delta_{l''}^{C1} := \sum_{s \in S_{l''}^{\text{nat}}(l)} v_{l''}(s)(\epsilon_{l''} - z_s)$ if $l'' \in G(w)$ and $:= 0$ otherwise

($|\Delta_{l''}^{C1}| = r_{l''}^{\text{nat}} \leq b_w$), $|\Delta_{l''}^{C2}| \leq b_w$, and $|\Delta_{l''}^H| \leq C_f(\text{Des}^2 \Lambda_w^2 + \text{Des}^{4/3} \beta_w^2)$; in particular $|u_{l''}(\hat{z}^\#) - u_{l''}(\hat{z})| \leq C_f \text{Des}^{1/6} \beta_w$;

- (c) *for every donor, $\text{vs}_m u_{c_m}(\hat{z}^\#) - \text{vs}_m u_{c_m}(\hat{z}) \in [\Lambda_w/(2\lambda_{c_m}), 4\Lambda_w/\lambda_{c_m}]$;*
- (d) *the total mass placed off F is at most $C_f \text{Des}^{1/3} \Lambda_w$; $\|a^\# - a\|_1 + \|e^\# - e\|_H + |\nu^\# - \nu| \leq C_f \text{Des}^{1/3} \Lambda_w$, and $q^*(A^\#) \leq 2$;*
- (e) *with $\delta := \hat{z}^\# - \hat{z}$ and $\theta_w := C_f \text{Des} T_{\text{lo}}(w) \log(e/T_{\text{lo}}(w))$: $\max_m \Delta_m(\delta) \leq C_f \Lambda_w$, $c(\delta) \leq \theta_w T_{\text{lo}}(w)^2$, $p^*(f_w^\# - f) \leq \theta_w T_{\text{lo}}(w)^2$, and for every block m*

$$\begin{aligned} & |C_m^\# - C_m| + |M_m^\# - M_m| + |\sigma_m^\# - \sigma_m| + |q_0^\# - q_0| + |A_m^\# - A_m| \\ & + \|D_m(w_m^\# - w_m)\|_2 + \sum_k \lambda_{k,m} |w_m^\#(k) - w_m(k)| \leq \theta_w T_{\text{lo}}(w)^2; \end{aligned}$$

moreover $\theta_w \leq C_f u_w^{20} \rightarrow 0$ as $l \rightarrow \infty$.

Proof. (a) (T1): by (C1) and (3), $z^2 = \epsilon_{l''}$ on $S_{l''} \setminus [1, s_{\max}(l)] \subseteq S_{l''}^{\text{nat}}(l)$ for $l'' \in L_0 \subseteq G(w)$; (C2) acts inside $[1, s_{\max}(l)]$ and (C3) on the sets S_{c_m} , $c_m \notin L_0$. (T2): $\text{supp } A^2 \subseteq F \cup \{s_m\}$. Also $\|A^2\|_1 \leq 1 + N \text{Des}^{1/3} \Lambda \leq 2$ and $\nu_2 \geq \nu$ (Lemma 4.14). For $l'' \in L_0$ (a kept carrier, not a donor), the value moves from f to $f^{(2)}$ by at most $r_{l''}^{\text{nat}} + b_w + C_f \text{Des}^2 \Lambda^2$ (as in the proof of Lemma 4.17); since $\|r\|_1 = 1$ and every component of $\Sigma_t(w)$ is tiny at f , $|V_{(r,m)}^{(2)}| \leq b_w \Phi_{\max}(r, m) + 2b_w + C_f \text{Des}^2 \Lambda^2 \leq C_f \beta_w$. Hence $\varepsilon_x \leq \|x\|_2 \leq \|R^+\|_2 (\#\Sigma_t)^{1/2} \max |V^{(2)}| \leq H_{\text{tune}}(l) C_f \beta_w \leq C_f \text{Des}^{1/6} \beta_w$, and $C_f \text{Des}^{1/6} \beta_w \leq c_f \text{Des}^{-1}$ for $l \geq l_f$ by Lemma 4.1(b).

(b) $\text{supp } u_{l''} \subseteq \text{supp } y_{l''} \cup S_{l''} \subseteq T(l) \cup S_{l''}$, and the sets $S_{l'}^{\text{nat}}(l)$, $l' \neq l''$, avoid it. So (C1) changes $u_{l''}(\hat{z})$ only through $S_{l''}^{\text{nat}}(l)$, and only if $l'' \in G(w)$; there the change is $\Delta_{l''}^{C1}$, of modulus $\sum v_{l''}(s)|\epsilon_{l''} - z_s| = r_{l''}^{\text{nat}} \leq b_w \mathfrak{m}_{l''} \leq b_w$. (C2) changes it by at most $\sum_{j \in T(l)} |u_{l''}(j)| b_w \leq b_w$. (C3) contributes at most $C_f \text{Des}^2 \Lambda^2$ (Lemma 4.17(iii)), and (C4) contributes $x_{l''}$ exactly if $l'' \in L_0$ (Lemma 4.15(a)) and at most $C_T \varepsilon_x^2 \leq C_f \text{Des}^{4/3} \beta_w^2$ otherwise (Lemma 4.15(b)).

(c) Lemma 4.17(ii) and Lemma 4.15(b) ($c_m \notin L_0$), together with $b_w + \text{Des}^2 \Lambda^2 + \text{Des}^{4/3} \beta_w^2 \ll \Lambda \leq \Lambda/\lambda_{c_m}$ (Lemma 4.1(b)).

(d) The masses are $\sum_m \mathfrak{d}_m \leq N \text{Des}^{1/3} \Lambda / 8$, $\sum \mathfrak{p}_{l''} + \sum \mathfrak{n}_{l''} \leq C_f \text{Des}^{4/3} \varepsilon_x \ll \Lambda$. With y the total mass, $q^*(A^\# - a) \leq (1 + \|U\|)y$, so $|q^*(A^\#) - 1| \leq 2y$ and $\|a^\# - a\|_1 \leq 4y$; Lemma 4.14(iii) with base a gives $\|e^\# - e\| \leq 2\|X\|/\nu \leq y/(8\nu)$; and $\nu^\# = \|U^* A^\# / q^*(A^\#)$.

(e) Let k be a carrier. If $k \leq l$ is not a donor, $|u_k(\delta)| \leq C_f \text{Des}^{1/6} \beta_w$ by (b) (for $k \notin \mathcal{C}(l)$ there is nothing to prove). For a donor, $|u_k(\delta)| \leq 4\Lambda/\lambda_k$ by (c). For $k > l$, $|u_k(\delta)| \leq \|u_k\|_1 \|z^\# - z\|_\infty + \|e^\# - e\| \leq 3$, and $\sum_{k>l} \lambda_k \leq b_w^2/3$. Hence $\Delta_m(\delta) = \sum_k \lambda_k |u_k(\delta)| \leq C_f \text{Des}^{1/6} \beta_w + 4N\Lambda + b_w^2 \leq C_f \Lambda$, and $\sum_k \min(\lambda_k, |u_k(\delta)|) \leq l C_f \text{Des}^{1/6} \beta_w + 4ND_r \Lambda + b_w^2 \leq C_f \text{Des}^{1/6} \Lambda$. As $x \log(e/x)$ is increasing on $(0, 1]$, $c(\delta) \leq C_f \text{Des}^{1/6} \Lambda \log(e/\Lambda) \leq \theta_w T_{10}^2$ (note $\log(e/\Lambda) \leq 3 \log(e/T_{10})$). For $l \geq l_f$, $\max_m \Delta_m(\delta) \leq c_f$, and Lemma 2.12 gives $p^*(f^\# - f) \leq (1 + \|U\|)(\|a^\# - a\|_1 + C_f c(\delta)) \leq \theta_w T_{10}^2$ and the block data bounds; the bound for $\sum_k \lambda_{k,m} |w_m^\#(k) - w_m(k)|$ is the one obtained in the proof of Lemma 2.12(ii), and $|A_m^\# - A_m| \leq \Delta_m(\delta)$. Finally $T \log(e/T) \leq 2T^{1/2}$ for $T \in (0, 1]$ and $T_{10} \leq (u_w/(4\text{Des}))^{40}$ (Lemma 4.1(b)) give $\theta_w \leq C_f \text{Des} T_{10}^{1/2} \leq C_f u_w^{20}$, and $u_w \rightarrow 0$ (Lemma 4.1(e)). $\square$

The donor raise ($\Lambda_w = T_{10}^3$) dominates all other changes of coarse values, which are of order $\beta_w \leq T_{10}^4/l$ (closings and tuning) or of second order ($\text{Des}^2 \Lambda_w^2$); this *threshold buffer* makes the status of every kept carrier stable.

Lemma 4.19 (Status table; stability under the threshold drift). *For $l \geq l_f$ and every sub-window w of level l that is clean for f , with $f^\# = f_w^\#$, $\varrho^\#$ the relative positions at $f^\#$ and $\bar{\vartheta}_m^\# := \bar{\vartheta}(\hat{\zeta}_m^\#)$:*

- (a) *for every block m , $E_m := \sum_{k \neq k(c_m)} |\hat{\zeta}_m^\#(k) - \hat{\zeta}_m(k)| \leq C_f \text{Des}^{1/6} \beta_w$ (the sum runs over all k if $m \notin I_D(w)$);*
- (b) *$c_f \Lambda_w \leq \bar{\vartheta}_m^\# - \bar{\vartheta}_m \leq C_f \Lambda_w$ for $m \in I_D(w)$, and $|\bar{\vartheta}_m^\# - \bar{\vartheta}_m| \leq C_f \text{Des}^{1/6} \beta_w$ for $m \notin I_D(w)$;*
- (c) *for every $l'' \in \mathcal{C}(l)$: if $\varrho_{l''} \geq 1 + u_w$ then $\varrho_{l''}^\# \geq 1 + u_w/2$; if $\varrho_{l''} \leq 1 - u_w$ then $\varrho_{l''}^\# \leq 1 - u_w/2$; if $\varrho_{l''} \in [u_w, 1 - u_w]$ then $\varrho_{l''}^\# \in [u_w/2, 1 - u_w/2]$; if $\varrho_{l''} \leq b_w$ then $\varrho_{l''}^\# \leq \frac{1}{2}$; and every (K4) carrier has $\varrho_{l''}^\# \leq 1 - c_f \Lambda_w$, so that it is a strict non-peak of $f^\#$ with $\text{gap}^\# \geq c_f M_m^\# \Lambda_w > 0$;*
- (d) *every kept carrier l'' is a strict non-peak of $f^\#$, with $\text{gap}^\# \geq M_m^\# u_w/2$ in cases (K1), (K2) and $\text{gap}^\# \geq M_m^\#/2$ in case (K3); for (K1), (K2), (K4), $\text{sign } w_m^\#(k(l'')) = \text{sign } w_m(k(l'')) \neq 0$; and $q_{l''}^\# := \epsilon_{l''} \Phi_{l''} w_m^\#(k(l'')) / (m C_m^\#) = \text{val}_{l''}^\# / A_m^\#$;*
- (e) *contacts of f in $T(l) \setminus F$ are contacts of $f^\#$ with the same sign, every contact-like coordinate is a contact of $f^\#$ with $z_j^\# = \text{sign } z_j$, free-robust coordinates have $z_j^\# = z_j$; for $l'' \in G(w)$, $z^\# = z^{(2)} = \epsilon_{l''}$ on $S_{l''}^{\text{nat}}(l)$ except at the pull coordinate $j_{l''}$ (if $l'' \in L_0$) and at s_m (if $l'' = c_m$).*

Proof. (a) $\hat{\zeta}_m(k) = \lambda_k u_k(\hat{z})$; use Lemma 4.18(b) for coarse non-donors, $\sum_{k>l} \lambda_k \cdot 3 \leq b_w^2$ for fine carriers, and note that block m contains at most one donor.

(b) Let $m \in I_D(w)$, $\zeta := \hat{\zeta}_m$, $\zeta' := \hat{\zeta}_m^\#$, $c := k(c_m)$, a peak of ζ . By Lemma 4.18(c), $|\zeta'(c)| = |\zeta(c)| + s$ with the same sign and $s \in [\Lambda/2, 4\Lambda]$. By (a), $E_m \leq C_f \text{Des}^{1/6} \beta_w \leq A_m s / (8(A_m + \bar{\vartheta}_m + 1))$ for $l \geq l_f$. Lemma 2.7(b) gives $\bar{\vartheta}_m^\# - \bar{\vartheta}_m \geq \min\{1, A_m s / (8\phi_m(A_m + \bar{\vartheta}_m + 1))\} \geq c_f \Lambda$, $\phi_m := \|\Phi_m\|_2^2$, and $\bar{\vartheta}_m^\# - \bar{\vartheta}_m \leq C_1(s + E_m) \leq C_f \Lambda$, where C_1, h_0 are the constants of Lemma 2.7

for ζ (they depend only on f) and $C_1(s + E_m) < h_0$ for $l \geq l_f$. For $m \notin I_D(w)$, Lemma 2.7(a) with $\|\zeta' - \zeta\|_1 = E_m$.

(c) For every k with $\zeta(k) \neq 0$, $\varrho_k(\zeta') = (|\zeta'(k)|/|\zeta(k)|)\varrho_k(\zeta)\bar{\vartheta}_m/\bar{\vartheta}_m^\#$, and $\varrho_{l''} = \varrho_{k(l'')}\hat{\zeta}_m$. If $\varrho_{l''} \geq u_w$ and l'' is not a donor, then $|\zeta(k)| = \varrho\Phi_{l''}^2\bar{\vartheta}_m \geq u_w\bar{\vartheta}_m/D_r^2$ and $|\zeta'(k) - \zeta(k)| \leq C_f\text{Des}^{1/6}\beta_w$, so the first factor is $1 + O(\text{Des}^{1/2}\beta_w/u_w)$; by (b), $\bar{\vartheta}_m/\bar{\vartheta}_m^\# \in [1 - C_f\Lambda, 1 + C_f\text{Des}^{1/6}\beta_w]$. Since $\Lambda + \text{Des}^{1/2}\beta_w/u_w \leq u_w^2$ for $l \geq l_f$ (Lemma 4.1(b)), $\varrho^\#/\varrho \in [1 - u_w/4, 1 + u_w/4]$, which gives the three robust cases (for $u_w \leq \frac{1}{4}$, $(1+u)(1-u/4) \geq 1+u/2$ and $(1-u)(1+u/4) \leq 1-u/2$). For a donor, $|\zeta'(c)| \geq |\zeta(c)|$ and $\bar{\vartheta}_m/\bar{\vartheta}_m^\# \geq 1 - C_f\Lambda$, so $\varrho^\# \geq (1+u_w)(1 - C_f\Lambda) \geq 1 + u_w/2$. If $\varrho_{l''} \leq b_w$, $\varrho_{l''}^\# = m|u_{l''}(\hat{z}^\#)|/(\Phi_{l''}\bar{\vartheta}_m^\#) \leq \varrho_{l''}\bar{\vartheta}_m/\bar{\vartheta}_m^\# + C_fD_r\text{Des}^{1/6}\beta_w \leq \frac{1}{2}$. For a (K4) carrier, $m \in I_D(w)$, $\varrho \leq 1 + b_w$, the first factor is at most $1 + C_f\text{Des}^{1/2}\beta_w$ and $\bar{\vartheta}_m/\bar{\vartheta}_m^\# \leq 1/(1 + c_f\Lambda/\bar{\vartheta}_m) \leq 1 - c'_f\Lambda$; as $b_w + C_f\text{Des}^{1/2}\beta_w \leq c'_f\Lambda/4$ for $l \geq l_f$, $\varrho^\# \leq 1 - c'_f\Lambda/2$. Finally $\text{gap}^\# = M_m^\#(1 - \varrho^\#)$ at strict non-peaks.

(d) Kept carriers are of types (K1)–(K4) and are strict non-peaks of $f^\#$ by (c). At a strict non-peak k of $f^\#$, [I, Lemma 1.7] gives $w_m^\#(k) = C_m^\#\hat{\zeta}_m^\#(k)/(\Phi_m(k)^2A_m^\#)$, so $\text{sign } w_m^\#(k) = \text{sign } u_{l''}(\hat{z}^\#)$ and $q_{l''}^\# = \epsilon_{l''}\lambda_{l''}u_{l''}(\hat{z}^\#)/(m\Phi_{l''}A_m^\#) = \text{val}_{l''}^\#/A_m^\#$. For (K1), (K2), (K4), $|u_{l''}(\hat{z})| = \varrho\Phi_{l''}\bar{\vartheta}_m/m \geq u_w\bar{\vartheta}_m/(mD_r)$ exceeds the change $C_f\text{Des}^{1/6}\beta_w$, so the sign is that of $u_{l''}(\hat{z})$, i.e. of $w_m(k)$.

(e) (C2) closes in the direction of $\text{sign } z_j$, and no other move acts on $T(l)$; (C1) sets $z^1 = \epsilon_{l''}$ on $S_{l''}^{\text{nat}}(l)$, and later moves act only on s_m and on pull coordinates (banks do not change z). $\square$

Lemma 4.20 (Exactness, no slaving, robust rates). *For $l \geq l_f$ and every sub-window w of level l that is clean for f :*

- (a) (No slaving.) *The change of the value of a coarse non-donor carrier from f to $f_w^\#$ is the sum of its own closing, the target closings, its own tuning and a second-order remainder (Lemma 4.18(b)); it does not depend on the closing of the room of any other carrier.*
- (b) *Every class-G room on $S_{l''}^{\text{nat}}(l)$ is exactly closed at $f_w^\#$ off the pull and donor coordinates, every contact-like target coordinate is an exact contact, and the zero-cost cone of the pattern read at $f_w^\#$ is $C(\kappa(w))$ (the types on $T(l) \setminus F$ are the same, Lemma 4.19(e)).*
- (c) *For every tiny component (r, m) of $\kappa(w)$, $D_{r,m}(f_w^\#) = 0$ exactly. For every robust component, $|D_{r,m}(f_w^\#)| \geq u_w\Phi_{\max}(r, m)/2$.*
- (d) *Free-robust target coordinates keep $z_j^\# = z_j$ (room $\geq u_w$), and robust relative positions stay robust in the sense of Lemma 4.19(c).*

Proof. (a), (b), (d): Lemma 4.18(b) and Lemma 4.19(c),(e). (c) If $(r, m) \in \Sigma_t(w)$, every $l'' \in \text{supp } r$ with $m(l'') = m$ lies in L_0 , so by Lemma 4.15(a), $D_{r,m}(f^\#) = \sum r(l'')(\text{val}_{l''}^{(2)} + x_{l''}) = V_{(r,m)}^{(2)} + (Rx)_{(r,m)} = 0$. For a robust component, $|D_{r,m}(f^\#) - D_{r,m}(f)| \leq \max_{l'' \in \text{supp } r} |\text{val}_{l''}^\# - \text{val}_{l''}| \leq C_f\text{Des}^{1/6}\beta_w$ (donors are not kept, hence not in $\text{supp } r$), while $u_w\Phi_{\max}(r, m) \geq u_w/D_r \geq 2C_f\text{Des}^{1/6}\beta_w$ for $l \geq l_f$. $\square$

Thus closing one object never reopens or moves another exactified object, the exactified set is closed under the moves, and no robust object moves by more than $C_f\text{Des}^{1/6}\beta_w$. We do not claim that the room of a donor stays robust at $f_w^\#$ (in case (b) of (C3) the closing at s_m may remove most of it); it is not used: donors are

pinned at f (Lemma 4.10), are not kept, and the data constructed below vanish at s_m .

Lemma 4.21 (Inward push of near-threshold coordinates). *Fix one block and drop its index. Let $\zeta \in \ell_1 \setminus \{0\}$, $\bar{\vartheta} := \bar{\vartheta}(\zeta)$, $\mathbf{a} := |\zeta|$, $0 < b \leq \frac{1}{2}$, let $\mathcal{N}$ be a nonempty set of coordinates with $|\mathbf{v}_k(\zeta) - \bar{\vartheta}| \leq \bar{\vartheta}b$ for $k \in \mathcal{N}$, and let $c \geq 4b(\mathbf{a} + \bar{\vartheta})$. Let $\zeta' \in \ell_1$ satisfy, for $k \in \mathcal{N}$, $\text{sign } \zeta'(k) = \text{sign } \zeta(k)$ and $|\zeta'(k)| = |\zeta(k)| - c_k \Phi(k)^2$ with $c_k \in [c, \bar{\vartheta}/2]$, and $E := \sum_{k \notin \mathcal{N}} |\zeta'(k) - \zeta(k)| \leq \bar{\vartheta}c \sum_{k \in \mathcal{N}} \Phi(k)^2 / (8(\mathbf{a} + \bar{\vartheta}))$. Then $\bar{\vartheta}(\zeta') > \bar{\vartheta}$, and $\mathbf{v}_k(\zeta') \leq \bar{\vartheta}(\zeta') - (c - \bar{\vartheta}b)$ for every $k \in \mathcal{N}$: every pushed coordinate is a strict non-peak of ζ' .*

Proof. Write $\mathbf{A}' = \mathbf{A}(\cdot; \zeta')$, $\mathbf{B}' = \mathbf{B}(\cdot; \zeta')$, $\Phi_{\mathcal{N}}^2 := \sum_{k \in \mathcal{N}} \Phi(k)^2$. For $k \in \mathcal{N}$, $|\zeta'(k)| - \bar{\vartheta}\Phi(k)^2 = \Phi(k)^2(\mathbf{v}_k - c_k - \bar{\vartheta}) < 0$, so these terms of $\mathbf{A}'(\bar{\vartheta})$ vanish, while their terms in $\mathbf{A}(\bar{\vartheta}; \zeta)$ are at most $\bar{\vartheta}b\Phi(k)^2$; the other terms change by at most $|\zeta'(k) - \zeta(k)|$. With Lemma 2.6(b), $\mathbf{A}'(\bar{\vartheta}) \geq \mathbf{a} - y$, $y := \bar{\vartheta}b\Phi_{\mathcal{N}}^2 + E$. In $\mathbf{B}'(\bar{\vartheta})$ the terms with $k \notin \mathcal{N}$ change by at most $2\bar{\vartheta}|\zeta'(k) - \zeta(k)|$; for $k \in \mathcal{N}$, $\min(\bar{\vartheta}, \mathbf{v}_k) \geq \bar{\vartheta}(1 - b)$ and $0 \leq \mathbf{v}'_k = \mathbf{v}_k - c_k \leq \bar{\vartheta}(1 + b) - c_k$, so the term drops by at least $\Phi(k)^2[\bar{\vartheta}^2(1 - b)^2 - (\bar{\vartheta}(1 + b) - c_k)^2] = \Phi(k)^2(c_k - 2\bar{\vartheta}b)(2\bar{\vartheta} - c_k) \geq \Phi(k)^2\bar{\vartheta}(c - 2\bar{\vartheta}b)$. Hence $\mathbf{B}'(\bar{\vartheta}) \leq \mathbf{a}^2 - \bar{\vartheta}(c - 2\bar{\vartheta}b)\Phi_{\mathcal{N}}^2 + 2\bar{\vartheta}E$, and, since $\mathbf{A}'(\bar{\vartheta})^2 \geq \mathbf{a}^2 - 2\mathbf{a}y$ (trivial if $\mathbf{a} < y$),

$$\Xi(\bar{\vartheta}; \zeta') \geq \bar{\vartheta}\Phi_{\mathcal{N}}^2(c - 2b(\mathbf{a} + \bar{\vartheta})) - 2(\mathbf{a} + \bar{\vartheta})E \geq \frac{1}{2}\bar{\vartheta}c\Phi_{\mathcal{N}}^2 - \frac{1}{4}\bar{\vartheta}c\Phi_{\mathcal{N}}^2 > 0.$$

By Lemma 2.6(d), $\bar{\vartheta}(\zeta') > \bar{\vartheta}$. Finally $\mathbf{v}_k(\zeta') = \mathbf{v}_k - c_k \leq \bar{\vartheta}(1 + b) - c < \bar{\vartheta}(\zeta') - (c - \bar{\vartheta}b)$. $\square$

Lemma 4.21 is not used in the construction of $f_w^\#$: a push that dominates the threshold drift would move the d -components of the rays through the pushed carriers by amounts between b_w and u_w and destroy the tiny/robust dichotomy on which (C4) rests, whereas the donor raise rescales the d -coefficients of untouched carriers of a block by one common factor (Corollary 2.8). Per carrier, a far pull at $j \in S_{l''}$ realizes a push of $\zeta(k(l''))$ by $2\lambda_{l''}v_{l''}(j)$, and by bounded gaps every size between c and $2^{G(l)}c$ is available; Lemma 4.21 is the appropriate tool when no tiny d -component passes through the pushed carriers.

4.4. Transplant to exact d -neutral two-piece data.

Lemma 4.22 (Split with respect to an arbitrary sign vector). *Let $(B_\pm, \Theta_\pm)$ be a two-sided decomposition of $g \in \mathcal{C}(f)$ at some scale $t > 0$, let $z' \in B_{\ell_\infty}$, $K' := \{j \notin F : |z'_j| = 1\}$, and suppose $\Delta B 1_{F^c} = V + e$ with $e \in \ell_1$ and $V \in \ell_1$ supported in K' and z' -signed. Then there are $\chi : K' \rightarrow [0, 1]$ and $\mathbf{e}_\pm \in \ell_1(F^c)$ with $B_+ 1_{F^c} = \chi V + \mathbf{e}_+$, $B_- 1_{F^c} = -(1 - \chi)V + \mathbf{e}_-$ and*

$$\|\mathbf{e}_\pm\|_1 \leq \|e\|_1 + \sum_{j \notin F} (\phi_{z'_j}(B_+(j)) + \phi_{-z'_j}(B_-(j))).$$

Proof. The proof of [I, Lemma 8.26], read with z' in place of z , gives the pointwise bound $|\mathbf{e}_\pm(j)| \leq |e(j)| + \phi_{z'_j}(B_+(j)) + \phi_{-z'_j}(B_-(j))$ for $j \notin F$; it uses only [I, Lemma 8.23(a),(b)]. Sum over $j \notin F$. $\square$

Lemma 4.23 (Ray removal). *Let E be a finite set, $r_1, \dots, r_p \in [0, \infty)^E$ with $\|r_i\|_1 = 1$, $\mu \in [0, \infty)^p$, $\tau := \sum_i \mu_i r_i$, $Q : \mathbb{R}^E \rightarrow \mathbb{R}$ linear, $Q_i := Q(r_i)$ and $Q_{\min} := \min\{|Q_i| : Q_i \neq 0\}$ ($:= \infty$ if all $Q_i = 0$). Then there is μ' with $0 \leq \mu'_i \leq \mu_i$ such that $\tau' := \sum_i \mu'_i r_i$ satisfies $Q(\tau') = 0$, $0 \leq \tau' \leq \tau$ coordinatewise, and $\|\tau - \tau'\|_1 = \sum_i (\mu_i - \mu'_i) \leq |Q(\tau)|/Q_{\min}$.*

Proof. If $Q(\tau) = 0$ take $\mu' = \mu$. Let $Q(\tau) = \mathbf{e} > 0$ (the case < 0 is symmetric). Then $S_+ := \sum_{Q_i > 0} \mu_i Q_i \geq \mathbf{e}$; put $\vartheta := \mathbf{e}/S_+ \in (0, 1]$, $\mu'_i := (1 - \vartheta)\mu_i$ if $Q_i > 0$ and $\mu'_i := \mu_i$ otherwise. Then $Q(\tau') = \mathbf{e} - \vartheta S_+ = 0$, and, since the r_i are nonnegative with $\|r_i\|_1 = 1$, $\|\tau - \tau'\|_1 = \vartheta \sum_{Q_i > 0} \mu_i \leq \vartheta S_+/Q_{\min} = \mathbf{e}/Q_{\min}$. $\square$

In the next proposition the two-piece data of [I, Definition 7.7] and the coordinate kinds [1]–[3] of Subsection 2.4 are read at the companion $f^\# = f_w^\#$, whose base support is $F^\# = F \cup \text{Ba} \cup \text{Pu}$; $\Gamma_w^\#$ denotes Γ_w computed at $f^\#$.

Proposition 4.24 (Transplant at the exactifying companion). *Let w be a sub-window of level $l \geq l_f$ that is clean for f and at which (SH_w) and (VR_w) hold, and let $f^\# = f_w^\#$. Let $g \in \mathcal{C}(f)$, $\eta \leq \eta_*$ with $\eta_\Gamma(\eta) \leq 1$, $t \in \mathcal{W}(w)$ with $t \leq \min(t_\eta, 1)$, and let $(B_\pm, \Theta_\pm)$ be a two-sided decomposition of g at scale t . Then there is $g_t \in X^*$ carrying d -neutral two-piece data $(b^\pm, \omega^\pm)$ at $f^\#$ such that:*

- (i) $b^\pm(\xi^\#) = 0$, $t\|b^\pm\|_1 \leq A_0$ with an f -constant A_0 , and $\Gamma_w^\#(b^\pm, \omega^\pm) \leq (\sqrt{1 + \eta_\Gamma(\eta)} + C_f \theta_w + K_w t)^2$;
- (ii) $p^*(g - g_t) \leq K_w t$ with $K_w := C_f \text{Des}(l) u_w^{-4}$;
- (iii) every $k \in \text{supp } \omega_m^\pm$ is, at $f^\#$, of kind [1], of kind [2] with $\gamma_B := \gamma(w) := u_w \min_m M_m/4$ and $A_2 := 22$, or of kind [3] for the side of the pair with $A_2 = 22$;
- (iv) $z_i^\# b^+(i) \geq 0 \geq z_i^\# b^-(i)$ for $i \in \text{Ba}$, and $t|b^\pm(i)| \leq |a^\#(i)|$ for $i \in \text{Pu}$.

Proof. Write $u := u_w$, $b := b_w$, $\Lambda := \Lambda_w$, and let $K_{\text{pin}} \leq C_f l^2 D_r^3 u^{-3}$ be as in Lemma 4.10.

Step 0 (pinning at f). By (SH_w) and Lemma 4.9, $|\Delta d_m| M_m \leq K'_d t$ for every m ; by Lemma 4.10, $|\Delta \theta_{l''}| \leq K_{\text{pin}} t$ for class R and $|\tau_{l''}| \leq K_{\text{pin}} t$ on $P(w)$, and Lemma 4.3 holds.

Step 1 (zero-cost projection). Let $\tau := (\tau_{l''})_{l'' \in G(w)}$ and $\beta_{l''} := 12\lambda_{l''}/t$. For $j \in T(l) \setminus F$, $\Delta B(j) = L_j(\tau) + e_0(j)$ with $L_j(\tau) := \sum_{G(w)} \epsilon_{l''} \tau_{l''} u_{l''}(j)$ and e_0 as in Lemma 4.3(b). We estimate the violations of the conditions defining $Z_{\kappa(w)}(\beta)$ by τ . Nonnegativity on $\text{Kp}(w)$: $\sum(\tau)_- \leq K_g t$. At a contact-like j , the condition is $\text{sign } z_j L_j \geq 0$, and $(\text{sign } z_j L_j(\tau))_- \leq (\text{sign } z_j \Delta B(j))_- + |e_0(j)| \leq \phi_{z_j}(\Delta B(j)) + |e_0(j)|$ (if $\text{sign } z_j x < 0$, then $\phi_{z_j}(x) = (1 + |z_j|)|x|$). At a free-robust j the condition is $L_j = 0$, and $|L_j(\tau)| \leq |\Delta B(j)| + |e_0(j)| \leq \phi_{z_j}(\Delta B(j))/u + |e_0(j)|$ by [I, Lemma 8.23(a)]. Summing with [I, Lemma 8.24], these violations are at most $(t/q_0)(1 + 1/u) + 2\|e_0\|_1$. The equations $\tau = 0$ on $P(w)$ are violated by at most $l K_{\text{pin}} t$, and the box conditions not at all ($|\tau| \leq 6\lambda/t$). So the total violation is $V_w \leq C_f l K_{\text{pin}} t$. As $0 \in Z_{\kappa(w)}(\beta)$, Definition 4.11 gives $\tau_0 \in Z_{\kappa(w)}(\beta)$ with $\|\tau - \tau_0\|_1 \leq G^{**}(l) V_w$.

Step 2 (d -rows at $f^\#$). For a block m and $\tau' \in \mathbb{R}^{G(w)}$ put

$$Q_m^\#(\tau') := \sum_{l'' \in \text{Kp}(w), m(l'')=m} q_{l''}^\# \tau'_{l''} = \frac{1}{A_m^\#} \sum_{l'' \in \text{Kp}(w), m(l'')=m} \text{val}_{l''}^\# \tau'_{l''}$$

(Lemma 4.19(d)). By [I, (17)],

$$\sum_{l'' \in G(w), m(l'')=m} q_{l''} \tau_{l''} = -\Delta d_m M_m + r_m + \frac{1}{m C_m} \sum \Phi w \Delta \theta,$$

the last sum over class R and fine carriers of block m ; hence, with Step 0, the sums over the kept carriers of block m satisfy $|\sum q\tau| \leq C_f l K_{\text{pin}} t$ and $|\sum q\tau_0| \leq$

$C_f(lK_{\text{pin}}t + G^{**}V_w)$. At a kept strict non-peak of f , $q = \text{val}/A_m$; at a kept (K4) peak of f , $q = \text{val}/(\varrho A_m)$ with $\varrho \in [1, 1+b]$ (Remark 4.12(c)). With Lemma 4.18(b), $|q_{l''}^\# - q_{l''} A_m/A_m^\#| \leq C_f \text{Des}^{1/6} \beta_w$, and $\sum \tau_0 \leq 12 \sum \lambda/t \leq 4/t$; since $\text{Des}^{1/6} \beta_w \leq t^2$ and $A_m/A_m^\# \leq 2$,

$$|Q_m^\#(\tau_0)| \leq K_Q t, \quad K_Q := C_f(lK_{\text{pin}} + G^{**}(l)V_w/t + 1) \leq C_f G^{**}(l) l^3 D_r^3 u^{-3}.$$

Step 3 (exact d-repair by ray removal). $\tau_0 \in C(\kappa(w))$, so $\tau_0 = \sum_{r \in \text{Ext}(\kappa(w))} \mu_r r$ with $\mu_r \geq 0$. Since $r = 0$ on $P(w)$, $Q_m^\#(r) = D_{r,m}(f^\#)/A_m^\#$. By Lemma 4.20(c) every component of r at $f^\#$ is 0 (tiny at f , or block not met) or has modulus $\geq u\Phi_{\max}(r, m)/2 \geq u/(2D_r)$ (robust at f); by (VR_w) , r has at most one nonzero component at $f^\#$, in a block $m(r)$ ($m(r)$ undefined if there is none). For each block m apply Lemma 4.23 to the rays with $m(r) = m$, the coefficients μ_r , and $Q := Q_m^\#$: as rays with $m(r) \neq m$ have $Q_m^\#(r) = 0$, $Q_m^\#(\sum_{m(r)=m} \mu_r r) = Q_m^\#(\tau_0)$, and we obtain $\mu'_r \in [0, \mu_r]$ with $Q_m^\#(\sum_{m(r)=m} \mu'_r r) = 0$ and $\sum_{m(r)=m} (\mu_r - \mu'_r) \leq |Q_m^\#(\tau_0)| 2D_r A_m^\# / u$. Put $\mu'_r := \mu_r$ when $m(r)$ is undefined and $\tau' := \sum_r \mu'_r r$. Then $\tau' \in C(\kappa(w))$, $0 \leq \tau' \leq \tau_0$ coordinatewise (so $\tau' \leq \beta$ on $\text{Kp}(w)$ and $\tau' = 0$ on $P(w)$), $Q_{m'}^\#(\tau') = 0$ for every block m' , and $\|\tau_0 - \tau'\|_1 \leq C_f N D_r K_Q t / u$. Put $V' := \sum_{l'' \in \text{Kp}(w)} \epsilon_{l''} \tau'_{l''} u_{l''} 1_{F^c}$. Then

$$\|\Delta B 1_{F^c} - V'\|_1 \leq \|e_0\|_1 + \|\tau - \tau'\|_1 \leq K_U t,$$

$$K_U := K_g + 1 + G^{**}V_w/t + C_f N D_r K_Q / u \leq C_f G^{**}(l) l^3 D_r^4 u^{-4}.$$

Zero cost. Let $z^{(2)}$ be the sign vector of $f^{(2)}$ and $K^{(2)} := \{j \notin F : |z_j^{(2)}| = 1\}$. We claim that V' is $z^{(2)}$ -signed and supported in $K^{(2)}$. Off F , $\text{supp } u_{l''} \subseteq T(l) \cup S_{l''}$ and $S_{l''} \setminus F \subseteq S_{l''}^{\text{nat}}(l) \cup T(l)$. On $S_{l''}^{\text{nat}}(l)$ only $u_{l''}$ is nonzero among the vectors of coarse carriers, $V'(s) = \epsilon_{l''} \tau'_{l''} v_{l''}(s)$ with $\tau'_{l''} \geq 0$, and $z_s^{(2)} = \epsilon_{l''}$ by (C1) (for kept l'' ; the coordinate s_m lies in S_{c_m} and $V'(s_m) = 0$, since donors are not kept). On $T(l) \setminus F$, $V'(j) = L_j(\tau')$, and the rows of $\kappa(w)$ give sign $z_j L_j(\tau') \geq 0$ at contact-like j , where $z_j^{(2)} = \text{sign } z_j$ (C2), and $L_j(\tau') = 0$ at free-robust j . This proves the claim.

Step 4 (split with respect to $z^{(2)}$). We bound $\Sigma := \sum_{j \notin F} (\phi_{z_j^{(2)}}(B_+(j)) + \phi_{-z_j^{(2)}}(B_-(j)))$. Where $z_j^{(2)} = z_j$, the terms sum to at most t/q_0 [I, Lemma 8.24]. Where $z_j^{(2)} = \pm 1$ and $z_j^{(2)} z_j \geq 0$ (the (C2) coordinates and the (C1) coordinates with $\epsilon_{l''} z_s \geq 0$), $\phi_{z_j^{(2)}} \leq 2\phi_{z_j}$ and $\phi_{-z_j^{(2)}} \leq 2\phi_{-z_j}$ (if $z_j^{(2)} x < 0$ then $\phi_{z_j^{(2)}}(x) = 2|x| \leq 2\phi_{z_j}(x)$), so these terms sum to at most $2t/q_0$. At the remaining modified coordinates (the *flipped* (C1) coordinates, $\epsilon_{l''} z_s < 0$, and the s_m), $\phi_{\pm z_j^{(2)}}(x) \leq 2|x|$ and $|B_+(j)| + |B_-(j)| \leq |\Delta B(j)| + \phi_{z_j}(B_+(j)) + \phi_{-z_j}(B_-(j))$ [I, Lemma 8.23(b)]. On $S_{l''}^{\text{nat}}(l)$, $\Delta B(s) = \epsilon_{l''} \tau_{l''} v_{l''}(s) - r(s)$ (proof of Lemma 4.3), and the flipped coordinates carry $v_{l''}$ -mass at most $\sum v_{l''}(s)(1 - \epsilon_{l''} z_s) = r_{l''}^{\text{nat}} \leq b$; so $\sum_{\text{flipped}} |\Delta B| \leq \sum_{l''} (6\lambda_{l''}/t)b + \frac{8b^2}{3t} \leq t^2$. At s_m only the donor c_m and fine carriers live, so $|\Delta B(s_m)| \leq |\Delta \theta_{c_m}| + t^7 \leq K_{\text{pin}} t + t^7$ (Lemma 4.10). Hence $\Sigma \leq C(1/q_0 + NK_{\text{pin}} + 1)t$, and Lemma 4.22 with $z' = z^{(2)}$, $V = V'$ and $e := \Delta B 1_{F^c} - V'$ gives $\chi : K^{(2)} \rightarrow [0, 1]$ and $\mathbf{e}_\pm$ with

$$B_+ 1_{F^c} = \chi V' + \mathbf{e}_+, \quad B_- 1_{F^c} = -(1 - \chi) V' + \mathbf{e}_-,$$

$$\|\mathbf{e}_\pm\|_1 \leq K_S t, \quad K_S := K_U + C(1/q_0 + NK_{\text{pin}} + 1).$$

Step 5 (data at $f^\#$). For a block m and a coordinate k define $\omega_m^\pm(k)$ as follows: $\omega_m^+(k) := \omega_{+,m}(k)$ if $k = k(l'')$ with $l'' \in \text{Kp}(w)$; $\omega_m^+(k) := 0$ if k is fine, or a peak of $f^\#$, or a peak of f and a strict non-peak of $f^\#$; otherwise (k coarse, not kept, a strict non-peak of f and of $f^\#$) $\omega_m^+(k) := \max\{-2\text{gap}_m^\#(k)/t, \min\{\omega_{+,m}(k), 2\text{gap}_m^\#(k)/t\}\}$ if $\text{gap}_m^\#(k) \geq t^2$ and $:= 0$ if $\text{gap}_m^\#(k) < t^2$. Put

$$\begin{aligned} \omega_m^- &:= \omega_m^+ + \sum_{l'' \in \text{Kp}(w), m(l'')=m} (\epsilon_{l''} \tau'_{l''} / \lambda_{l''}) e_{k(l'')}, \\ b^+ &:= B_+ 1_F + \chi V' - \varkappa a / a(\hat{z}^\#), \quad \varkappa := (B_+ 1_F + \chi V')(\hat{z}^\#), \\ b^- &:= b^+ - \sum_{l'' \in \text{Kp}(w)} \epsilon_{l''} \tau'_{l''} u_{l''}, \quad g_t := b^+ + \sum_m R_m^* (\omega_m^+ - d_m^\#(\omega_m^+) w_m^\#). \end{aligned}$$

Here $a(\hat{z}^\#) = \|a\|_1 + \langle U^* a, e^\# \rangle$, and by (5) with base a , $\langle U^* a, e^\# \rangle = \nu^2 / (\nu^2 + \|X\|^2)^{1/2} \in [\nu - \|X\|^2 / (2\nu), \nu]$, so $a(\hat{z}^\#) \in [\frac{1}{2}, 1]$. Finally, at every kept (K4) coordinate k (with $\varsigma_k := \text{sign } w_m(k) = \text{sign } w_m^\#(k)$) apply the *shift*: replace $\omega_m^\pm(k)$ by $\omega_m^\pm(k) + o_k$, where $\varsigma_k o_k := (-\frac{3}{2} \text{gap}_m^\#(k)/t - \varsigma_k \omega_m^-(k))_+$. The shift does not change $\omega_m^- - \omega_m^+$.

Admissibility. Off $F^\#$ we have $z^\# = z^{(2)}$, $b^+ = \chi V'$ and $b^- = -(1 - \chi)V'$; these are $z^\#$ -signed, resp. $(-z^\#)$ -signed, and supported in $K^\#$ by Step 3. The vectors $\omega_m^\pm$ are finitely supported in the strict non-peak set of $f^\#$ (kept carriers are strict non-peaks of $f^\#$ by Lemma 4.19(d)). Since $R_m^* e_k = \lambda_{k,m} u_{k,m}$, $\sum_m R_m^* (\omega_m^- - \omega_m^+) = \sum_{\text{Kp}(w)} \epsilon \tau' u = b^+ - b^-$, and

$$d_m^\#(\omega_m^-) - d_m^\#(\omega_m^+) = \sum_{l'' \in \text{Kp}(w), m(l'')=m} \frac{\epsilon_{l''} \tau'_{l''}}{\lambda_{l''}} \frac{\Phi_{l''}^2 w_m^\#(k(l''))}{C_m^\#} = Q_m^\#(\tau') = 0.$$

So (b^-, ω^-) also represents g_t , and the data are d -neutral two-piece data at $f^\#$. Since $b^+(\hat{z}^\#) = \varkappa - \varkappa = 0$, $b^+(\xi^\#) = 0$, and $b^-(\xi^\#) = g_t(\xi^\#) = b^+(\xi^\#) = 0$ by [I, Lemma 3.2] at $f^\#$.

(iv). At a pull coordinate $j = j_{l''}$, $V'(j) = \epsilon_{l''} \tau'_{l''} v_{l''}(j)$, so $t|b^\pm(j)| \leq t \tau'_{l''} v_{l''}(j) \leq 12 \lambda_{l''} v_{l''}(j) = \mathfrak{p}_{l''}/2 \leq |a^\#(j)|$, because $q^*(A^\#) \leq 2$. At a tuning bank $j'_{l''}$, $z_j^{(2)} = z_{j'}^\# = \epsilon_{l''}$, so $b^+(j') = \chi V'(j')$ and $b^-(j') = -(1 - \chi)V'(j')$ have the required signs; at a donor bank s_m , $V'(s_m) = 0$, so $b^\pm(s_m) = 0$.

Estimates. First, $B_+ - b^+ = \mathfrak{e}_+ + \varkappa a / a(\hat{z}^\#)$, and $\varkappa = (B_+ - \mathfrak{e}_+)(\hat{z}^\#)$ with

$$B_+(\hat{z}^\#) = B_+(\hat{z}) + \sum_{j \notin F} B_+(j)(z_j^\# - z_j) + \langle U^* B_+, e^\# - e \rangle.$$

Here $|B_+(\hat{z})| \leq t/(2q_0)$ [I, Lemma 8.7(a)]; at the (C2) coordinates and the (C1) coordinates with $\epsilon z_s \geq 0$, $|z_j^\# - z_j| = 1 - |z_j|$ and $(1 - |z_j|)|B_+(j)| \leq \phi_{z_j}(B_+(j))$, in total $\leq t/(2q_0)$; at flipped coordinates, at the s_m and at pulls, $|z_j^\# - z_j| \leq 2$ and $|B_+(j)| \leq |\Delta B(j)| + \phi_{z_j}(B_+(j)) + \phi_{-z_j}(B_-(j))$, where $\sum_{\text{pulls}} |\Delta B(j)| \leq \sum_{l''} 6 \lambda_{l''} v_{l''}(j_{l''})/t + t^7 \leq 6 \cdot 2^{G(l)} \varepsilon_x / t + t^7 \leq t^2$; and $|\langle U^* B_+, e^\# - e \rangle| \leq \|U\| \|B_+\|_1 \|e^\# - e\| \leq C_f \text{Des}^{1/3} \Lambda / t \leq t$, using $t \|B_+\|_1 \leq \eta \leq 1$ [I, Lemma 8.6] and Lemma 4.18(d). With $\|\hat{z}^\#\|_\infty \leq 1 + \|U\|$, $|\varkappa| \leq C(K_S + N K_{\text{pin}} + 1/q_0 + 1)t$, and $\|B_+ - b^+\|_1 \leq K_S t + 2|\varkappa|$.

Next, for the blocks put $X_m := \Theta_{+,m} - \omega_m^+ + d_m^\#(\omega_m^+) w_m^\#$, so that $g - g_t = (B_+ - b^+) + \sum_m R_m^* X_m$ and $\|R_m^* X_m\|_1 \leq \sum_k \lambda_{k,m} |X_m(k)|$. Drop m ; since $\Theta_+ =$

$$\omega_+ - d_+ w,$$

$$X = (\omega_+ - \omega^+) + (d^\#(\omega^+) - d_+)w + d^\#(\omega^+)(w^\# - w).$$

Let $\delta_k^\omega := |\omega_+(k) - \omega^+(k)|$. Recall from the claim in the proof of [I, Proposition 8.15(c)] (which uses only [I, Lemma 8.8]): at a peak k of f , $\lambda_k|\omega_+(k)| \leq \lambda_k(|\omega_+(k)| + |\omega_-(k)|) \leq |\Delta\theta_k| + \lambda_k|\Delta d|M$ ([I, (16)]); at a strict non-peak k of f , $|\omega_+(k)| \leq 2\text{gap}(k)/t + |\Delta\theta_k|/\lambda_k + |\Delta d|M$, and the clamp of $\omega_+(k)$ with radius $2\text{gap}(k)/t$ (and with 0 if $\text{gap}(k) < t^2$) differs from $\omega_+(k)$ by at most $2t + |\Delta\theta_k|/\lambda_k + |\Delta d|M$. Replacing gap by $\text{gap}^\#$ in the clamp changes it by at most $2t + 2|\text{gap}^\#(k) - \text{gap}(k)|/t$. Coordinates that change status between f and $f^\#$ and are not kept satisfy $|\varrho - 1| \leq b$ (Lemma 4.19(c)); there $\omega^+ = 0$ and the bounds above apply with $\text{gap} \leq bM \leq b$. At kept coordinates $\delta_k^\omega = |o_k|$, and $\lambda_k|o_k| \leq |\tau_{l''} - \tau_{l'''}| + \lambda_k|\Delta d|M$: indeed, by [I, Lemma 8.8(c),(f)], $\varsigma_k\omega_-(k) \geq -\frac{3}{2}\text{gap}(k)/t \geq -\frac{3}{2}\text{gap}^\#(k)/t$ (as $\text{gap}(k) \leq bM \leq \text{gap}^\#(k)$ for (K4)), and $\varsigma_k\omega_-(k) = \varsigma_k\omega_+(k) + \tau_{l''}/\lambda_k = \varsigma_k\omega_-(k) + (\tau_{l''} - \tau_{l''})/\lambda_k + \Delta d|w(k)|$ (from $\omega_- - \omega_+ = -\Delta\Theta - \Delta d w$ and $\varsigma_k\epsilon_{l''} = 1$). Summing, with $\sum \lambda \leq 1$, Step 0 and Steps 1–3,

$$\begin{aligned} \sum_{\text{coarse}} \lambda_k \delta_k^\omega &\leq K_{\text{pin}}t + lK_{\text{pin}}t + \|\tau - \tau'\|_1 + K_d't + 4t + \frac{2b}{t} \\ &\quad + \frac{2}{t} \sum_k \lambda_k |\text{gap}^\#(k) - \text{gap}(k)| \leq C_f(K_U + lK_{\text{pin}})t, \end{aligned}$$

because $\sum_k \lambda_k |\text{gap}^\# - \text{gap}| \leq |M^\# - M| + \sum_k \lambda_k |w^\#(k) - w(k)| \leq 2\theta_w T_{\text{lo}}^2 \leq 2\theta_w t^2$ (Lemma 4.18(e)). At fine coordinates $\omega^+ = 0$ and $\sum_{\text{fine}} \lambda_k |\omega_+(k)| \leq (4/t) \sum_{l' > l} \lambda_{l'} \leq t^7$ ([I, Lemma 8.8(a),(e)]). Next $|d^\#(\omega^+) - d_+| \leq |d^\#(\omega^+) - d(\omega^+)| + \|D(\omega^+ - \omega_+)\|_2 + |d(\omega_+) - d_+|$, where $|d(\omega_+) - d_+| \leq t/\sigma$ [I, Lemma 8.8(d)], $\|D(\omega^+ - \omega_+)\|_2 \leq \sum_k \lambda_k \delta_k^\omega + t^7$, and $|d^\#(\omega) - d(\omega)| \leq C_f \|D\omega\|_2 (\|D(w^\# - w)\|_2 + |C^\# - C|) \leq C_f \theta_w t$ for $\omega = \omega^+$, as $\|D\omega^+\|_2 \leq \|\Phi\|_2 \|\omega^+\|_\infty \leq 11/t$ (by (iii) below). Finally $|d^\#(\omega^+)| \leq 11/t$ and $\sum_k \lambda_k |w^\#(k) - w(k)| \leq \theta_w t^2$. Altogether

$$p^*(g - g_t) \leq (1 + \|U\|)\|g - g_t\|_1 \leq C_f(K_U + lK_{\text{pin}} + 1)t \leq K_w t,$$

since $K_U + lK_{\text{pin}} + 1 \leq C_f G^{**}(l) l^3 D_r^4 u^{-4} \leq C_f (l D_r G^{**}(l))^4 u^{-4} \leq C_f \text{Des}(l)^{2/3} u^{-4}$ (l , D_r and $G^{**}(l)$ are distinct factors of the bracket of $\text{Des}(l)$). This is (ii).

(i). $\|V'\|_1 \leq \sum \tau' \leq 4/t$ and $t\|B_+\|_1 \leq 1$, so $t\|b^+\|_1 \leq 1 + 4 + 2t|\varkappa|$ and $t\|b^-\|_1 \leq t\|b^+\|_1 + 4$: an f -constant A_0 bounds both. $\sqrt{\Gamma_w^\#}$ is a seminorm, and $H_m^\#$ vanishes on multiples of $w_m^\#$, so $\sqrt{\Gamma_w^\#(b^+, \omega^+)} \leq \sqrt{\Gamma_w^\#(B_+, \Theta_+)} + \sqrt{\Gamma_w^\#(B_+ - b^+, X)}$. The second term is at most $C_f(\|B_+ - b^+\|_1 + \sum_m \sum_k \lambda_k |X_m(k)|) \leq K_w t$, using $q_0^\# h^\#(y) \leq \|U\|^2 \|y\|_1^2 / \nu^\#$ and $\sigma_m^\# H_m^\#(X) \leq (\sum_k \lambda_k |X(k)|)^2 / C_m^\#$. For the first term, $q_0^\# h^\#(B_+) = q_0^\# \|P_{e^\#}^\perp U^* B_+\|^2 / \nu^\#$ with $\|P_{e^\#}^\perp U^* B_+\| \leq \|P_e^\perp U^* B_+\| + 2\|e^\# - e\| \|U\| \|B_+\|_1$, $\|P_e^\perp U^* B_+\|^2 = \nu h(B_+) \leq 2\nu/q_0$ [I, Lemma 8.7(d)], $\|e^\# - e\| \|B_+\|_1 \leq C_f \text{Des}^{1/3} \Lambda/t \leq C_f \theta_w$, and $|q_0^\# - q_0| + |\nu^\# - \nu| \leq C_f \text{Des}^{1/3} \Lambda$; so $q_0^\# h^\#(B_+) \leq q_0 h(B_+) + C_f \theta_w$. For the blocks, $H_m(\Theta) = (\|D_m \Theta\|_2^2 - \langle D_m \Theta, D_m w_m \rangle^2 / C_m^2) / C_m$, so $|H_m^\#(\Theta_+) - H_m(\Theta_+)| \leq C_f \|D_m \Theta_+\|_2^2 (\|D_m(w_m^\# - w_m)\|_2 + |C_m^\# - C_m|) \leq C_f t^{-2} \theta_w T_{\text{lo}}^2 \leq C_f \theta_w$ (as $\|D_m(t\Theta_+)\|_2 \leq \eta \leq 1$), and $|\sigma_m^\# - \sigma_m| \leq C_f \theta_w$. With [I, Lemma 8.7(d)], $\Gamma_w^\#(B_+, \Theta_+) \leq 1 + \eta_\Gamma(\eta) + C_f \theta_w$. The $-$ side is compared with (B_-, Θ_-) : off F , $B_- - b^- = \mathfrak{e}_-$; on F , by [I, (15)], $(B_- - b^-)1_F = \sum_{\text{KP}(w)} \epsilon(\tau' - \tau) u 1_F + \sum' \Delta \theta u 1_F + \varkappa a/a(\hat{z}^\#)$, the sum $\sum'$ over all carriers that

are not kept; and $\Theta_- - \omega^- + d^\#(\omega^-)w^\# = X - \Delta\Theta - \sum_{Kp,m}(\epsilon\tau'/\lambda)e_k(l'')$ (as $d^\#(\omega^-) = d^\#(\omega^+)$), whose last two terms have λ -weighted ℓ_1 -norm at most $\|\tau - \tau'\|_1 + \sum' |\Delta\theta|$. All these are at most $K_w t$.

(iii). Clamped coordinates are of kind [1] at $f^\#$. Kept carriers of types (K1), (K2) have $\text{gap}^\# \geq M^\# u/2 \geq \gamma(w)$ and type (K3) has $\text{gap}^\# \geq M^\#/2 \geq \gamma(w)$ (Lemma 4.19(d), $M_m^\# \geq M_m/2$); and $|\omega^+(k)| = |\omega_+(k)| \leq |\Theta_+(k)| + |d_+|M \leq (3 + \eta)/t \leq 4/t$ [I, Lemma 8.8(a),(e)], $|\omega^-(k)| \leq 4/t + \tau'/\lambda \leq 16/t$: kind [2] with $A_2 = 22$. At a (K4) coordinate, after the shift, $\varsigma_k \omega^- \geq -\frac{3}{2} \text{gap}^\# / t$, and $\varsigma_k \omega^+ = \varsigma_k \omega^- - \tau'/\lambda \leq \varsigma_k \omega^-$; before the shift $\varsigma_k \omega^+ = \varsigma_k \omega_+(k) \leq \frac{3}{2} \text{gap}(k)/t \leq \frac{3}{2} \text{gap}^\#(k)/t$, and the shift moves $\varsigma_k \omega^\pm$ up only when $\varsigma_k \omega^- < -\frac{3}{2} \text{gap}^\# / t$, in which case afterwards $\varsigma_k \omega^+ \leq -\frac{3}{2} \text{gap}^\# / t$. So for the $+$ pair either $\varsigma_k \omega^+ \leq 0$ (kind [3], side $+$) or $0 < \varsigma_k \omega^+ \leq \frac{3}{2} \text{gap}^\# / t$ (kind [1]); for the $-$ pair either $\varsigma_k \omega^- \geq 0$ (kind [3], side $-$) or $|\omega^-| \leq \frac{3}{2} \text{gap}^\# / t$ (kind [1]). Sizes: $|o_k| \leq |\omega^-(k)| + \frac{3}{2}/t \leq 17.5/t$, so $|\omega^+| \leq 21.5/t$ and $|\omega^-| \leq 16/t$ after the shift; $A_2 = 22$ suffices. Peaks of $f^\#$ carry no data. $\square$

4.5. Master Theorem II.

Theorem 4.25 (Master Theorem II). *Let $T = T_{\text{final}}$ with base U_{final} , $N \geq 1$, and let $f \in S_{p_N}^*$ have finite base support F . Suppose that for infinitely many levels l there is a sub-window w of level l which is clean for f and at which the following two conditions hold:*

- (SH $_w$) *every block $m \leq N$ has an upper and a lower shift source at w (Definition 4.5), or the shift cost of the shift pattern of f at w is robust: $c_{\pi(w)}(f) \geq u(w)$ (Definition 4.8);*
- (VR $_w$) *every extreme ray of the zero-cost cone $C(\kappa(w))$ of the pattern of f at w has at most one robust d -component (Definition 4.11).*

Then $f \in \mathcal{R}$: $(f, g) \in \text{clNA}((c_0, p_N), \ell_2^2)$ for every $g \in \mathcal{C}(f)$.

Proof. Fix $g \in \mathcal{C}(f)$, $\rho \in (0, 1)$ and $\eta_0 \in (0, 2]$ with $\rho^2(1 + \eta_0) \leq (1 + \rho^2)/2$, and choose $\eta \leq \eta_*$ with $\eta_\Gamma(\eta) \leq \min(1, \eta_0/8)$. Let $l_1 < l_2 < \dots$ be levels $\geq l_f$ with sub-windows w_j of level l_j that are clean for f and satisfy (SH $_{w_j}$) and (VR $_{w_j}$), and put $f_j := f_{w_j}^\#$ (Definition 4.16). The rows f_j depend only on f and on the w_j . Their base supports $F_j = F \cup \text{Ba}_j \cup \text{Pu}_j$ are finite and contain F , and $p^*(f_j - f) \leq \theta_{w_j} T_{\text{lo}}(w_j)^2 \rightarrow 0$ (Lemma 4.18(e)). Let A_0 be the f -constant of Proposition 4.24(i), and let t_1, j_0, c_0 be given by Lemma 2.18 for the sequence (f_j) , $\epsilon_{\text{tr}} := \eta_0/2$ and A_0 . Put $T_j := T_{\text{hi}}(w_j)$, $n_j := n(w_j)$, so that $\mathcal{S}(T_j, n_j)$ is the set of dyadic scales of $\mathcal{W}(w_j)$ and $T_j 2^{-n_j} = T_{\text{lo}}(w_j)$; $A_{2,j} := 22$, $\gamma_j := \gamma(w_j)$, $K_j := \max(K_{w_j}, 1)$, $c_j := c_0 \gamma_j / 22$. Discarding finitely many j , we may assume $T_j \leq \min(t_\eta, 1)$. For $t \in \mathcal{S}(T_j, n_j)$ choose a two-sided decomposition of g at scale t [I, Lemma 8.5] and let $g_{j,t}$ and $(b_{j,t}^\pm, \omega_{j,t}^\pm)$ be given by Proposition 4.24 at w_j . We verify the hypotheses of Theorem 2.20.

(E-a): $b^\pm(\xi_j) = 0$, $t\|b^\pm\|_1 \leq A_0$, and $\Gamma_w^{(j)} \leq (\sqrt{1 + \eta_0/8 + C_f \theta_{w_j}} + K_{w_j} T_j)^2 \leq 1 + \eta_0/2$ for large j , because $\theta_{w_j} \rightarrow 0$ and $K_{w_j} T_j \rightarrow 0$ (below). (E-b): Proposition 4.24(iii) gives the kinds; for $i \in F_j \setminus F = \text{Ba}_j \cup \text{Pu}_j$, Proposition 4.24(iv) gives $z_j(i)b^+(i) \geq 0 \geq z_j(i)b^-(i)$ on Ba_j and $t|b^\pm(i)| \leq |a_j(i)|$ on Pu_j . (E-c): Proposition 4.24(ii). (E-d): write $u := u(w_j)$, $\text{Des} := \text{Des}(l_j)$, $Q := Q(w_j) \geq (4\text{Des}/u)^{20} \geq \text{Des} u^{-5}$; then $K_j T_j \leq C_f \text{Des} u^{-4} 2^{-l_j^3} / (l_j Q) \leq C_f 2^{-l_j^3} / l_j \rightarrow 0$, and, as $c_j \geq c_f u$ and $n_j \geq l_j 2^{l_j^3} Q$, $n_j c_j / K_j \geq c_f l_j 2^{l_j^3} Q u^5 / \text{Des} \geq c_f l_j 2^{l_j^3} \rightarrow \infty$;

so $n_j \geq 48\rho^2 K_j / (c_j(1 - \rho^2))$ for large j . (E-e): $\varepsilon_j := p^*(f_j - f) \leq \theta_{w_j} T_{10}(w_j)^2 \leq C_f u^{20} T_{10}(w_j)^2$, while $\theta_* c_j^2 (T_j 2^{-n_j})^2 \geq c_f u^2 T_{10}(w_j)^2$; since $u(w_j) \rightarrow 0$ (Lemma 4.1(e)), (E-e) holds for large j . After discarding finitely many j , Theorem 2.20 gives $(f, \rho g) \in \text{clNA}((c_0, p_N), \ell_2^2)$. As $\rho < 1$ was arbitrary and clNA is closed, $(f, g) \in \text{clNA}$. $\square$

The order of choices is that of Remark 3.21: the design and T_{final} ; then N and f ; then the levels, the clean sub-windows and the companions $f_{w_j}^\#$ (which depend on f only); then g and ρ ; then the data at each scale. Every constant entering the window arithmetic is an f -constant, a design number of level l bounded by a power of $\text{Des}(l)$, or a power of $1/u(w)$ coming from robust rates.

Corollary 4.26 (One block). *Let $N = 1$ and let $f \in S_{p_1}^*$ have finite base support. If for infinitely many levels l some sub-window of level l that is clean for f satisfies (SH_w) , then $f \in \mathcal{R}$.*

Proof. For $N = 1$ every component of every ray lies in the single block, so (VR_w) holds at every w (Remark 4.12(b)); apply Theorem 4.25. $\square$

Corollary 4.27 (Every block has a bank donor). *Let $N \geq 1$, let $f \in S_{p_N}^*$ have finite base support, and let w be a sub-window of level $l \geq l_f$ that is clean for f . Then every block $m \leq N$ has a donor c_m : a coarse peak with $\varrho_{c_m} \geq 1 + u(w)$, which is of class R or dropped, and which is pinned when (SH_w) holds. For every block $m \in I_D(w)$, the move (C3) at c_m raises $\text{vs}_m u_{c_m}(\hat{z})$ by an amount in $[\Lambda_w, 3\Lambda_w]/\lambda_{c_m}$ up to $2b_w + C_f \text{Des}(l)^2 \Lambda_w^2$, either by a move of the sign vector alone or, when the room of S_{c_m} at s_m toward vs_m is too small, by closing z_{s_m} to vs_m and placing a bank of mass at most $\text{Des}(l)^{1/3} \Lambda_w$ at s_m ; it changes the value of every other carrier $\leq l$ by at most $C_f \text{Des}(l)^2 \Lambda_w^2$. At $f_w^\#$ the threshold of block m has risen by at least $c_f \Lambda_w$, and every (K4) carrier of block m (weak or degenerate swallowing-type peak, or swallowing-type strict non-peak with tiny gap) is a strict non-peak of $f_w^\#$ with gap at least $c_f M_m^\# \Lambda_w$. In particular no donor hypothesis is needed in Theorem 4.25, also when every far signature coordinate of every robust-margin peak is a contact with the sign of that peak.*

Proof. Lemmas 4.7, 4.10, 4.17 and 4.19(b),(c). $\square$

Remark 4.28 (The aligned corner). A move of the sign vector at a coordinate $s \in S_c$ raises $|u_c(\hat{z})|$ only if $z_s \neq \text{vs}_c$. The configuration in which, for every peak that could serve as a donor, every far signature coordinate is a contact with the sign of that peak (the *aligned corner*) therefore defeats every companion that keeps the base part fixed. A bank at such a contact (mass with the contact sign, sign vector unchanged) acts through the Hilbert part: by (5), $\text{vs}_c u_c(\hat{z})$ increases by $\mathfrak{d}(\mu_s^*)^2 v_c(s)/\nu$ at first order, while every other coarse value changes only at second order, because the base is diagonal ($\langle U^* u_k, k_s \rangle = \mu_s^* u_k(s) = 0$ for $k \neq c$). Banks are admissible in Theorem 2.20, since only the sign of a_j at the coordinates of $F_j \setminus F$ enters (Remark 2.21(c)); the bank coordinates need not be contacts of f . The smallness of the bank masses (Lemma 4.17(i)) is genuinely used: it keeps the second-order effects below the raise.

Corollary 4.29 (Residual configurations). *Let $N \geq 1$ and let $f \in S_{p_N}^*$ have finite base support. If $f \notin \mathcal{R}$, then there is l_0 such that for every level $l \geq l_0$ and every sub-window w of level l that is clean for f at least one of the following holds:*

- (B_w) some extreme ray r of $C(\kappa(w))$ has robust d -components in two different blocks $m_1 \neq m_2$: $|D_{r,m_i}(f)| \geq u(w)\Phi_{\max}(r, m_i)$, $i = 1, 2$;
- (C_w) some block has no upper or no lower shift source at w , and the shift cost of the shift pattern of f at w is tiny: $c_{\pi(w)}(f) \leq b(w)$.

For $N = 1$, (B_w) never occurs.

Proof. At a clean sub-window every rate object is tiny or robust, so (SH_w) fails iff (C_w) holds, and (VR_w) fails iff (B_w) holds. Clean sub-windows exist at every level (Theorem 3.19). If the conclusion failed, there would be infinitely many levels with a clean sub-window satisfying (SH_w) and (VR_w), and Theorem 4.25 would give $f \in \mathcal{R}$. The last sentence is Remark 4.12(b). $\square$

Remark 4.30 (Scope and comments). (a) Theorem 4.25 imposes no hypothesis on donors, on compensation or one-signedness of blocks, on nearly neutral carriers, and no rate or growth condition: rooms of signature sets, target rooms, margins (including degenerate peaks of either type), gaps, d -coefficients and ray d -sums are arbitrary, and so are the contact set, the number of swallowed carriers and the alignment of the far signature tails. Every f -dependent rate enters only through the tiny/robust dichotomy at a clean sub-window: tiny objects are made exactly zero at the companion (closed rooms and target rooms, neutralized ray components, raised thresholds), and robust objects enter the constants as powers of $1/u(w)$.

(b) Earlier versions of this argument assumed shift sources in every block, donors given by moves of the sign vector, every block compensated or one-signed, and no wrong-sign nearly neutral kept carrier. The first is the case $I_{\text{up}}(w) \cup I_{\text{lo}}(w) = \emptyset$ of (SH_w); the second is removed by Corollary 4.27; the last two are removed by the exact two-sided tuning (C4) together with ray removal (Step 3 of Proposition 4.24), within (VR_w). Rays with robust components in two blocks that are compensated by fixed repair directions are not covered by Theorem 4.25 as stated.

(c) Item (C_w) is not exactified by the companion: the shift cost $c_{\pi(w)}(f)$ is computed from the decomposition at f , and no move at the companion changes that decomposition. For item (B_w) the obstruction is not that tiny joint objects of several blocks (smallest singular values or minors of the ray d -matrix, rate objects (O7), (O8)) cannot be made zero by moving the values: these objects are polynomials in the values with design coefficients. The tuning (C4) is *linear* (one least-norm solve), and it neutralizes single components only. A nonlinear exactification through the determinantal data $\delta_{\text{comb}}(l)$, $\text{Lip}(l)$, $C_L(l)$, $N_L(l)$ of Definition 3.6(c) is not carried out in this section; it is the subject of Theorem 5.7 and Corollary 5.8.

(d) The base enters the constants of this section only through the bank efficiency $(\mu_{s'}^*)^2 v_c(s')$ at coordinates $s' \leq s_t(l)$ (Lemma 4.1(d)), through $\|U\|$, and through the identities of Lemma 3.2(b); pull coordinates lie far beyond $s_t(l)$, but their Hilbert footprints are only bounded from above.

5. DETERMINANTAL EXACTIFICATION, THE SHIFT DICHOTOMY AND THE RESIDUAL (C*)

This section treats first rows with *finite* base support. It removes two of the residual configurations left by the assembly of Section 4 (nearly dependent d -vectors of several blocks, and f -dependent d -row Hoffman constants), reduces the third one (an unpinned uniform shift of the d -coefficients) to a single residual property (C*), and collects what is proved about (C*).

Standing conventions. Throughout, $T = T_{\text{final}}$ and $U = U_{\text{final}}$ (Section 3), except in Subsection 5.6, where a modification of T_{final} is used, and except where a statement says “any admissible T ”. $N \geq 1$, $I = \{1, \dots, N\}$, $p = p_N$, and $f \in S_{p^*}$ has forced data $(\xi, q_0, a, w, F, z, \hat{z}, e, \nu)$ with $F = \text{supp } a$ finite. We use the notation of Sections 2 and 3; in particular for a carrier $l = j(k, m)$

$$\text{val}_l := \epsilon_l u_l(\hat{z}), \quad A_m := |\hat{\zeta}_m|_m = \sigma_m/q_0, \quad \bar{\vartheta}_m := \bar{\vartheta}(\hat{\zeta}_m), \quad \varrho_l := \varrho_k(\hat{\zeta}_m),$$

where ϵ_l is a sign attached to l (the swallowing sign when l is swallowed), and ϱ_l is the relative position (it equals the rate object (O4) of Definition 3.7). The d -weight of l is $q_l = \epsilon_l \Phi_l w_m(k)/(mC_m)$ (Definition 4.2); at a strict non-peak, by [I, Lemma 1.7] and $\hat{\zeta}_m(k) = m\Phi_l u_l(\hat{z})$,

$$(6) \quad q_l = \text{val}_l/A_m.$$

For a clean sub-window w of level l (Definition 3.18) we write $b = b(w)$, $u = u(w)$, $\text{Des} = \text{Des}(l)$, and we use the classification of the coarse carriers (classes R and G, swallowing and anti type, kept and dropped carriers, the closed pattern $\kappa(w)$), the upper and lower shift sources, the sets $I_{\text{up}}(w)$, $I_{\text{lo}}(w)$, the shift pattern $\pi(w)$ with its shift cost $c_{\pi(w)}(f)$ (rate object (O6)), the shift condition (SH $_w$), the robust-margin peaks of Lemma 4.7, the pinning bounds of Lemma 4.3, the assembled companion $f_w^\#$ with its moves (C1)–(C4), the tuning Lemma 4.15, the status Lemma 4.19, the transplant Proposition 4.24 and Master Theorem II (Theorem 4.25), all from Section 4. Two conventions for shifts occur: the *data convention* $\Delta d_m = d_m(\omega_m^-) - d_m(\omega_m^+)$ of [I, Definition 7.7] for two-piece data, and the *decomposition convention* $\Delta d_m = d_{+,m} - d_{-,m}$ of [I, Lemma 8.27] for two-sided decompositions at a scale t ; for the latter we write $\Delta\theta_l := \lambda_l(\Theta_+ - \Theta_-)_m(k(l))$ and, for class-G carriers, $\tau_l := -\epsilon_l \Delta\theta_l$. Constants C_f depend on f (and on g, ρ where said) but not on the level, the sub-window or the scale; they change from line to line. l_f is a level threshold depending on f only.

Organization. Subsection 5.1: Hoffman constants through minors and simultaneous exactification. Subsection 5.2: exactification of all tiny minors at a companion (Theorem 5.7). Subsection 5.3: the shift dichotomy (Theorem 5.10), recovery with shifted data and Master Theorem III' (Theorem 5.15), whose only residual is (C*) (Definition 5.16). Subsection 5.4: rigidity of shifted data. Subsection 5.5: realizations of (C*)-type resonances at block-tame rows, and genericity facts. Subsection 5.6: absorber pairs and Master Theorem IV' under the lever hypothesis (KN). Subsection 5.7: nested-tuning rows and further tools. Subsection 5.8: what remains open.

5.1. Hoffman constants through minors; simultaneous exactification. For a matrix $A \in \mathbb{R}^{p \times n}$ and index sets $J \subseteq \{1, \dots, p\}$, $K \subseteq \{1, \dots, n\}$ we write A_J for the rows in J and $A_{J,K}$ for the submatrix. J is *independent* if $J \neq \emptyset$ and the rows A_i ($i \in J$) are linearly independent; $\sigma_{\min}(A_J) > 0$ is then the least singular value of A_J . Put

$$\delta(A) := \min\{|\det A_{J,K}| : |J| = |K| \geq 1, \det A_{J,K} \neq 0\}$$

(the least *nonzero* minor; $\delta(A) := 1$ if $A = 0$).

Lemma 5.1 (Hoffman constants through nonzero minors). *Let $A \in \mathbb{R}^{p \times n}$, $\beta \in \mathbb{R}^p$, and $\mathcal{P} := \{x \in \mathbb{R}^n : Ax \leq \beta\} \neq \emptyset$. Then for every $x \in \mathbb{R}^n$*

$$(7) \quad \text{dist}_2(x, \mathcal{P}) \leq \|(Ax - \beta)_+\|_2 \cdot \max\{\sigma_{\min}(A_J)^{-1} : J \text{ independent}\}$$

(the maximum over the empty family is 0), and for every independent J

$$(8) \quad \sigma_{\min}(A_J) \geq \max_{|K|=|J|} |\det A_{J,K}| / \|A_J\|_2^{|J|-1},$$

where the maximum is positive. Consequently

$$(9) \quad \begin{aligned} \text{dist}_2(x, \mathcal{P}) &\leq \max(1, \|A\|_2)^{n-1} \delta(A)^{-1} \|(Ax - \beta)_+\|_2, \\ \text{dist}_1(x, \mathcal{P}) &\leq \sqrt{n} \max(1, \|A\|_2)^{n-1} \delta(A)^{-1} \|(Ax - \beta)_+\|_1. \end{aligned}$$

Equality constraints $Ex = e$ are included by writing them as two rows $Ex \leq e$, $-Ex \leq -e$; their residual is $|Ex - e|$, and δ is computed for the matrix with rows A and E . The bounds do not depend on β .

Proof. Let x^* be the Euclidean projection of x onto the closed convex set $\mathcal{P}$ and assume $x \neq x^*$. Then $x - x^*$ lies in the normal cone of $\mathcal{P}$ at x^* , which for a polyhedron is the cone generated by the rows A_i^T with $A_i x^* = \beta_i$ (Farkas' lemma). By the conic Carathéodory theorem $x - x^* = \sum_{i \in J} v_i A_i^T$ with $v_i > 0$ and J independent. As $A_i x^* = \beta_i$ for $i \in J$,

$$\|x - x^*\|_2^2 = \sum_{i \in J} v_i A_i(x - x^*) = \sum_{i \in J} v_i (A_i x - \beta_i) \leq \|v\|_2 \|(A_J x - \beta_J)_+\|_2,$$

and $\|v\|_2 \leq \|A_J^T v\|_2 / \sigma_{\min}(A_J) = \|x - x^*\|_2 / \sigma_{\min}(A_J)$; this gives (7). For (8) let $s_1 \geq \dots \geq s_k > 0$ ($k = |J|$) be the singular values of A_J . By the Cauchy–Binet formula $s_1^2 \dots s_k^2 = \det(A_J A_J^T) = \sum_K \det(A_{J,K})^2 \geq \max_K \det(A_{J,K})^2$, which is positive because A_J has rank k ; hence $s_k \geq (s_1 \dots s_k) / s_1^{k-1} \geq \max_K |\det A_{J,K}| / \|A_J\|_2^{k-1}$. Every positive maximum in (8) is $\geq \delta(A)$, $\|A_J\|_2 \leq \|A\|_2$ and $|J| \leq n$, which gives the first bound of (9); the second follows from $\|y\|_1 \leq \sqrt{n} \|y\|_2$ on $\mathbb{R}^n$ and $\|y\|_2 \leq \|y\|_1$. An independent set never contains a row together with its negative, so for equality rows only $|Ex - e|$ enters. $\square$

Thus a system whose minors are either exactly zero or bounded below has a controlled Hoffman constant, however degenerate its combinatorics; only minors that are small *but nonzero* are harmful.

Remark 5.2 (Nearly dependent d -vectors). Linear tuning of single ray components (rate objects (O5)) does not control the d -row of several blocks. Let $\mathcal{C} = \{\mu_1 r_1 + \mu_2 r_2 : \mu \in [0, \infty)^2\}$ with extreme rays r_1, r_2 of disjoint supports and $\|r_i\|_1 = 1$, and let the d -map D of two blocks take the values $D(r_1) = (1, 1)$ and $D(r_2) = (-1, -1 + \delta)$, $0 < \delta < 1$. Every component of $D(r_1), D(r_2)$ has modulus $\geq 1 - \delta$. But $\mathcal{C} \cap \ker D = \{0\}$, and $\tau := r_1 + r_2$ has $D(\tau) = (0, \delta)$, $\text{dist}_1(\tau, \mathcal{C} \cap \ker D) = \|\tau\|_1 = 2$: the Hoffman constant of the system “ $\tau \in \mathcal{C}$, $D\tau = 0$ ” is at least $2/\delta$. The responsible quantity is the minor $\det \begin{pmatrix} 1 & 1 \\ -1 & -1 + \delta \end{pmatrix} = \delta$, which is quadratic in the entries; making it vanish is not a linear problem.

Lemma 5.3 (Simultaneous exactification). *Let $\pi_1, \dots, \pi_s$ be real polynomials in n variables and $\mathcal{Q} := [-2, 2]^n$. There are $C_L \geq 1$ and an integer $N_L \geq 1$, depending only on $(\pi_1, \dots, \pi_s)$, such that for every $S \subseteq \{1, \dots, s\}$, every $\beta \in [0, 1]$ and every $v \in \mathcal{Q}$ with $\max_{i \in S} |\pi_i(v)| \leq \beta$, at least one of the following holds:*

- (i) $\beta \geq 1/C_L$;
- (ii) there is $v' \in \mathbb{R}^n$ with $\pi_i(v') = 0$ for all $i \in S$ and $\|v' - v\|_2 \leq C_L \beta^{2/N_L}$.

Proof. Fix S , put $\Pi_S := \sum_{i \in S} \pi_i^2$ and $Z_S := \Pi_S^{-1}(0)$, and let $g_S(v) := \text{dist}_2(v, Z_S)$ if $Z_S \neq \emptyset$, $g_S \equiv 1$ otherwise. Both Π_S and g_S are continuous semialgebraic functions on the compact semialgebraic set $\mathcal{Q}$, and $\Pi_S^{-1}(0) \cap \mathcal{Q} \subseteq g_S^{-1}(0)$. By the Łojasiewicz inequality [1, Corollary 2.6.7] there are an integer $N_S \geq 1$ and $C_S > 0$ with $g_S^{N_S} \leq C_S \Pi_S$ on $\mathcal{Q}$. If $Z_S = \emptyset$ this reads $1 \leq C_S \Pi_S(v) \leq C_S |S| \beta^2$, i.e. (i) for every $C_L \geq (C_S |S|)^{1/2}$. Otherwise a nearest point v' of Z_S has $\|v' - v\|_2 = g_S(v) \leq (C_S |S| \beta^2)^{1/N_S}$. Take $N_L := \max_S N_S$ and $C_L := \max_S \max\{(C_S |S|)^{1/2}, (C_S |S|)^{1/N_S}, 1\}$; for $\beta \leq 1$, $\beta^{2/N_S} \leq \beta^{2/N_L}$. $\square$

The design constants $N_L(l)$, $C_L(l)$ of Definition 3.6(c) are the constants of Lemma 5.3 for the finite family of determinantal polynomials $\pi_{\kappa, J, K}$ of level l ; only their existence is used, and the sub-window parameter $b(w)$ absorbs the non-explicit exponent (Proposition 3.14(d)). The exponent cannot be 1 in general: for $\pi_1(v) = v_1^2$, $|\pi_1(v)| \leq \beta$ allows $|v_1| = \beta^{1/2}$, while the zero set is $\{v_1 = 0\}$.

Lemma 5.4 (Ray components are minors). *Let $\mathcal{C} = \{\tau \in \mathbb{R}^n : A\tau \leq 0\}$ be a pointed polyhedral cone (equalities written as two rows), r an extreme ray with $\|r\|_1 = 1$, and $d \in \mathbb{R}^n$ a row vector. There is a set J' of $n - 1$ linearly independent rows of A , tight at r , such that*

$$\det \begin{pmatrix} A_{J'} \\ d \end{pmatrix} = \pm \|c\|_1 (d \cdot r), \quad c_i := (-1)^{n+i} \det A_{J', \{1, \dots, n\} \setminus \{i\}}, \quad c \neq 0.$$

Proof. The rows tight at r have rank $n - 1$ since r is extreme in a pointed cone; choose J' independent among them. Laplace expansion along the last row gives $\det[A_{J'}; d] = d \cdot c$ for every d ; for d a row of $A_{J'}$ the determinant vanishes, so $c \in \ker A_{J'} = \mathbb{R}r$, and $c \neq 0$ because $A_{J'}$ has rank $n - 1$. Hence $c = \pm \|c\|_1 r$. $\square$

Applied to the zero-cost cone $\mathcal{C}(\kappa)$ of a pattern and to the value row of a block m in the matrix $A_\kappa(v)$ of Definition 3.6(c), Lemma 5.4 shows that every ray component $D_{r,m}$ (rate object (O5)) is $\pm \|c\|_1^{-1}$ times a minor $\pi_{\kappa, J, K}(\text{val})$ (rate object (O8)), where $\|c\|_1$ is a sum of moduli of minors free of value rows, hence lies between $\delta_{\text{comb}}(l)$ and $(r(l)c'(l))^l$ times a design number. The converse fails: in Remark 5.2 all components are robust and only the 2×2 minor is small, and there are three-block configurations with all ray components and all 2×2 value minors robust and a single small 3×3 minor.

Lemma 5.5 (Threshold buffer through a peak). *Let $\zeta \in \ell_1 \setminus \{0\}$ be a vector of one block, $\bar{\vartheta} := \bar{\vartheta}(\zeta)$, $\mathbf{a} := |\zeta|$, $\phi := \|\Phi\|_2^2$, and let c be a peak of ζ , $s > 0$, and $\zeta' \in \ell_1$ with $\text{sign } \zeta'(c) = \text{sign } \zeta(c)$, $|\zeta'(c)| = |\zeta(c)| + s$ and*

$$E := \sum_{k \neq c} |\zeta'(k) - \zeta(k)| \leq \frac{\mathbf{a}s}{8(\mathbf{a} + \bar{\vartheta} + 1)}.$$

Let L be any set of coordinates, $R(s) := \min\{1, \mathbf{a}s/(8\phi(\mathbf{a} + \bar{\vartheta} + 1))\}$ and $X := \sup_{k \in L \setminus \{c\}} |\zeta'(k) - \zeta(k)|/\Phi(k)^2$. Then:

- (a) $\bar{\vartheta}(\zeta') \geq \bar{\vartheta} + R(s)$;
- (b) every $k \in L \setminus \{c\}$ with $\mathbf{v}_k(\zeta) < \bar{\vartheta} + R(s) - X$ is a strict non-peak of ζ' ;
- (c) if k_0 is a peak of ζ with $\mathbf{v}_{k_0}(\zeta) > \bar{\vartheta}$ ($k_0 = c$ is allowed), $C_1 := (1 + 2(\bar{\vartheta} + 1)/\mathbf{a})/\Phi(k_0)^2$, $h_0 := \min(1, \mathbf{v}_{k_0}(\zeta) - \bar{\vartheta})$, and $C_1(s + E) < h_0$, then $\bar{\vartheta}(\zeta') \leq \bar{\vartheta} + C_1(s + E)$, and every $k \in L \setminus \{c\}$ with $\mathbf{v}_k(\zeta) > \bar{\vartheta} + C_1(s + E) + X$ is a peak of ζ' with $\text{sign } \zeta'(k) = \text{sign } \zeta(k)$.

Proof. (a) is the lower bound of Lemma 2.7(b). (b) $v_k(\zeta') \leq v_k(\zeta) + X < \bar{\vartheta} + R(s) \leq \bar{\vartheta}(\zeta')$. (c) The upper bound of Lemma 2.7(a) applies to every peak k_0 with $v_{k_0} > \bar{\vartheta}$, the pushed peak c included, with $\|\zeta' - \zeta\|_1 = s + E$ (the c -term of $\mathbf{A}$ grows by s , which the proof of Lemma 2.7(a) accounts for as part of the ℓ_1 distance). Then $v_k(\zeta') \geq v_k(\zeta) - X > \bar{\vartheta}(\zeta')$, and $|\zeta'(k) - \zeta(k)| \leq X\Phi(k)^2 < v_k(\zeta)\Phi(k)^2 = |\zeta(k)|$ gives the sign. $\square$

The point of allowing $k_0 = c$ is that a block may have no non-degenerate peak other than the one that is pushed. In applications L is the set of coarse carriers of the block; fine coordinates may change their status, and enter only through E .

Lemma 5.6 (Buffer peaks). *There is l_f such that for every level $l \geq l_f$, every sub-window w of level l and every block m there is a coarse peak $c = c_m$ (a carrier $\leq l$) with $|\alpha_m(c)| \geq 1/(2l)$. Its relative margin satisfies*

$$\varrho_c - 1 = \frac{A_m |\alpha_m(c)|}{\Phi_c^2 \bar{\vartheta}_m} \geq \frac{A_m}{2l \Phi_c^2 \bar{\vartheta}_m} \geq \frac{A_m}{2l \bar{\vartheta}_m}.$$

Proof. By [I, Lemma 1.7], $\sum_{k \in P_m} |\alpha_m(k)| = 1$, and by [I, (2)], $\sigma_m |\alpha_m(k)| = \lambda_{k,m} \mu_{k,m}$ with $\mu_{k,m} \leq |u_{k,m}(\xi)| \leq q_0 \|\hat{z}\|_\infty \leq 2q_0$. Hence the peaks of carriers $> l$ carry at most $2q_0 \sum_{l' > l} \lambda_{l'}/\sigma_m \leq 2q_0 T_{lo}(w)^8/\sigma_m$ (Lemma 3.22(GW3)) of the α -mass, which is $\leq \frac{1}{2}$ for $l \geq l_f$; and block m has at most l coarse carriers. The identity for the margin is Lemma 2.6(b) applied to $\hat{\zeta}_m$ ($A_m |\alpha_m(c)| = |\hat{\zeta}_m(c)| - \bar{\vartheta}_m \Phi_c^2$), and $\Phi_c^2 \leq 1$. $\square$

Since $A_m/\bar{\vartheta}_m$ depends on f only, while $u(w) = b(w^-) \leq T_{lo}(w^-)^4$ for $w \neq (1, 1)$ decays faster than any power of $1/l$ (Proposition 3.14), the buffer peak is robust ($\varrho_c \geq 1 + u(w)$) at every sub-window of every large level. Pushing it outward by a controlled amount raises the block threshold (Lemma 5.5) and protects the strict non-peaks of the block; with the diagonal base such a push is available in either direction on far coordinates of its signature set, by z -moves (for a class-R peak, as in (C3)) or by pulls and banks (for a class-G peak after (C1), as in Lemma 4.15), so no donor is needed for this purpose.

5.2. Exactification of all tiny minors at a companion. Let w be a clean sub-window of level $l \geq l_f$ for f and let $\kappa = (E, P, \epsilon, F', \text{type})$ be the closed pattern of w , with kept set $Kp = E \setminus P$ (Definition 3.6(a); E is the set of coarse class-G carriers, P the dropped ones). The transplant of Proposition 4.24 projects the switching amplitudes $\tau = (\tau_{l''})_{l'' \in E}$ of the decompositions of a mate at the scales of w onto the solution set of the following system $\Sigma^\#$, whose coefficients are taken at a companion $f^\#$ of f :

- (Z1) $\tau_{l''} \geq 0$ ($l'' \in Kp$);
- (Z2) $\text{type}(j)L_j(\tau) \geq 0$ for $j \in T(l) \setminus F'$ with $\text{type}(j) = \pm 1$, and (Z3) $L_j(\tau) = 0$ if $\text{type}(j) = 0$, where $L_j(\tau) = \sum_{l'' \in E} \epsilon_{l''} \tau_{l''} u_{l''}(j)$;
- (Z4) $\tau_{l''} = 0$ for $l'' \in P$ (dropped carriers);
- (Z5) $\sum_{l'' \in Kp, m(l'')=m} (\text{val}_{l''}^\# / A_m^\#) \tau_{l''} = 0$ for every block m meeting E ;
- (box) $\tau_{l''} \leq 12\lambda_{l''}/t$ ($l'' \in E$).

By (6) the rows (Z5) are the d -rows of the kept carriers at $f^\#$, which are strict non-peaks of $f^\#$ (Lemma 4.19). The matrix of (Z1)–(Z5) is $A_\kappa(v)$ of Definition 3.6(c) evaluated at $v = \text{val}^\#$, except that the value row of block m is multiplied by $1/A_m^\#$; the rows (Z4) are unit rows whatever the status of the dropped carrier at $f^\#$, so

the statuses of dropped carriers do not enter $\Sigma^\#$. Recall $\eta(w) = T_{\text{lo}}(w)^4/(l \text{Des})$ and $\beta(w) = (1 + \text{Lip}(l)C_*(l))b(w)$ from Proposition 3.14(d).

Theorem 5.7 (Exactification of all tiny minors). *Let $T = T_{\text{final}}$, $N \geq 1$, and let $f \in S_{p^*}$ have F finite, let w be a clean sub-window of level $l \geq l_f$ for f , κ its closed pattern, and suppose an assembly is given in the following form: first z -moves on coarse coordinates which close the rooms of the class- G carriers on their natural signature sets and the tiny target rooms on $T(l)$, producing a row f_1 ; then finitely many further moves $\mathcal{M}_{\text{ass}}$ (donor raises by z -moves or banks; pulls converting tiny-margin peaks) at coordinates beyond $s_{\text{max}}(l)$ of carriers not in Kp . Assume:*

- (A1) $|\text{val}_{l''}(f_1) - \text{val}_{l''}(f)| \leq C_*(l)b(w)$ for $l'' \in \text{Kp}$, and $p^*(f_1 - f) \leq C_f \text{Des} b(w) \log(e/b(w))$;
- (A2) after $\mathcal{M}_{\text{ass}}$ every kept carrier ($l'' \in \text{Kp}$) is a strict non-peak, every coarse peak of f with $\varrho \geq 1 + u(w)$ is a peak with the same sign and relative margin $\geq u(w)/2$, and every kept carrier either has gap $\geq c_f \Lambda$ ($\Lambda = T_{\text{lo}}(w)^3$, in a block with a donor raise) or has $\varrho \leq 1 - u(w)/2$, $\varrho \leq C_*(l)b(w)$ or $|\varrho - 1| \leq C_*(l)b(w)$ (in a block without donor raise).

Then there is a companion $f^\#$ of f , obtained from f_1 by the moves $\mathcal{M}_{\text{ass}}$, in each block without donor raise one outward push of the buffer peak c_m of Lemma 5.6, and the exact tuning of the values of Kp , such that:

- (a) $f^\#$ has the carriers E , the kept set Kp , the signs ϵ and the contacts on $T(l)$ of κ , and $\text{supp } a^\#$ is finite; every kept carrier is a strict non-peak of $f^\#$, every coarse peak of f with $\varrho \geq 1 + u(w)$ is a peak of $f^\#$ with the same sign, and every coarse carrier with $|\varrho - 1| \leq C_*(l)b(w)$ at f (kept or dropped; in particular every tiny-margin peak) is a strict non-peak of $f^\#$. Nothing else is asserted about dropped carriers; they enter $\Sigma^\#$ only through the unit rows (Z4);
- (b) $\text{val}_{l''}(f^\#) = \text{val}_{l''}(f_1) + x_{l''}$ exactly for $l'' \in \text{Kp}$, with $\|x\|_2 \leq \eta(w)$; every minor $\pi_{\kappa, J, K}$ which is tiny at f (value $\leq b(w)$) vanishes at $\text{val}(f^\#)$, and every minor which is robust at f has modulus $\geq u(w)/2$ at $\text{val}(f^\#)$;
- (c) $p^*(f^\# - f) \leq p^*(f_1 - f) + C_f(\text{cost of } \mathcal{M}_{\text{ass}}) + C_f \text{Des}^4 \eta(w) \log(e/\eta(w))$, which is $o(T_{\text{lo}}(w)^2)$ when the cost of $\mathcal{M}_{\text{ass}}$ is;
- (d) the ℓ_1 -Hoffman constant of $\Sigma^\#$ at $f^\#$ (all rows, box rows included) satisfies $C_H^\# \leq C_f^l \text{Des}^2/u(w)$, for every configuration of the zero-cost cone (single- or multi-block rays, compensated, mixed or one-signed blocks, any face structure).

Proof. Step 1 (minors at f_1). Every minor $\pi_{\kappa, J, K}$ with a value row is $\text{Lip}(l)$ -Lipschitz on $[-2, 2]^{[1, l]}$, so by (A1) it moves by at most $\text{Lip}(l)C_*(l)b$ from f to f_1 . Hence tiny minors have modulus $\leq \beta(w)$ at $\text{val}(f_1)$, robust ones $\geq u - \beta(w) \geq 3u/4$ (Proposition 3.14(d) and (a)).

Step 2 (Łojasiewicz point). Apply Lemma 5.3 with the family of level l , S the tiny minors of κ , and $v := \text{val}(f_1) \in [-2, 2]^{[1, l]}$ ($|u_{l''}(\hat{z})| \leq q^*(u_{l''}) \|\hat{z}\|_\infty \leq 2$). Alternative (i) is excluded because $\beta(w) < 1/C_L(l)$; so there is v' with $\pi(v') = 0$ for every tiny minor and $\|v' - v\|_2 \leq C_L(l)\beta(w)^{2/N_L(l)} \leq \eta(w)$ (Proposition 3.14(d)). The polynomials of κ involve only the coordinates in Kp ; put $x := (v' - v)|_{\text{Kp}}$. Robust minors satisfy $|\pi(v')| \geq 3u/4 - \text{Lip}(l)\eta(w) \geq u/2$ (as $\|v' - v\|_\infty \leq \|v' - v\|_2 \leq \eta(w)$ and $\text{Lip}(l)$ is a constant for the maximum norm), since $\text{Lip}(l)\eta(w) \leq T_{\text{lo}}(w)^4 \leq u/8$ (Proposition 3.14(e)).

Step 3 (one combined realization). First, in each block m without donor raise, realize an outward push of the buffer peak $c_m \notin \text{Kp}$ of a size $s \in [s_m, 2s_m]$, with s_m the number fixed in Step 4 from a priori bounds: for a class-G buffer peak by a pull or bank on far coordinates of S_{c_m} (which satisfy (T1) of Lemma 4.15 after (C1)), for a class-R buffer peak by a z-move (or a close-and-bank) on far coordinates of its robust room, as in (C3) (Lemma 4.17). Then apply Lemma 4.15, with $L_0 = \text{Kp}$, at the row obtained from f_1 by the moves $\mathcal{M}_{\text{ass}}$ and the pushes: their masses and z-changes are part of the base data (A^2, z^2) of that lemma; their coordinates lie in signature sets $S_{l''}$ with $l'' \notin \text{Kp}$, so (T2) is unaffected, and they are distinct from the pull and bank coordinates of Kp , which lie in the signature sets $S_{l''}$, $l'' \in \text{Kp}$, beyond $s_{\max}(l)$. The target values are $v'_{l''}$. Since Lemma 4.15 solves for exact values given all other masses, the second-order side effects of $\mathcal{M}_{\text{ass}}$ and of the pushes on the values of Kp are absorbed, and $\text{val}_{l''}(f^\#) = v'_{l''}$ exactly. The required increments are $\leq 2\eta(w)$, far below the range of Lemma 4.15 (an f -constant times a negative power of Des). Other coarse carriers move by at most $C_f \eta(w)^2$ beyond the effect of $\mathcal{M}_{\text{ass}}$ and of the pushes; pulls and banks lie beyond $s_{\max}(l)$, so $T(l)$, the contact pattern on $T(l)$ and all swallowing signs are unchanged.

Step 4 (statuses). In a block with a donor raise, Lemma 4.19 applies verbatim with the additional value moves $|x| + C_f \eta^2 \leq 2\eta(w) \ll c_f \Lambda$ and $\ll u$, so every kept carrier stays a strict non-peak and every coarse peak of f with $\varrho \geq 1 + u$ stays a peak with its sign (Lemma 4.19(c),(d)); and every coarse carrier with $|\varrho - 1| \leq C_* b$ becomes a strict non-peak, since the threshold rises by at least $c_f \Lambda$ (Lemma 4.19(b)) while its value moves by at most $C_f \text{Des}^{1/6} \beta_w + 2\eta \ll \Lambda$, and $C_* b \ll \Lambda$. In a block m without donor raise put $\zeta := \hat{\zeta}_m(f_1)$, $\zeta^\# := \hat{\zeta}_m(f^\#)$ (both with the moves of the other blocks included), and let X , E be the quantities of Lemma 5.5 for $L :=$ all coarse carriers of block m (kept or dropped) and the pushed peak $c = c_m$. By Step 3 (Lemma 4.15(a),(b) and the bound $2\eta(w)$ on the increments), $X \leq \bar{X} := m \cdot 2\eta(w)/\Phi_{\min}(l) \leq C \text{Des} \eta(w)$ and $E \leq \bar{E} := \phi \bar{X} + 2^{-l} \eta(w)$, where $\Phi_{\min}(l)$ is the least weight of a carrier $\leq l$; these a priori bounds do not depend on the push. With $r_m := C_*(l)b(w)\bar{\vartheta}_m$ the push size of Step 3 is fixed as

$$s_m := \frac{16(a + \bar{\vartheta} + 1)}{a} \max\{\bar{E}, \phi(\bar{X} + r_m)\} \quad (a, \bar{\vartheta}, \phi \text{ of } \zeta),$$

and the realized push has size $s \in [s_m, 2s_m]$. Then $E \leq as/(8(a + \bar{\vartheta} + 1))$ and $R(s) \geq R(s_m) \geq X + r_m$ (for $l \geq l_f$, $R(s_m) < 1$), so Lemma 5.5(a),(b) make every coarse carrier of block m other than c_m with $\varrho \leq 1 + C_*(l)b$ (kept or dropped; in particular every tiny-margin peak) a strict non-peak of $\zeta^\#$. With $k_0 := c_m$ in Lemma 5.5(c) ($\Phi(c_m) \geq 1/\text{Des}$, and $h_0 \geq \min(1, A_m/(2l))$) because Lemma 5.6 gives $v_{c_m} - \bar{\vartheta}_m = \bar{\vartheta}_m(\varrho_{c_m} - 1) \geq A_m/(2l)$, we have $C_1(s + E) + X \leq C_f \text{Des}^4 \eta(w) \leq u \bar{\vartheta}_m/4$; in particular $C_1(s + E) < h_0$ for $l \geq l_f$, because $C_f \text{Des}^4 \eta(w) \leq C_f T_{10}(w)^4 \text{Des}^3/l$ tends to 0 while A_m is an f -constant. So every coarse peak of f with $\varrho \geq 1 + u(w)$ stays a peak with its sign, and kept carriers with $\varrho \leq 1 - u/2$ stay strict non-peaks. Kept near-threshold carriers of such a block become strict non-peaks with $\text{gap}^\# \geq \text{gap}$, since the threshold does not decrease (Lemma 5.5(a)). The buffer peak stays a robust peak. The value rows of $\Sigma^\#$ are rescaled by $1/A_m^\#$, which changes no statement in (b). This proves (a) and (b).

Step 5 (cost). By Lemma 4.15(c) the pull and bank masses are $\leq C_f \text{Des} 2^{G(l)} \eta(w)$; with the masses of $\mathcal{M}_{\text{ass}}$ and of the pushes, Lemma 2.12

gives, exactly as in the proof of Lemma 4.18(e), $p^*(f^\# - f_1) \leq C_f(\text{cost of } \mathcal{M}_{\text{ass}}) + C_f(2\eta) \log(e/(2\eta)) + C_f(s/\lambda_{c_m}) \log(e\lambda_{c_m}/s)$ (summed over the blocks without donor raise, $s \in [s_m, 2s_m]$ the realized push), and $s/\lambda_{c_m} \leq 2s_m/\lambda_{c_m} \leq C_f \text{Des}^4 \eta(w)$. Since $\eta(w) \leq T_{\text{lo}}(w)^4$, the added terms are $o(T_{\text{lo}}(w)^2)$.

Step 6 (Hoffman constant). The minors of the matrix of $\Sigma^\#$ at $f^\#$ that contain value rows equal the product of the row factors $1/A_m^\#$ of the value rows used times $\pi_{\kappa,J,K}(\text{val}(f^\#))$; since $A_m^\# \leq m2^{-m} \leq \frac{1}{2}$ ($A_m \leq \|\hat{\zeta}_m\|_1 \leq m \sum_k \Phi_m(k) \leq m2^{-m}$), these factors are ≥ 2 , so by (b) such a minor vanishes or has modulus $\geq u/2$. Minors without value rows are design numbers, zero or $\geq \delta_{\text{comb}}(l)$. The entries of the matrix are bounded by C_f (value entries are $\text{val}/A_m^\#$, and $A_m^\#$ is within $C_f p^*(f^\# - f)$ of A_m by Lemma 2.12). Lemma 5.1 with at most $c'(l)$ columns and $r(l)$ rows gives $C_H^\# \leq \sqrt{c'(l)} (C_f r(l) c'(l))^{c'(l)} / \min\{\delta_{\text{comb}}(l), u/2\} \leq C_f^l \text{Des}^2/u$, using Proposition 3.16(e) and $c'(l) \leq l$. $\square$

The assembly of Section 4 has the form required by Theorem 5.7: its moves (C1), (C2) produce f_1 and satisfy (A1) by Lemma 4.18(b) and Lemma 2.12 (cost bound), its donor raise (C3) is $\mathcal{M}_{\text{ass}}$, and (A2) holds by Lemma 4.19(c),(d) for kept carriers and robust peaks; the statuses of dropped carriers are not needed. The linear tuning (C4) is replaced by the combined solve of Step 3, whose target, the Łojasiewicz point, absorbs the second-order effects of the donor banks on the kept values.

Corollary 5.8 (Multi-block rays are not residuals). *Let w be a clean sub-window of level $l \geq l_f$ for f , and suppose that the shift bound $|\Delta d_m| M_m \leq K_d t$ holds for every block m and every scale $t \in \mathcal{W}(w)$, with $K_d \leq C_f \text{Des}^C / u(w)^C$. Then the conclusions of Proposition 4.24 hold at the companion $f^\#$ of Theorem 5.7 without the hypothesis (VR_w), with a window constant $K_w \leq C_f^l \text{Des}^{C'} / u(w)^{C'}$. In particular Master Theorem II (Theorem 4.25) holds with (VR_w) deleted from its hypotheses.*

Proof. In Proposition 4.24, (VR_w) is used only to bound the distance of the projected amplitudes τ_0 to the solution set of the d -rows (blockwise ray removal). Replace this step by the ℓ_1 -Hoffman projection of τ_0 onto the solution set of $\Sigma^\#$, box rows included: Theorem 5.7(d) gives τ' with all (Z5) rows exactly zero and $\|\tau_0 - \tau'\|_1 \leq C_H^\# \cdot (\text{violations of } \Sigma^\# \text{ at } \tau_0)$. The violations of (Z1)–(Z4) and (box) are those already bounded in Proposition 4.24; the violation of (Z5) is bounded by the shift term $C_f K_d t$ (through [I, (17)]) plus $C_f \eta(w) \sum_{l''} |\tau_{l''}| \leq C_f \eta(w) \cdot 12/t \leq C_f T_{\text{lo}}(w)^3$. To see this, use the homogeneity of (Z5): multiplying the row of block m by $A_m^\# / A_m \in [\frac{1}{2}, 2]$ changes neither its solution set nor (up to a factor ≤ 2) its violation, and turns its coefficients into $\text{val}_{l''}^\# / A_m = q_{l''} + x_{l''}^\# / A_m$ with $q_{l''} = \text{val}_{l''} / A_m$ the coefficients at f (Lemma 4.19(d)) and $|x_{l''}^\#| \leq C_*(l)b(w) + \eta(w) \leq 2\eta(w)$; so $|q_{l''}^\# A_m^\# / A_m - q_{l''}| \leq 2\eta(w) / A_m \leq C_f \eta(w)$, and the violation is at most $2(|\sum_{l''} q_{l''} \tau_{l''}| + C_f \eta(w) \sum_{l''} |\tau_{l''}|)$, where we used $\sum_{l''} \lambda_{l''} \leq 1$ and the box rows. (A bound $|q_{l''}^\# - q_{l''}| \leq C \eta(w)$ itself would be false in blocks with a donor raise, where $|A_m^\# - A_m|$ is of order $\Lambda_w = T_{\text{lo}}(w)^3$.) The remaining steps of Proposition 4.24 use only $\|\tau_0 - \tau'\|_1$, the sign rows and the box, not the additional property $\tau' \leq \tau_0$ of the ray removal. All new factors are of the form $C_f^l \text{Des}^C u^{-C}$, which the window arithmetic absorbs (Lemma 3.22(GW2)). The extra cost $C_f \text{Des}^4 \eta \log(e/\eta)$ of Theorem 5.7(c) is $o(T_{\text{lo}}(w)^2)$. $\square$

Consequently the f -dependent d -row Hoffman constants of mixed or uncompensated blocks, and nearly dependent d -vectors of several blocks (Remark 5.2), are not obstructions for T_{final} and finite F . What Theorem 5.7 does not give is the shift bound itself: the violation of (Z5) by the actual amplitudes contains the uniform shift Δd_m , which must be pinned at f . This is the subject of the next subsection.

5.3. The shift dichotomy and Master Theorem III'. Fix a clean sub-window w of level $l \geq l_f$ for f . The *enlarged shift pattern* of f at w is the index $\kappa^{\text{sh}}(w) = (\kappa, N_0, R_7, I_{\text{sh}}, \text{src}, G_{\text{pk}})$ of the rate object (O9) of Definition 3.7 given by: $\kappa = \kappa(w)$ the closed pattern; N_0 the kept carriers with $\varrho \leq b$ (nearly neutral; they are exactified to value 0 at the companion); R_7 the anti-type class-G strict non-peaks with $|\varrho - 1| \leq b$; $I_{\text{sh}} := I_{\text{up}}(w) \cup I_{\text{lo}}(w)$ the source-deficient blocks; $\text{src}(m)$ equal to “ $\leq$ ” if block $m \in I_{\text{sh}}$ has an upper source of type (U1), “ $\geq$ ” if it has a lower source of type (L1), and “free” otherwise; and G_{pk} the class-G coarse carriers that are peaks of f with $\varrho \geq 1 + u(w)$ (tiny-margin peaks, with $\varrho \in [1, 1 + b(w)]$, belong to $E \setminus G_{\text{pk}}$; by cleanness every class-G peak is of one of these two kinds). Since $\kappa^{\text{sh}}(w)$ records all window classifications entering the system Σ^{sh} of (O9), the shift-pinning rate $\rho^{\text{sh}}(\kappa^{\text{sh}}(w), f)$ is a number depending on f only, and it is tiny or robust at w . We use the decomposition convention $\delta_m := \Delta d_m M_m$ for two-sided decompositions at scales $t \in \mathcal{W}(w)$, and put $K_g := (1/q_0 + 1)\text{Des}/u$.

Lemma 5.9 (Relations satisfied by the decompositions). *Let $g \in \mathcal{C}(f)$, $t \in \mathcal{W}(w)$, and let $(B_{\pm}, \Theta_{\pm})$ be a two-sided decomposition of g at scale t ([I, Lemma 8.5]). Then the vector (δ, τ) , $\delta_m = \Delta d_m M_m$, $\tau_{l''} = -\epsilon_{l''} \Delta \theta_{l''}$ ($l'' \in E$), satisfies the rows (S1)–(S7) of $\Sigma^{\text{sh}}(\kappa^{\text{sh}}(w), f)$ up to a total ℓ_1 -violation $\text{viol}(\delta, \tau) \leq C_f \text{Des}^2 K_g t / u$.*

Proof. (S1) at strict non-peaks and at peaks (in the form $\tau_{l''} \geq -K_g t$) is the one-sided pinning of class-G carriers, Lemma 4.3. (S2): by [I, (16)], at a peak $k = k(l'')$ of block m with $\varsigma_k = \text{vs}_{l''}$, $-\Delta \theta_{l''} = \text{vs}_{l''} \lambda_{l''} (\delta_m + e_k)$ with $0 \leq e_k \leq t/(\sigma_m |\alpha_m(k)|) = t/(\lambda_{l''} \mu_k)$; at $l'' \in G_{\text{pk}}$ (relative margin $\geq u$) $\mu_k \geq c_f \Phi_{l''} u \geq c_f u / \text{Des}$, so $|\tau_{l''} - \epsilon_{l''} \text{vs}_{l''} \lambda_{l''} \delta_m| \leq C_f \text{Des} t / u$. (S3): $\Delta B 1_{F^c} = V(\tau) + e_0$ with $V(\tau) = \sum_{l'' \in E} \epsilon_{l''} \tau_{l''} u_{l''} 1_{F^c}$ and $\|e_0\|_1 \leq K_g t + 4b^2/t$ (Lemma 4.3); the switching budget [I, Lemma 8.24] and the cost of the closed rooms (Step 1 of the proof of Proposition 4.24) bound $\sum_{j \notin F} \phi_{z_j^\#}(\Delta B(j))$ by $2t/q_0 + CK_g t$, so by [I, Lemma 8.23(c)] the cost of $V(\tau)$ is at most $2t/q_0 + CK_g t + 2\|e_0\|_1$; at a contact $\phi_z(x) = 2(zx)_-$, and at a free coordinate of κ , $\phi_{z_j}(x) \geq (1 - |z_j|)|x| \geq u|x|$. (S4): by [I, (17)], $\delta_m = (mC_m)^{-1} \sum_k \Phi_m(k) w_m(k) \Delta \theta_{k,m} + r_m$, $|r_m| \leq 2t/\sigma_m$, and for $l'' \in E$ the term equals $-q_{l''} \tau_{l''}$; class-R and fine carriers contribute at most $(K_g t + 6t^2)/(mC_m)$ (Lemma 4.3, Lemma 3.22(GW3)), and carriers of N_0 contribute $|q_{l''} \tau_{l''}| \leq b \Phi M 6\lambda / (mCt) \leq t^3$ by the box bound. (S5), (S6): Lemma 4.6 (an upper source gives $\delta_m \leq K_d t$, a lower one $\delta_m \geq -K_d t$, $K_d \leq C_f \text{Des}^3 / u$). (S7): for $l'' \in R_7$, [I, Lemma 8.8] gives $\tau_{l''} \leq \lambda_{l''} (3 \text{gap} / t - \Delta d_m |w_m(k)|)$ with $\text{gap} \leq bM_m$, so the defect is $\leq 3\lambda b / t \leq t^3$. Summing, and using $t \geq T_{\text{lo}}(w)$, $b \leq T_{\text{lo}}(w)^4$, gives the bound. $\square$

Any further relation satisfied by the decompositions up to the same order may be added to Σ^{sh} ; this can only increase ρ^{sh} . $\rho^{\text{sh}}(\kappa^{\text{sh}}, f) = 0$ means that Σ^{sh} has exact or asymptotically exact solutions with $\delta \neq 0$: a *d-constrained coherent shift resonance*.

Theorem 5.10 (Shift dichotomy). *Let $T = T_{\text{final}}$, $N \geq 1$, and let $f \in S_{p^*}$ have F finite and let w be a clean sub-window of level $l \geq l_f$ for f . Then exactly one of the following holds.*

- (I) $\rho^{\text{sh}}(\kappa^{\text{sh}}(w), f) \geq u(w)$. Then for every $g \in \mathcal{C}(f)$, every scale $t \in \mathcal{W}(w)$, every two-sided decomposition at scale t and every block m ,

$$|\Delta d_m| M_m \leq K_{\text{sh}} t, \quad K_{\text{sh}} := C_f \text{Des}^2 K_g / u(w)^2 \leq C_f \text{Des}^3 / u(w)^3.$$

- (II) $\rho^{\text{sh}}(\kappa^{\text{sh}}(w), f) \leq b(w)$.

If $I_{\text{up}}(w) \cup I_{\text{lo}}(w) = \emptyset$, then $\rho^{\text{sh}} \geq 1$, so case (I) holds.

Proof. The rate object $\rho^{\text{sh}}(\kappa^{\text{sh}}(w), \cdot)$ belongs to $\mathcal{O}(l)$, so its value at f is tiny or robust (Theorem 3.19). viol is positively homogeneous of degree one, so $\text{viol}(\delta, \tau) \geq \|\delta\|_1 \rho^{\text{sh}}$ for every (δ, τ) . In case (I), Lemma 5.9 gives $\|\delta\|_1 \leq C_f \text{Des}^2 K_g t / u^2$. If $I_{\text{sh}} = \emptyset$, (S5) forces $\text{viol} \geq \|\delta\|_1$. $\square$

In case (I) the shift is pinned with a constant of the form $C_f \text{Des}^C u^{-C}$, which the window arithmetic absorbs (Lemma 3.22(GW2)); every use of a shift bound in Section 4 holds at w with this constant.

We use the *combinatorial shift cost* of Definition 3.6(e). For the enlarged shift pattern $\kappa^{\text{sh}} = \kappa^{\text{sh}}(w)$ of f at w take the signs $\text{vs}_{l''} := \text{sign } w_{m(l'')}(k(l''))$ ($l'' \in G_{\text{pk}}$) of (O9)(S2), and write $c^{\text{comb}}(\delta; \kappa^{\text{sh}}) := c^{\text{comb}}(\delta; \kappa^{\text{sh}}, \text{vs})$, $c_*^{\text{comb}}(\kappa^{\text{sh}}) := c_*^{\text{comb}}(\kappa^{\text{sh}}, \text{vs})$, with $G_{\text{np}} = E \setminus G_{\text{pk}}$ (dropped carriers included) and $\Pi_m, L, \text{coef}_{l''}$ as there. Since $l \geq l_f \geq \max F$ and $F' = F \cap T(l)$, we have $F \subseteq [1, l]$, so $1_{F^c} = 1$ on $T(l) \setminus F'$ and on every set $S_{l''} \setminus ([1, l] \cup T(l))$; hence these design quantities coincide with the costs computed with $\Pi_m 1_{F^c}$ in place of Π_m , which is the form in which they arise from the decompositions below. By Definition 3.6(e), $c_*^{\text{comb}}(\kappa^{\text{sh}}) > 0$ implies $c_*^{\text{comb}}(\kappa^{\text{sh}}) \geq \delta_{\text{sh}}(l)$.

Corollary 5.11 (Combinatorial shift cost). *In the situation of Theorem 5.10, write $c_* := c_*^{\text{comb}}(\kappa^{\text{sh}}(w))$.*

- (a) *For every (δ, τ) : if $c_* > 0$, then $\|\delta\|_1 \leq (C \text{Des} / c_* + 2) \text{viol}(\delta, \tau)$. Consequently $c_* > 0$ implies $\rho^{\text{sh}} \geq \delta_{\text{sh}}(l) / (C \text{Des}) > b(w)$, i.e. case (I).*
 (b) *In case (II), $c_* = 0$, and the minimum is attained: there are δ in the sign-constrained unit sphere of $\mathbb{R}^{I_{\text{sh}}}$ and $x \geq 0$ such that $\sum_m \delta_m \Pi_m + \sum_{G_{\text{np}}} x \epsilon u$ is admissible for the contact pattern on $T(l)$ and has no sign defect on the natural signature sets: an exact combinatorial coherent resonance.*

Proof. (a) Given (δ, τ) , take $x := (\tau)_+$ on G_{np} . On the natural part of $S_{l''}$, $\epsilon_{l''} \text{coef}_{l''} = x_{l''} \geq 0$ for $l'' \in G_{\text{np}}$, and for $l'' \in G_{\text{pk}}$ its sign defect is at most $(\tau_{l''})_- + |\tau_{l''} - \epsilon \text{vs} \lambda \delta_m|$, i.e. (S1) and (S2) defects. On $T(l)$, $|L(j) - V_j(\tau)|$ is bounded by $\max |u|$ times the (S2) defects and $\sum_{G_{\text{np}}} (\tau)_-$, and $2(\text{type}(j)V_j)_-$, $|V_j|$ are (S3) defects. Hence $c^{\text{comb}}(\delta_{I_{\text{sh}}}; \kappa^{\text{sh}}) \leq C \text{Des} \text{viol}(\delta, \tau)$. Let δ' be the ℓ_1 -nearest point of the sign cone to $\delta_{I_{\text{sh}}}$; $\|\delta' - \delta_{I_{\text{sh}}}\|_1$ is at most the (S6) violation, and $c^{\text{comb}}(\cdot; \kappa^{\text{sh}})$ is $C \text{Des}$ -Lipschitz in δ (L is linear in δ with $\|\Pi_m\|_1 \leq 1$, ϕ_z is 2-Lipschitz, and the defect terms are λ -Lipschitz), so $c_* \|\delta'\|_1 \leq c^{\text{comb}}(\delta') \leq C \text{Des} \text{viol}$, and $\|\delta\|_1 \leq \|\delta'\|_1 + ((S5) \text{ and } (S6) \text{ violations})$. If $c_* > 0$ then $c_* \geq \delta_{\text{sh}}(l) \geq 1/\text{Des}$ (Proposition 3.16(e)) and $b(w) \leq T_{\text{lo}}(w)^4 \leq 2^{-4 \text{Des}}$. (b) follows from (a) and compactness. $\square$

So a shift-pinning rate that is positive but smaller than every design quantity cannot occur at a clean sub-window; only an exact combinatorial resonance together with a tiny ρ^{sh} blocks the pinning.

Remark 5.12 (Two independent pinning criteria). Lemma 4.9 pins the shift under (SH_w) , i.e. when $c_{\pi(w)}(f) \geq u(w)$ or $I_{\text{sh}}(w) = \emptyset$. Neither of $c_{\pi(w)} \geq u$ and $\rho^{\text{sh}} \geq u$ is known to imply the other: $c_{\pi(w)}$ uses z on all of F^c and charges terms proportional to tiny rooms that the rows (S1)–(S7), written for the closed pattern, do not see (for the actual decomposition these terms are $\leq 6b\lambda/t$, but a near-solution of Σ^{sh} may have large $\|\tau\|_1$); conversely $c_{\pi(w)}$ sees only the peak traces and zero cost, not the d -identity (S4), so ρ^{sh} can be robust while $c_{\pi(w)}$ is tiny. Both are therefore kept.

The recovery engines for data that follow an unpinned shift are the following variants of Theorem 2.20.

Theorem 5.13 (Windowed recovery with shifted data). *Let $f, g, \rho, \eta_0, f_j, (T_j, n_j), K_j, A_{2,j}, \gamma_j$ be as in Theorem 2.20, and suppose all its hypotheses hold except that the two-piece data $(b_{j,t}^\pm, \omega_{j,t}^\pm)$ at f_j are not required to be d -neutral.*

- (a) *If $\Delta d_m^{(j)} := d_m^{(j)}(\omega_m^-) - d_m^{(j)}(\omega_m^+) \geq 0$ for every m, j and t , then $(f, \rho g) \in \text{clNA}((c_0, p), \ell_2^2)$.*
- (b) *If for every j there is a set $I_-^{(j)}$ of blocks satisfying the scrambling condition (SC) at f_j ([I, Definition 7.16]) such that $\{m : \Delta d_m^{(j)} < 0\} \subseteq I_-^{(j)}$ for every t , then $(f, \rho g) \in \text{clNA}((c_0, p), \ell_2^2)$.*

In both cases, if the hypotheses hold for every $\rho < 1$, then $(f, g) \in \text{clNA}((c_0, p), \ell_2^2)$.

Proof. In the proof of Theorem 2.20, d -neutrality is used only in the verification of (e) of Theorem 2.19: (i) it is preserved by the averaging over $t \in S_j$, and (ii) [I, Corollary 7.21] is applied at f_j to the averaged data. Lemma 2.18 is applied to each side separately and does not involve Δd . (a) $d_m^{(j)}$ is linear, so the averaged data have $\Delta \bar{d}_m^{(j)}$ equal to the average of the $\Delta d_m^{(j)}$, hence ≥ 0 ; [I, Corollary 7.21] needs exactly $\Delta d_m \geq 0$ and $\kappa_w \leq 1$. (b) The averaged data have $\{m : \Delta \bar{d}_m^{(j)} < 0\} \subseteq I_-^{(j)}$, and (SC) passes to subsets of $I_-^{(j)}$ (the maximum over fewer blocks is smaller). The data $\rho \bar{D}_j$ of $\rho \bar{g}_j$ have $\kappa_w \leq 1$, so [I, Theorem 7.18] at f_j (finite I , finite F_j) gives $(f_j, \rho \bar{g}_j) \in \text{clNA}$ whenever $\rho \bar{g}_j \in \mathcal{C}(f_j)$. $\square$

Lemma 5.14 (Averaging over a subset of scales). *Theorems 2.19, 2.20 and 5.13 remain valid if, for each j , the set $\mathcal{S}(T_j, n_j)$ is replaced by a subset $\mathcal{S}'_j \subseteq \mathcal{S}(T_j, n_j)$ with $n'_j := |\mathcal{S}'_j|$, all averages are taken over $\mathcal{S}'_j$, and the conditions on n_j are replaced by $n'_j \geq 48\rho^2 K_j / (c_j(1 - \rho^2))$ and $K_j T_j / n'_j \rightarrow 0$ (conditions (d) of Theorem 2.19 and (E-e) unchanged).*

Proof. In the proof of Theorem 2.19 the set of scales enters at three places. For $|r| \leq r_0$: the set $\mathcal{S}_r$ of scales $t \in \mathcal{S}'_j$ with $ct < \rho|r|$ satisfies $\sum_{\mathcal{S}_r} t < 2\rho|r|/c$ (dyadic scales), so the averaged error is $\leq 2\rho^2 K r^2 / (c n'_j) \leq (1 - \rho^2) r^2 / 24$; if $\mathcal{S}_r \neq \emptyset$, then $\rho|r| > c \min \mathcal{S}'_j \geq 2c T_j 2^{-n_j}$, which is all that the use of (d) requires. For $|r| > r_0$: $\frac{1}{n'_j} \sum_{\mathcal{S}'_j} t \leq 2T_j / n'_j$, and $2\rho K_j T_j / n'_j \rightarrow 0$. Finally $p^*(\rho \bar{g}_j - \rho g) \leq 2\rho K_j T_j / n'_j \rightarrow 0$. Averaging over $\mathcal{S}'_j$ preserves exactness, side conditions, the sign of Δd_m and a common shift. $\square$

Theorem 5.15 (Master Theorem III'). *Let $T = T_{\text{final}}$ with base U_{final} , let $N \geq 1$, and let $f \in S_{p_N}^*$ have finite base support. Suppose that for infinitely many levels l some clean sub-window w of level l satisfies*

$$\rho^{\text{sh}}(\kappa^{\text{sh}}(w), f) \geq u(w) \quad \text{or} \quad (\text{SH}_w).$$

Then $f \in \mathcal{R}$, i.e. $(f, g) \in \text{clNA}((c_0, p_N), \ell_2^2)$ for every $g \in \mathcal{C}(f)$.

Proof. Either alternative gives $|\Delta d_m| M_m \leq Kt$ for every block m and every scale $t \in \mathcal{W}(w)$, with $K \leq C_f \text{Des}^3/u^3$ (Theorem 5.10(I)) resp. $K \leq C_f l \text{Des}^3/u^2$ (Lemma 4.9). Master Theorem II uses (SH_w) only through this bound, and (VR_w) only in the step of Proposition 4.24 that Corollary 5.8 replaces. Hence the proof of Theorem 4.25 applies along the infinitely many sub-windows in question, with the companion of Theorem 5.7 and the window constants of Corollary 5.8; the remaining window arithmetic is that of Theorem 4.25. $\square$

Definition 5.16 (The residual (C^*)). *Let $T = T_{\text{final}}$ and $N \geq 1$. A first row $f \in S_{p_N}^*$ with finite base support is a (C^*) row if for all but finitely many levels l , every clean sub-window w of level l satisfies*

$$\rho^{\text{sh}}(\kappa^{\text{sh}}(w), f) \leq b(w) \quad \text{and} \quad c_{\pi(w)}(f) \leq b(w);$$

in particular $I_{\text{up}}(w) \cup I_{\text{lo}}(w) \neq \emptyset$ at these sub-windows.

Corollary 5.17. *Let $T = T_{\text{final}}$ and $N \geq 1$. If $f \in S_{p_N}^*$ has finite base support and $f \notin \mathcal{R}$, then f is a (C^*) row.*

Proof. At a clean sub-window both ρ^{sh} and $c_{\pi(w)}$ are tiny or robust, and (SH_w) fails exactly when $I_{\text{sh}}(w) \neq \emptyset$ and $c_{\pi(w)} \leq b(w)$. Apply Theorem 5.15. $\square$

In words: a (C^*) row has, at every clean sub-window of every large level, a block lacking an upper or a lower shift source in which the shift traces of the robust peaks are compensated, exactly at the combinatorial level (Corollary 5.11(b)) and up to $b(w)$ in the d -identity, by zero-cost switching of class-G strict non-peaks.

5.4. Structure of shifted data. By Corollary 5.17 the only first rows with finite base support that may fail to be in $\mathcal{R}$ are (C^*) rows. This subsection records what exact two-piece data with a nonzero shift look like. We use the notation $W_\Delta = \sum_m \Delta d_m \psi_m$, $\psi_m = R_m^* w_m$, and $\Sigma(w)$ of Subsection 2.6 (data convention).

Lemma 5.18 (Shape of a source-deficient block). *Let w be a clean sub-window of level $l \geq l_f$ for f (F finite) and m a block. Block m has a coarse peak with $\varrho \geq 1 + u(w)$. If block m has no upper source at w , then every coarse peak of block m is a class-G peak of swallowing type. If block m has no lower source at w , then every coarse peak of block m with $\varrho \geq 1 + u(w)$ is a class-G peak of anti type.*

Proof. The first claim is Lemma 4.7. At a clean sub-window every coarse carrier is of class R or of class G. A class-R peak is an upper source (U1), and a lower source (L1) if $\varrho \geq 1 + u$; a class-G anti-type peak is an upper source (U2); a class-G swallowing-type peak with $\varrho \geq 1 + u$ is a lower source (L2). Peaks have $|w_m(k)| = M_m \neq 0$, so their type is defined. $\square$

Thus in a source-deficient block ($m \in I_{\text{up}}(w) \cup I_{\text{lo}}(w)$) of a (C^*) row every robust coarse peak has, at that clean sub-window, a tiny room on its natural signature set,

and all of them have one type: swallowing type in an upper-deficient block (“configuration (i)”; there the data shift is negative, $\Delta d_m < 0$ in the data convention) and anti type in a lower-deficient block (“configuration (ii)”, $\Delta d_m > 0$).

Proposition 5.19 (Rigidity of shifted data). *Let $(b^\pm, \omega^\pm)$ be two-piece data at a first row f with F finite, for some functional.*

- (a) *For $j \notin \Sigma(\omega)$: $W_\Delta(j) = 0$ if j is a free coordinate, and $z_j W_\Delta(j) \leq 0$ if j is a contact.*
- (b) *Let l be a carrier of block m with $l \notin \Omega$ (the carriers in the supports of $\omega^\pm$), $S_l \cap F = \emptyset$, $w_m(k(l)) \neq 0$ and $\Delta d_m \neq 0$. Then there is s_0 such that every $s \in S_l \setminus \Sigma(\omega)$ with $s \geq s_0$ is a contact with $z_s = -\text{sign}(\Delta d_m) \text{sign } w_m(k(l))$.*
- (c) *For every coordinate j and every block m , $u_{k,m}(j) \neq 0$ for infinitely many k (any admissible T). Hence at every free coordinate $j \notin \Sigma(\omega)$ the identity $W_\Delta(j) = 0$ of (a) is an infinite series over carriers of every block m with $\Delta d_m \neq 0$.*

Proof. (a) is Lemma 2.24(b),(c). (b) For $s \in S_l$, $W_\Delta(s) = \Delta d_m \lambda_l w_m(k(l)) v_l(s) + \sum_{l'' > l} \Delta d_m \lambda_{l''} w(l'') y_{l''}(s) / n_{l''}$: earlier carriers do not meet S_l (allowedness (a) of Definition 3.5), and signature sets are disjoint. A later carrier l'' meets $s \in S_l$ only through its target, and then $c_{l''} \leq c_l 2^{-2s} c_l \delta_l / 2$ by (W4); with $c_{l''+1} \leq c_{l''} / 4$ the later terms total at most $C \max_{m'} |\Delta d_{m'}| 2^{-2s} c_l \delta_l$, while the own term has modulus $|\Delta d_m| |w_m(k(l))| \lambda_l \delta_l 2^{-s} / n_l$. For $s \geq s_0$ the own term dominates, so $W_\Delta(s) \neq 0$ with the sign of $\Delta d_m w_m(k(l))$, and (a) forces s to be a contact with the stated sign. (c) By (T-d), $e_j^* / q^*(e_j^*) \in S_{q^*}$ is a q^* -limit of a subsequence of $(u_{k,m})_k$, so $u_{k,m}(j) \rightarrow 1/q^*(e_j^*) \neq 0$ along it. $\square$

Remark 5.20. (a) Proposition 5.19(c) shows that shifted exact data exist only at rows satisfying one exact cancellation identity per free coordinate outside the finite set $\Sigma(\omega)$. That this is a non-generic property at partial contact is a *heuristic* reading only; it is not claimed as a theorem. (b) The identities are void at maximal contact. There the dichotomy is as follows. If $z = \epsilon_0$ is constant on $\mathbb{N} \setminus F$, then f is not a (C^*) row (Proposition 5.21). If $|z| = 1$ on $\mathbb{N} \setminus F$ with signs that are not eventually constant, the cited results do not exclude (C^*) : the type of a peak l is $\epsilon_l \text{sign } w(k(l))$, and signs on the signature sets can be prescribed carrier by carrier so that every coarse peak of a block is *self-aligned* ($z = \text{sign } u_l(\hat{z})$ on S_l , hence swallowing type), while some strict non-peak of robust relative position is anti-aligned (anti type, $q < 0$); then (U1), (U2), (U3) fail and the block lacks an upper source (Subsection 5.5).

Proposition 5.21 (Constant-sign maximal contact). *Let $T = T_{\text{final}}$, $N \geq 1$, $f \in S_{p_N}^*$ with F finite, and suppose $z_j = \epsilon_0$ for all $j \notin F$, for a fixed sign ϵ_0 . Then there is l_1 such that at every clean sub-window of every level $l \geq l_1$ every block has an upper source of type (U2) and a lower source of type (L2). Hence $I_{\text{up}}(w) = I_{\text{lo}}(w) = \emptyset$, $\rho^{\text{sh}}(\kappa^{\text{sh}}(w), f) \geq 1$, f is not a (C^*) row, and $f \in \mathcal{R}$.*

Proof. Since $\hat{z} \neq 0$ there is $y \in S_{q^*}$ with $y(\hat{z}) =: 2c > 0$. By (T-d), every block m has carriers $k_i \rightarrow \infty$ with $q^*(u_{k_i,m} - y) < c$, and carriers $k'_i \rightarrow \infty$ with $q^*(u_{k'_i,m} + y) < c$; then $u_{k_i,m}(\hat{z}) > c$ and $u_{k'_i,m}(\hat{z}) < -c$, because $|u(\hat{z}) - y(\hat{z})| \leq q^*(u - y)$. As $\Phi_m(k) \rightarrow 0$, their relative positions $\varrho = m|u(\xi)| / (\Phi \cdot m\vartheta_m)$ tend to infinity; in each block fix one carrier l_+ of the first kind and one carrier l_- of the second kind with $S_{l_\pm} \cap F = \emptyset$ and $\varrho_{l_\pm} \geq 2$ (so both are peaks). Their signature sets carry $z \equiv \epsilon_0$, so

their rooms (rate object (O1)) vanish and they are of class G with swallowing sign ϵ_0 at every clean sub-window of every level $l \geq l_1 :=$ the largest of these carriers over all blocks. Their types are $\epsilon_0 \text{ sign } u(\hat{z})$, i.e. one is of swallowing type and one of anti type, and both have $\varrho \geq 2 \geq 1 + u(w)$. The anti-type one is an upper source (U2), the swallowing-type one a lower source (L2). The remaining claims follow from Theorem 5.10 and Theorem 5.15. $\square$

Remark 5.22. The first part of this proof (existence of carriers $l_{\pm}$ in every block with $\varrho \geq 2$, $S_{l_{\pm}} \cap F = \emptyset$ and $z \equiv \epsilon_0$ on their signature sets) uses only (T-d) and the disjointness of the signature sets, so it holds for every admissible operator with pairwise disjoint signature sets. The conclusions about clean sub-windows, sources, ρ^{sh} , (C^*) and $\mathcal{R}$ use the objects and theorems of T_{final} .

Proposition 5.23 (Oscillating profiles force shifted data). *Let f' be a first row with F' finite, l a peak carrier of block m of f' with $\text{vs} := \text{sign } w'_m(k(l))$, and let h carry two-piece data $(b^{\pm}, \omega^{\pm})$ at f' . Let $S_1, S_2 \subseteq S_l \setminus \Sigma(\omega)$ be finite, nonempty, disjoint sets of contacts with $z'_s = \epsilon_l$ for $s \in S_1 \cup S_2$. Put $V_i := \sum_{s \in S_i} v_l(s)$, $A_i(h) := V_i^{-1} \sum_{s \in S_i} \epsilon_l h(s)$, $d_{m'}^{\pm} := d'_{m'}(\omega_{m'}^{\pm})$ and*

$$\text{tgt} := \max_{m', \pm} |d_{m'}^{\pm}| \sum_{s \in S_1 \cup S_2} \sum_{l'' > l} \lambda_{l''} |u_{l''}(s)|.$$

If $\epsilon_l \text{vs} = +1$, then $A_i(h) \in [-d_m^+ \lambda_l M'_m - \text{tgt}/V_i, -d_m^- \lambda_l M'_m + \text{tgt}/V_i]$; if $\epsilon_l \text{vs} = -1$ the same holds with d_m^+ and d_m^- interchanged. In both cases

$$|A_1(h) - A_2(h)| \leq |\Delta d_m| \lambda_l M'_m + \text{tgt} (1/V_1 + 1/V_2).$$

Consequently, if g is a functional and g_t carry two-piece data at f' with $\|g - g_t\|_1 \leq Kt$, then

$$|\Delta d_m(g_t)| \geq (|A_1(g) - A_2(g)| - (Kt + \text{tgt})(1/V_1 + 1/V_2))/(\lambda_l M'_m).$$

Proof. At $s \in S_l \setminus \Sigma(\omega)$ the $+$ representation gives $h(s) = b^+(s) - \sum_{m'} d_{m'}^+ \psi'_{m'}(s)$, and $\psi'_{m'}(s)$ consists of the own term $\lambda_l w'_m(k(l)) v_l(s) = \lambda_l \text{vs} M'_m v_l(s)$ (for $m' = m$) and target terms of carriers $l'' > l$: earlier targets avoid S_l by allowedness (a), and signature sets are disjoint. Side admissibility gives $\epsilon_l b^+(s) = z'_s b^+(s) \geq 0$, so $\epsilon_l h(s) \geq -d_m^+ \lambda_l \epsilon_l \text{vs} M'_m v_l(s) - \sum_{m'} |d_{m'}^+| \sum_{l'' > l} \lambda_{l''} |u_{l''}(s)|$; sum over S_i and divide by V_i . The $-$ side gives the other inequality with $\epsilon_l b^-(s) \leq 0$. The last claim follows from $|A_i(g) - A_i(g_t)| \leq \|g - g_t\|_1 / V_i$. $\square$

The decompositions of a mate g at a scale t satisfy the same sandwich up to the budget, so the profile $\epsilon_l g(s)/v_l(s)$ on the signature set of a robust class-G peak may oscillate inside an interval of length $|\Delta d_m| \lambda_l M'_m$ (decomposition convention). Proposition 5.23 shows that such an oscillation is seen by every approximating datum at every row f' sharing the peak: d -neutral data ($\Delta d = 0$) cannot follow a mate whose profile oscillates by more than $(Kt + \text{tgt})(1/V_1 + 1/V_2)$. In case (II) of Theorem 5.10 the exact-data route therefore needs shifted data.

Lemma 5.24 (Completion of the fine tail). *Let $f^{\#} \in S_{p^*}$ have finite base support $F^{\#}$, let l be a level, $J_{\text{fine}} := \mathbb{N} \setminus (F^{\#} \cup T(l) \cup \bigcup_{l'' \leq l} S_{l''})$, and $\Delta \in \mathbb{R}^l$. For $z' \in [-1, 1]^{J_{\text{fine}}}$ let $f(z') := f[a^{\#}, z^{\#}(z')]$, where $z^{\#}(z')$ equals z' on J_{fine} and $z^{\#}$ elsewhere, and put $W(z') := \sum_m \Delta_m R_m^* w_m(z')$, $w_m(z')$ the block functionals of $f(z')$. Then there is $z'^* \in [-1, 1]^{J_{\text{fine}}}$ with*

$$W(z'^*)(j) = 0 \quad \text{if } |z'^*_j| < 1, \quad z'^*_j W(z'^*)(j) \leq 0 \quad \text{if } |z'^*_j| = 1 \quad (j \in J_{\text{fine}}).$$

Moreover the values $u_{l''}(\hat{z})$ of all carriers $l'' \leq l$ are unchanged, and, with $\varepsilon_l := \sum_{l'' > l} \lambda_{l''}$ and $\max_m 2\varepsilon_l \leq c_{f\#}$, $p^*(f(z'^*) - f^\#) \leq C_{f\#}\varepsilon_l \log(e/\varepsilon_l)$.

Proof. The data $(a^\#, z^\#(z'))$ are admissible ($J_{\text{fine}} \cap F^\# = \emptyset$), so $f(z')$ is defined (Subsection 2.1). The set $\mathcal{K} := [-1, 1]^{J_{\text{fine}}}$ with the product topology is a compact, convex, metrizable subset of the locally convex space $\mathbb{R}^{J_{\text{fine}}}$. Define $\Psi(z')_j := \text{clamp}_{[-1, 1]}(z'_j - W(z')(j))$. Continuity: $\hat{z}(z') = z^\#(z') + Ue^\#$ depends coordinatewise continuously on z' ; $R_m^{**}\hat{z}(z') = (\lambda_{k,m}u_{k,m}(\hat{z}(z')))_k$ is ℓ_1 -continuous by dominated convergence ($\sum_k \lambda_{k,m}\|u_{k,m}\|_1 < \infty$, $\|\hat{z}(z')\|_\infty \leq 2$); it is never 0; the norming functional of the smooth norm $|\cdot|_m$ is unique ([I, Lemma 1.7]), so the duality map is norm-to-weak* continuous, and each $w_m(z')(k)$ is continuous and bounded by 1; finally $W(z')(j) = \sum_m \Delta_m \sum_k \lambda_{k,m}w_m(z')(k)u_{k,m}(j)$ is continuous by dominated convergence. By the Schauder–Tychonoff theorem Ψ has a fixed point z'^* , and $\text{clamp}(z_j - W_j) = z_j$ means $W_j = 0$ if $|z_j| < 1$, $W_j \leq 0$ if $z_j = 1$ and $W_j \geq 0$ if $z_j = -1$. Carriers $l'' \leq l$ have $\text{supp } u_{l''} \subseteq T(l) \cup S_{l''}$, which misses J_{fine} , so their values are unchanged; with $\delta := \hat{z}(z'^*) - \hat{z}^\#$ one has $u_{l''}(\delta) = 0$ for $l'' \leq l$ and $|u_{l''}(\delta)| \leq 2$ otherwise, so $\Delta_m(\delta) \leq 2\varepsilon_l$, $c(\delta) \leq C\varepsilon_l \log(e/\varepsilon_l)$, and Lemma 2.12 gives the cost bound. $\square$

Remark 5.25 (One ray only). W is linear in Δ , so the completion for Δ also completes $c\Delta$, $c > 0$. It does not complete two shift vectors $\Delta^{(1)}$, $\Delta^{(2)}$ spanning a two-dimensional cone at the same time: at a free coordinate both $W_{\Delta^{(1)}}$ and $W_{\Delta^{(2)}}$ would have to vanish. If the shift of the decompositions is unpinned, its direction may change from scale to scale inside a window, while the averaging of Theorems 2.20 and 5.13 uses one companion for all scales; this is resolved in Subsection 5.6 by normalizing all scales of a class onto one shift ray (Proposition 5.54). Lemma 5.24 also does not act on the coarse coordinates $T(l) \cup \bigcup_{l'' \leq l} S_{l''}$, where the fine contributions to W_Δ must be absorbed exactly by the coarse data.

For one block the existence of shifted exact data is a property of the row and of the set of switching carriers alone.

Corollary 5.26 (One block: shifted exact data). *Let $I = \{1\}$, $f \in S_{p^*}$ with F finite, $\psi := R_1^*w_1$, $\Omega \subseteq Q$ finite, and $\Delta \neq 0$.*

- (a) *If two-piece data with $\Delta d = \Delta$ and block parts supported in Ω exist for some functional, then outside $\Sigma(\omega)$: ψ vanishes at every free coordinate, and $z_j\Delta\psi(j) \leq 0$ at every contact.*
- (b) *Conversely, let $\gamma \in c_{00}$ be supported in Ω with $d(\gamma) = \Delta$ (possible iff $w(k) \neq 0$ for some $k \in \Omega$), and suppose that $v := R_1^*\gamma - \Delta\psi$ is z -signed at every contact and vanishes at every free coordinate off F (outside $\Sigma(\omega)$ this is the condition of (a); on $\Sigma(\omega) \setminus F$ it is a condition on γ). Then for every ω^+ supported in Ω , every $\chi : K \rightarrow [0, 1]$ and every β supported in F , the pairs (b^+, ω^+) , (b^-, ω^-) with $\omega^- := \omega^+ + \gamma$, $b^+ := \beta + \chi v|_K$, $b^- := b^+ - v$ are two-piece data with $\Delta d = \Delta$ for $h := b^+ + R_1^*(\omega^+ - d(\omega^+)w_1)$.*

Proof. (a) is Lemma 2.24 with $W_\Delta = \Delta\psi$. (b) $d(e_k) = \Phi(k)^2 w(k)/C$, so $d(\gamma) = \Delta$ is solvable iff some $w(k) \neq 0$; $\mathcal{V}(\gamma) = R_1^*(\gamma - \Delta w_1) = v$, and Lemma 2.22(c) applies. $\square$

With several blocks ψ is replaced by W_Δ , and blocks with $\Delta d_m = 0$ contribute nothing at the coordinates they own: every coordinate owned by a carrier of an

unshifted block is a *dead zone* for the shifted blocks, where the sign of W_Δ is decided by later carriers.

Remark 5.27 (An earlier route to (C^*) ; sketch only). The following route was proposed for (C^*) rows and is recorded here *as a sketch*, not as a result. Suppose that at infinitely many levels a clean sub-window w in case (II) has a coarse shifted exact cone at the companion (the system $\Sigma^\#$ together with the shift columns $\Delta_m R_m^* w_m^\#$ restricted to coarse coordinates and the coupling $\Delta_m = \sum q_l^\# \tau_l$) with a Slater point of robust margin, and that every shifted block has $\Delta_m \geq 0$ (data convention), or $\Delta_m < 0$ together with (SC) at the companions. Then one would transplant with shift columns, complete the fine tail by Lemma 5.24, absorb the coarse fine perturbations by the Slater condition, and conclude by Theorem 5.13. The gaps of this route are: (g1) exact adjustment of the block scalars entering the coupling; (g2) a conversion of near coordinates of coarse signature sets into support coordinates preserving the values; (g3) (SC) at the companions in configuration (i); (g4) shift directions that vary with the scale when two or more blocks are shifted (Remark 5.25); (g5) completion and coarse absorption must be solved jointly. The conversion in (g2) is *false* as stated for super-decreasing signature weights: the achievable signed sums of the near weights form a Cantor set, and the discrepancy is of the order of the first converted weight. Subsection 5.6 replaces (g2) by zero-value absorber pairs, and, under the hypotheses of Master Theorem IV' (Theorem 5.68), closes (g3) in configuration (i), (g4) and (g5); it reduces (g1) to one scalar per block, which is handled there by the lever hypothesis (KN); whether a second, independent lever is necessary is not decided (Remark 5.47).

5.5. Coherent shift resonance at block-tame rows; genericity. In this subsection $T = T_{\text{final}}$, whose features (SF_l) , (SF^*) , (b') , (Z_0) and the diagonal base are listed in Proposition 3.15; the propositions on genericity hold for every admissible T with a diagonal base, as stated. For $p = p_N$ the *present* carriers are $L_N := \{l : m(l) \leq N\}$. For a diagonal base with entries μ_j^* , $\nu = \|U^*a\|$ and $(Ue)_j = (\mu_j^*)^2 a_j / \nu$, so $\hat{z}_j = \text{sign } a_j + (\mu_j^*)^2 a_j / \nu$ on F and $\hat{z}_j = z_j$ off F .

Definition 5.28 (Owners and the owner recursion). The *owner* of a coordinate j is $o_N(j) := \min\{l \in L_N : u_l(j) \neq 0\}$ (it exists by Proposition 5.19(c) applied to block 1); if $j \in S_l$ with $l \in L_N$, then $o_N(j) = l$ by allowedness (a). Given a with finite support F and a sign ϵ_l for every $l \in L_N$ chosen along the recursion below, the *owner recursion* assigns z off F as follows. Initially the coordinates of F are *assigned*. At step $l \in L_N$ (in increasing order) let

$$A_l := \sum_{j \in \text{supp } y_l \text{ assigned}} y_l(j) \hat{z}_j, \quad B_l := \sum_{j \in \text{supp } y_l \text{ unassigned}} |y_l(j)|;$$

then put $z_j := \epsilon_l \text{sign } y_l(j)$ at the unassigned $j \in \text{supp } y_l$ and $z := \epsilon_l$ on S_l , and declare these coordinates assigned. The *owner sign* is $\epsilon_l := \text{sign } A_l$ (+1 if $A_l = 0$).

At step l the set S_l is unassigned (earlier targets avoid S_l by allowedness (a), signature sets are disjoint, and $F \cap S_l = \emptyset$ if $F \subseteq Z_0$); every coordinate off F is eventually assigned; and the data (a, z) are admissible, so they define a unique first row. When $S_l \cap F = \emptyset$ and $z = \epsilon_l$ on S_l , the value formula

$$(10) \quad n_l u_l(\hat{z}) = A_l + \epsilon_l (B_l + \delta_l H_l)$$

holds, because $u_l(\hat{z}) = u_l(z) + \langle U^* u_l, e \rangle$ and $\langle U^* u_l, e \rangle = \sum_{j \in F} (\mu_j^*)^2 u_l(j) a_j / \nu$. For the owner sign, $|n_l u_l(\hat{z})| = |A_l| + B_l + \delta_l H_l$. All owner notions are taken relative to L_N : with the recursion over all carriers, a coordinate owned by an absent carrier would receive that carrier's sign, while every quantity of p_N is dominated there by the first present carrier.

Let $F_\star := \{3, 5\} \subseteq Z_0$ and recall the target $y^* = (e_3^* - e_5^*) / q^*(e_3^* - e_5^*)$ of Definition 3.4(iv); put $c^* := 1/q^*(e_3^* - e_5^*)$. Since y^* is supported in Z_0 it is allowed at every stage, so every block has infinitely many carriers with target y^* . For a block m let $\delta_l^\circ := \delta_l H_l / n_l$ as in Definition 3.5.

Theorem 5.29 (Self-aligned rows). *Let $N \geq 1$, $m_0 \leq N$, and let l_c be the first carrier of block m_0 . Let $l_- > l_c$ be a carrier of block m_0 with target y^* and $\delta_{l_-} H_{l_-} < c^* \mu_5^* / 2$, and let $0 < \eta < \min\{c^* \mu_5^* / 2, n_{l_-} \theta_0 \Phi_{l_-} / (2m_0)\}$, $\theta_0 := \lambda_{l_c} \delta_{l_c}^\circ / 2$. Then there are $a_3, a_5 > 0$ such that the first row $f_{SA} \in S_{p_N}^*$ with $a = (a_3 e_3^* + a_5 e_5^*) / q^*(a_3 e_3^* + a_5 e_5^*)$ and with z given by the owner recursion over L_N , with owner signs except $\epsilon_{l_-} := +1$, has the following properties.*

- (a) Every $l \in L_N$ is exactly swallowed with sign ϵ_l ; for $l \neq l_-$, $\text{sign } u_l(\hat{z}) = \epsilon_l$ and $|u_l(\hat{z})| \geq \delta_l^\circ$.
- (b) Every $l \in L_N \setminus \{l_-\}$ is a non-degenerate peak of swallowing type; if l is not the first carrier of its block, then $\varrho_l \geq 2^{10}$ and $\mu_l \geq q_0 \delta_l^\circ / 2$.
- (c) $n_{l_-} u_{l_-}(\hat{z}) = -\eta$; l_- is a strict non-peak of anti type with $\varrho_{l_-} < \frac{1}{2}$, $\text{gap} \geq M_{m_0} / 2$ and $q_{l_-} < 0$, and $u_{l_-} 1_{F^c} = v_{l_-}$ is z -signed.
- (d) (Dominance.) For each of the vectors

$$W := \sum_{l \in L_N} \lambda_l \epsilon_l u_l 1_{F^c}, \quad \sum_{l \in L_N, l \leq L} \lambda_l \epsilon_l u_l 1_{F^c} \quad (L \in \mathbb{N}),$$

$$\sum_{l \in L_N} c_l \lambda_l \epsilon_l u_l 1_{F^c} \quad (c_l \in [\frac{1}{2}, 1]) :$$

at every $j \notin F$ where the sum contains the term of $o_N(j)$, the sum is nonzero with sign z_j .

- (e) Every coordinate off F is a contact; $Q_{m_0} = \{k(l_-)\}$ and $Q_m = \emptyset$ for $m \neq m_0$; there is no degenerate peak; every block satisfies $\text{Scr}_m(\Upsilon s) = o(s)$ as $s \rightarrow 0$ for every $\Upsilon \geq 1$; and f_{SA} satisfies (BT).
- (f) There is l_1 such that at every clean sub-window w of every level $l \geq l_1$: block m_0 has no upper source and a lower source (L2), every other block has both sources, and $c_{\pi(w)}(f_{SA}) = 0$.

Proof. Step 1 (tuning). $\text{supp } y^* = F_\star$ and S_{l_-} is unassigned at step l_- , so $n_{l_-} u_{l_-}(\hat{z}) = y^*(\hat{z}_{F_\star}) + \delta_{l_-} H_{l_-}$, independently of the recursion. With $\hat{z}_j = 1 + (\mu_j^*)^2 a_j / \nu$ on $F_\star$, $y^*(\hat{z}_{F_\star}) = c^*((\mu_3^*)^2 a_3 - (\mu_5^*)^2 a_5) / \nu$ runs continuously over $(-c^* \mu_5^*, c^* \mu_3^*)$ as a_3/a_5 runs over $(0, \infty)$. By the choice of l_- and η and the intermediate value theorem there is a ratio with $n_{l_-} u_{l_-}(\hat{z}) = -\eta$; fix it.

Step 2 ((a)). $S_l \cap F = \emptyset$ because $F \subseteq Z_0$, so each l is exactly swallowed with sign ϵ_l , and (10) with the owner sign gives $|n_l u_l(\hat{z})| = |A_l| + B_l + \delta_l H_l \geq \delta_l H_l$, $\text{sign } u_l(\hat{z}) = \epsilon_l$.

Step 3 (threshold bound). For a block vector $\zeta = \hat{\zeta}_m$: $A(x; \zeta) \geq \|\zeta\|_1 - x \|\Phi_m\|_2^2 \geq \|\zeta\|_1 - x/4$ and $B(x; \zeta) \leq x \|\zeta\|_1$, so $\Xi(\|\zeta\|_1/2; \zeta) \geq (\frac{49}{64} - \frac{1}{2}) \|\zeta\|_1^2 > 0$ and $\vartheta_m > \|\zeta\|_1/2$ (Lemma 2.6(a)). Hence $M_m = \vartheta_m / (\vartheta_m + A_m) > \frac{1}{3}$ for every block of every first row, and $\bar{\vartheta}_{m_0} > \lambda_{l_c} |u_{l_c}(\hat{z})|/2 \geq \theta_0$.

Step 4 ((b)). Let $l \neq l_-$ be a carrier of block m and $x := v_l = m|u_l(\hat{z})|/\Phi_l$. For an earlier carrier k of block m , $\Phi_l/\Phi_k \leq c_l/c_{l-1} \leq \delta_l 2^{-\min S_l - l - 10}/(1 + \|U\|) \leq |u_l(\hat{z})|2^{-l-9}/(1 + \|U\|)$ by (W5) and $\delta_l^\circ \geq \frac{4}{5}\delta_l 2^{-\min S_l}$; as $|u_k(\hat{z})| \leq 1 + \|U\|$, the term $(|\zeta(k)| - x\Phi_k^2)_+ = m\Phi_k(|u_k(\hat{z})| - |u_l(\hat{z})|\Phi_k/\Phi_l)_+$ vanishes. Later carriers contribute at most $(1 + \|U\|)\sum_{l'' > l} \lambda_{l''} \leq 2^{-10}\lambda_l|u_l(\hat{z})|$ by (SF_l), and the own term vanishes at $x = v_l$. So $A(v_l) \leq 2^{-10}m\Phi_l|u_l(\hat{z})| < m|u_l(\hat{z})| \leq B(v_l)^{1/2}$, hence $\Xi(v_l) < 0$ and $\vartheta_m < v_l$: l is a non-degenerate peak, of swallowing type since $w_m(k(l)) = \text{sign } u_l(\hat{z})M_m = \epsilon_l M_m$. If l_1 is the first carrier of block m , $\vartheta_m < v_{l_1} \leq m(1 + \|U\|)/\Phi_{l_1}$, while for later l the second half of (SF*) gives $v_l \geq m\delta_l^\circ/\Phi_l \geq 2^{10}m(1 + \|U\|)/\Phi_{l_1}$; so $\varrho_l \geq 2^{10}$ and $\mu_l = q_0|u_l(\hat{z})|(1 - 1/\varrho_l) \geq q_0\delta_l^\circ/2$.

Step 5 ((c)). $v_{l_-} = m_0\eta/(n_{l_-}\Phi_{l_-}) < \theta_0/2 < \bar{\vartheta}_{m_0}/2$: a strict non-peak with $\varrho_{l_-} < \frac{1}{2}$ and gap $= M(1 - \varrho_{l_-}) \geq M/2$; $w(k(l_-))$ has the sign of $u_{l_-}(\hat{z}) < 0 = -\epsilon_{l_-}$, so l_- is of anti type and $q_{l_-} < 0$; $u_{l_-}1_{F^c} = v_{l_-}$ lives on S_{l_-} , where $z = +1$.

Step 6 ((d)). Let $j \notin F$ and $o := o_N(j)$, with $m := m(o)$, $k := k(o)$. Later carriers meet j only through their targets, with $|u_{l''}(j)| \leq \frac{4}{3}$. If $j \in S_o$, then $z_j = \epsilon_o$ and the own term is $\lambda_o\delta_o2^{-j}/n_o$. If $j \leq \min S_o + m + k + 4$, (SF_l) bounds the later terms by $\frac{4}{3}\sum_{l'' > o} \lambda_{l''} \leq \frac{4}{3}2^{-10}\lambda_o\delta_o2^{-(m+k+\min S_o)}/n_o \leq 2^{-5}\lambda_o\delta_o2^{-j}/n_o$; otherwise (W4) gives $c_{l''} \leq \iota_o2^{-2j}c_o\delta_o/2$ for every later l'' with $j \in \text{supp } y_{l''}$, so their total is $\leq \frac{4}{9}2^{-2j}c_o\delta_o$ against an own term $\geq \frac{4}{5}2^{-m-k}c_o\delta_o2^{-j}$, ratio $\leq \frac{5}{9}2^{m+k-j} < 2^{-5}$. If $j \notin S_o$, then j was unassigned at step o , $z_j = \epsilon_o \text{sign } y_o(j)$, and $|\lambda_o y_o(j)/n_o| \geq \lambda_o y_o/n_o \geq 2^9 \cdot \frac{4}{3}\sum_{l'' > o} \lambda_{l''}$ by (SF_l). In both cases the term of o has sign z_j and exceeds twice the sum of the moduli of all later terms, also after multiplication of the terms by factors in $[\frac{1}{2}, 1]$.

Step 7 ((e)). Every coordinate off F is assigned ± 1 . By Steps 4, 5 the only strict non-peak is $k(l_-)$, and there is no degenerate peak. For the scrambling sums: finitely many carriers (the first carrier of each block, and l_-) have fixed positive margins or gaps; every other carrier is a peak with $\mu_l \geq q_0\delta_l^\circ/2$ and contributes to $\text{Scr}_m(\Upsilon s)$ only if $\delta_l^\circ \leq 2\Upsilon s/q_0$, in which case (W5) gives $\Phi_l \leq c_l \leq Cc_{l-1}2^{-l}\Upsilon s/q_0$; if $l(s)$ is the least contributing carrier, the contributions are $\leq \sum_{l \geq l(s)} \Phi_l \leq 2\Phi_{l(s)} = o(s)$, since $l(s) \rightarrow \infty$. So (MS) holds, and f_{SA} satisfies (BT).

Step 8 ((f)). All carriers are exactly swallowed, so every room (rate object (O1)) vanishes and every coarse carrier is of class G. By (b), (c), every coarse peak is of swallowing type with $\varrho \geq 2^{10} \geq 1 + u$ (beyond the first carriers), and l_- is a class-G carrier with $q < 0$ and $\varrho_{l_-} \in (0, \frac{1}{2})$, which at a clean sub-window of large level is neither $\leq b$ nor $\geq 1 - b$. Hence (U1), (U2) fail in every block, (U3) fails in block m_0 and holds vacuously in the other blocks (no carrier with $q < 0$), and (L2) holds in every block. So $I_{\text{up}}(w) = \{m_0\}$, $I_{\text{lo}}(w) = \emptyset$, and in the shift cost $c_{\pi(w)}$ the admissible δ is $\delta_{m_0} = 1$. Take $x_{l''} := \lambda_{l''}$ for $l'' \in B_f(w)$. Since swallowing-type peaks have $vs_{l''} = \epsilon_{l''}$, the vector $\Pi_{m_0} + \sum_{B_f} x\epsilon u 1_{F^c}$ is the partial sum of W over the coarse carriers, which by (d) is z -signed wherever nonzero; there are no free coordinates; so every summand $\phi_{z_j}(\cdot)$ vanishes and $c_{\pi(w)}(f_{\text{SA}}) = 0$. $\square$

So exact coherent shift resonance occurs at block-tame rows of T_{final} , and the shift cost of (O6) vanishes there. By [I, Corollary 7.22] these rows are nevertheless in $\mathcal{R}$.

Corollary 5.30. *The rows f_{SA} of Theorem 5.29 belong to $\mathcal{R}$.*

Proof. Theorem 5.29(e) and [I, Corollary 7.22]. $\square$

Proposition 5.31 (Explicit coherent-shift mates). *Let $N = 1$, $f = f_{\text{SA}}$, $k_- := k(l_-)$, $A := |\hat{\zeta}_1|_1$, $\psi := R_1^* w_1$, and*

$$V := u_{l_-} - \frac{u_{l_-}(\hat{z})}{A} \psi.$$

Let $\alpha \in \mathbb{R}$, $\Delta_\alpha > 0$, $\omega^+ := \alpha e_{k_-}$, $\omega^- := (\alpha + \Delta_\alpha) e_{k_-}$.

- (i) $\Delta d = d(\omega^-) - d(\omega^+) = \Delta_\alpha \lambda_{l_-} u_{l_-}(\hat{z})/A < 0$.
- (ii) $V1_{F^c}$ is z -signed and nonzero at every coordinate off F .
- (iii) For every $\chi : F^c \rightarrow [0, 1]$ and every β^+ supported in F with $b^+(\xi) = 0$, i.e. $\beta^+(\xi) = -q_0 \Delta_\alpha \lambda_{l_-} \sum_{j \notin F} \chi_j |V_j|$ (such β^+ exist, since $\xi \neq 0$ on F), the pairs $b^+ := \chi \Delta_\alpha \lambda_{l_-} V1_{F^c} + \beta^+$, $b^- := b^+ - \Delta_\alpha \lambda_{l_-} V$ with $\omega^\pm$ are two-piece data for $g := b^+ + R_1^*(\omega^+ - d(\omega^+) w_1)$.
- (iv) $cg \in \mathcal{C}(f)$ for all small $c > 0$, and $(f, cg) \in \text{clNA}((c_0, p_1), \ell_2^2)$.
- (v) For $s \in S_l$, $l \neq l_-$: $g(s) = \psi(s)(-d(\omega^+) + \chi_s \Delta_\alpha \lambda_{l_-} |u_{l_-}(\hat{z})|/A)$; in particular the profile g/ψ is not constant on S_l whenever χ is not.

Proof. (i) Off the peaks, $\Phi(k)^2 w(k)/C = \hat{\zeta}(k)/A$ ([I, Lemma 1.7]), so $d(e_{k_-}) = \hat{\zeta}(k_-)/A = \lambda_{l_-} u_{l_-}(\hat{z})/A$. (ii) All carriers other than l_- are peaks with $w(k) = \epsilon_k M$, so $\psi = MW' + \lambda_{l_-} w(k_-) u_{l_-}$ with $W' := \sum_{l \neq l_-} \lambda_l \epsilon_l u_l$, and $w(k_-) = -M \varrho_{l_-}$. Hence $V = c_1 u_{l_-} + c_2 W'$ with $c_2 := M |u_{l_-}(\hat{z})|/A > 0$ and $c_1 := 1 - \lambda_{l_-} |u_{l_-}(\hat{z})| M \varrho_{l_-}/A \in (0, 1]$ (as $\lambda |u(\hat{z})| \leq A$ and $M \varrho < 1$). Off $F \cup S_{l_-}$, $V = c_2 W'$, which is z -signed and nonzero by Theorem 5.29(d); on S_{l_-} , $V(s) \geq (c_1 - 2^{-5} c_2 \lambda_{l_-}) v_{l_-}(s) > 0$, so V is z_s -signed there, since $c_1 \geq \frac{1}{2}$ (as $\varrho_{l_-} < \frac{1}{2}$ and $M \leq 1$) and $2^{-5} c_2 \lambda_{l_-} \leq \frac{1}{32}$ (as $\lambda_{l_-} |u_{l_-}(\hat{z})| \leq A$; the factor 2^{-5} is from Step 6 of that proof). (iii) $b^+ - b^- = \Delta_\alpha \lambda_{l_-} V = R_1^*(\omega^- - \omega^+) - \Delta d \psi$, since $R_1^* e_{k_-} = \lambda_{l_-} u_{l_-}$; so both pairs represent g . Off F , $b^+ = \chi \Delta_\alpha \lambda_{l_-} V$ is z -signed and $b^- = (\chi - 1) \Delta_\alpha \lambda_{l_-} V$ is $(-z)$ -signed; every coordinate off F is a contact, and $\text{supp } \omega^\pm \subseteq Q = \{k_-\}$. Tangency: off F we have $\hat{z}_j = z_j$ (the vector Ue is supported in F), every coordinate is a contact and $V1_{F^c}$ is z -signed, so $(\chi \Delta_\alpha \lambda_{l_-} V1_{F^c})(\xi) = q_0 \Delta_\alpha \lambda_{l_-} \sum_{j \notin F} \chi_j |V_j|$, and the normalization of β^+ gives $b^+(\xi) = 0$. Moreover $V(\hat{z}) = u_{l_-}(\hat{z}) - (u_{l_-}(\hat{z})/A) \psi(\hat{z}) = 0$ because $\psi(\hat{z}) = w_1(\hat{\zeta}_1) = A$, hence $b^-(\xi) = b^+(\xi) = 0$; and $R_1^*(\omega^+ - d(\omega^+) w_1)(\xi) = 0$, so $g(\xi) = 0$. (iv) [I, Proposition 7.9] on both sides gives $p^*(f + rg) \leq 1 + \frac{r^2}{2}(\kappa_w + 1)$ for $|r| \leq r_0(g)$. For $c \leq r_0$ with $c^2(\kappa_w + 1) \leq \frac{3}{4}$ and $cp^*(g) \leq \frac{1}{3}$: if $|r| \leq 1$ then $p^*(f + rcg) \leq 1 + \frac{3}{8} r^2 \leq s(r)$, and if $|r| \geq 1$ then $p^*(f + rcg) \leq 1 + |r| cp^*(g) \leq s(r)$, by $s(r) - 1 \geq \min(r^2, |r|)/3$ ([I, Lemma 3.6(a)]). So $cg \in \mathcal{C}(f)$, and Corollary 5.30 applies. (v) At $s \in S_l$, $l \neq l_-$, $u_{l_-}(s) = 0$ and $R_1^* \omega^+(s) = 0$, so $g(s) = \chi_s \Delta_\alpha \lambda_{l_-} V(s) - d(\omega^+) \psi(s)$ with $V(s) = (|u_{l_-}(\hat{z})|/A) \psi(s)$. $\square$

These mates switch through the anti-type strict non-peak k_- and carry the uniform shift through all peaks at every scale: exactly the coherent shift that windows cannot follow. The next proposition realizes the hypotheses of Theorem 2.28 (compare Remark 2.29).

Proposition 5.32 (Realization of Theorem 2.28). *Let $N = 1$ and $f = f_{\text{SA}}$. There are first rows $f_n \rightarrow f$ satisfying (BT) such that, for every carrier $l \neq l_-$, all large n and all $s_1, s_2 \in S_l$, the hypotheses of Theorem 2.28 hold. Consequently, for every mate cg of Proposition 5.31 with $\chi_{s_1} \neq \chi_{s_2}$ and every $\rho \in (0, 1]$, $\liminf_n \text{dist}_{p^*}(pcg, \mathcal{C}(f_n)) > 0$, although f and every f_n belong to $\mathcal{R}$.*

Proof. Step 1 (the rows). By Step 1 of Theorem 5.29, $y^*(\hat{z}_{F_*}) = -(\eta + \delta_{l_-} H_{l_-}) < 0$. Choose carriers $o_n \rightarrow \infty$, $o_n > l_-$, with target y^* and $\delta_{o_n} H_{o_n} < (\eta + \delta_{l_-} H_{l_-})/2$. At f , $B_{o_n} = 0$ and $A_{o_n} < 0$, so $\epsilon_{o_n} = -1$ and $z = -1$ on S_{o_n} . Let f_n have the same a and $z^{(n)} := z$, except $z^{(n)} := +1$ on S_{o_n} and the coordinates owned by carriers $l > o_n$ are re-assigned by the *hysteretic* recursion: with the signs ϵ_l of f as reference signs, keep ϵ_l unless $\epsilon_l A_l \leq -(B_l + \delta_l H_l)/2$, in which case $\epsilon_l := \text{sign } A_l$. Coordinates owned by carriers $< o_n$ are unchanged, so $z^{(n)} \rightarrow z$ coordinatewise and $f_n \rightarrow f$ (Subsection 2.1).

Step 2 (f_n satisfies (BT)). Values of carriers $< o_n$ are unchanged. $n_{o_n} u_{o_n}(\hat{z}) = A_{o_n} + \delta_{o_n} H_{o_n} \leq -(\eta + \delta_{l_-} H_{l_-})/2$ while $z^{(n)} = +1$ on S_{o_n} : o_n is an exactly swallowed anti-type peak with margin bounded below independently of n (its relative position tends to infinity, the threshold being bounded). Every re-run carrier has $n_l |u_l(\hat{z})| \geq (B_l + \delta_l H_l)/2$ with sign ϵ_l , hence (Step 4 of Theorem 5.29 with $|u_l(\hat{z})| \geq \delta_l^\circ/2$) is a swallowing-type peak with margin $\geq q_0 \delta_l^\circ/4$. As $\hat{z}^{(n)} \rightarrow \hat{z}$ in ℓ_1 , the threshold converges, and the finitely many carriers whose status is not robust by Step 4 (the first carrier and l_-) keep their statuses for large n . So $Q^{(n)} = \{k_-\}$, there is no degenerate peak, and (MS) holds as in Step 7.

Step 3 (alignment). $\Sigma_n = F \cup \text{supp } u_{l_-} = F \cup S_{l_-}$, and every coordinate off F is a contact of f_n . For $s \in S_c$, $c \neq l_-$ the owner, $\psi_n(s) = \lambda_c w^{(n)}(c) v_c(s) + \sum_{l'' > c} \lambda_{l''} w^{(n)}(l'') y_{l''}(s) / n_{l''}$, and by Step 6 of Theorem 5.29 (which bounds moduli, $|w| \leq 1$) the sum is $\leq 2^{-5} \lambda_c v_c(s) < \lambda_c M^{(n)} v_c(s)$ since $M^{(n)} > \frac{1}{3}$. So $\text{sign } \psi_n(s) = \text{sign } w^{(n)}(c) = \text{sign } u_c(\hat{z}^{(n)})$: the contacts of S_c are aligned for every swallowing-type peak c , and those of S_{o_n} are anti-aligned ($z^{(n)} = +1$, $w^{(n)}(o_n) = -M^{(n)}$). All of them lie outside Σ_n . For $s_1, s_2 \in S_l$, $l \neq l_-$, and $o_n > l$, the hypotheses of Theorem 2.28 hold. By Proposition 5.31(v), $\pi_{cg}(s_1) \neq \pi_{cg}(s_2)$ when $\chi_{s_1} \neq \chi_{s_2}$, and Theorem 2.28 applies. $\square$

Hence membership in $\mathcal{R}$ cannot be propagated along an arbitrary dense sequence of block-tame rows: approximants for mates with oscillating profiles must either bank the oscillation coordinates, as the engineered approximants of [I, Section 7] do, or keep every peak of a shifted block exactly swallowed and aligned outside the data supports, as the hysteretic re-alignments below do.

Genericity in fixed- z fibres. For the next results let T be any admissible operator and U diagonal with entries $\mu_j > 0$ ($U^* e_j^* = \mu_j k_j$); for $T = T_{\text{final}}$, $\mu_j = \mu_j^*$. Fix a finite set F with $d := |F| \geq 2$, signs $\sigma \in \{-1, 1\}^F$ and $z \in B_{\ell_\infty}$ with $z = \sigma$ on F , and put

$$\mathcal{A}_\sigma := \{a \in \ell_1 : \text{supp } a = F, \text{ sign } a = \sigma, q^*(a) = 1\}, \quad f_a := f[a, z] \quad (a \in \mathcal{A}_\sigma).$$

For $a \in \mathcal{A}_\sigma$ let $\nu(a) := \|U^* a\|$, $b_j(a) := \mu_j a_j / \nu(a)$ and $\Phi(a) := (\sigma_j + \mu_j b_j(a))_{j \in F}$. Then $\hat{z}(a)_j = \Phi(a)_j$ on F and $\hat{z}(a)_j = z_j$ off F , so $y(\hat{z}(a)) = \langle y, \Phi(a) \rangle$ for y supported in F ; and $a \mapsto b(a)$ is a homeomorphism of $\mathcal{A}_\sigma$ onto the open orthant Ω_σ of the unit sphere S^{d-1} with signs σ .

Proposition 5.33 (Near-threshold carriers are generic). *In this setting assume that z has a contact off F , let m be a block and (ε_l) any sequence of positive numbers. Then*

$$\{a \in \mathcal{A}_\sigma : |v_l(f_a) - \bar{v}_m(f_a)| < \varepsilon_l \text{ for infinitely many carriers } l \text{ of block } m\}$$

($\mathbf{v}_l = m|u_l(\hat{z})|/\Phi_l$) is a dense G_δ subset of $\mathcal{A}_\sigma$. If $\varepsilon_l \rightarrow 0$, the row f_a fails (BT) for every a in this set.

Proof. Work on Ω_σ . For $y \in \ell_1$, $\psi_y(b) := y(\hat{z}) = \sum_{j \notin F} y_j z_j + \sum_{j \in F} y_j (\sigma_j + \mu_j b_j)$ is affine in b ; $\hat{\zeta}_m(b)$ is ℓ_1 -continuous, so $\bar{\vartheta}_m(b)$ is continuous and positive. Fix L , $b_0 \in \Omega_\sigma$ and a connected open $O \ni b_0$ with compact closure in Ω_σ . Choose $n \in \mathbb{R}^F$ with $(n_j \mu_j)_j$ not parallel to b_0 ($d \geq 2$), $c := -\sum_{j \in F} n_j (\sigma_j + \mu_j b_{0,j})$, a contact $j_1 \notin F$, and $y := (n + c z_{j_1} e_{j_1}^*)/q^*(\cdot)$. Then $\psi_y(b_0) = 0$ and ψ_y has nonzero differential along the sphere at b_0 , so $\psi_y(b_\pm) = \pm c_0$ for some $b_\pm \in O$, $c_0 > 0$. By (T-d) there are carriers $l_i \rightarrow \infty$ of block m with $q^*(u_{l_i} - y) \rightarrow 0$, and $\sup_b |u_{l_i}(\hat{z}(b)) - \psi_y(b)| \leq q^*(u_{l_i} - y)$. For large i , $u_{l_i}(\hat{z}(b_\pm))$ has sign $\pm$ and modulus $\geq c_0/2$; the function $G_i := \mathbf{v}_{l_i} - \bar{\vartheta}_m$ is continuous on O , positive at b_+ (as $\Phi_{l_i} \rightarrow 0$), and equal to $-\bar{\vartheta}_m < 0$ at a zero of $u_{l_i}(\hat{z})$ on a path from b_+ to b_- in O . Hence $\{b \in O : |G_i(b)| < \varepsilon_{l_i}\}$ is nonempty and open, and $V_L := \bigcup_{l \geq L} \{|\mathbf{v}_l - \bar{\vartheta}_m| < \varepsilon_l\}$ is open and dense. The set in question is $\bigcap_L V_L$, a dense G_δ by Baire's theorem. Infinitely many carriers with $|\mathbf{v}_l - \bar{\vartheta}_m| < \varepsilon_l$ are infinitely many strict non-peaks (Q_m infinite) or infinitely many peaks with $\mu_l = q_0 \Phi_l (\mathbf{v}_l - \bar{\vartheta}_m)/m < s_l := q_0 \Phi_l \varepsilon_l/m$, and $\Phi_l/s_l = m/(q_0 \varepsilon_l) \rightarrow \infty$ (when $\varepsilon_l \rightarrow 0$) contradicts (MS). $\square$

So no admissible design can ensure that only finitely many carriers are near the threshold, and (BT) is meagre in every fixed- z fibre with a contact. Nevertheless (BT) rows are dense among all rows with finite base support (Proposition 5.40 below): the owner-type rows escape Proposition 5.33 because their z depends on a . The next two statements show that (C*) is not stable under tilts of a at fixed z .

Proposition 5.34 (Tilts create both kinds of robust peaks). *In this setting let $a_0, a_1 \in \mathcal{A}_\sigma$ with $\Phi(a_0), \Phi(a_1)$ not positively proportional. Then there is $\eta > 0$ such that every block contains infinitely many carriers k with $u_k(\hat{z}(a_0)) \geq \eta/2$ and $u_k(\hat{z}(a_1)) \leq -\eta/2$, and infinitely many carriers k' with $u_{k'}(\hat{z}(a_0)) \geq \eta/2$ and $u_{k'}(\hat{z}(a_1)) \geq \eta/2$. Moreover, at most one $a_1 \neq a_0$ has $\Phi(a_1)$ positively proportional to $\Phi(a_0)$.*

Proof. $\Phi(a)_j = \sigma_j(1 + \mu_j|b_j(a)|)$, so all $\Phi(a)$ lie in the open orthant of the signs σ and $\langle \Phi(a_0), \Phi(a_1) \rangle > 0$. Two nonzero vectors that are not positively proportional and not opposite are linearly independent, so there is $c \in \mathbb{R}^F$ with $\langle c, \Phi(a_0) \rangle > 0 > \langle c, \Phi(a_1) \rangle$. Put $y_0 := c/q^*(c)$, $y'_0 := \Phi(a_0)/q^*(\Phi(a_0))$ (supported in F) and let η be the least of $\langle y_0, \Phi(a_0) \rangle$, $-\langle y_0, \Phi(a_1) \rangle$, $\langle y'_0, \Phi(a_0) \rangle$, $\langle y'_0, \Phi(a_1) \rangle$. By (T-d) every block has carriers arbitrarily far out in every q^* -ball around y_0 and y'_0 , and $|u(\hat{z}(a)) - y(\hat{z}(a))| \leq q^*(u - y)$. For the last claim, $\Phi(a_1) = \lambda \Phi(a_0)$ together with $\|b(a_1)\|_2 = 1$ is a quadratic equation in λ with the root $\lambda = 1$. $\square$

Corollary 5.35 (Tilts of aligned rows leave (C*)). *Let $T = T_{\text{final}}$, F , σ , z as above, and let $a_0 \in \mathcal{A}_\sigma$ be such that in every block all but finitely many carriers k are exactly swallowed with the sign of their value: $z = \text{sign } u_k(\hat{z}(a_0))$ on S_k . Then for every $a_1 \in \mathcal{A}_\sigma$ with $\Phi(a_1)$ not positively proportional to $\Phi(a_0)$, the row f_{a_1} has in every block infinitely many exactly swallowed robust peaks of anti type and infinitely many of swallowing type. Consequently, at every clean sub-window of every sufficiently large level, every block of f_{a_1} has an upper source (U2) and a lower source (L2), $\rho^{\text{sh}} \geq 1$, f_{a_1} is not a (C*) row, and $f_{a_1} \in \mathcal{R}$.*

Proof. For large i the carriers k_i of Proposition 5.34 satisfy $z = +1$ on S_{k_i} (hypothesis, $u_{k_i}(\hat{z}(a_0)) > 0$) and $u_{k_i}(\hat{z}(a_1)) \leq -\eta/2$: at f_{a_1} they are exactly swallowed

(rooms 0, class G at every level) with $w = -M$, i.e. of anti type, and, as $\Phi_{k_i} \rightarrow 0$, their relative positions tend to infinity. The carriers k'_i are exactly swallowed robust peaks of swallowing type in the same way. A fixed carrier is coarse at all later levels, so for all large levels every clean sub-window has, in every block, a class-G anti-type peak (U2) and a class-G swallowing-type peak with $\varrho \geq 1 + u$ (L2). Theorem 5.10 and Theorem 5.15 give the rest. $\square$

Remark 5.36. (a) The hypothesis of Corollary 5.35 (aligned rows) matters. For a general (C^*) row the argument shows only that the source-deficient blocks acquire both sources at f_{a_1} (by Lemma 5.18 all robust coarse peaks of such a block are of one type); a block that has both sources at f_{a_0} through carriers of both types may lose one type at f_{a_1} . So the correct reading is: *aligned* (C^*) rows are isolated in their fixed- z fibre, up to at most one other point. (b) Along $f_{a_1} \rightarrow f_{a_0}$ the oscillating coherent-shift mates of f_{a_0} are lost whenever f_{a_1} is block-tame (Proposition 2.27, since the anti-type peaks provide anti-aligned contacts outside the data supports). For non-block-tame f_{a_1} the analogous statement is only heuristic.

Exact data with negative shift. We now show that exact two-piece data with a negative shift in every block are rigid, that they are recovered when they have no dead zones, and that they exist only on a meagre set. In the rest of this subsection $T = T_{\text{final}}$, $p = p_N$ and F is finite. For two-piece data $(b^\pm, \omega^\pm)$ at f put (data convention)

$$D := b^+ - b^- = \sum_k \gamma_k u_k, \quad \gamma_k := \lambda_k [(\omega_m^- - \omega_m^+)(k) - \Delta d_m w_m(k)] \quad (m = m(k))$$

(Lemma 2.22), K_ω the finite set of carriers in the supports of $\omega^\pm$, $E := \bigcup_{k \in K_\omega} \text{supp } y_k \setminus F$ (finite), and $C_1 := \max_m |\Delta d_m|$. A carrier k is *neutral* if $\gamma_k = 0$ and *negligible* if $0 < |\gamma_k| < 2\iota_k C_1 \lambda_k$, with the tolerance sequence $\iota_k = 2^{-k}$ of Definition 3.4.

Lemma 5.37 (Dominance with design thresholds). *Let $j \notin F$, $o := o_N(j)$, and suppose $|\gamma_k| \leq C_1 \lambda_k$ for every $k > o$ with $u_k(j) \neq 0$. If $|\gamma_o| \geq \iota_o C_1 \lambda_o$, then $\text{sign } D(j) = \text{sign}(\gamma_o u_o(j))$ and $|D(j)| \geq \frac{31}{32} |\gamma_o u_o(j)|$.*

Proof. As in Step 6 of Theorem 5.29, with the factor ι_o carried along: later carriers meet j only through targets, $|u_k(j)| \leq \frac{4}{3}$. If $j \in S_o$ and $j \leq \min S_o + m(o) + k(o) + 4$, (SF _{ι}) bounds the later terms by $\frac{4}{3} C_1 \sum_{k>o} \lambda_k \leq \frac{4}{3} 2^{-10} \iota_o C_1 \lambda_o \delta_o 2^{-(m+k+\min S_o)}/n_o$, at most 2^{-5} times the own term $\geq \iota_o C_1 \lambda_o \delta_o 2^{-j}/n_o$. If $j \in S_o$ is larger, (W4) gives $c_k \leq \iota_o 2^{-2j} c_o \delta_o / 2$ for each later k at j , so the later total is $\leq \frac{4}{9} C_1 \iota_o 2^{-2j} c_o \delta_o$ against an own term $\geq \frac{4}{5} \iota_o C_1 2^{-m-k} c_o \delta_o 2^{-j}$. If $j \in \text{supp } y_o \setminus S_o$, the later total is $\leq \frac{4}{3} C_1 \sum_{k>o} \lambda_k \leq 2^{-9} \iota_o C_1 \lambda_o y_o / n_o \leq 2^{-9} |\gamma_o u_o(j)|$. $\square$

Proposition 5.38 (Exact data with negative shift force owner alignment). *Let $(b^\pm, \omega^\pm)$ be two-piece data at f with $C_1 > 0$. For every $j \notin F \cup E$ whose owner $o = o_N(j)$ is neither neutral nor negligible, $D(j) \neq 0$, j is a contact, and $z_j = \text{sign}(\gamma_o u_o(j))$; if moreover $o \notin K_\omega$, then $z_j = -\text{sign}(\Delta d_{m(o)} \text{sign}(u_o(\hat{z})) \text{sign}(u_o(j)))$.*

Proof. If $o \notin K_\omega$, a carrier of K_ω meeting j would have $j \in S_k$ (as $j \notin E$) and would be the owner; so every carrier at j lies outside K_ω and has $|\gamma_k| = \lambda_k |\Delta d_m| |w_m(k)| \leq C_1 \lambda_k$. If $o \in K_\omega$, then $j \in S_o$ and the later carriers at j are again outside K_ω .

Lemma 5.37 gives $D(j) \neq 0$ with the sign of $\gamma_o u_o(j)$, and Lemma 2.22(b) (D is z -admissible off F) gives the rest; for $o \notin K_\omega$, $\gamma_o = -\lambda_o \Delta d_m w_m(k(o))$ and $\text{sign } w_m(k) = \text{sign } u_k(\hat{z})$. $\square$

Thus exact data with $\Delta d < 0$ in a block force $z_j = \text{sign } u_o(\hat{z}) \text{sign } u_o(j)$ (owner alignment) at every coordinate outside $F \cup E$ owned by a carrier of that block that does not carry ω and is not neutral or negligible. The coordinates owned by K_ω -carriers, neutral or negligible carriers are the *dead zones*.

Proposition 5.39 (Exact all-negative data are non-generic). *In the setting of Proposition 5.33 with $T = T_{\text{final}}$, the set of $a \in \mathcal{A}_\sigma$ for which some $g \in \mathcal{C}(f_a)$ carries two-piece data with $\Delta d_m < 0$ for every $m \in I$ is meagre in $\mathcal{A}_\sigma$.*

Proof. For a carrier l of block m let O_l be the set of $b \in \Omega_\sigma$ with $2\iota_l \bar{\vartheta}_m \Phi_l / (m M_m) < |u_l(\hat{z})| < \bar{\vartheta}_m \Phi_l / (2m)$ such that some $j \in S_l \setminus F$ has $z_j \neq \text{sign } u_l(\hat{z})$; it is open. $V_L^m := \bigcup_{l \geq L, m(l)=m} O_l$ is dense. Given O , take y and carriers l_i as in the proof of Proposition 5.33 (with $S_{l_i} \cap F = \emptyset$). Let ϵ_i be the value of z on S_{l_i} if z is constant there and $\epsilon_i := 1$ otherwise, and let $\kappa_i(b)$ be the midpoint of the interval in the definition of O_{l_i} ; it is continuous, positive and $O(\Phi_{l_i})$, and the interval is nonempty for large i ($\iota_{l_i} \rightarrow 0$, $M_m > \frac{1}{3}$). Then $h := u_{l_i}(\hat{z}) + \epsilon_i \kappa_i$ has opposite signs at b_+ and b_- , hence a zero $b \in O$, where $|u_{l_i}(\hat{z})| = \kappa_i(b)$ and $\text{sign } u_{l_i}(\hat{z}) = -\epsilon_i$; in both cases some $j \in S_{l_i}$ has $z_j \neq -\epsilon_i$, so $b \in O_{l_i}$. *Conclusion.* $\mathcal{G} := \bigcap_m \bigcap_L V_L^m$ is a dense G_δ . Let $b(a) \in \mathcal{G}$ and suppose data with all $\Delta d_m < 0$ exist; let $|\Delta d_{m_*}| = C_1$. Pick l of block m_* with $b \in O_l$, $l \notin K_\omega$, $S_l \cap (F \cup E) = \emptyset$ (all but finitely many l qualify). Then $\nu_l < \bar{\vartheta}/2$, so l is a strict non-peak with $|w(l)| = M\nu_l/\bar{\vartheta} > 2\iota_l$, and $|\gamma_l| = \lambda_l C_1 |w(l)| > 2\iota_l C_1 \lambda_l$; l is neither neutral nor negligible. For the coordinate $j \in S_l \setminus F$ with $z_j \neq \text{sign } u_l(\hat{z})$, $o_N(j) = l$ and $j \notin F \cup E$, and Proposition 5.38 gives $z_j = \text{sign } u_l(\hat{z})$ (as $\Delta d_{m_*} < 0$ and $u_l(j) > 0$), a contradiction. $\square$

Proposition 5.40 (Block-tame rows are dense). *Let $f \in S_{p_N^*}$ have finite base support. For every $\varepsilon > 0$ there is $f_* \in S_{p_N^*}$ with $a_* = a$, satisfying (BT), with $p^*(f_* - f) < \varepsilon$.*

Proof. Fine re-alignment. For $L \geq 1$ let $z^{(L)} := z$ on F and on the supports of the carriers $l \leq L$, and let $z^{(L)}$ be given on the remaining coordinates by the owner recursion over $L_N \cap (L, \infty)$ (owner signs). Put $f^{(L)} := f[a, z^{(L)}]$. With $\delta := \hat{z}^{(L)} - \hat{z}$, $u_k(\delta) = 0$ for $k \leq L$ and $|u_k(\delta)| \leq 2$ otherwise, so $c(\delta) \leq C\varepsilon_L \log(e/\varepsilon_L)$, $\varepsilon_L := \sum_{l > L} \lambda_l$, and $p^*(f^{(L)} - f) \leq C_f \varepsilon_L \log(e/\varepsilon_L)$ by Lemma 2.12. Carriers $\leq L$ keep their values; every carrier $l > L$ with $S_l \cap F = \emptyset$ is an exactly swallowed non-degenerate swallowing-type peak, with margin $\geq q_0 \delta_l^\circ / 2$ beyond the first peak of its block (Steps 2 and 4 of Theorem 5.29 use only the owner rule at l , $|u_k(\hat{z})| \leq 1 + \|U\|$ and (SF*)).

Levers. Let K_0 be the finite set of carriers $l \leq L$ or with $S_l \cap F \neq \emptyset$; only they can be degenerate peaks or strict non-peaks. Choose carriers $l_a(1) < \dots < l_a(N)$, $l_a(m)$ of block m , beyond K_0 , put $s_m := \min S_{l_a(m)}$, and replace $z_{s_m} = \epsilon_{l_a(m)}$ by $t_m \epsilon_{l_a(m)}$, $t_m \in [\frac{3}{4}, 1]$; afterwards re-run the recursion beyond $l_a(m)$ *hysteretically* (reference signs kept unless $\epsilon_l A_l \leq -(B_l + \delta_l H_l)/2$, then $\epsilon_l := \text{sign } A_l$), so that every re-run carrier has $n_l |u_l(\hat{z})| \geq (B_l + \delta_l H_l)/2$ and is a peak with margin $\geq q_0 \delta_l^\circ / 4$. The lever changes $|u_{l_a(1)}(\hat{z})|$ at the exact rate $\delta_{l_a(1)} 2^{-s_1} / n_{l_a(1)}$ in $1 - t_1$ and keeps it $\geq \frac{3}{4} \delta_{l_a(1)}^\circ$; the values of K_0 are unchanged. Write $\hat{\zeta}_1(t) = \zeta^{\text{fix}} + (\text{own lever term}) +$

$\zeta^{\text{later}}(t)$ with $\|\zeta^{\text{later}}(t) - \zeta^{\text{later}}(1)\|_1 \leq e_1 := 2(1 + \|U\|) \sum_{l > l_a(1)} \lambda_l$, and $\varphi(t) := \bar{\vartheta}(\zeta^{\text{fix}} + \text{own term} + \zeta^{\text{later}}(1))$. By Lemma 2.7(b) (applied between two values of t , the lever carrier being a peak) $|\varphi(t) - \varphi(t')| \geq c_\Theta \lambda_{l_a(1)} \delta_{l_a(1)} 2^{-s_1} |t - t'|/n_{l_a(1)}$, and by Lemma 2.7(a) $|\bar{\vartheta}_1(t) - \varphi(t)| \leq L_\Theta e_1$, with $c_\Theta, L_\Theta > 0$ uniform over the family. Hence for each $k \in K_0$ the set $\{t : \bar{\vartheta}_1(t) = v_k\}$ lies in an interval of length $\leq 2L_\Theta e_1 n_{l_a(1)} / (c_\Theta \lambda_{l_a(1)} \delta_{l_a(1)} 2^{-s_1}) \leq (L_\Theta / c_\Theta) 2^{-8-k(l_a(1))}$ by (SF _{ℓ}). For $l_a(1)$ large these $|K_0|$ intervals cover less than a quarter of $[\frac{3}{4}, 1]$; choose t_1 outside them, so $\text{dgap}_1 := \min_{k \in K_0} |\bar{\vartheta}_1 - v_k| > 0$. Choose $l_a(2)$ with $2L_\Theta(1 + \|U\|) \sum_{l \geq l_a(2)} \lambda_l < \text{dgap}_1/2$, then t_2 in the same way, and so on. The resulting row f_* has no degenerate peak, $Q_m \subseteq K_0$, and (MS) holds as in Step 7 of Theorem 5.29: it satisfies (BT). Its cost is $\leq C_f \varepsilon \log(e/\varepsilon)$ with $\varepsilon := \sum_{l > \min(L, l_a(1))} \lambda_l$, which is small for L large. $\square$

Lemma 5.41 (Transplant of exact data). *Let $f, f^\# \in S_{p^*}$, where F is finite and $F^\# = F \cup B$ with a finite set B of contacts of f at which $\text{sign } a_j^\# = z_j$ (banks), and $z^\# = z$ on B . Let $(b^\pm, \omega^\pm)$ be two-piece data at f for some g with $g(\xi) = 0$, $D := b^+ - b^-$, and assume:*

- (T1) *every $k \in \text{supp } \omega_m^\pm$ is a strict non-peak of $f^\#$;*
- (T2) *$D^\# := \sum_m R_m^*(\omega_m^- - \omega_m^+) - \sum_m \Delta d_m^\# \psi_m^\#$, $\Delta d_m^\# := d_m^\#(\omega_m^-) - d_m^\#(\omega_m^+)$, vanishes at the free coordinates of $f^\#$ and satisfies $z_j^\# D^\#(j) \geq 0$ at the contacts of $f^\#$ and at $j \in B$.*

Then there are two-piece data $(b_\#^\pm, \omega^\pm)$ at $f^\#$, with $z_j b_\#^+(j) \geq 0 \geq z_j b_\#^-(j)$ for $j \in B$, for the functional $G^\# := b_\#^+ + \sum_m R_m^(\omega_m^+ - d_m^\#(\omega_m^+) w_m^\#)$, such that*

$$\|b_\#^\pm - b^\pm\|_1 \leq 2(\|D^\# - D\|_1 + \|D 1_{\text{Fl}}\|_1 + \|D^\# 1_{\text{Fl}}\|_1) + C_f |c|, \quad \text{Fl} := \{j : z_j^\# z_j = -1\},$$

where the rebalancing scalar c satisfies $|c| \leq C_f(\|D^\# - D\|_1 + \|D 1_{\text{Fl}}\|_1 + \|D^\# 1_{\text{Fl}}\|_1 + p^(f^\# - f))$. Consequently, along rows $f^\# \rightarrow f$ for which the bracket tends to 0, $\Delta d_m^\# \rightarrow \Delta d_m$, $\|G^\# - g\|_1 \rightarrow 0$ and the Γ_w of the transplanted pairs converge to those at f .*

Proof. Off F , at each j with $D^\#(j) \neq 0$ (a contact of $f^\#$ or a bank) write $D^\#(j) = z_j^\# s_j$, $s_j > 0$, and put $b_\#^+(j) := z_j^\# x_j$, $b_\#^-(j) := -z_j^\# y_j$ with $x_j, y_j \geq 0$, $x_j + y_j = s_j$, choosing (x_j, y_j) nearest to $(z_j^\# b^+(j), -z_j^\# b^-(j))$; both entries are ≥ 0 when $z_j^\# = z_j$, by side admissibility at f , and then the distance is $\leq |D^\#(j) - D(j)|$; at flips it is $\leq |D(j)| + |D^\#(j)|$. Off $F \cup \text{supp } D^\#$ put $b_\#^\pm := 0$. On F keep b^+ and put $b_\#^- := b^+ - D^\#$. Finally add $c a 1_F$ to both $b_\#^\pm$, with c chosen such that $b_\#^+(\xi^\#) = 0$; this is possible because $a 1_F(\xi^\#) = q_0^\# a(\hat{z}^\#) \rightarrow q_0 a(\hat{z}) = q_0 \neq 0$, and it keeps $b_\#^+ - b_\#^- = D^\#$ and the contact-like split on B . Then $(b_\#^-, \omega^-)$ represents the same functional as $(b_\#^+, \omega^+)$, by the definition of $D^\#$; side admissibility holds off $F^\#$ by construction and by (T2); the block parts are supported in $Q^\#$ by (T1). The convergence statements follow from Lemma 2.12 ($R_m^* w_m^\# \rightarrow R_m^* w_m$ in ℓ_1 , and $d_m^\# \rightarrow d_m$ on the fixed finite supports), and $b_\#^+(\xi^\#) - b^+(\xi) \rightarrow 0$ with $b^+(\xi) = g(\xi) = 0$. $\square$

Hypotheses for the recovery of exact negative data. Let $g \in \mathcal{C}(f)$ carry two-piece data with $\kappa_w \leq 1$ and $\Delta d_m < 0$ for every block m , and assume

- (H1) no carrier is neutral, and only finitely many carriers (the set Neg) are negligible;
 (H2) $D(j) \neq 0$ for every $j \in E' := E \cup \{j \notin F : o_N(j) \in K_\omega \cup \text{Neg}, |\gamma_{o_N(j)} u_{o_N(j)}(j)| \leq 32 \sum_{k > o_N(j)} |\gamma_k u_k(j)|\}$.

E' is finite: for $o \in K_\omega \cup \text{Neg}$ the later terms on $S_o \setminus E$ are $\leq C2^{-2j}$ by (W4), while $|\gamma_o|v_o(j) \geq c2^{-j}$, and o has finitely many target coordinates.

Lemma 5.42 (Scope). *Under (H1), (H2), $|z_j| = 1$ for every $j \notin F$.*

Proof. Let $j \notin F$. If $j \in E'$, $D(j) \neq 0$ by (H2). If $o_N(j) \notin K_\omega \cup \text{Neg}$ and $j \notin E$, $D(j) \neq 0$ by (H1) and Proposition 5.38. If $o_N(j) \in K_\omega \cup \text{Neg}$ and $j \notin E'$, the owner term exceeds 32 times the later terms, so $D(j) \neq 0$. By Lemma 2.22(b), D vanishes at free coordinates. $\square$

So the recovery below concerns rows of maximal contact (typically sign mixed), exactly where Remark 5.20(b) does not exclude (C*); by Proposition 5.39, however, rows carrying such data form a meagre set in every fixed- z fibre.

Lemma 5.43 (Fine re-alignment carries exact data). *Assume (H1), (H2). There is L_0 such that for every $L \geq L_0$ the fine re-alignment $f^{(L)}$ of the proof of Proposition 5.40 satisfies (T1) and (T2) (with $B = \emptyset$), and the transplanted data of Lemma 5.41 at $f^{(L)}$ represent functionals $G_L \rightarrow g$ with $\Delta d^{(L)} \rightarrow \Delta d$ and Γ_w converging, as $L \rightarrow \infty$.*

Proof. Choose L_0 beyond K_ω , Neg, E' and all carriers l with $S_l \cap F \neq \emptyset$, and so large that $\iota_{L_0} \leq \min_m |\Delta d_m| M_m / (4C_1)$. Coarse values are unchanged and $\|\hat{\zeta}^{(L)} - \hat{\zeta}\|_1 \leq 2(1 + \|U\|)\varepsilon_L$; $w_m(k)$ is Lipschitz in $\hat{\zeta}_m$ uniformly in $k \leq L$ (it is $\text{sign } \zeta(k)M$ on peaks and $M\varrho_k \text{sign } \zeta(k)$ off peaks, continuous across $\varrho = 1$, with $\bar{\vartheta}$, M , C , $|\zeta|$ Lipschitz by Lemmas 2.6, 2.7); so $\sup_{k \leq L} |w^{(L)}(k) - w(k)| \leq C_f \varepsilon_L \leq \iota_L / 4$ for large L by (SF _{ι}), and $\Delta d^{(L)} \rightarrow \Delta d$. (T1): the ω -carriers are coarse strict non-peaks with fixed values, and the threshold converges. (T2) at $j \notin F$ (all such j are contacts of f by Lemma 5.42, and of $f^{(L)}$ since the owner recursion assigns ± 1): (i) $o_N(j) > L$: $z_j^{(L)} = \epsilon_o^{(L)} \text{sign } u_o(j)$ and $w^{(L)}(o) = \epsilon_o^{(L)} M^{(L)}$, so $\text{sign } \gamma_o^{(L)} = \epsilon_o^{(L)}$ and $|\gamma_o^{(L)}| \geq \iota_o C_1^{(L)} \lambda_o$ by the choice of L_0 ; fine coordinates meet no K_ω -carrier; Lemma 5.37 gives $z_j^{(L)} D^{(L)}(j) > 0$. (ii) $o_N(j) \leq L$, $o \notin K_\omega \cup \text{Neg}$, $j \notin E$: $z_j = \text{sign}(\gamma_o u_o(j))$ by Proposition 5.38 and $|\gamma_o| \geq 2\iota_o C_1 \lambda_o$; the perturbation of γ_o is $\leq |\Delta d^{(L)}| \lambda_o \iota_L / 4 + |\Delta d^{(L)} - \Delta d| \lambda_o$, so $\gamma_o^{(L)}$ keeps its sign and $|\gamma_o^{(L)}| \geq \iota_o C_1^{(L)} \lambda_o$; Lemma 5.37 applies (it needs only $|\gamma_k^{(L)}| \leq C_1^{(L)} \lambda_k$ for later carriers, true also for the re-aligned fine ones). (iii) $o_N(j) \in K_\omega \cup \text{Neg}$, $j \notin E'$: $\gamma_o^{(L)} \rightarrow \gamma_o \neq 0$; on S_o beyond a finite set the bound $C2^{-2j}$ of the later terms against $|\gamma_o|v_o(j) \geq c2^{-j}$ gives dominance uniformly in L ; on the finitely many remaining coordinates the fine contributions are $\leq \frac{8}{3} C_1 \varepsilon_L \rightarrow 0$ and the coarse ones converge, so the factor-32 dominance at f persists. (iv) $j \in E'$: $z_j D(j) > 0$ by (H2) and $D^{(L)}(j) \rightarrow D(j)$. Finally $\|D^{(L)} - D\|_1 \leq \frac{5}{3} \sum_k \lambda_k |\Delta d^{(L)} w^{(L)}(k) - \Delta d w(k)| \rightarrow 0$ by dominated convergence, and Fl consists of fine coordinates, where D and $D^{(L)}$ are sums over fine carriers, so $\|D1_{\text{Fl}}\|_1 + \|D^{(L)}1_{\text{Fl}}\|_1 \leq \frac{10}{3} C_1 \varepsilon_L \rightarrow 0$. Lemma 5.41 applies. $\square$

Lemma 5.44 (Bank levers). *Assume (H1), (H2), let $L \geq L_0$ and $f' := f^{(L)}$ with the transplanted data. For each block m choose a carrier $l_b(m) > L$ of block m ,*

$l_b(1) < \dots < l_b(N)$, put $s_m := \min S_{l_b(m)}$ (a contact with $z = \epsilon_{l_b(m)}$), and for $\mu \in [0, \infty)^N$ let $f^\#(\mu)$ have $a^\#(\mu) := (a + \sum_m \mu_m \epsilon_{l_b(m)} e_{s_m}^*)/q^*(\cdot)$, z unchanged up to a hysteretic re-run of the owner recursion beyond $l_b(1)$. For suitable small μ , chosen block after block, $f^\#(\mu)$ has base support $F \cup \{s_1, \dots, s_N\}$, satisfies (BT), carries transplanted two-piece data with $\Delta d_m^\# < 0$ for every m and with the contact-like split at the banks, and $p^*(f^\#(\mu) - f') + \|G^\# - G'\|_1 \rightarrow 0$ as $\mu \rightarrow 0$.

Proof. By Lemma 3.2(b), $\nu(\mu)^2 = \nu^2 + \sum_m (\mu_{s_m}^* \mu_m)^2$ before normalization, so $\hat{z}$ moves by $O(|\mu|^2)$ on F , and $\hat{z}_{s_m} = \epsilon_{l_b(m)}(1 + (\mu_{s_m}^*)^2 \mu_m / \nu) + O(|\mu|^2)$. Hence $|u_{l_b(m)}(\hat{z})|$ increases at the first-order rate $v_{l_b(m)}(s_m)(\mu_{s_m}^*)^2 / \nu$; the other carriers containing s_m are later carriers, of total weight $\leq 2^{-10} \lambda_{l_b(m)} v_{l_b(m)}(s_m) 2^{-(m+k(l_b(m)))}$ by (SF_L); carriers depending on F move by $O(|\mu|^2)$; and forced flips of the hysteretic re-run occur only beyond a level tending to infinity as $\mu \rightarrow 0$, with total weight $o(|\mu|)$. By Lemma 2.7(b), $\bar{v}_m(\mu) - \bar{v}_m \geq \kappa_m \mu_m / 2$ for small μ_m , with $\kappa_m > 0$, while each coarse v_k ($k \leq L$, the only candidates for degeneracy) moves by $O(|\mu|^2)$. So a small $\mu_1 > 0$, avoiding finitely many values, makes block 1 free of degenerate peaks with a gap $\text{dgap}_1 > 0$; then $l_b(2)$, μ_2 are chosen so that block 1 moves by less than $\text{dgap}_1/2$, and so on. (BT): $F^\#$ is finite, $Q_m^\#$ lies among the coarse carriers, there is no degenerate peak, and (MS) holds by the margins of the re-run peaks. (T1) holds since raising the threshold keeps strict non-peaks; (T2) as in Lemma 5.43, with $|\gamma| \leq C_1 \lambda$ for the re-run carriers; at s_m the owner $l_b(m)$ dominates, so $z_{s_m} D^\#(s_m) > 0$ and the split is contact-like there. The cost bound follows from Lemma 2.12 and Lemma 5.41. $\square$

Theorem 5.45 (Exact negative data without dead zones are recovered). *Let $N \geq 1$, let $f \in S_{p_N}^*$ have finite base support, and let $g \in \mathcal{C}(f)$ carry two-piece data with $\kappa_w \leq 1$ and $\Delta d_m < 0$ for every block m , satisfying (H1) and (H2). Then $(f, g) \in \text{clNA}((c_0, p_N), \ell_2^2)$.*

Proof. Let $f_i := f^\#(\mu_i)$ be the rows of Lemma 5.44 built on $f^{(L_i)}$, with $L_i \rightarrow \infty$ and then $\mu_i \rightarrow 0$ so fast that, by Lemmas 5.43 and 5.44, $f_i \rightarrow f$ and the transplanted data at f_i represent $G_i \rightarrow g$ with $\Delta d_m^{(i)} < 0$ and $\kappa_w^{(i)} \rightarrow \kappa_w$. Every f_i satisfies (BT), hence (SC) for every set of blocks. Fix $\rho < 1$. The data have fixed block supports (strict non-peaks whose gaps at f_i are bounded below), fixed $\|b\|_1$ -bounds and contact-like splits at the banks; Lemma 2.18, applied with $t := t_1$ and constants A_0, A_2, γ_B fixed by these data, gives $r_1 > 0$ and i_0 with $p^*(f_i + rG_i) \leq 1 + \frac{r^2}{2}(\kappa_w^{(i)} + \varepsilon)$ for $|r| \leq r_1$, $i \geq i_0$. With ε so small that $\rho^2(1 + \varepsilon) < 1$, and after decreasing r_1 so that $\rho^2(1 + \varepsilon)r^2/2 \leq r^2/2 - r^4/8$ for $|r| \leq r_1/\rho$, we get $p^*(f_i + r\rho G_i) \leq s(r)$ for $|r| \leq r_1/\rho$; for larger $|r|$, $p^*(f_i + r\rho G_i) \leq s(\rho r) + p^*(f_i - f) + |r|\rho p^*(G_i - g) \leq s(r)$ for i large, since $s(r) - s(\rho r) \geq (1 - \rho^2) \min(r^2, |r|)/3$ ([I, Lemma 3.6(a)]). So $\rho G_i \in \mathcal{C}(f_i)$, with two-piece data of $\kappa_w \leq 1$ and $\Delta d < 0$, and [I, Theorem 7.18] at f_i gives $(f_i, \rho G_i) \in \text{clNA}$. Let $i \rightarrow \infty$, then $\rho \rightarrow 1$. $\square$

At the self-aligned rows of Theorem 5.29 with finitely many exceptional carriers tuned through F , the hypotheses (H1), (H2) hold for data supported on the exceptional strict non-peaks with nonzero coefficients there (every non-exceptional carrier is a robust peak with $|w| = M$, and every coordinate off F is dominated by its owner); these rows are, however, block-tame and recovered directly (Corollary 5.30). In the other direction, by Proposition 5.39 the exact-data route through Theorem 5.45 acts only on a meagre set of every fixed- z fibre: the mates of a generic

row with finite base support must be handled by window data, and for these the residual is (C^*) .

5.6. Absorber pairs and Master Theorem IV'. This subsection constructs exact *shifted* two-piece data at companions of a row with finite base support, at the scales of a clean sub-window, under a hypothesis (KN) on the availability of a second lever in each shifted block. We use the data convention, and for a block m of a first row f' we write $\Delta'_m := \Delta d_m M'_m$. By Lemma 2.22, exact data with switching difference $\Delta\omega$ exist iff the single vector $\mathcal{V}(\Delta\omega) = \sum_l \gamma_l u_l$ is z' -admissible off F' , where $\gamma_l = \lambda_l(\Delta\omega(k(l)) - \Delta d_{m(l)} w'(k(l)))$; by Lemma 2.9, if $\Delta\omega_m$ is supported in a finite set $\Omega_m \subseteq Q'_m$, then

$$(11) \quad \Delta'_m \varkappa'_m = \sum_{l \in \Omega_m} u_l(\hat{z}') \gamma_l, \quad \varkappa'_m := \varkappa(\hat{\zeta}'_m; \Omega_m),$$

so the only f' -dependent quantities of block m entering the coarse exact system are the values $u_l(\hat{z}')$, $l \in \Omega_m$, and the single scalar $\varkappa'_m$.

Why one scalar requires two levers.

Proposition 5.46 (Status rigidity in ratio space). *Let $\zeta = \hat{\zeta}'_m$ be the block vector of a first row, P its peak set, $\Omega \subseteq Q$ finite, $\varkappa := \varkappa(\zeta; \Omega)$, and put $r_l := u_l(\hat{z}')/\varkappa$ for $l \in \Omega$.*

- (i) *After division by $\varkappa_m$, (11) reads $\Delta'_m = \sum_{l \in \Omega_m} r_l \gamma_l$: the coarse exact system depends on block m only through the ratios r_l . By Corollary 2.11 the relative positions ϱ_l , $l \in \Omega$, are functions of $(|r_l|)_{l \in \Omega}$, Φ_P^2 and s_Q ; more precisely (a, k) of Corollary 2.11 depend smoothly on (R^2, Φ_P^2, s_Q) near any given point.*
- (ii) *An outward push of a peak changes $(\bar{\vartheta}, |\zeta|, \varkappa)$ at the rates $(C/\Phi_P^2, M, 1 - X)$ (Lemma 2.10(b)(i)); with the values on Ω fixed it multiplies all ratios of the block by one factor. If the ratios are restored afterwards, every ϱ_l , $l \in \Omega$, returns to its old value up to the change of Φ_P^2 and s_Q .*
- (iii) *A push of an Ω -carrier d_0 changes $\varkappa$ only at the rate $\varrho_{d_0} MX/C$ and $\bar{\vartheta}$ at the rate $-\varrho_{d_0} M/\Phi_P^2$ (Lemma 2.10(b)(ii),(iii)). So an inward push of d_0 , followed by a restoration of $\varkappa$ and of the ratios r_l , $l \neq d_0$, raises $\bar{\vartheta}$ at first order and lowers every ϱ_l , $l \in \Omega \setminus \{d_0\}$.*

Proof. (i) With $k = a + \Phi_P^2 + s_Q$, the identity $a^2 = \Phi_P^2 + k^2 R^2 + s_Q$ is an equation $G(a; R^2, \Phi_P^2, s_Q) = 0$ with $\partial_a G = 2a - 2kR^2 = 2(a - kR^2) > 0$, since $a > kR \geq kR^2$ ($R < 1$); the implicit function theorem gives smooth dependence, and $\varrho_l = r_l k / \Phi(l)$ with $r_l = m|r_l|$. (ii) The ratios are $u_l(\hat{z}')/\varkappa$; restoring them restores R^2 , hence (a, k) and ϱ_l by (i), up to the change of (Φ_P^2, s_Q) . (iii) The rates are those of Lemma 2.10; at fixed $\varkappa$ the relation $\varrho_l = r_l \varkappa / (\bar{\vartheta} \Phi(l))$ shows that raising $\bar{\vartheta}$ at fixed ratios lowers ϱ_l . $\square$

Remark 5.47 (The gap of a one-lever exactification). Suppose one tries to realize a Łojasiewicz point of the tiny minors of the coarse shifted system in the variables $(u_l)_{l \in \Omega}$ and $(\varkappa_m)_m$ by the assembly of Section 4, the tuning Lemma 4.15, and one peak push per block that sets $\varkappa_m$. The donor raise (C3) of the assembly raises the donor peak by an amount in $[\Lambda, 3\Lambda]$, $\Lambda = T_{10}(w)^3$, so it moves $\varkappa_m$ by at least $\frac{3}{4}\Lambda$, which is far larger than $\text{Des } b(w)$. If $\varkappa_m$ is pushed back to a Łojasiewicz value near its value before the raise, then by Proposition 5.46(ii) the threshold returns as well

and the raise is undone, so near-threshold switching carriers lose the strict-non-peak status that the raise provided. If instead the Łojasiewicz point is computed after the raise, the tiny minors are only $\leq b + \text{Lip}(l)\Lambda$ there, and Lemma 5.3 gives a distance $C_L(\text{Lip}\Lambda)^{2/N_L}$, which is not $\leq T_{\text{lo}}(w)^4$ when $N_L > 2$. Thus exactness and threshold protection cannot be achieved independently by peak pushes; by Proposition 5.46(iii) the missing ingredient is a lever that moves the threshold without moving $\varkappa$ or the active ratios. No claim is made that the conclusion of a one-lever exactification fails.

The design $D^{U1'}$.

Definition 5.48 (Absorber targets and the design $D^{U1'}$). *Absorber targets.* For a coordinate s , a sign $\beta \in \{\pm 1\}$ and a stage p , an *absorber target* is $y := x/q^*(x)$ with

$$x := \beta e_s^* + e_{p_0}^* + 4 \sum_{r=1}^R 2^{-r} e_{p_r}^*,$$

where $p_0 < p_1 < \dots < p_R$ are fresh odd integers (in no signature set and in no earlier target) and R is the least integer with $2^{3-R} \leq (\mu_{p_0}^*)^2 (c_p^{\text{low}})^2 / 4$; here c_p^{low} is the minimum of $c_{p-1}/4$, $b(p-1, M(p-1))^2$, $(\delta_p^{\text{max}} H_p / 2)^2$ and the bounds (W4'') at s , (W5) with δ_p replaced by $\delta_p^{\text{max}}/2$, and (W7) below, which do not depend on R or on p_0 . Moreover p_0 is chosen so large that

$$(\mu_{p_0}^*)^{2^{2p_0}-1} < 2^{-R}$$

(possible: $2^{-R} \geq (\mu_{p_0}^*)^2 (c_p^{\text{low}})^2 / 64$ by the minimality of R , and c_p^{low} does not depend on p_0). Then $|y(s)| \in [\frac{1}{8}, 1]$; p_0 is the *tuning coordinate* and $p_1, \dots, p_R$ the *binary coordinates*. Two consecutive carriers of block 1 with absorber targets for $(s, +1)$ and $(s, -1)$ form a *biased pair* for s .

Stages. $D^{U1'}$ is obtained from T_{final} by the following changes of the schedule and of the weights; everything else is kept, the ladder bijection being determined by the schedule. The stages are grouped in rounds $r = 1, 2, \dots$; round r serves a block m_r from a fixed sequence in which every block occurs infinitely often, and a target y for m_r , namely the scheduled target of T_{final} if it satisfies allowedness (a) and (c) with margin $(\text{supp } y \cap S_{l'} \subseteq [1, p-2|\text{supp } y| - 1])$, p the stage at which y will be placed), and the fresh-dominated target otherwise. Round r consists of

PRE: for every $s \in \text{supp } y \cap \bigcup_{l'} S_{l'}$ a biased pair for s (block 1, consecutive stages);

MAIN: the stage p at which y is placed (block m_r), called a *main stage*;

CLUSTER: a biased pair for every $s \in E^{\text{ch}}(p) := T(p) \cup \bigcup_{l'' \leq p} (S_{l''} \cap [1, p])$ (block 1).

Weights. At PRE and MAIN stages, c_l is the minimum of (W1)–(W5), (W7) of Definition 3.5, with (W6) omitted at absorber stages, and with (W4) strengthened to

$$(W4'') \quad c_l \leq \iota_{l'} 2^{-2s} c_{l'} \delta_{l'} T_{\text{lo}}(l-1, M(l-1))^3 / 2 \quad (l' < l, s \in \text{supp } y_l \cap S_{l'});$$

at the i -th stage of the cluster of a main stage p , $c_{p+i} := c_{p+1}^{\text{low}} 4^{1-i}$, where in (W4'') the factor $T_{\text{lo}}(\cdot)^3$ is read with the main stage p . Sub-window data (step (6) of Definition 3.5) are formed at every stage, but windows are used only at main stages. At a main stage L the design factor $\text{Des}(L)$ additionally contains $4^{R_{\text{cl}}(L)} (R_{\text{cl}}(L) := 2|E^{\text{ch}}(L)|, 2^{k(L+R_{\text{cl}}(L))})$, the reciprocal of the least nonzero modulus of a target entry; the rate scheme contains the minors of Definition 5.52. Put

$R_\Gamma(L) := 2|T(L)| + 11L$ (a bound for the number of rows, equations counted twice, of the systems defining the cones $\Gamma^\#(a)$ of Definition 5.52 at main stage L , after the aggregation in (X1)), $P(L) := 2^{R_\Gamma(L)}$ (a bound for the number of their extreme rays), and, for a sub-window w of a main stage L ,

$$\bar{D}(w) := (2L + 1) 3^{2L} \left(\frac{\text{Des}(L)}{u(w)} \right)^{C_\Gamma(P(L)+2L+2)},$$

where C_Γ is a numerical constant, fixed once and for all, so large that every bound stated in the form “ $\leq C_f \bar{D}(w)$ ” in the second half of this subsection holds. Then $\bar{D}(w)$ is a design number of the sub-window (it is formed from L , $T(L)$, $\text{Des}(L)$ and $u(w)$), and at main stages $Q(w)$ is replaced by $\max\{Q(w), \bar{D}(w)^3\}$ (the value of Definition 3.5(6) and $\bar{D}(w)^3$; both are known when $Q(w)$ is formed). The window arguments of Sections 3–5 use only the lower bound $Q(w) \geq (4\text{Des}(L)/u(w))^{\omega(L)+20}$ and the definitions of $n(w)$, $T_{\text{hi}}(w)$, $T_{\text{lo}}(w)$, $\eta(w)$ and $b(w)$ in terms of $Q(w)$, so they are not affected.

Lemma 5.49 (Properties of $D^{U1'}$). (a) $D^{U1'}$ is admissible (conditions (T-a)–(T-d) of [I, Definition 1.3]) and does not depend on N . (b) At every main stage L , $\sum_{l' > L} \lambda_{l'} \leq b(L, M(L))^2 \leq T_{\text{lo}}(w)^8$ for every sub-window w of L . (c) Every result of Sections 2–4 and of Subsections 5.1–5.3 that is used below holds for $D^{U1'}$ with “level” read as “main stage”.

Proof. (a) (T-a) is immediate. (T-b), (T-c) follow as for T_{final} , since that argument uses only allowedness (a) and (W4): absorber targets satisfy (a) (their coordinates are fresh or the coordinate s , which lies in Z_0 or in a signature set of an earlier carrier), cluster weights satisfy (W4) by the choice of c^{low} , and (W4'') implies (W4). (T-d): every target of the dense family is placed infinitely often in every block, because every block is served in infinitely many rounds and every request of a round is finite. The definition is well founded since R is determined from c_p^{low} , which does not depend on R . (W3) at cluster stages: $c_{p+i} \leq b(p, M(p))^2 \leq 2^{-8n(p, M(p))}$, while $(\delta_{p+i} H_{p+i})^2 \geq 2^{-2(p+i)-6 \cdot 2^{p+i}-4}$ and $p+i \leq 3p + 2s_{\text{max}}(p)$, $n(p, M(p)) \geq 2^{6s_{\text{max}}(p)}$. Absorbers live in block 1, which is present for every N . (b) The cluster weights sum to $\leq 2c_{L+1}^{\text{low}}$ and the later weights to $\leq \frac{4}{3}c_{L+R_{\text{cl}}}+1$, so $\sum_{l' > L} c_{l'} \leq 2c_{L+1} \leq 2b(L, M(L))^2$, and $\lambda_{l'} \leq c_{l'}/4$. (c) The window arguments use one sub-window’s dyadic scales, the design factors of its level, the box bound (which needs (b)) and $T_{\text{hi}}(w') \leq T_{\text{lo}}(w)$ for later w' (Lemma 3.22); all hold at main stages. The non-window arguments use (T-a)–(T-d), allowedness, bounded gaps and the diagonal base. The features (W5), (W6), (W7) are lost at absorber stages; they are not used by the results invoked in this subsection. $\square$

Lemma 5.50 (First touchers). Let w be a sub-window of a main stage L , let $s_{\text{far}}(w)$ be the least s with $2^{-s} \leq c_{L+1}^2$, and

$$E_c(w) := T(L) \cup \bigcup_{l'' \leq L} (S_{l''} \cap [1, s_{\text{far}}(w)]).$$

For every $s \in E_c(w)$, either no carrier at a stage $> L$ meets s , or the first two carriers at stages $> L$ whose vectors meet s form a biased pair for s .

Proof. If $s \in E^{\text{ch}}(L)$, the cluster of L follows L immediately. Otherwise $s \in S_{l''}$ with $l'' \leq L < s$. After L , the carriers meeting s are absorber pairs for s (pre-pairs

and cluster pairs) and main targets containing s , each of which is preceded by its pre-pair for s ; fresh-dominated and absorber targets meet only fresh coordinates and their own coordinate s' , and signature sets are disjoint. Pairs are placed at consecutive stages. $\square$

Lemma 5.51 (Zero-value tuning of an absorber). *Let a be an absorber carrier with target $y_a = x/q^*(x)$ for (s, β, p) , let $s_{\text{far}}(a)$ be the least s' with $2^{-s'} \leq \Phi_a^2$, and let $f^{(1)}$ be a first row with finite base support containing none of $p_0, \dots, p_R$, none of $S_a \cap [1, s_{\text{far}}(a)]$, and not s . Let $0 < \theta_a \leq \lambda_a$. Put $z := +1$ at p_0 and on S_a , make p_r ($r \geq 1$) and the coordinates of $S_a \cap [1, s_{\text{far}}(a)]$ support coordinates with masses $\theta_a \sigma_r$ (binary signs $\sigma_r \in \{\pm 1\}$, $z_{p_r} := \sigma_r$), resp. θ_a , and p_0 a support coordinate with mass $m_0 \in [\theta_a, \theta_a + c_p^2]$.*

- (i) *For every $\theta_a \in (0, \lambda_a]$ there are binary signs σ_r (depending on θ_a , chosen in the proof) and $m_0 \in [\theta_a, \theta_a + c_p^2]$ with $u_a(\hat{z}) = 0$ exactly; then a is a strict non-peak with $w(a) = 0$ and gap M .*
- (ii) *Finitely many absorbers can be tuned to value 0 simultaneously (the signs of each absorber chosen as in (i) at the row where every tuning mass equals its θ_a). Here and in (i) the absorber is at a stage p so late that $2\mu_{p_0}^* \leq \nu_{(1)}/4$ and $160\lambda_a^2 \leq \nu_{(1)}^2/2$, where $\nu_{(1)}$ is the ν of $f^{(1)}$; this holds for all absorbers at stages beyond an f -dependent stage, which is how the lemma is used.*
- (iii) *A tuned absorber has no d -effect and no effect on $\varkappa$: $u_a(\hat{z})\gamma_a = 0$ in (11), and $\mathbf{v}_a = 0$.*
- (iv) *The cost is $\leq C(\lambda_a + \text{sum of the masses}) \log$, and the values of other carriers move only by the second-order effect $\leq C(\sum \text{masses} \cdot \mu^*)^2$ of the diagonal base, except for carriers at later stages whose targets meet the new support coordinates.*

Proof. (i) We have $q^*(x)n_a u_a(\hat{z}) = \beta \hat{z}_s + \hat{z}_{p_0} + 4 \sum_r 2^{-r} \hat{z}_{p_r} + q^*(x) \delta_a h_a(\hat{z}) =: \Psi(m_0)$, and at a support coordinate p with mass m_p and sign z_p , $\hat{z}_p = z_p(1 + (\mu_p^*)^2 m_p / \nu)$, while $\hat{z}_j = z_j$ off the support (Lemma 3.2(b)). Since U is diagonal, $\nu^2 = \sum_j (\mu_j^*)^2 a_j^2$ depends only on the moduli of the masses, not on the signs σ_r ; and s , p_0 and S_a carry no binary mass. Hence the quantity

$$\text{Tg} := -(\beta \hat{z}_s + \hat{z}_{p_0} + q^*(x) \delta_a h_a(\hat{z})),$$

evaluated at the row with $m_0 = \theta_a$, does not depend on the signs σ_r , and at that row $\hat{z}_{p_r} = \sigma_r(1 + e_r)$ with $e_r := (\mu_{p_r}^*)^2 \theta_a / \nu \geq 0$. As $\nu \geq \mu_{p_0}^* \theta_a$ and $\mu_{p_r}^* \leq \mu_{p_0+1}^* = (\mu_{p_0}^*)^{2^{2^{p_0}}}$ for $r \geq 1$, we have $e_r \leq e := (\mu_{p_0}^*)^{2^{2^{p_0}}-1} < 2^{-R}$ (Definition 5.48). Moreover $|\text{Tg}| \leq 1 + (1 + \mu_{p_0}^*) + \frac{3}{2}(1 + \mu_{p_0}^*) \leq 3.6$, because $|\hat{z}_s| = |z_s| \leq 1$ and $q^*(x) \delta_a \|h_a\|_1 \leq \frac{3}{2}$ by the choice of $\delta^{\max}$. The exact binary sums $4 \sum_r 2^{-r} \sigma_r$ run through the odd multiples of 2^{2-R} in $(-4, 4)$, so there are signs with $|(\text{Tg} - 2^{3-R}) - 4 \sum_r 2^{-r} \sigma_r| \leq 2^{2-R}$; for these signs $|\sum_r 4 \cdot 2^{-r} e_r \sigma_r| \leq 4e$, hence

$$\Psi(\theta_a) = -\left(\text{Tg} - \sum_r 4 \cdot 2^{-r} (1 + e_r) \sigma_r\right) \in [-3 \cdot 2^{2-R} - 4e, -2^{2-R} + 4e] \subseteq [-2^{4-R}, 0).$$

Raising m_0 from θ_a by $\Theta \in [0, c_p^2]$ raises $\hat{z}_{p_0}$ by at least $(\mu_{p_0}^*)^2 \Theta / (2\nu)$: ν is at least the value $\nu_{(1)}$ of $f^{(1)}$ minus a negligible amount, and $\mu_{p_0}^* (\theta_a + c_p^2) \leq 2\mu_{p_0}^* \leq \nu_{(1)}/4$ for absorbers of late stages, so $d(m_0/\nu)/dm_0 = (\nu^2 - (\mu_{p_0}^* m_0)^2)/\nu^3 \geq 1/(2\nu)$. The other terms of Ψ depend on m_0 only through ν : they are sums $\sum_j c_j (1 + (\mu_j^*)^2 m_j / \nu)$

over the binary coordinates and $S_a \cap [1, s_{\text{far}}(a)]$ with $\sum_j |c_j|(\mu_j^*)^2 m_j \leq 5\theta_a$, and $|\Delta(1/\nu)| \leq (\mu_{p_0}^*)^2 m_0 \Theta / \nu^3$ (as $\Delta(\nu^2) \leq 2(\mu_{p_0}^*)^2 m_0 \Theta$); so their total change is at most $10\theta_a m_0 / \nu^2$ times the increase $(\mu_{p_0}^*)^2 \Theta / (2\nu)$, and $10\theta_a m_0 / \nu^2 \leq 160\lambda_a^2 / \nu_{(1)}^2 \leq \frac{1}{2}$ for absorbers of late stages. So Ψ is continuous and increases by at least $(\mu_{p_0}^*)^2 c_p^2 / (4\nu) \geq 4(\mu_{p_0}^*)^2 c_p^2$ ($\nu \leq \|U\| = \frac{1}{16}$) over $[\theta_a, \theta_a + c_p^2]$. Finally $c_p \geq c_p^{\text{low}}$: at a cluster stage $p+i$, $i \geq 2$, by the term $c_{p+i-1}/4 = c_{p+i}$ of c_{p+i}^{low} , at the first cluster stage by definition, and at PRE stages because every term of c_p^{low} is at most the corresponding term of the minimum defining c_p ($\delta_p \geq \delta_p^{\text{max}}/2$). Hence the increase is at least $4(\mu_{p_0}^*)^2 (c_p^{\text{low}})^2 \geq 2^{7-R} > 2^{4-R}$ by the choice of R , and the intermediate value theorem gives $m_0 \in [\theta_a, \theta_a + c_p^2]$ with $\Psi(m_0) = 0$, i.e. $u_a(\hat{z}) = 0$; by [I, Lemma 1.7], $\zeta(a) = 0 < \bar{\vartheta}\Phi_a^2$ and $w(a) = 0$. (ii) Choose the signs of every absorber as in (i) at the row where all tuning masses equal their θ_a . Each absorber's value depends at first order only on its own m_0 ; the other masses enter through ν at second order (bounded as in (i)), so the joint system is solved by the explicit fixed point of Lemma 4.15, each coordinate staying in its interval by the margins of (i). (iii) Lemma 2.9 and the second expression of $\varkappa$: a strict non-peak with $\mathbf{v} = 0$ contributes nothing. (iv) Lemma 2.12 and Lemma 3.2(b). $\square$

For a biased pair (a^+, a^-) for s and a number r (the *residue* at s) put $c_0 := \lambda_{a^-} 2^{-s_{\text{far}}(a^-)}$ and

$$\gamma_{a^+} := c_0 + \frac{(-r)_+}{|u_{a^+}(s)|}, \quad \gamma_{a^-} := c_0 \frac{|u_{a^+}(s)|}{|u_{a^-}(s)|} + \frac{r_+}{|u_{a^-}(s)|}.$$

Since $u_{a^\pm}(s)$ has the sign $\pm$, $\gamma_{a^+} u_{a^+}(s) + \gamma_{a^-} u_{a^-}(s) = -r$; both coefficients are $\geq c_0/4$ and Lipschitz in r . On the far parts of $S_{a^\pm}$ the own term $\gamma_a v_a(s')$ dominates the terms of all later carriers (by (W4'') the ratio is $\leq C_f 2^{s_{\text{far}}(a) - s'} T_{10}^3 2^{1+k(a)} < \frac{1}{2}$), so $z = +1$ there is admissible whatever r is. A tuned absorber that is not used to cancel a residue receives the constant coefficient $c_0(a)$; with the coefficient 0 the far part of its signature set, where $z = +1$, would carry only later contributions of uncontrolled sign.

The coarse shifted cone, activity classes, and one shift ray. In the rest of this subsection the design is $D^{U1'}$, N is fixed, $f \in S_{p_N^*}$ has finite base support, $g \in \mathcal{C}(f)$, $\rho < 1$, and w is a clean sub-window of a main stage $L \geq l_f$ for f . K denotes quantities of the form $C_f(\text{Des}(L)/u(w))^C$ with a numerical constant C ; for $L \geq l_f$ every constant of this subsection is at most $C_f \bar{D}(w)$ (Definition 5.48), and such constants are absorbed directly by the window, because $T_{\text{hi}}(w) \leq 2^{-L^3}/(LQ(w))$, $n(w) \geq L2^{L^3}Q(w)$ and $Q(w) \geq \bar{D}(w)^3$. The shift components of all data considered below are bounded by an f -constant Δ_{max} (Lemma 2.23 and the remark following it; a non-degenerate peak of f persists at the companions).

Definition 5.52 (The coarse shifted cone). Let $f^\#$ be the companion of Theorem 5.7 for the closed pattern $\kappa(w)$ (statuses as in Theorem 5.7(a)). Let Ω be the set of coarse strict non-peaks of $f^\#$ other than absorbers (kept, clamped and class-R ones; nearly neutral kept carriers have value 0 after Theorem 5.7), Ω_m its part in block m , and P_m^c the coarse peaks of block m , with signs $\text{vs}_l := \text{sign } w_m^\#(k(l))$.

For $(\Delta', \gamma) \in \mathbb{R}^I \times \mathbb{R}^\Omega$ put

$$\mathcal{L}(\Delta', \gamma) := \left(\sum_{l \in \Omega} \gamma_l u_l - \sum_m \Delta'_m \sum_{l \in P_m^c} \text{vs}_l \lambda_l u_l \right) \Big|_{E_c(w) \setminus F^\#}.$$

For an *activity class* a (Lemma 5.53), $\Gamma^\#(a)$ is the polyhedral cone of all (Δ', γ) with:

- (X1) $\mathcal{L}(\Delta', \gamma)$ is $z^\#$ -admissible on $E_c(w) \setminus F^\#$; on each set $S_l \cap (L, s_{\text{far}}(w)]$, $l \leq L$, where only l lives among the coarse carriers, the rows of (X1) are positive multiples of one sign row and are replaced by that row with the coefficient $\|v_l 1_{S_l \cap (L, s_{\text{far}}(w)]}\|_1$;
- (X3) $\Delta'_m \varkappa_m^\# = \sum_{l \in \Omega_m} u_l(\hat{z}^\#) \gamma_l$ for every m , where $\varkappa_m^\# = \varkappa(\hat{z}^\#; \Omega_m)$; in the ratio variables $r_l := u_l(\hat{z}^\#) / \varkappa_{m(l)}^\#$ this reads $\Delta'_m = \sum_{l \in \Omega_m} r_l \gamma_l$;
- (X4) $\text{vs}_l \gamma_l / \lambda_l + \Delta'_{m(l)} \geq 0$ for every $l \in \Omega$ with $\text{gap}^\#(k(l)) < t^2$ for some $t \in \mathcal{W}(w)$ (near-threshold carriers);
- (X5) $\gamma_l = 0$ and $\Delta'_m = 0$ for the components declared inactive in a .

The rows other than (X3) have design coefficients (the numbers $\lambda_l u_l(j)$), the aggregated coefficients, signs, $1/\lambda_l$). The minors of the matrix of $\Gamma^\#(a)$ in the ratio variables, and the minors of the matrices g_J formed by the Δ' -parts of the cofactor vectors of its $(n-1)$ -row subsystems, are polynomials in r with design coefficients; for every pattern and every class of level L they are added to the rate scheme of $D^{U1'}$, with values computed at the ratios of f .

For the actual decompositions at a scale $t \in \mathcal{W}(w)$ we read off the coarse data (data convention, compare Lemma 5.9): $\Delta'_{\text{dec}, m}(t) := -\Delta d_m M_m$ for $m \in I_{\text{sh}}(w)$ and $:= 0$ otherwise (for $m \notin I_{\text{sh}}(w)$ the shift is $\leq K_d t$ by Lemma 4.6), and $\gamma_{\text{dec}, l}(t) := -\Delta \theta_l$ for $l \in \Omega$.

Lemma 5.53 (Activity classes). *Let $J := |\Omega| + N$; since $L \geq l_f$, every block $m \leq N$ has a carrier $\leq L$, so $N \leq L$ and $J \leq 2L$. Let $H(L) \leq C \text{Des}(L)^C / u(w)$ be the bound for the ℓ_1 -Hoffman constants of all cones $\Gamma^\#(a)$ at the exactified point (Proposition 5.54(b), Proposition 5.56), let $K_* \geq \max\{2\text{Des}(L), 4C_f \text{Des}(L) J H(L)\}$ with $K_* \leq (\text{Des}(L) / u(w))^C$, where $C_f \text{Des}(L)$ bounds the entries of the rows of $\Gamma^\#(a)$, and put $A_1 := K_*$, $A_{i+1} := K_* A_i$. For every scale t one of the $J+1$ intervals $[A_i K t, A_{i+1} K t)$, $1 \leq i \leq J+1$, contains the modulus of no component of $X(t) := (\Delta'_{\text{dec}}(t), \gamma_{\text{dec}}(t))$; let $i(t)$ be the least such i . The class of t is $a(t) := (i(t), \text{the set of components with modulus } \geq A_{i(t)+1} K t, \text{ their signs})$; the other components are inactive and are set to 0, at an ℓ_1 -cost $\leq J A_{i(t)} K t$. There are at most $(J+1)3^J$ classes. Zeroing the inactive components and projecting onto $\Gamma^\#(a(t))$ moves no active component below $A_{i(t)+1} K t / 2$. For $L \geq l_f$ the number of classes, K_*^{J+2} , and all constants A_i ($i \leq J+2$) are at most $\bar{D}(w)$.*

Proof. Pigeonhole (J moduli, $J+1$ disjoint intervals). Zeroing the inactive components changes the violation of the rows of $\Gamma^\#(a(t))$ by at most $C_f \text{Des}(L) \cdot J A_{i(t)} K t$, so the violation after zeroing is at most $(1 + C_f \text{Des}(L) J A_{i(t)}) K t$ (Proposition 5.54(a)). The projection moves components by at most $H(L)$ times this violation, i.e. by at most $H(L)(1 + C_f \text{Des}(L) J A_{i(t)}) K t \leq \frac{1}{2} K_* A_{i(t)} K t = \frac{1}{2} A_{i(t)+1} K t$, because $A_{i(t)} \geq 1$ and $K_* \geq 4C_f \text{Des}(L) J H(L)$. The bounds by $\bar{D}(w)$ follow from $J \leq 2L$, $(J+1)3^J \leq (2L+1)3^{2L}$ and $K_* \leq (\text{Des}/u)^C$, with C_Γ large. The inactive

components have modulus $< A_{i(t)}Kt \leq A_{i(t)+1}Kt/K_*$. An inward row (X4) coupling a small γ_d with a large Δ'_m would force $\lambda_d \varrho_d K_* \geq 1$, which is excluded by $K_* \geq 2\text{Des}(L) \geq 2/\lambda_d$. $\square$

A class is *one-signed* if all its active shift components Δ'_m have the same sign or none is active, and *mixed* otherwise (possible only for $N \geq 2$). For a one-signed class with active shifts, *configuration (i)* means $\Delta' < 0$ (the active blocks lack an upper source) and *configuration (ii)* means $\Delta' > 0$.

Proposition 5.54 (Projection and one shift ray). *Suppose that at the companion $f^\#$ every minor of Definition 5.52 is 0 or $\geq u/(2\text{Des}^C)$. Then for every scale $t \in \mathcal{W}(w)$, with $a = a(t)$:*

- (a) $X(t)$ violates (X1), (X3), (X4) at the coefficients of $f^\#$ by at most Kt , and after the inactive components are set to 0 by at most $(1 + C_f \text{Des} J A_{i(t)})Kt$;
- (b) the ℓ_1 -projection $X_{\text{pr}}(t)$ onto $\Gamma^\#(a)$ of $X(t)$ with inactive components set to 0 satisfies $\|X_{\text{pr}}(t) - X(t)\|_1 \leq (1 + C_f \text{Des} J A_{i(t)})Kt$ (after enlarging K by the factor $H(L)$), and no active component is moved below $A_{i(t)+1}Kt/2$ (last sentence of Lemma 5.53);
- (c) (one ray) let $r_1, \dots, r_p$ be the extreme rays of $\Gamma^\#(a)$, $\|r_j\|_1 = 1$, and g_j their nonzero Δ' -parts; write $\Delta'_{\text{pr}}(t) = \sum_j a_j(t)g_j$ by a fixed rule (a basic solution), $a(t) \in [0, A_{\max}]^p$ with $A_{\max} \leq K$. For a cube $\mathcal{Q}$ of side ε in $[0, A_{\max}]^p$ with corner $a^{\max}$ (coordinatewise maximum) and every scale t with $a(t) \in \mathcal{Q}$ put

$$\Delta'^* := \sum_j a_j^{\max} g_j, \quad \text{incr}(t) := \sum_j (a_j^{\max} - a_j(t))r_j \in \Gamma^\#(a);$$

then $\Delta'_{\text{pr}}(t) + \Delta'(\text{incr}(t)) = \Delta'^*$ and $\|\text{incr}(t)\|_1 \leq p\varepsilon$;

- (d) for $\varepsilon := c_f(u/\text{Des})^C$ the number of pairs (class, cube) is at most $D_{\text{cls}}(w) := (J+1)3^J(A_{\max}/\varepsilon)^p$, and $D_{\text{cls}}(w) \leq \bar{D}(w)$ for $L \geq l_f$;
- (e) $|\Delta'_m| \geq |\Delta'_{\text{pr},m}(t)|$ with the same sign, and the signs $\epsilon_l \gamma_l \geq 0$ of the class- G components are preserved by the increments.

Proof. (a) As in Steps 1–2 of Proposition 4.24: by the switching budget [I, Lemma 8.24], $\sum_{j \notin F} \phi_{z_j}(\Delta B(j)) \leq t/q_0$ with $\Delta B = -\sum_l \Delta \theta_l u_l$, and on $E_c(w)$ the coarse part of ΔB equals $\mathcal{L}(X(t))$ up to: the peak errors $e_k \text{vs} \lambda \leq C_f \text{Des} t/u$ ([I, (16)]; tiny-margin peaks of f are strict non-peaks of $f^\#$, hence in Ω); fine carriers ($\leq 3b^2/t$); the difference between z and $z^\#$ on $E_c(w)$ (closed tiny rooms, as in Proposition 4.24); and the change of the coefficients from f to $f^\#$ (values and $\varkappa$ move by $\leq C \text{Des} b + \eta(w)$, times $\|\gamma\|_1 \leq 12 \sum_l \lambda_l/t$, which is $\leq T_{\text{lo}}(w)^3$). (X3) is [I, (17)] with the peak traces, as in the proof of Lemma 2.9. (X4): for a near-threshold strict non-peak of f , [I, Lemma 8.8] gives $\text{vs}(\gamma/\lambda) + \Delta' \geq -(4 \text{gap}(f)/t + C_f b)$, and at a peak of f , [I, (16)] gives $\text{vs}(\omega_- - \omega_+)(k) = e_k \geq 0$; $\text{gap}(f) \leq bM$. A class-R near-threshold strict non-peak (or a dropped anti-type near-threshold strict non-peak) of a block pins $\Delta'_{\text{dec}} \geq -C_f \text{Des} K_g t$ ([I, Lemma 8.8] with $|\Delta \theta| \leq K_g t$), so it does not occur in an active block with $\Delta' < 0$, and in an active block with $\Delta' > 0$ its row holds with $\gamma = 0$. (b) Lemma 5.1: after the aggregation of (X1) the nonzero minors are $\geq u/(2\text{Des}^C)$ and the entries are $\leq C_f \text{Des}$, so the Hoffman constant is $\leq C \text{Des}^C/u$. (c) By Lemma 4.7 every active block has a robust coarse peak;

it is of class G (a class-R peak is a source of both kinds), of swallowing type in a block with $\Delta' < 0$ and of anti type in a block with $\Delta' > 0$ (Lemma 5.18), and (X1) on its near signature set fixes $\text{sign } \Delta'_m$ on the whole cone; (X1) on natural signature sets fixes the signs of the class-G components of γ , and (X1) at free coordinates forces $\gamma = 0$ for class-R components. So $\Gamma^\#(a)$ lies in an orthant, is pointed, and is the cone generated by its extreme rays (Minkowski–Weyl); its Δ' -projection is the cone generated by $g_1, \dots, g_p$. A basic representation uses linearly independent g_J with $a_J = g_J^+ \Delta'_{\text{pr}}(t)$; by Cauchy–Binet $\sigma_{\min}(g_J)$ is bounded below by a robust minor of g_J over a polynomial bound of its entries, so $A_{\max} \leq C_f(\text{Des}/u)^C \Delta_{\max}$. $\text{incr}(t)$ is a nonnegative combination of extreme rays. (d) Count. The number p of extreme rays is at most $P(L)$ (a pointed cone in $\mathbb{R}^n$ cut out by at most $R_\Gamma(L)$ inequalities has at most $\binom{R_\Gamma(L)}{n-1} \leq 2^{R_\Gamma(L)}$ extreme rays), so $(A_{\max}/\varepsilon)^p \leq (C_f \Delta_{\max} (\text{Des}/u)^{2C})^{P(L)} \leq (\text{Des}/u)^{(2C+1)P(L)}$ as soon as $C_f \Delta_{\max} \leq \text{Des}(L)/u(w)$, which holds for $L \geq l_f$; together with $(J+1)3^J \leq (2L+1)3^{2L}$ this gives $D_{\text{cls}}(w) \leq \bar{D}(w)$ for $C_\Gamma \geq 2C+1$. (e) follows from the orthant structure in (c). $\square$

Exactification with threshold protection under (KN). An Ω -carrier is *near-threshold* if $|\varrho - 1| \leq b(w)$ at f , and *active in the class a* if its component γ_l is active.

Definition 5.55 (The lever hypothesis $(\text{KN}_{w,a})$). $(\text{KN}_{w,a})$: every block which is active in a and contains an Ω -carrier that is near-threshold and active in a also contains an Ω -carrier d_0 that is inactive in a and has $\varrho_{d_0} \geq u(w)$ at f .

Proposition 5.56 (Exactification with threshold protection). *Let w be a clean sub-window of a main stage $L \geq l_f$ for f , and let a be a class satisfying $(\text{KN}_{w,a})$. Then there is a companion $f^{(1a)}$ of f with:*

- (a) *the ratios r_l of the active Ω -carriers equal a point r' at which every minor of Definition 5.52 that is tiny at f vanishes and every robust one is $\geq u/2$;*
- (b) *every active near-threshold Ω -carrier is a strict non-peak with gap $\geq c_f \eta(w) \text{Des}(L)M$; every inactive near-threshold Ω -carrier is a strict non-peak with gap $\geq c_f \eta(w)M$; every other coarse carrier keeps its robust status;*
- (c) $p^*(f^{(1a)} - f) \leq C_f \text{Des} T_{l_0}(w)^4 \log(1/T_{l_0}(w)) = o(T_{l_0}(w)^2)$.

Proof. Step 0. Apply only the closing moves (C1), (C2) of the assembly (no donor raise in active blocks); by Lemma 4.19 values move by $\leq C_*(l)b$, and $\varkappa$ by $\leq C \text{Des } b$ (it is Lipschitz in the block data, Lemma 2.10). *Step 1.* The tiny minors are $\leq \beta(w)$ at the ratios $r(f_1)$; Lemma 5.3 in the ratio variables gives r' with $|r' - r(f_1)| \leq C_L \beta^{2/N_L} \leq \eta(w)$, and robust minors stay $\geq u/2$. *Step 2 (realization).* Apply Lemma 4.15 to the active Ω -carriers with target values $u_l := r'_l \varkappa_m$ ($\varkappa_m$ held at its current value). In every active block m with an active near-threshold carrier, push the lever d_0 of $(\text{KN}_{w,a})$ inward (pulls or z-moves on its far signature coordinates, two-sided by Lemma 4.15) by $s_0 := C \text{Des}(L) \eta(w) \bar{\vartheta}_m \Phi_{P_m}^2 / (\varrho_{d_0} M_m)$ in units of $\hat{\zeta}_m$; then push a peak to restore $\varkappa_m$ exactly (it is continuous, Lemma 2.10(a), and the required push is $O(X \cdot s_0)$, with $X \leq C \text{Des}^2 \sum_{\text{fine}} \Phi^2$); then re-solve the tuning. The Jacobian of (ratios, $\varkappa$) with respect to (tuning parameters, pushes) is block triangular with identity and diagonal blocks, so the explicit fixed point of Lemma 4.15 extended by one scalar equation per block converges. This gives (a). (b). By Proposition 5.46(iii) and Lemma 2.10, $\bar{\vartheta}_m$ rises by $\geq c \varrho_{d_0} M s_0 / \Phi_P^2 \geq$

$C'\text{Des } \eta \bar{\vartheta}_m$, while the active values move by $\leq \eta$ in u -units, i.e. by $\leq m\eta\text{Des}$ in units of v ; so an active carrier with $\varrho \leq 1+b$ ends with $\varrho^\# \leq (1+b+C\text{Des } \eta)(1-C'\text{Des } \eta) < 1-c\text{Des } \eta$ for C' large ($b \leq \eta$). Robust carriers move by $O(\text{Des } \eta) \ll u$. The capacity of d_0 suffices: $|\hat{\zeta}_m(d_0)| = \varrho_{d_0} \bar{\vartheta}_m \Phi_{d_0}^2 \geq u \bar{\vartheta}_m / \text{Des}^2 > s_0$. Inactive near-threshold carriers are not variables of the minors; they are pushed inward individually by η , and $\varkappa$ is restored as above. (c) Lemma 4.15 and the cost bound of Lemma 2.12, with all moves $\leq C\text{Des } \eta$, $\eta = T_{\text{lo}}^4 / (L\text{Des})$. $\square$

In a class without active shift no lever is needed: only values enter, and Theorem 5.7 applies.

The companion and its fine structure. Fix a class a with active blocks $A \subseteq I_{\text{sh}}(w)$, signs $\sigma_m := \text{sign } \Delta'_m$ ($m \in A$), active Ω -carriers Ω_{act} and the normalized reference data $(\Delta'(t), \gamma(t)) \in \Gamma^\#(a)$ of Proposition 5.54 for the scales t of one (class, cube) set $\mathcal{S}$.

Definition 5.57 (The companion $f^\#$ of a class). $f^\#$ is obtained from f by the following moves, which act on pairwise distinct coordinates.

- (1a) Proposition 5.56 (for a class without active shift: Theorem 5.7), producing $f^{(1a)}$ with the exact ratios r' and $\varkappa'_m := \varkappa_m(f^{(1a)})$.
- (1b) For every $s \in E_c(w) \setminus F^{(1a)}$ the first pair $P(s)$ of Lemma 5.50, and all absorbers of the cluster of L , are tuned to value 0 (Lemma 5.51).
- (1c) $\varkappa_m$ and the ratios of Ω_{act} are restored to $(\varkappa', r')$ and the tuned absorbers returned to value 0 (joint explicit fixed point; each target carrier is controlled at first order by its own masses, cross effects are of second order; $\varkappa$ is continuous, Lemma 2.10(a)).
- (2) *Release and fine structure.* For every coarse carrier l'' that is not an owner candidate below (inactive Ω -carriers, and peaks of inactive blocks), the far part $S_{l''} \cap (s_{\text{far}}(w), \infty)$ is released. Let J_{fine} consist of the coordinates outside $F^{(1b)} \cup E_c(w)$ not fixed in (1a), (1b), together with the released ones. On J_{fine} , z is defined by the recursion of Lemma 5.60 in configuration (i), and by the fixed point of Lemma 5.63 otherwise.

Step (2) does not change the base part, hence not $e^\#$. The release changes the values of the released carriers by $\leq 4c_{L+1}^2$ (these are not variables of $\Gamma^\#(a)$), and $\varkappa$ by $\leq C\text{Des } c_{L+1}^2$.

The data at $f^\#$ use the switching coefficients $\gamma_l(t)$ on Ω_{act} , 0 on the inactive Ω -carriers, the biased coefficients on the used pairs $P(s)$, and the constant coefficient $c_0(a)$ on the other tuned absorbers. By Lemma 2.22 the forced coefficient of every other carrier k of block m is $-\Delta''_m \lambda_k w_m^\#(k)$, where the *true shift* $\Delta''(t)$ is given by (11) at $f^\#$; tuned absorbers contribute nothing to it (Lemma 5.51(iii)).

Lemma 5.58 (The true shift). (a) $|\Delta''_m(t) - \Delta'_m(t)| \leq C_f(|\varkappa_m^\# - \varkappa'_m| + \|u^\# - u'\|_{\ell_\infty(\Omega_{\text{act}})} \|\gamma(t)\|_1) \leq C_f \text{Des } c_{L+R_{\text{cl}}(L)+1} + C_f c_{L+1}^2 / t$. After (1c) the values of Ω_{act} are not changed by (2), so $\Delta''(t) = \Delta'(t) \varkappa'_m / \varkappa_m^\#$; for the normalized data of a set $\mathcal{S}$ this is the same vector Δ''^* for all $t \in \mathcal{S}$. (b) $\mathcal{L}(\Delta''(t), \gamma(t)) - \mathcal{L}(\Delta'(t), \gamma(t)) = -\sum_m (\Delta''_m - \Delta'_m) \mathcal{P}_m$, $\mathcal{P}_m := \sum_{l \in P_m^c} v_{sl} \lambda_l u_l|_{E_c(w)}$. On the near signature sets of the coarse peaks this keeps the sign of the peak trace, so $\mathcal{L}(\Delta'', \gamma)$ is admissible there; on $T(L)$ the difference is absorbed by the cluster pairs (Proposition 5.59). (c) The signs and the activity of $\Delta''(t)$ are those of $\Delta'(t)$.

Proof. (a) Subtract (11) at $(u', \varkappa')$ from (11) at $(u^\#, \varkappa^\#)$ and divide by $\varkappa^\# \geq A^\# \geq c_f$; $\sum u' \gamma = \Delta' \varkappa'$. After (1c), $|\varkappa^\# - \varkappa'|$ is bounded by the effect of later moves: only carriers at stages $> L + R_{\text{cl}}$ can change status (cluster carriers are tuned to value 0 with gap M , coarse carriers have protected margins and gaps), step (2) changes the block data by $\leq C(\sum_{l>L+R_{\text{cl}}} \lambda_l + 2^{-s_{\text{far}}(w)})$, and $\varkappa$ is Lipschitz in the block data with constant $\leq \text{Des}/u$. The values u_l , $l \in \Omega_{\text{act}}$, move in (1b) by second-order effects $\leq Cc_{L+1}^2$ and are restored in (1c); their vectors meet J_{fine} only in their own far signature parts, which are owned by l with sign fixed since (C1). (b) $\mathcal{L}$ is linear in Δ' , and on the natural signature set of a coarse peak only that peak lives among coarse carriers; $|\Delta'' - \Delta'| < |\Delta'|/2$ for active components by (c). (c) Active components have modulus $\geq A_{i+1}Kt/2$, much larger than the bound in (a), since $c_{L+1} \leq T_{\text{io}}(w)^8$. $\square$

Proposition 5.59 (Exact absorption on the coarse coordinates). *For $s \in E_c(w) \setminus F^\#$ let the residue $\varrho(s)$ be $\mathcal{V}(\Delta\omega)(s) - \mathcal{L}(\Delta'', \gamma)(s)$ minus the contribution of $P(s)$. Then:*

- (a) $\varrho(s)$ is the sum of the terms of the carriers after $P(s)$, so $|\varrho(s)| \leq C_f \sum_{k \text{ after } P(s)} \lambda_k$; on $T(L)$ the correction of Lemma 5.58(b), of size $\leq C_f \text{Des} c_{L+1}$, is added to it;
- (b) with the biased coefficients for $r := \varrho(s)$ (plus correction), $\mathcal{V}(\Delta\omega)(s) = \mathcal{L}(\Delta'', \gamma)(s)$ exactly, which is $z^\#$ -admissible;
- (c) (capacity) $|\Delta\omega(a)| = |\gamma_a|/\lambda_a \leq A_2/t$ for every used absorber a , which is a strict non-peak with $w(a) = 0$ and gap M (kind [2]);
- (d) at the support coordinates of the absorbers the data satisfy $t|b| \leq$ the mass, and the masses cost $\leq C_f c_{L+1} T_{\text{hi}}(w) + Cc_{L+1}^2$;
- (e) the pairs have no d -effect.

Proof. (a) Lemma 5.50 and Lemma 2.22(a); the coefficients of later carriers are $\leq C_f \lambda$ ($|\Delta''| \leq \Delta_{\text{max}}$, and later absorbers have $|\gamma| \leq C_f \lambda$). (b) The biased pair. (c) Cluster pairs have $\lambda_a \geq 2^{-1-k(a)} 4^{1-R_{\text{cl}}} c_{L+1}^{\text{low}} \geq c_{L+1}/\text{Des}(L)$, while the residues are $\leq 2c_{L+R_{\text{cl}}+1} \leq 2b(L + R_{\text{cl}})^2 \ll c_{L+1}/\text{Des}$ plus the correction $\leq C_f \text{Des} c_{L+1}$, so $|\gamma_a|/\lambda_a \leq C_f \text{Des}^2 \ll A_2/T_{\text{hi}}(w)$. A pre-pair at a stage $p > L$ sees residues only from carriers after it, of total weight $\leq 2c_{p+1} \leq 2b(p, M(p))^2 \ll \lambda_a$ (by (W2) and $n(p) \geq \text{Des}(p) \geq 2^{1+k(a)}/c_p$), and $s \notin T(L)$. (d) $b(p_r) = \gamma_a u_a(p_r)$ and $t|\gamma_a| \leq C_f \lambda_a T_{\text{hi}} \leq$ the mass. (e) Lemma 5.51(iii). $\square$

Lemma 5.60 (Ownership and the robust recursion). *Call a carrier an owner candidate if its coefficient in $\mathcal{V}(\Delta\omega)$ is nonzero with a sign known in advance:* (a) $l \in \Omega_{\text{act}}$ (sign ϵ_l); (b) coarse peaks of active blocks (coefficient $-\Delta_m'' \text{vs}_l \lambda_l$); (c) all tuned absorbers (coefficients $\geq c_0/4 > 0$); (d) carriers k at stages $> L$ of active blocks other than tuned absorbers (coefficient $-\Delta_m'' \lambda_k w(k)$, sign $-\sigma_m \varsigma_k$, $\varsigma_k := \text{sign } w(k)$ to be chosen). For $j \in J_{\text{fine}}$ let $\text{own}(j)$ be the least owner candidate meeting j ; if there is none, $\mathcal{V}(\Delta\omega)(j) = 0$ and z_j is free. For carriers k of type (d), in increasing order, let $O(k) := \{j \in J_{\text{fine}} : \text{own}(j) = k\} \supseteq S_k$, let Y_k be $u_k(\hat{z})$ computed with the coordinates of $\text{supp } u_k \setminus O(k)$ only, $\varsigma_k := \text{sign } Y_k$ (+1 if $Y_k = 0$), and $z_j := \text{sign}(-\sigma_m \varsigma_k u_k(j))$ on $O(k)$; on the sets owned by (a)–(c) put $z_j := \text{sign}(\text{coefficient} \cdot u_{\text{own}}(j))$. In configuration (i) ($\sigma \equiv -1$) this gives $z_j =$

$\varsigma_k \text{sign } u_k(j)$ on $O(k)$ and

$$|u_k(\hat{z}^\#)| = |Y_k| + \sum_{j \in O(k)} |u_k(j)| \geq \delta_k^\circ,$$

so every carrier of type (d) is a peak of sign ς_k with margin $\geq q_0^\# \delta_k^\circ / 2$.

Proof. The recursion is well founded, since Y_k uses only coordinates fixed before k . On J_{fine} , $\hat{z}_j^\# = z_j$ (diagonal base, $J_{\text{fine}} \cap F^\# = \emptyset$). With $\sigma = -1$, $z_j u_k(j) = \varsigma_k |u_k(j)|$ on $O(k)$. By (W3), $\Phi_k \leq c_k \leq (\delta_k H_k)^2$, so $\bar{\nu} \Phi_k / m \ll \delta_k^\circ$, and [I, Lemma 1.7] gives the peak and the margin. $\square$

Lemma 5.61 (Dominance on the fine coordinates). *In configuration (i), and for the owners of types (a)–(c) in every class, for every $j \in J_{\text{fine}}$ owned by k*

$$|\text{coef}_k| |u_k(j)| \geq 2 \sum_{k' > k, u_{k'}(j) \neq 0} |\text{coef}_{k'}| |u_{k'}(j)|;$$

hence $\text{sign } \mathcal{V}(\Delta\omega)(j) = \text{sign}(\text{coef}_k u_k(j))$ and $z_j \mathcal{V}(\Delta\omega)(j) > 0$. The bound holds for every shift and every switching of the class.

Proof. Later coefficients are $\leq C_f \lambda$. (i) $j \in S_k$: later carriers meet j only through targets, and (W4'') gives $c_{k'} \leq \iota_k 2^{-2j} c_k \delta_k T_{\text{lo}}(k-1)^3 / 2$; their total is $\leq C_f 2^{-2j} c_k \delta_k T_{\text{lo}}(k-1)^3$, while the own term is $\geq |\text{coef}_k| \delta_k 2^{-j} / n_k$ with $|\text{coef}_k| \geq c A_2 K T_{\text{lo}}(w) \lambda_k$ for types (a), (b), (d) and $\geq c_0 / 4$ for type (c). With $T_{\text{lo}}(k-1) \leq T_{\text{lo}}(w)$, $\lambda_k = m 2^{-m-k(k)} c_k$ and $j \geq 3 \cdot 2^{k(k)}$, the ratio is $< \frac{1}{2}$; for type (c), $j > s_{\text{far}}(a)$ and $c_0 = \lambda_a 2^{-s_{\text{far}}(a)}$ give a ratio $\leq C_f 2^{m+k-(j-s_{\text{far}}(a))} T_{\text{lo}}^2 < \frac{1}{2}$. (ii) $j \in \text{supp } y_k \setminus S_k$: the own term is $\geq |\text{coef}_k| \underline{y}_k / n_k$, the later terms are $\leq C_f \sum_{k' > k} \lambda_{k'} \leq C_f c_{k+1} \leq C_f b(k, M(k))^2 \leq C_f 2^{-8n(k)}$, and $n(k) \geq \text{Des}(k) \geq 1/(\underline{y}_k \lambda_k)$. If k is an absorber inside a cluster or a pre-sequence, only its partner meets j inside the block of consecutive block-1 stages, and λ decreases by a factor ≥ 8 between consecutive block-1 stages, so the partner's term is $\leq \frac{1}{4}$ of k 's; the first later non-absorber stage p' has $c_{p'} \leq b(p'-1, M(p'-1))^2$, and all later carriers together give a ratio $\leq \frac{1}{4} + C_f 2^{-n(p'-1)} / (T_{\text{lo}}(w) \underline{y}_k \lambda_k) < \frac{1}{2}$. Owners of types (a), (b) have no target coordinates in J_{fine} . $\square$

Lemma 5.62 (Scrambling in configuration (i)). *In configuration (i) the set A of active blocks satisfies (SC) at $f^\#$, along any sequence $s_i \downarrow 0$.*

Proof. Fix $m \in A$ and $\Upsilon \geq 1$. Coarse carriers are finitely many, peaks have relative margin $\geq u/2$, strict non-peaks have gaps bounded below (Proposition 5.56(b)), and there is no degenerate peak; tuned absorbers are finitely many strict non-peaks with gap M . For s below a positive bound none of them contributes to $\text{Scr}_m(\Upsilon s)$. Every other carrier of block m is a peak with margin $\geq q_0^\# \delta_k^\circ / 2$ (Lemma 5.60) and contributes only if $\delta_k H_k \leq 2n_k \Upsilon s / q_0^\# =: X$, and then at most $\Phi_k \leq c_k \leq (\delta_k H_k)^2$ by (W3). The numbers $\delta_k H_k$ decrease super-exponentially, so the sum of their squares over $\{\delta_k H_k \leq X\}$ is $\leq 2X^2$. Hence $\text{Scr}_m(\Upsilon s) \leq C(\Upsilon s / q_0)^2 = o(s)$. $\square$

Lemma 5.63 (Exact completion by a fixed point). *Let the data of the class carry the common true shift Δ''^* and let $J_{\text{free}} \subseteq J_{\text{fine}}$ be the complement of the sets owned by the owners of types (a)–(c). For $z' \in [-1, 1]^{J_{\text{free}}}$ let $f(z')$ be the row with these coordinates replaced, $\Delta''(z')$ the true shift at $f(z')$, and $\mathcal{V}(z')$ the fine part of $\mathcal{V}(\Delta\omega)$ at $f(z')$. Then $\Psi(z')_j := \text{clamp}_{[-1, 1]}(z'_j + \mathcal{V}(z')(j))$ has a fixed point z'^* , and at*

$f^\# := f(z'^*)$ the vector $\mathcal{V}(\Delta\omega)$ is $z^\#$ -admissible on J_{fine} . This holds in every class, one-signed or mixed.

Proof. As in Lemma 5.24: the product topology makes $[-1, 1]^{J_{\text{free}}}$ a compact convex metrizable set; $z' \mapsto \mathcal{V}(z')(j)$ is continuous (dominated convergence, norm-to-weak* continuity of the duality maps, continuity of $\varkappa^\#$ and of the absorber coefficients, which are Lipschitz in the residues); Schauder–Tychonoff gives z'^* , and $\text{clamp}(z_j + \mathcal{V}_j) = z_j$ means $\mathcal{V}_j = 0$ if $|z_j| < 1$ and $z_j \mathcal{V}_j \geq 0$ if $|z_j| = 1$. On the sets owned by (a)–(c) use Lemma 5.61. Signs of the shifts are not used. $\square$

Lemma 5.64 (Cost and statuses). $p^*(f^\# - f) \leq C_f \text{Des } T_{\text{lo}}(w)^4 \log \frac{1}{T_{\text{lo}}(w)} + C_f c_{L+1} \log \frac{1}{c_{L+1}} =: \theta_w T_{\text{lo}}(w)^2$ with $\theta_w \rightarrow 0$; every coarse carrier has the status prescribed by the closed pattern, with the gaps of Proposition 5.56(b); and every robust minor of $\Gamma^\#(a)$ stays $\geq u/(2\text{Des}^C)$.

Proof. (1a): Proposition 5.56(c). (1b), (1c): $\lambda_a \leq c_{L+1}$ and masses $\leq C_f c_{L+1} T_{\text{hi}}$. (2): with $\delta := \hat{z}^\# - \hat{z}^{(1c)}$ supported in J_{fine} , coarse carriers meet J_{fine} only beyond $s_{\text{far}}(w)$, so $|u_l''(\delta)| \leq 4\delta_l'' 2^{-s_{\text{far}}(w)} \leq 4c_{L+1}^2$, and carriers at stages $> L$ contribute $\leq \sum \lambda \leq c_{L+1}$; Lemma 2.12. $c_{L+1} \leq b(L, M(L))^2 \leq T_{\text{lo}}(w)^8$. The drifts $C c_{L+1}$ are $\ll c_f \eta(w) M$. $\square$

Mixed classes. In a mixed class put $A_- := \{m \in A : \sigma_m = -1\}$ and $A_+ := \{m \in A : \sigma_m = +1\}$, both nonempty. [I, Theorem 7.18] needs (SC) exactly for the blocks with $\Delta d_m < 0$, i.e. for A_- . By Lemma 5.63, exact completions exist in every class; what may fail in a mixed class is (SC) for A_- . For a carrier l of block m write $\text{thr}_l := \vartheta_m \Phi_l / m$, so that l is a peak iff $|u_l(\hat{z})| \geq \text{thr}_l$.

Lemma 5.65 (States of a negative fine carrier). *Let z be a point at which $\mathcal{V}(\Delta\omega)$ is admissible on J_{fine} , and l a carrier of a block $m \in A_-$ at a stage $> L + R_{\text{cl}}(L)$ which is not a tuned absorber. Let $O(l)$ be the set of $j \in \text{supp } u_l \cap J_{\text{fine}}$ at which no earlier carrier has a nonzero coefficient ($O(l) \supseteq S_l$), $\text{own}_l := \sum_{j \in O(l)} |u_l(j)| \geq \delta_l^\circ$, $v_l := u_l(\hat{z})$ and $Y_l := v_l - \sum_{j \in O(l)} u_l(j) \hat{z}_j$. Then exactly one of the following holds:*

- (Z') $|w_m(l)| < M_m/2$, i.e. $|v_l| < \text{thr}_l/2$: a strict non-peak with gap $> M_m/2$;
- (R) $|w_m(l)| \geq M_m/2$ and $\text{sign } w_m(l) = \text{sign } Y_l$ (or $Y_l = 0$): then $v_l = \text{sign}(w_m(l))(|Y_l| + \text{own}_l)$, a peak with margin $\geq \text{own}_l/2$ in units of $|u(\hat{z})|$;
- (W) $|w_m(l)| \geq M_m/2$ and $\text{sign } w_m(l) = -\text{sign } Y_l$: then $|Y_l| < \text{own}_l$ and $|v_l| = \text{own}_l - |Y_l|$.

Proof. If $|w(l)| \geq M/2$, the coefficient $|\Delta_m''| \lambda_l w(l)$ of l dominates every later term at every $j \in O(l)$ (the estimates of Lemma 5.61 with $|\text{coef}| \geq |\Delta_m''| \lambda_l M/2$), so $\mathcal{V}_j \neq 0$ and admissibility forces $z_j = \text{sign}(w(l)) \text{sign}(u_l(j))$ on $O(l)$ (as $\Delta_m'' < 0$ and $\hat{z}_j = z_j$ there). Hence $v_l = Y_l + \text{sign}(w(l)) \text{own}_l$, and $\text{sign } v_l = \text{sign } w(l)$ by the clamp formula (Lemma 2.6(b)); the two sign cases give (R) and (W). If $|w(l)| < M/2$, the clamp formula $|w(l)| = M|v_l|/\text{thr}_l$ gives the bound, and gap $= M - |w(l)|$. Finally $\text{own}_l \geq \delta_l^\circ \geq 2\text{thr}_l$ by (W3). $\square$

Lemma 5.66 (A scrambling criterion that tolerates wrong branches). *Call a carrier l of a block of A_- at a stage $> L + R_{\text{cl}}(L)$ bad if $|u_l(\hat{z}^\#)|$ lies in the band $B_l := (\text{thr}_l(1 - \Phi_l^{1/2}/M), \text{thr}_l + \Phi_l^{1/2})$. If at $f^\#$ no such carrier is bad, then A_- satisfies (SC) at $f^\#$, along a sequence common to all blocks of A_- .*

Proof. Fix $\Upsilon \geq 1$. Coarse carriers and tuned absorbers do not contribute for small s (proof of Lemma 5.62). A peak that is not bad has $\mu_l = q_0^\#(|u_l(\hat{z}^\#)| - \text{thr}_l) \geq q_0^\# \Phi_l^{1/2}$, and a strict non-peak that is not bad has $\text{gap}_l = M(1 - |u_l(\hat{z}^\#)|/\text{thr}_l) \geq \Phi_l^{1/2}$. So a contribution to $\text{Scr}_m(\Upsilon s)$ requires $\Phi_l^{1/2} \leq \Upsilon s/q_0^\#$, or $\Phi_l \text{gap}_l \leq \Upsilon s$ with $\text{gap}_l \geq M/2$ (then $\Phi_l \leq 2\Upsilon s/M$), or $\text{gap}_l < M/2$ and $\Phi_l^{3/2} \leq \Upsilon s$. In all cases $\Phi_l \leq X(s) := \max\{(\Upsilon s/q_0^\#)^2, 2\Upsilon s/M, (\Upsilon s)^{2/3}\}$. Choose main stages $L_i \rightarrow \infty$ and s_i with $X(s_i) = (\Phi_{L_i} \Phi_{L_i-1})^{1/2}$. By (W2) at L_i , $\Phi_{L_i} \leq b(L_i - 1, M(L_i - 1))^2 \leq T_{10}^8$ of the preceding stage, which is $\leq \Phi_{L_i-1}^7$ for large i (as $\log_2(1/T_{10}) \geq n \geq \text{Des} \geq 1/\Phi_{L_i-1}$); carriers before L_i have $\Phi_l \geq 2^{-N-l} c_{L_i-1} > X(s_i)$, and carriers after L_i have total weight $\leq c_{L_i+1} \ll \Phi_{L_i}$. So the contributions are $\leq 2\Phi_{L_i}$, and, as $s_i \geq X(s_i)^{3/2}/\Upsilon$, $\Phi_{L_i}/s_i \leq \Upsilon \Phi_{L_i}^{1/4} \Phi_{L_i-1}^{-3/4} \leq \Upsilon \Phi_{L_i-1} \rightarrow 0$. $\square$

Carriers in the state (R) are never bad ($\text{own}_l \geq \delta_l^\circ$ while $\text{thr}_l + \Phi_l^{1/2} \leq Cc_l^{1/2}$ and $c_l \leq T_{10}(l-1)^3 \ll (\delta_l^\circ)^8$), and (Z') carriers are not bad; only carriers in the wrong branch (W) with $\text{own}_l - |Y_l| \in B_l$ can be bad. In configuration (i) the recursion of Lemma 5.60 never produces (W).

Lemma 5.67 (The recursion in a mixed class). *Run the recursion of Lemma 5.60 in a mixed class with the rule: a carrier of type (d) of a block of A_- takes $\varsigma := \text{sign } Y$ (self-aligned); a carrier l of type (d) of a block of A_+ takes $\varsigma := \text{sign } Y_l$ and anti-aligned own coordinates $z_j := -\varsigma \text{sign } u_l(j)$ on $O(l)$. Call such a positive carrier frustrated if $|Y_l| < \text{own}_l + \text{thr}_l/2$. If no positive carrier is frustrated, the recursion defines z on J_{fine} at which $\mathcal{V}(\Delta\omega)$ is admissible on J_{fine} and every negative carrier is in the state (R); hence A_- satisfies (SC) at $f^\#$.*

Proof. For a non-frustrated positive carrier, $v_l = \varsigma(|Y_l| - \text{own}_l)$ with $|v_l| \geq \text{thr}_l/2$, so $|w(l)| \geq M/2$ with sign ς ; its coefficient $-\Delta_m''|\lambda_l w(l)|$ has the sign $-\varsigma$ and dominates on $O(l)$, so $\text{sign } \mathcal{V}_j = -\varsigma \text{sign } u_l(j) = z_j$. Negative carriers are treated as in Lemma 5.60; every $j \in J_{\text{fine}}$ is owned or carries $\mathcal{V}_j = 0$. Lemma 5.66 gives (SC). $\square$

A frustrated positive carrier has no consistent contact state: anti-aligned own coordinates give the wrong sign of v_l , aligned ones the wrong sign of $\mathcal{V}$ on $O(l)$. It must take a value of modulus $< \text{thr}_l/2$, which needs free coordinates or cancellations among later carriers, and downstream negative carriers may then be forced into (W).

Master Theorem IV'.

Theorem 5.68 (Master Theorem IV'). *Let $N \geq 1$, let the design be $D^{U1'}$, and let $f \in S_{p_N}^*$ have finite base support. Suppose that for infinitely many main stages L some clean sub-window w of L has at least $n(w)/\bar{D}(w)$ dyadic scales whose activity class a is of one of the following kinds:*

- (α) *a has no active shift;*
- (β) *a is one-signed and satisfies $(\text{KN}_{w,a})$;*
- (γ) *a is mixed, satisfies $(\text{KN}_{w,a})$, and the recursion of Lemma 5.67 meets no frustrated positive carrier.*

Then $f \in \mathcal{R}$: $(f, g) \in \text{clNA}((c_0, p_N), \ell_2^2)$ for every $g \in \mathcal{C}(f)$.

Proof. Fix $g, \rho < 1$ and one such pair (L, w) with $L \geq l_f$. By pigeonhole over the $D_{\text{cls}}(w) \leq \bar{D}(w)$ pairs (class, cube) of Proposition 5.54(d) there is a set $\mathcal{S}$ of at least $n(w)/\bar{D}(w)^2$ scales with one class a of the admitted kinds and one cube; $n(w) \geq L2^{L^3}Q(w)$ and $Q(w) \geq \bar{D}(w)^3$ give $|\mathcal{S}| \geq L2^{L^3}\bar{D}(w) \geq L2^{L^3}(\text{Des}/u)^C$, which exceeds $48\rho^2K/(c(1-\rho^2))$ with $KT_{\text{hi}}(w)/|\mathcal{S}| \rightarrow 0$ along the main stages, for K and c^{-1} of the form $C_f(\text{Des}/u)^C$. Build $f^\#$ by Definition 5.57: in configuration (i) with the recursion of Lemma 5.60, in configuration (ii) and in mixed classes with the fixed point of Lemma 5.63 (in case (γ) the recursion of Lemma 5.67), and for (α) with the fixed owners only.

Data. For $t \in \mathcal{S}$ let ω^+ be the clamped decomposition ω_+ (as in Proposition 4.24), let $\Delta\omega(t)$ have the components $\gamma_{\text{pr}}(t) + \gamma(\text{incr}(t))$ on Ω_{act} , the shift Δ''^* (Lemma 5.58), the absorber coefficients of Proposition 5.59, and 0 on inactive Ω -carriers; $\omega^- := \omega^+ + \Delta\omega(t)$; $b^+ := \beta + \chi\mathcal{V}_{\text{pr}} + \chi_f\mathcal{V}_{\text{fine}}$, where χ is the split of the $+$ side of the decomposition with respect to the projected coarse vector, $\chi_f \in [0, 1]$ is arbitrary on the fine contacts, and β is supported in $F^\#$ and chosen with $b^+(\xi^\#) = 0$ (possible since $\xi^\# \neq 0$ on $F^\#$; this normalization is needed for tangency, compare Proposition 5.31(iii)); $b^- := b^+ - \mathcal{V}(\Delta\omega(t))$. By Lemma 2.22(c) these are exact two-piece data at $f^\#$ as soon as $\mathcal{V}(\Delta\omega(t))$ is $z^\#$ -admissible off $F^\#$. On $E_c(w)$ it equals $\mathcal{L}(\Delta''^*, \gamma(t))$ exactly (Proposition 5.59), which is admissible (Proposition 5.54, Lemma 5.58(b)); on J_{fine} it is admissible by Lemmas 5.61, 5.60, 5.63 or 5.67; the active far parts of Ω -carriers are dominated by their own coefficients. The $+$ side is that of Proposition 4.24 up to Kt (projection Kt , inactive components $JA_{i(t)}Kt$, fine part and absorber coefficients $\leq C_f c_{L+1}$), so $p^*(g - g_t) \leq Kt$. The $-$ side differs from that of Proposition 4.24 by $\mathcal{V}(\text{incr}(t))|_{E_c}$ and $\Delta\omega(\text{incr}(t))$, so, $\sqrt{\Gamma_w}$ being a seminorm, $\sqrt{\Gamma_w^\#(b^-, \omega^-)} \leq \sqrt{\Gamma_w(B_-, \Theta_-)} + Kt + C_f p\varepsilon$; with $\varepsilon = c_f(u/\text{Des})^C$ small, $\kappa_w \leq 1 + \eta_0/2$. The block parts are of kinds [1]–[3]: increments satisfy the inward rows (X4), so near-threshold carriers are of kind [3]; absorbers are of kind [2] (Proposition 5.59(c)). All data of $\mathcal{S}$ have the same shift Δ''^* .

Statuses and (SC). Lemma 5.64. In configuration (i), A satisfies (SC) at $f^\#$ (Lemma 5.62); in case (γ) , A_- does (Lemma 5.67); in configuration (ii) and in case (α) , $\Delta d_m \geq 0$ for all m and no (SC) is needed.

Recovery. $p^*(f^\# - f) \leq \theta_w T_{\text{lo}}(w)^2$ with $\theta_w \rightarrow 0$ (Lemma 5.64). Theorem 5.13 (part (a) or (b)), with the banked, pulled and absorber supports allowed by Theorem 2.20(E-b) (at absorber coordinates $t|b| \leq$ the mass, so no flips), run on the subsets $\mathcal{S}$ (Lemma 5.14) along the infinitely many sub-windows, gives $(f, \rho g) \in \text{cl NA}$; let $\rho \rightarrow 1$. $\square$

Corollary 5.69 (Intrinsic form). *In the setting of Theorem 5.68, suppose that for infinitely many main stages L some clean sub-window w of L has $I_{\text{up}}(w) = \emptyset$ or $I_{\text{lo}}(w) = \emptyset$, and that $(\text{KN}_{w,a})$ holds for every activity class a of w . Then $f \in \mathcal{R}$.*

Proof. An upper source gives $\Delta'_{\text{dec},m} \geq -K_d t$ and a lower source $\Delta'_{\text{dec},m} \leq K_d t$ (Lemma 4.6, data convention). A block of $I_{\text{up}}(w)$ has a lower source ($I_{\text{up}} \cap I_{\text{lo}} = \emptyset$, Lemma 4.9), so its active shift can only be negative; a block of $I_{\text{lo}}(w)$ can only have a positive active shift. Zeroing and projection move components by at most $(1 + C_f \text{Des} JA_{i(t)})Kt$ and no active component below $A_{i(t)+1}Kt/2$ (Proposition 5.54(b), Lemma 5.53), and components below $A_1 Kt = K_* Kt > (K + K_d)t$ are inactive; so the sign of an active shift component is that of the decomposition data. If $I_{\text{up}}(w)$

or $I_o(w)$ is empty, every class of w is one-signed, and some class carries at least $n(w)/\bar{D}(w)$ scales (there are at most $(J+1)3^J \leq \bar{D}(w)$ classes). $\square$

Remark 5.70 (What is not proved). (a) The statement of Theorem 5.68 *without* $(KN_{w,a})$ in case (β) is not proved; in particular it is *not* proved that every row with finite base support belongs to $\mathcal{R}$ for $N = 1$, although for $N = 1$ every class is one-signed. The reason is Remark 5.47: near-threshold switching carriers of an active block with negative shift are forced to be active, so (KN) is a genuine restriction even for one block. Without a lever, the residual is an active block whose inactive Ω -carriers all have $\varrho \leq b$ and which has an active near-threshold switching carrier d such that, near the ratios of f , the exact zero set of the tiny minors lies in $\{\varrho_d \geq 1\}$. (b) The hypotheses of Theorem 5.68 depend on N ; since [I, Lemma 1.5] transfers density and not membership of individual rows, the theorem gives no statement for Martín's norm p .

5.7. Aligned rows, nested-tuning rows and further tools. In this subsection $N = 1$, $T = T_{\text{final}}$, and we drop the block index. We show that (C^*) occurs at explicit rows, among them rows that are not block-tame, and that the exact-data mates of such rows are recovered.

Definition 5.71 (Aligned rows). A first row f is *aligned with exceptional carriers* l_- and Λ if $F \subseteq Z_0$ is finite, z is given by the owner recursion (Definition 5.28) with some signs ϵ_l , and:

- (A1) l_- has its target supported in F , $\epsilon_{l_-} = +1$, $u_{l_-}(\hat{z}) < 0$, and l_- is a strict non-peak with $\varrho_{l_-} \in (0, \frac{1}{2}]$;
- (A2) every carrier $l \neq l_-$ is *aligned*: $\text{sign } u_l(\hat{z}) = \epsilon_l$;
- (A3) the carriers of the set Λ are strict non-peaks with $\varrho \geq \frac{1}{2}$; every carrier outside $\Lambda \cup \{l_-\}$ is a peak, and all but finitely many of them have $\varrho \geq 2$.

Every coordinate off F is then a contact, all carriers are exactly swallowed, l_- is of anti type ($q_{l_-} < 0$) and all other carriers are of swallowing type. The rows f_{SA} of Theorem 5.29 (with $N = 1$) are aligned with $\Lambda = \emptyset$. The dominance property of Theorem 5.29(d) holds for every aligned row, since Step 6 of that proof uses only the owner recursion, (SF_l) and $(W4)$.

Proposition 5.72 (Coherent resonance at aligned rows). *Let f be aligned with exceptional carriers l_- and Λ . There is l_1 such that at every clean sub-window w of every level $l \geq l_1$: every coarse carrier is of class G , the block has no upper source and has a lower source (L2),*

$$\rho^{\text{sh}}(\kappa^{\text{sh}}(w), f) = 0 \quad \text{and} \quad c_{\pi(w)}(f) = 0.$$

In particular f is a (C^) row.*

Proof. All rooms vanish, so every coarse carrier is of class G and there is no class- R carrier. All peaks are of swallowing type, so (U1), (U2) fail; l_- is a class- G carrier with $q < 0$ and $\varrho \in (0, \frac{1}{2}]$, which for large l is neither $\leq b$ nor $\geq 1 - b$, so (U3) fails; the peaks with $\varrho \geq 2 \geq 1 + u$ give (L2). Hence $I_{\text{sh}} = I_{\text{up}} = \{1\}$, $\text{src}(1)$ is “free” (no class- R source), the set N_0 is empty (Λ and the peaks have $\varrho \geq \frac{1}{2}$, and ϱ_{l_-} is fixed), and $R_7 = \emptyset$ (the only anti-type carrier is l_-). Take $\delta := 1$, $\tau_c := \lambda_c$ for every coarse carrier $c \neq l_-$, and $\tau_{l_-} := (1 + \sum_{c \neq l_-} q_c \lambda_c) / |q_{l_-}| > 0$. (S1) holds; (S2) holds since $\epsilon_c \text{vs}_c = +1$ at peaks; (S4): $\delta + \sum_c q_c \tau_c = 1 + \sum_{c \neq l_-} q_c \lambda_c - (1 + \sum_{c \neq l_-} q_c \lambda_c) = 0$;

(S5), (S6), (S7) are void. (S3): $V(\tau) = \sum_{c \neq l_-} \lambda_c \epsilon_c u_c 1_{F^c} + \tau_{l_-} u_{l_-} 1_{F^c}$. A coordinate $j \in T(l) \setminus F$ is owned by a coarse carrier o . If $o \neq l_-$, the term $\lambda_o \epsilon_o u_o(j)$ has the sign z_j and dominates the later coarse terms (Step 6 of Theorem 5.29, which bounds moduli), which never include l_- because $\text{supp } u_{l_-} \subseteq F \cup S_{l_-}$. If $o = l_-$, then $j \in S_{l_-}$, $z_j = +1$, and $\tau_{l_-} v_{l_-}(j)$ dominates a fortiori, as $\tau_{l_-} \geq 1/|q_{l_-}| \geq \lambda_{l_-}$. So all contact rows hold and there are no free rows: $\text{viol}(\delta, \tau) = 0$ with $\|\delta\|_1 = 1$. For $c_{\pi(w)}$: $I_{\text{up}} = \{1\}$ forces $\delta = 1$; with $x := \lambda$ on $B_f(w)$ the vector $\Pi_1 + \sum_{B_f} x \epsilon u 1_{F^c}$ is the partial sum of $\sum_c \lambda_c \epsilon_c u_c 1_{F^c}$ over the coarse carriers, which is z -signed wherever it is nonzero (dominance) and vanishes at coordinates touched by no coarse carrier; so every summand $\phi_{z_j}(\cdot)$ vanishes. $\square$

So (C*) is a property of rows that may well belong to $\mathcal{R}$: the rows f_{SA} are (C*) rows and block-tame. The following rows are not block-tame.

Proposition 5.73 (Nested-tuning rows). *Let f be aligned with exceptional carriers l_- and an infinite set $\Lambda = \{l_1 < l_2 < \dots\}$ whose relative gaps $g_i := 1 - \varrho_{l_i}$ satisfy $0 < g_i < \Phi_{l_{i+1}}/2$, and assume that f has no degenerate peak. Then Q is infinite, f does not satisfy (BT), the block fails (SC): $\text{Scr}(\Upsilon s) \geq s$ for every $\Upsilon \geq 1$ and every $0 < s \leq \Phi_{l_1}$, and f is a (C*) row.*

Proof. $Q \supseteq \{k(l_i)\}$ is infinite. For $\Phi_{l_{i+1}} \leq s \leq \Phi_{l_i}$ the strict non-peak l_i has gap $= Mg_i < \Phi_{l_{i+1}} \leq s$ and contributes $\min(\Phi_{l_i}, \Upsilon s) \geq s$ to $\text{Scr}(\Upsilon s)$; so $\text{Scr}(\Upsilon s)/s \not\rightarrow 0$. Proposition 5.72. $\square$

Remark 5.74 (Existence of nested-tuning rows; sketch only). The existence of rows as in Proposition 5.73 is supported by the following construction, which is *a sketch and not a proof*. Let $F = \{p, p', p''\} \subseteq Z_0$, $\sigma = (+, +, +)$, and assume the design addition (Z0+): for $F_0 := F$ the target family contains $c/q^*(c)$ for a countable dense set of directions $c \in \mathbb{R}^{F_0}$, each used infinitely often (in T_{final} this is compatible with allowedness, since Z_0 meets no signature set). Parametrize a by b in the open positive orthant of S^2 ; for y supported in F , $y(\hat{z}) = \langle y, \Phi(b) \rangle$. Non-special carriers follow a *hysteretic* owner rule (reference signs kept unless $\epsilon_l A_l \leq -(B_l + \delta_l H_l)/2$), so they are aligned with $n_l |u_l(\hat{z})| \geq (B_l + \delta_l H_l)/2$; their hysteresis margins decrease super-exponentially. Specials have targets $c_i/q^*(c_i)$ from (Z0+), so $B = 0$ and $A = \langle c_i, \Phi(b) \rangle / q^*(c_i)$. One holds at stage i a closed ball B_i , a level L_i , specials $l_1 < \dots < l_i$, and numbers $\varepsilon_j, \kappa_i > 0$ such that: (J1) all assignments of carriers $\leq L_i$ are constant on B_i ; (J2) for $j < i$ the gap of l_j relative to the threshold of the carriers $\leq L_i$ lies in $[\varepsilon_j, \Phi_{l_{j+1}}/2 - \varepsilon_j]$ on B_i , with margin against the threshold perturbation of later carriers; (J3) l_- and the specials keep their signs and the sign conditions with margin; (J4) the gap function G_i of l_i takes values $\geq \kappa_i$ and $\leq -\kappa_i$ in the interior of B_i . The step $i \rightarrow i+1$ picks a zero of G_i , a direction c_{i+1} nearly orthogonal to Φ there with independent tangential gradient, moves along the curve $\{G_i = \Phi_{l_{i+1}}/4\}$ to a zero of the gap of a late carrier l_{i+1} with that target (intermediate value theorem; theta is Lipschitz, but not differentiable on $\{v = \bar{v}\}$, so the argument must be topological), takes a ball of radius $\sim \Phi_{l_i} \Phi_{l_{i+1}}$ around that point, and only then chooses L_{i+1} . The limit point of the nested balls gives the row. (An earlier version of this induction confined the new gap to an interval bounded away from 0 and is inconsistent: the region required at the next stage is empty.)

Proposition 5.75 (Explicit coherent-shift mates of aligned rows). *Let f be aligned with exceptional carriers l_- and Λ , $\psi := R^*w$, $k_- := k(l_-)$.*

- (i) $\psi 1_{F^c}$ is z -signed and nonzero at every coordinate off $F \cup S_{l_-}$, and $z\psi < 0$ on S_{l_-} .
- (ii) Let $\Omega \subseteq Q$ be finite with $k_- \in \Omega$, and assume either $\Omega = \{k_-\}$, or that the targets of all carriers of Ω are supported in F . Let ω^+ and $\Delta\omega$ be supported in Ω with $\epsilon_k \Delta\omega(k) \geq 0$ for $k \in \Omega \setminus \{k_-\}$, $\Delta := d(\Delta\omega) < 0$ and $\Delta\omega(k_-) > \Delta w(k_-)$; let $\chi : F^c \rightarrow [0, 1]$ and β be supported in F with $(\beta + \chi v 1_{F^c})(\hat{z}) = 0$, where $v := R^*\Delta\omega - \Delta\psi$. Then $\omega^- := \omega^+ + \Delta\omega$, $b^+ := \chi v 1_{F^c} + \beta$, $b^- := b^+ - v$ are two-piece data ($\Delta d = \Delta < 0$) for $g := b^+ + R^*(\omega^+ - d(\omega^+)w)$, and $v \neq 0$ off F .
- (iii) $cg \in \mathcal{C}(f)$ for all small $c > 0$.
- (iv) For $s \in S_l$ with $l \notin \Omega \cup \{l_-\}$ and $s \notin \Sigma(\omega)$, $g(s)/\psi(s) = -d(\omega^+) + \chi_s |\Delta|$; the profile is not constant whenever χ is not.

Proof. (i) At a coordinate owned by $o \neq l_-$, $|w(o)| \geq M/2$ (M at peaks, $M\varrho_o \geq M/2$ on Λ) and $M > \frac{1}{3}$, while the later terms of ψ have moduli $\leq 2^{-5}$ times the own term computed with $|w| = 1$ (Step 6 of Theorem 5.29); so $\text{sign } \psi(j) = \text{sign}(w(o)u_o(j)) = \epsilon_o \text{sign } u_o(j) = z_j$, by (A2). On S_{l_-} , $w(k_-) < 0$ and $z = +1$. (ii) Off F , $v = R^*\Delta\omega - \Delta\psi$. Under either assumption the carriers of Ω meet F^c only in their own signature sets. On S_k , $k \in \Omega \setminus \{k_-\}$: $v(j) = \lambda_k(\Delta\omega(k) - \Delta w(k))v_k(j) - \Delta \sum_{l>k} \lambda_l w(l)u_l(j)$, and $\epsilon_k(\Delta\omega(k) - \Delta w(k)) \geq |\Delta||w(k)| \geq |\Delta|M/2$ (as $\text{sign } w(k) = \epsilon_k$), while the later terms are $\leq |\Delta|2^{-5}\lambda_k v_k(j)$; so $z_j v(j) \geq (M/2 - 2^{-5})|\Delta|\lambda_k v_k(j) > 0$. On S_{l_-} : $\Delta = d(\Delta\omega) = \sum_{k \in \Omega} \Phi_k^2 w(k) \Delta\omega(k)/C$, the terms with $k \neq k_-$ are ≥ 0 , and $w(k_-) < 0$, so $\Delta\omega(k_-) \geq |\Delta|C/(\Phi_{k_-}^2 |w(k_-)|) \gg |\Delta|$; the own coefficient $\lambda_{l_-}(\Delta\omega(k_-) - \Delta w(k_-))$ is positive and dominates the later terms. Elsewhere off F , $v = -\Delta\psi = |\Delta|\psi$ is z -signed and nonzero by (i). Hence b^+ is z -signed, $b^- = (\chi - 1)v 1_{F^c} + (\beta - v 1_F)$ is $(-z)$ -signed off F , there are no free coordinates, both pairs represent g since $b^+ - b^- = v = R^*(\omega^- - \omega^+) - \Delta\psi$, and $g(\xi) = q_0 b^+(\hat{z}) = 0$. (iii) As in Proposition 5.31(iv). (iv) At such s , $R^*\Delta\omega(s) = 0$, so $g(s) = \chi_s v(s) - d(\omega^+)\psi(s)$ and $v(s) = -\Delta\psi(s)$. $\square$

Without one of the two assumptions in (ii) the conclusion can fail: a target coordinate off F of a carrier of Ω that is owned by another carrier receives the term $\lambda_k \Delta\omega(k) y_k(j)/n_k$, which is not controlled by Δ .

Corollary 5.76 (Exact-data mates of aligned rows are recovered). *Let f and the data be as in Proposition 5.75(ii), with $\kappa_w \leq 1$ and $\gamma_k := \lambda_k(\Delta\omega(k) - \Delta w(k)) \neq 0$ for $k \in \Omega$. Then the functional g satisfies $(f, g) \in \text{clNA}((c_0, p_1), \ell_2^2)$ whenever $g \in \mathcal{C}(f)$.*

Proof. We verify (H1), (H2) of Theorem 5.45. Every carrier has $|w(k)| \geq \min(M\varrho_{l_-}, M/2) > 0$, so no carrier outside Ω is neutral ($\gamma_k = -\lambda_k \Delta w(k) \neq 0$), and a carrier outside Ω is negligible iff $|w(k)| < 2\iota_k$, which happens for finitely many k since $\iota_k \rightarrow 0$; carriers of Ω are not neutral by assumption. By Proposition 5.75(ii), $D = b^+ - b^- = v$ is nonzero at every coordinate off F , so (H2) holds. Theorem 5.45 applies. $\square$

So the dangerous mates of the nested-tuning rows (oscillating profiles on the signature sets of robust class-G peaks, forced shift $\Delta < 0$, no (SC) at f) are

recovered, through block-tame companions, not through windows. Whether every mate of such a row is recovered is not proved.

Banks and pinning. The following two tools are used in the discussion of the remaining steps.

Lemma 5.77 (Quadratic cost of banks). *Let f have finite base support, let E be a finite set of contacts, $m_j > 0$ ($j \in E$), and let f^b have the data $a^b := (a + \sum_{j \in E} m_j z_j e_j^*)/q^*(\cdot)$ and z (so $F^b = F \cup E$).*

- (a) *For $x \in \mathbb{R}$ and $m > 0$, $2(x-m)_+ \leq (x_+)^2/(2m)$. Consequently the summand of the base excess of [I, Lemma 7.1] at $j \in E$ for a perturbation tB , namely $2(-\text{sign}(a_j^b)tB_j - |a_j^b|)_+$, is at most $(t^2/2)((z_j B_j)_-)^2/|a_j^b|$ for $t > 0$; for a side-+ pair it vanishes unless $z_j b_j^+ < 0$, for a side-− pair unless $z_j b_j^- > 0$, and for the average piece $b^\theta = \frac{1}{2}(b^+ + b^-)$ it is at most $(t^2/2)(b_j^\theta)^2/|a_j^b|$.*
- (b) *Suppose that at every $j \in E$, $\max(|b_j^+|, |b_j^-|, |b_j^\theta|) \leq C_E e_j$ with $e_j := (z_j b_j^+)_- + (z_j b_j^-)_+$, and put $\varepsilon := \sum_{j \in E} e_j$. For $\beta_0 > 0$ let $m_j := (C_E^2 q_0 \varepsilon / \beta_0) e_j$ and $m := \sum_j m_j = C_E^2 q_0 \varepsilon^2 / \beta_0$. Then $q_0 \sum_{j \in E} (C_E e_j)^2 / |a_j^b| \leq \beta_0(1+m)$, and $p^*(f^b - f) \leq C_f m \log(e/m)$.*

Proof. (a) $x^2/(2m) - 2(x-m) = (x-2m)^2/(2m) \geq 0$, and the left side vanishes for $x \leq m$. At $j \in E$, $\text{sign } a_j^b = z_j$, so the flip occurs only when $z_j t B_j < 0$. (b) $|a_j^b| = m_j/q^*(a + \sum m_j z_j e_j^*) \geq m_j/(1+m)$, so $q_0 \sum C_E^2 e_j^2 / |a_j^b| \leq (1+m) q_0 C_E^2 \sum e_j^2 / m_j = (1+m)\beta_0$. The cost: Lemma 2.12 with $\|a^b - a\|_1 \leq 2m$, and $\hat{z}$ moves at first order only on E (diagonal base). $\square$

A violation of ℓ_1 -mass ε on finitely many contacts thus costs a companion move of order ε^2 ; the hypothesis $|b_j^\pm| \leq C_E e_j$ holds when the violations are of fine origin and the split at j is clamped.

Lemma 5.78 (Pinned strict non-peaks bound the shift from above). *Let f be any first row, m a block, $k \in Q_m$ with $w_m(k) \neq 0$, and let $(B_\pm, \Theta_\pm)$ be a two-sided decomposition at scale $t \leq \min(t_\eta, 1)$ as in [I, Lemma 8.8]. Then, in the decomposition convention,*

$$\Delta d_m |w_m(k)| \leq 3 \text{gap}_m(k)/t + |\Delta \Theta_m(k)|.$$

Proof. With $\varsigma := \text{sign } w_m(k)$, [I, Lemma 8.8] gives $\varsigma \omega_+(k) \leq (1 - d_+ t) \text{gap}/t$ and $\varsigma \omega_-(k) \geq -(1 + d_- t) \text{gap}/t$; with $|d_\pm t| \leq \frac{1}{2}$, $\varsigma(\omega_+ - \omega_-)(k) \leq 3 \text{gap}/t$. Since $\Theta_\pm = \omega_\pm - d_\pm w$, $\omega_+ - \omega_- = \Delta \Theta + \Delta d w$, so $\varsigma(\omega_+ - \omega_-)(k) = \varsigma \Delta \Theta(k) + \Delta d |w(k)|$. $\square$

In particular a class-R strict non-peak with $\text{gap} \leq t^2$ (for which $|\Delta \theta| \leq K_g t$) pins the upward shift at scale t : it acts as an additional upper source.

Proposition 5.79 (Exact coupling for one shifted block). *Let w be a clean sub-window of level $l \geq l_f$, $I_{\text{sh}}(w) = \{m\}$, and let k be a class-G strict non-peak of block m with $q_k < 0$ whose column $e_k u_k 1_{F^c}$ is a free ray of the combinatorial zero-cost rows (at every $j \in T(l) \setminus F$ with $u_k(j) \neq 0$, j is a contact whose row is satisfied for every value $x_k \geq 0$ of the coefficient, the other coefficients being fixed, e.g. because the owner of j dominates; in particular this holds if u_k vanishes on $T(l) \setminus F$). Let (δ, x) satisfy the combinatorial rows of Corollary 5.11 exactly, and let $r := \delta_m + \sum_l q_l x_l$ be the coupling residual. If $r \geq 0$, or if $r < 0$ and $x_k \geq |r|/|q_k|$, then $(\delta, x + (r/|q_k|)e_k)$ satisfies the combinatorial rows and the coupling exactly, and*

$|r|/|q_k| \leq C\text{Des}(l)|r|/u(w)$. If $r < 0$, $x_k < |r|/|q_k|$, and all other carriers in the support of x have $q \geq 0$, then $\delta_m < 0$.

Proof. Changing x_k within $[0, \infty)$ keeps the combinatorial rows (free ray), and changes the coupling by q_k times the change; in the case $r < 0$ the new coefficient $x_k - |r|/|q_k|$ is ≥ 0 by hypothesis. At a clean sub-window ϱ_k is robust (it is not nearly neutral, since $q_k \neq 0$ is used with $q'_k = q_k$), so $|q_k| = \Phi_k|w(k)|/(mC) \geq \Phi_k Mu/(mC) \geq u/(C\text{Des})$. In the last case $\delta_m = r - \sum_{l \neq k} q_l x_l - q_k x_k \leq r + |q_k| x_k < r + |r| = 0$. $\square$

For one shifted block with such a carrier, the coupling row (the d -identity) can therefore be satisfied exactly inside the exact combinatorial cone with a constant of the form Des/u ; in the last case of the proposition the sign of the shift is the one excluded, up to $K_d t$, by a lower source (Lemma 4.6), so that the shift is pinned. This is the situation of the aligned rows, where l_- has its target in F .

5.8. What remains open for finite base support. For T_{final} (and $D^{U1'}$) and first rows with finite base support, the results of this section reduce Lemma Z ([I, Problem 8.21]) to the following question.

Problem 5.80. Let $N \geq 1$. Is every (C^*) row $f \in S_{p_N}^*$ (Definition 5.16) in $\mathcal{R}$?

By Corollary 5.17 an affirmative answer gives $\mathcal{R} \supseteq \{f : F \text{ finite}\}$ for p_N . (C^*) rows exist (Proposition 5.72): some of them are block-tame and therefore in $\mathcal{R}$ (Corollary 5.30); the nested-tuning rows of Proposition 5.73, whose existence is only sketched (Remark 5.74), are not block-tame, and only their mates with exact data are known to be recovered (Corollary 5.76). What is proved about (C^*) leaves the following precise gaps. Every item below is open; where a route is indicated, it is a sketch or a heuristic and is labelled as such.

Remark 5.81 (Status coherence without (KN)). Master Theorem IV' needs the lever hypothesis $(\text{KN}_{w,a})$ in every class with active shift that is used. Without it the residual is: an active block whose inactive Ω -carriers all have $\varrho \leq b(w)$, with an active near-threshold switching carrier d , such that the exact zero set Z of the tiny minors of Definition 5.52, in the ratio variables and near the ratios of f , lies in $\{\varrho_d \geq 1\}$ (Proposition 5.46); if d ends as a peak, its excess $|\omega_+(d)| + |\omega_-(d)|$ must be $O(Kt/\lambda_d)$, which nothing proved implies. *Sketch:* near-threshold carriers whose inward excess is inactive can be removed from Ω with the continuous coefficient $-\Delta' \lambda_d \min(\varrho_d, 1) \text{vs}_d$, which needs no status, so only carriers with active excess need a lever. *Heuristic:* if the design were generic in the sense that the map $(\varrho_d)_d$ restricted to every stratum of every zero set Z_S is transversal to $(1, \dots, 1)$, then Z would enter $\{\varrho_d < 1\}$ near $\{\varrho_d = 1\}$ with design-controlled depth; this needs a construction of such a design (only the parameters of stage l are free at stage l), quantitative transversality near singular strata, and a quantitative treatment of the case $\dim Z_S < |\{d\}|$. The absorbers and the one-ray normalization are not affected by this gap; it concerns only the joint exactness of the ratios and the protection of the thresholds.

Remark 5.82 (Mixed classes (C_{mix})). In a mixed class (possible only for $N \geq 2$) exact completions exist (Lemma 5.63); what is missing is a completion at which no carrier of a block of A_- is bad in the sense of Lemma 5.66. Carriers in the state (R) are never bad, configuration (i) never produces the state (W), and a bad carrier in

the state (W) requires the coincidence $\text{own}_l - |Y_l| \in B_l$, an interval of length about $\Phi_l^{1/2}$. Finite lexicographic models show that mixing can force the state (W) at every exact completion (*numerical evidence only*). Conceivable routes, none carried out: a selection among the Schauder fixed points by continuation from vanishing positive shift; a genericity argument in finitely many coarse parameters; or tolerating the sign violations of frustrated positive carriers (which have ℓ_1 -mass $\leq C_f c_{L+1}$, of fine origin) by Lemma 2.30, which would reduce (C_{mix}) to item (S2) below (*sketch* of a reduction).

Remark 5.83 (A second route: exact coupling and uniform composition). A different route to Problem 5.80 works at the assembled companion of Section 4 and tolerates fine-origin violations instead of absorbing them. Its proved ingredients are: the exact combinatorial resonance (Corollary 5.11(b)); the characterization of shifted exact data (Proposition 5.19, Corollary 5.26); the quadratic cost of banks (Lemma 5.77); the violation tolerance of a single engineered stage (Lemma 2.30); and exact coupling for one shifted block with a free-ray carrier (Proposition 5.79). The remaining steps are:

- (S1) *Exact coupling in general*: with two or more shifted blocks, or without a free-ray carrier with $q < 0$, satisfy the coupling rows $\Delta d_m = d_m^\#(\omega^-) - d_m^\#(\omega^+)$ exactly inside the exact combinatorial cone with constants of the form $C_f(\text{Des}/u)^C$. (Proposition 5.46 and Remark 5.47 explain why a single push per block does not obviously suffice; whether two independent levers per block are necessary is not decided — *heuristic*.)
- (RT*c) *Coarse exactness* at the companion modulo the coupling rows and the contributions of carriers beyond the level (class-R and unused class-G strict non-peaks of shifted blocks must be neutralized as switching carriers, with sizes controlled by Lemma 5.78) — *sketch*.
- (S2) *Uniform composition*: (S2a') engineered approximants that are *d-consistent*, i.e. at which the normalized values of the finitely many coarse switching carriers are re-tuned exactly to their values at the row (so that $d' = d$ on every switching vector), with per-piece bounds $T_0 \geq c_b t$ uniform in the scale — *sketch*; (S2b) the averaging of Theorem 2.20 carried out at one engineered approximant for all pieces of a window — *sketch*; (S2c) a comparison of the violation mass with the late threshold of the companion — *heuristic*.

The *d*-consistency in (S2a') appears to be needed for this route (*heuristic*): without it, the window masses of an engineered approximant change the coarse block data at first order, and for window data with switching of size $1/t$ this produces a mismatch between the θ -piece and the $\pm$ pieces of order s_1/t^2 (s_1 the stage scale); an upper bound of this order follows from the proof of [I, Theorem 7.18], and finite models show the scaling with equality up to a constant (*numerical evidence*). Rebalancing such a mismatch would require transfer peaks of inefficiency $\lesssim t^2$ at weights that super-fast ladders do not provide (*heuristic*). This does not affect Theorems 2.20 and 5.13, which apply [I, Theorem 7.18] at a fixed row to fixed exact data.

Remark 5.84 (Further open points). (a) The general mates of the nested-tuning rows: a route through exact window data at the row with a shift column, transplant to block-tame companions with inward moves of near-threshold special carriers, and $(Z0+)$, is plausible but is only a *sketch*; Lemma Z at these rows is not proved. (b)

The existence of the nested-tuning rows themselves is a *sketch* (Remark 5.74). (c) Rows that are given by the owner rule above some level and carry an *exactly* degenerate swallowing-type peak are not known to exist (the threshold has a jump part from fine sign changes which may skip the degenerate value); this concerns only the illustration of configurations, not the residual. (d) The results of this section are proved for p_N with N fixed; their hypotheses are not independent of N , so they give no row-wise statement for Martín's norm p . (e) Rows with infinite base support are not treated here.

Lemma Z ([I, Problem 8.21]) and the density of $\text{NA}((c_0, p_N), \ell_2^2)$ and of $\text{NA}((c_0, p), \ell_2^2)$ remain open for every admissible operator, including T_{final} ; nothing proved in this section points to a counterexample.

6. INFINITE BASE SUPPORT

This section treats first rows f whose base support $F = \text{supp } a$ is *infinite*. In Part I this case was open except for a sketch [I, Remark 8.46]; the missing step was the transfer of peaks to a first row with $a \notin c_{00}$. We prove that step (Theorem 6.3), so that [I, Remark 8.46] becomes a theorem (Theorem 6.21), and then develop the tools that are specific to infinite base support: at a scale t a support coordinate j behaves like a contact of sign $\text{sign } a_j$ with a free anti-sign allowance (*cushion*) of size $|a_j|/t$ on each side, and the cost of exceeding the cushions is governed by the *flip profile* of the data. The main results are the following.

- Engineered recovery at arbitrary F needs sparsity of the flips only for the one-sided parts of the data, because the support may be *raised* where the average data need it (Theorem 6.14).
- Finitely many exactly swallowed carriers whose swallowing by the support is *cushion sparse* are harmless for every signature-ladder operator (Theorem 6.29, under (W^*) , (H2), (H3-inf)); box-dominated support ($F = \mathbb{N}$) is harmless for *every* admissible operator (Theorem 6.31, under (SC_I) and (W_M)).
- For the designed operator T_{final} the raise-transfer lemma (Lemma 2.15) removes every profile condition: support swallowing of any profile by finitely many bad carriers is harmless (Theorems 6.34, 6.38, 6.40, under (W^*) , (H2), (H3-inf)), and infinitely many bad carriers are harmless under an explicit rate condition (Theorem 6.42).

The residual at infinite base support is described in Subsection 6.8.

Standing conventions. $N \geq 1$, $I = \{1, \dots, N\}$, $p = p_N$, and $f \in S_{p^*}$ has forced data $(\xi, q_0, a, w, F, z, \hat{z}, e, \nu)$ [I, Proposition 1.9] with $F = \text{supp } a$ *arbitrary* (finite or infinite). We write $s_j := \text{sign } a_j = z_j$ for $j \in F$ and $\alpha(M) := \sum_{j \in F, j > M} |a_j|$. Subsections 6.1–6.3 and 6.6 hold for every admissible operator T . In Subsections 6.4 and 6.5, T is a *signature-ladder operator*: either the operator of [I, Definition 8.1] with its windows $\mathcal{W}(l)$, or $T = T_{\text{final}}$ with a fixed choice of one sub-window w_l of each level l , used as the l -th window (Lemma 3.22(b)); in both cases (T-a)–(T-d), (P1), (P2), (P3) of [I, Theorem 8.2] and allowedness (a), (b) of [I, Definition 8.1] hold, and these are the only properties of the operator used there. In Subsections 6.7 and 6.8, $T = T_{\text{final}}$ and $U = U_{\text{final}}$ (Section 3). For a first row f' with prescribed data we use the notation $f[a', z']$ of Subsection 2.1; forced data of other first rows carry primes, superscripts or indices.

For a two-sided decomposition $(B_\pm, \Theta_\pm)$ of a mate at scale t [I, Lemma 8.5] we use $\Delta B = B_+ - B_-$, $\Delta\theta_{k,m}$ and [I, (15)], and the flip quantities of [I, Lemma 8.28],

$$f_j^+ := \left(-s_j B_+(j) - \frac{|a_j|}{t} \right)_+, \quad f_j^- := \left(s_j B_-(j) - \frac{|a_j|}{t} \right)_+ \quad (j \in F),$$

which satisfy $\sum_{j \in F} f_j^\pm \leq t/(4q_0)$ for every $t > 0$ and every F [I, Lemma 8.28]. The *box profile* of T is $\text{Bx}(j) := \sum_{m \in I} \sum_k \lambda_{k,m} |u_{k,m}(j)|$; since $\|u_{k,m}\|_1 \leq 1$ and $\sum_k \lambda_{k,m} \leq m2^{-m}$, $\|\text{Bx}\|_1 < 2$.

6.1. Transfer peaks at a base vector outside c_{00} . The proof of [I, Theorem 5.8] uses $a \in c_{00}$ only through the canonical truncations, which keep the base vector. For $a \notin c_{00}$ we truncate the base vector as well.

Definition 6.1 (Truncated canonical approximants). For $N' \in \mathbb{N}$ with $a1_{[1,N']} \neq 0$ put

$$\begin{aligned} c_{N'}^a &:= q^*(a1_{[1,N']}), & a_{N'} &:= a1_{[1,N']} / c_{N'}^a, \\ \nu_{N'} &:= \|U^* a_{N'}\|_H, & e_{N'} &:= U^* a_{N'} / \nu_{N'}, & z^{N'} &:= z1_{[1,N']}, \\ \tilde{x}_{N'} &:= z^{N'} + Ue_{N'}, & x'_{N'} &:= \tilde{x}_{N'} / p(\tilde{x}_{N'}), & f'_{N'} &:= \nabla p(x'_{N'}), \end{aligned}$$

and $c_{N'} := q(x'_{N'})$.

Lemma 6.2 (Truncated canonical approximants). *Let T be admissible and $f \in S_{p^*}$.*

- (a) $f'_{N'} \in \text{NA}((c_0, p), \mathbb{R}) \cap S_{p^*}$, with forced data $a_{N'}$, $e_{N'}$, $z^{N'}$ and $\hat{z}'_{N'} = \tilde{x}_{N'}$.
- (b) As $N' \rightarrow \infty$: $a_{N'} \rightarrow a$ in ℓ_1 , $e_{N'} \rightarrow e$, $f'_{N'} \rightarrow f$ in norm, all forced data of $f'_{N'}$ converge to those of f as in [I, Proposition 1.11], $x'_{N'} \rightarrow \xi$ weak* and $c_{N'} \rightarrow q_0$.
- (c) Let $P_{N'}^\perp$ be the orthogonal projection onto $e_{N'}^\perp$ and $\Psi_{N'}(h) := \|e_{N'} + h\| - 1 - \langle e_{N'}, h \rangle$. If $B \in \ell_1$ is supported in $F \cap [1, N']$ and $\text{sign}(a_{N'}(j) + tB_j) = \text{sign } a_{N'}(j)$ for $j \in F \cap [1, N']$, then

$$q^*(a_{N'} + tB) = 1 + tB(\tilde{x}_{N'}) + \nu_{N'} \Psi_{N'}(tU^*B/\nu_{N'}),$$

$$\text{and } \Psi_{N'}(h) \leq \|P_{N'}^\perp h\|^2 / (2(1 - \|h\|)) \text{ for } \|h\| \leq \frac{1}{2}.$$

Proof. (a) $a_{N'} \in c_{00}$, $q^*(a_{N'}) = 1$, $z^{N'} \in c_{00}$, $\|z^{N'}\|_\infty \leq 1$ and $z^{N'} = z = \text{sign } a = \text{sign } a_{N'}$ on $\text{supp } a_{N'} = F \cap [1, N']$; apply [I, Proposition 1.10(c)] and [I, Proposition 1.9] at $x'_{N'}$. (b) $c_{N'}^a \rightarrow q^*(a) = 1$, U^* is bounded, and $z^{N'} \rightarrow z$ coordinatewise; [I, Proposition 1.12] ((ii) $\Rightarrow$ (i)) and [I, Proposition 1.11] give the convergence of $f'_{N'}$ and of its forced data, and $c_{N'} = q(x'_{N'}) = 1/p(\tilde{x}_{N'}) \rightarrow 1/p^{**}(\hat{z}) = q_0$. (c) No sign change on $F \cap [1, N']$ gives $\|a_{N'} + tB\|_1 = \|a_{N'}\|_1 + tB(z^{N'})$, and $\|U^* a_{N'} + tU^* B\| = \nu_{N'} + t\langle U^* B, e_{N'} \rangle + \nu_{N'} \Psi_{N'}(tU^* B/\nu_{N'})$; add, using $\|a_{N'}\|_1 + \nu_{N'} = 1$ and $B(z^{N'}) + \langle U^* B, e_{N'} \rangle = B(\tilde{x}_{N'})$. The bound on $\Psi_{N'}$ is [I, Lemma 3.3]. $\square$

Theorem 6.3 (Transfer peaks at an arbitrary base vector). *Let T be admissible, $f \in S_{p^*}$ (F arbitrary), and let $g = b + \sum_m R_m^*(\omega_m - d_m(\omega_m)w_m)$ with $b \in \ell_1$, $\text{supp } b \subseteq F$, $b(\xi) = 0$,*

$$\|b/a\|_\infty := \sup_{j \in F} |b_j|/|a_j| < \infty,$$

and $\omega_m \in c_{00}$, $\text{supp } \omega_m \subseteq Q_m$, such that $g \in \mathcal{C}(f)$ and $\Gamma_w(g) \leq 1$. Then for every $\rho \in (0, 1)$ there are $g'_{N'} \in X^$ with $g'_{N'} \rightarrow g$ and $\rho g'_{N'} \in \mathcal{C}(f'_{N'})$ for all large N' , where $f'_{N'}$ are the truncated canonical approximants. In particular $(f, g) \in \text{cl NA}((c_0, p), \ell_2^2)$.*

Proof. We follow the proof of [I, Theorem 5.8], whose only use of $a \in c_{00}$ is that its canonical truncations keep the base vector a ; we use the truncated canonical approximants instead, and indicate the changes. Steps 2 and 3 there (transfer peaks at ξ , built from $\tau_{*,m} = (\hat{\pi}_m(\xi)/q_0)a - \hat{\pi}_m$ and the density of the tails of $(u_{k,m})_k$, and the limiting levels) do not involve the approximants and hold verbatim for $a \in \ell_1$. The block estimates of Steps 1, 4, 5 and Step 6 use only [I, Lemmas 1.7 and 3.4] at $x'_{N'}$, the convergence of the forced data (Lemma 6.2(b)), [I, Lemma 1.6] and $g \in \mathcal{C}(f)$; they are unchanged.

Step 1 (base). Put $\kappa_{N'} := (b1_{[1,N']})(\tilde{x}_{N'})$ and $b'_{N'} := b1_{[1,N']} - \kappa_{N'}a_{N'}$. Since $\tilde{x}_{N'} \rightarrow \hat{z}$ weak* boundedly (Lemma 6.2(b)) and $z^{N'} = z$ on $[1, N']$, $\kappa_{N'} \rightarrow b(\hat{z}) = b(\xi)/q_0 = 0$, so $b'_{N'} \rightarrow b$ in ℓ_1 ; $\text{supp } b'_{N'} \subseteq F \cap [1, N']$ and $b'_{N'}(\tilde{x}_{N'}) = \kappa_{N'} - \kappa_{N'}a_{N'}(\tilde{x}_{N'}) = 0$, because $a_{N'}(\tilde{x}_{N'}) = 1$. For large N' , $c_{N'}^a \leq 2$ and $|\kappa_{N'}| \leq \frac{1}{4}$; then for $|t| \leq r_b := \min(1, 1/(4\|b/a\|_\infty))$ and $j \in F \cap [1, N']$,

$$|tb'_{N'}(j)| \leq |t||b_j| + |t\kappa_{N'}||a_{N'}(j)| \leq \frac{1}{4}|a_j| + \frac{1}{4}|a_{N'}(j)| \leq \frac{3}{4}|a_{N'}(j)|,$$

since $|a_j| = c_{N'}^a|a_{N'}(j)| \leq 2|a_{N'}(j)|$. So no sign changes occur, and Lemma 6.2(c) gives

$$q^*(a_{N'} + tb'_{N'}) \leq 1 + \frac{t^2}{2}h_{N'}\left(1 + \frac{2|t|\|U^*b'_{N'}\|}{\nu_{N'}}\right),$$

$$h_{N'} := \frac{\|P_{N'}^\perp U^*b'_{N'}\|^2}{\nu_{N'}} = \frac{\|P_{N'}^\perp U^*(b1_{[1,N']})\|^2}{\nu_{N'}},$$

and $h_{N'} \rightarrow h(b) = H_b$ because $P_{N'}^\perp \rightarrow P^\perp$ in norm and $\nu_{N'} \rightarrow \nu$. This is the base bound of Step 1 of [I, Theorem 5.8] with a replaced by $a_{N'}$: $q^*(a_{N'} + tb'_{N'}) \leq 1 + \frac{t^2}{2}H_b\theta(t)$ with $\theta(t) \rightarrow 1$ as $t \rightarrow 0$, uniformly in large N' .

Step 4. Replace $\beta_{m,N}$ by $\beta_{m,N'} := ((R_m^*y_{m,N'})(x'_{N'})/c_{N'})a_{N'}$ and $e_{m,N'} := R_m^*y_{m,N'} - \beta_{m,N'}$; then $e_{m,N'} \rightarrow e_{\infty,m}$, because $R_m^*y_{m,N'} \rightarrow R_m^*y_m$ in ℓ_1 , $x'_{N'} \rightarrow \xi$ weak* boundedly, $c_{N'} \rightarrow q_0$ and $a_{N'} \rightarrow a$ in ℓ_1 . The bookkeeping identity [I, (6)] is an identity at $x'_{N'}$ and holds unchanged.

Step 5. With $A(t) := a_{N'} + t\rho b'_{N'} + \sum_m \varepsilon_m R_m^*y_{m,N'} = (1+E)a_{N'} + t\rho b'_{N'} + \sum_m \varepsilon_m e_{m,N'}$ one has, by homogeneity and Step 1,

$$q^*(A(t)) \leq (1+E)q^*\left(a_{N'} + \frac{t\rho}{1+E}b'_{N'}\right) + \rho^2 t^2 \sum_m |\tau_m| q^*(e_{m,N'}),$$

which is the base bound of Step 5 of [I, Theorem 5.8] with a replaced by $a_{N'}$. Put $g'_{N'} := b'_{N'} + \sum_m R_m^*(\omega_m - d'_{m,N'}w'_{m,N'})$ with $d'_{m,N'}$ as there. Then $g'_{N'} \rightarrow g$ and $g'_{N'}(x'_{N'}) = 0$ ($b'_{N'}(\tilde{x}_{N'}) = 0$, and [I, Lemma 3.2] at $x'_{N'}$, the support of ω_m consisting of strict non-peaks of $w'_{m,N'}$ for large N'). Step 6 uses only $p^*(f'_{N'} - f) \rightarrow 0$ and $p^*(g'_{N'} - g) \rightarrow 0$. Hence $\rho g'_{N'} \in \mathcal{C}(f'_{N'})$ for large N' . As $f'_{N'}$ attains its norm, $(f'_{N'}, \rho g'_{N'})$ attains its norm [I, Proposition 1.13(d)] and tends to $(f, \rho g)$; let $\rho \rightarrow 1$. $\square$

Corollary 6.4 (Windowed averaging at arbitrary base support). *Let T be admissible and $f \in S_{p^*}$ (F arbitrary). Then [I, Theorem 8.17] holds for f provided every certificate c_t in its hypotheses has a base part b_t with $\|b_t/a\|_\infty < \infty$ (for instance $\text{supp } b_t$ finite).*

Proof. The proof of [I, Theorem 8.17] up to its last sentence uses only the local validity (i), the closeness (ii), $g \in \mathcal{C}(f)$, the convexity of p^* and of Γ_w , and not

the finiteness of F . It produces an averaged certificate c whose base part is a finite average of the b_t , hence has $\|b/a\|_\infty < \infty$, with $\rho g_c \in \mathcal{C}(f)$, $\Gamma_w(\rho c) \leq 1$ and $\rho g_c \rightarrow \rho g$. Theorem 6.3 applied to the mate ρg_c gives $(f, \rho g_c) \in \text{clNA}$, and clNA is closed. $\square$

Remark 6.5. (a) The bounded ratio $\|b/a\|_\infty < \infty$ is used only to obtain a no-flip radius r_b independent of N' . (b) The approximants $f'_{N'}$ do not depend on g : all mates with bounded-ratio base parts and $\Gamma_w \leq 1$ are recovered simultaneously along one sequence. (c) Pure base mates ($g \in \ell_1$ with $q^*(a + tg) \leq s(t)$ for all t) are recovered along every truncation of a as well, by a direct computation with the base identity; we do not need this.

6.2. Cushions and flip profiles.

Definition 6.6 (Flip profiles). For $\beta \in \ell_1$ with $\beta \geq 0$ put

$$m_\beta(x) := \sum \{\beta_j : j \in F, |a_j| < x\beta_j\} \quad (x > 0),$$

$$\text{Fl}_\beta(r) := \sum_{j \in F} 2(r\beta_j - |a_j|)_+ \quad (r \geq 0),$$

and $\kappa_\beta(x) := m_\beta(x)/x$. The profile of β is *sparse* if $\kappa_\beta(x) \rightarrow 0$ as $x \rightarrow 0^+$, *super-critical* if $\kappa_\beta(x) \rightarrow \infty$, and *critical* if $0 < \liminf_{x \rightarrow 0^+} \kappa_\beta(x) \leq \limsup_{x \rightarrow 0^+} \kappa_\beta(x) < \infty$.

For finite F every profile is sparse ($m_\beta(x) = 0$ for $x < \min_F |a_j|/\|\beta\|_\infty$).

Lemma 6.7 (Flip calculus). *Let $\beta, \beta_1, \beta_2 \in \ell_1$ be nonnegative.*

- (a) $\text{Fl}_\beta(r) = 2 \int_0^r m_\beta(s) ds$; it is convex and nondecreasing in r , and $\text{Fl}_\beta(0) = 0$.
- (b) $\beta \mapsto \text{Fl}_\beta(r)$ is convex and coordinatewise nondecreasing; $\beta \leq \beta'$ implies $m_\beta \leq m_{\beta'}$; $m_{c\beta}(x) = c m_\beta(cx)$ for $c > 0$; and $m_{\beta_1 + \beta_2}(x) \leq 2m_{\beta_1}(2x) + 2m_{\beta_2}(2x)$.
- (c) For $b \in \ell_1$ and $r > 0$, $\sum_{j \in F} (|a_j + rb_j| - |a_j| - s_j rb_j) = \text{Fl}_{(sb)_-}(r)$, where $(sb)_- := ((s_j b_j)_-)_{j \in F}$; for $r < 0$ the left side equals $\text{Fl}_{(sb)_+}(|r|)$.
- (d) If $m_\beta(x) \leq \kappa x$ for $0 < x \leq x_0$, then $\text{Fl}_\beta(r) \leq \kappa r^2$ for $0 \leq r \leq x_0$; the same with $\geq$. In particular $\text{Fl}_\beta(r) = o(r^2)$ iff the profile of β is sparse.

Proof. (a) For each j , $r \mapsto 2(r\beta_j - |a_j|)_+$ is convex, vanishes at 0 and has derivative $2\beta_j 1[r\beta_j > |a_j|]$ off one point; sum by monotone convergence. (b) Each summand is convex and nondecreasing in $\beta_j \geq 0$. The scaling identity is a substitution. For the sum, split $\{|a_j| < x(\beta_1 + \beta_2)_j\}$ according to which of $\beta_{1,j}, \beta_{2,j}$ is larger: on the first part $(\beta_1 + \beta_2)_j \leq 2\beta_{1,j}$ and $|a_j| < 2x\beta_{1,j}$. (c) For $x \neq 0$ and $y \in \mathbb{R}$, $|x + y| - |x| - \text{sign}(x)y = 2(-\text{sign}(x)y - |x|)_+$; apply it with $x = a_j$, $y = rb_j$. (d) Integrate (a). If $\text{Fl}_\beta(r) \leq \varepsilon r^2$ for $r \leq r_0$, then, m_β being nondecreasing, $\varepsilon(2r)^2 \geq \text{Fl}_\beta(2r) \geq 2 \int_r^{2r} m_\beta \geq 2rm_\beta(r)$, i.e. $\kappa_\beta(r) \leq 2\varepsilon$. $\square$

Lemma 6.8 (Critical profiles oscillate). *Let $S = \{s_1 < s_2 < \dots\} \subseteq F$ be infinite, $\beta := v1_S$ with $v(s) = c2^{-s}$ ($c > 0$), and suppose that $\mathbf{a}(s) := |a_s|/v(s)$ is nonincreasing along S with $\mathbf{a}(s_i) \rightarrow 0$. Then $\kappa_\beta(y^+) \geq 2\kappa_\beta(y)$ at every point $y = \mathbf{a}(s_i)$ with $\mathbf{a}(s_{i+1}) < \mathbf{a}(s_i)$, where $\kappa_\beta(y^+) := \lim_{x \downarrow y} \kappa_\beta(x)$. Consequently $\limsup_{x \rightarrow 0^+} \kappa_\beta(x) \geq 2 \liminf_{x \rightarrow 0^+} \kappa_\beta(x)$.*

Proof. Put $R_i := \sum_{i' \geq i} v(s_{i'})$. As $s_{i'+1} \geq s_{i'} + 1$, $R_{i+1} \leq 2v(s_{i+1}) \leq v(s_i)$, so $R_i = v(s_i) + R_{i+1} \geq 2R_{i+1}$. At $y = \mathbf{a}(s_i)$ with $\mathbf{a}(s_{i+1}) < y$, monotonicity gives

$m_\beta(y) = R_{i+1} > 0$ and $m_\beta(y^+) \geq R_i$. Since $\mathbf{a}(s_i) \rightarrow 0$, such points y accumulate at 0, and $\kappa_\beta(y^+) \geq 2\kappa_\beta(y) \geq 2\inf_{x \leq y} \kappa_\beta(x)$ there. $\square$

For signature vectors $v_l = \delta_l h_l / n_l$ of a signature-ladder operator whose signature sets have bounded gaps (as for T_{final} , $S_l = \{2^l(2i+1) : i \geq 1\}$), and $|a_s| \approx v_l(s)^{1+\beta_0}$ on $S_l \cap F$ ($\beta_0 > 0$) with $S_l \setminus F$ finite, the profile of $v_l 1_{S_l \cap F}$ is sparse for $\beta_0 < 1$ (this part holds for every S_l), critical for $\beta_0 = 1$ and super-critical for $\beta_0 > 1$; in the critical case the local constant κ oscillates by a factor at least 2. For lacunary $S_l \cap F$ the cases $\beta_0 \geq 1$ need not be critical or super-critical (for $\beta_0 = 1$ and lacunary $S_l \cap F$ one has $\liminf_y m(y)/y = 0$).

Lemma 6.9 (Cushions). *Let T be admissible, $g \in \mathcal{C}(f)$, $t > 0$, and let $(B_\pm, \Theta_\pm)$ be a two-sided decomposition of g at scale t . For $j \in F$,*

$$(s_j B_+(j))_- \leq \frac{|a_j|}{t} + \mathfrak{f}_j^+, \quad (s_j B_-(j))_+ \leq \frac{|a_j|}{t} + \mathfrak{f}_j^-,$$

$$(s_j \Delta B(j))_- \leq \frac{2|a_j|}{t} + \mathfrak{f}_j^+ + \mathfrak{f}_j^-,$$

and $\sum_{j \in F} (\mathfrak{f}_j^+ + \mathfrak{f}_j^-) \leq t/(2q_0)$.

Proof. The first two inequalities are the definitions of $\mathfrak{f}_j^\pm$; the third follows from $(x - y)_- \leq x_- + y_+$; the sum is [I, Lemma 8.28]. $\square$

Thus at scale t a support coordinate is a contact of sign s_j whose anti-sign use is free up to the *cushion* $|a_j|/t$ on each side. The cushion grows as $t \rightarrow 0$; far support coordinates ($|a_j| \ll t^2$) behave like contacts, near ones are two-sided.

We next extend the one-sided transfer expansion of Lemma 2.18 to arbitrary base supports, allowing flips on the support whose profile is sparse uniformly along the sequence. For a first row f_j we write $s_i^{(j)} := \text{sign } a_j(i)$ on $F_j := \text{supp } a_j$, and use the coordinate kinds [1], [2], [3] of Subsection 2.4.

Lemma 6.10 (One-sided expansion at arbitrary base support). *Let T be admissible, $f \in S_{p^*}$, and let $f_j \rightarrow f$ in S_{p^*} (the supports F, F_j arbitrary; the constant sequence $f_j = f$ is allowed). Let $C_+ \geq 1$, $C_* \geq 0$ and $U_* \in \ell_1$, $U_* \geq 0$, and suppose that*

$$\bar{m}(y) := \sup_j \sum \{U_*(i) : i \in F_j, |a_j(i)| < yU_*(i)\} = o(y) \quad (y \rightarrow 0^+).$$

For every $\varepsilon_{\text{tr}} > 0$ and $A_0 \geq 1$ there are $t_1 > 0$ and $c_0 \in (0, \frac{1}{8}]$, depending only on $f, \varepsilon_{\text{tr}}, A_0, C_+, C_$ and $\bar{m}$, and $j_0 \in \mathbb{N}$, depending in addition on the sequence (f_j) , with the following property. Let $A_2 \geq 1$, $\gamma_B \in (0, 1]$ and $c_b := c_0 \gamma_B / A_2$. Let $j \geq j_0$, $t \in (0, t_1]$, and let (b, ω) , $b \in \ell_1$, $\omega_m \in c_{00}$, satisfy:*

- (OS1) $b(\xi_j) = 0$, $t\|b\|_1 \leq A_0$ and $\Gamma_w^{(j)}(b, \omega) \leq 2$;
- (OS2') $b(i) = 0$ for $i \notin F_j \cup K_j$, $z_j(i)b(i) \geq 0$ for $i \in K_j$, and $(s_i^{(j)}b(i))_- \leq C_+|a_j(i)|/t + C_*U_*(i)$ for $i \in F_j$;
- (OS3) every $k \in \text{supp } \omega_m$ is of kind [1], [2] (with A_2, γ_B) or [3] (side +) at f_j .

Then $G := b + \sum_m R_m^(\omega_m - d_m^{(j)}(\omega_m)w_{j,m})$ satisfies*

$$p^*(f_j + rG) \leq 1 + \frac{r^2}{2} (\Gamma_w^{(j)}(b, \omega) + \varepsilon_{\text{tr}}) \quad (0 < r \leq c_b t).$$

The same holds for side— pairs (in (OS2') and [3] the signs reversed, $(s_i^{(j)}b(i))_+$ in place of $(s_i^{(j)}b(i))_-$ and $-c_b t \leq r < 0$.

Proof. We run the proof of Lemma 2.18 and record the changes; side +. In Step 0 the number $a_{\min}$ and the condition on $\min_{i \in F} |a_j(i)|$ are dropped (they served only to exclude flips); all other limits and transfer data are as there. Transfer data [I, Lemmas 7.4, 7.5] exist at every first row; their construction does not involve F . In that proof the sequence enters only through the index j_0 beyond which the forced data of f_j are within the factors of Step 0 of those of f and the transfer data persist; every condition on c_b and t_1 involves only quantities of f , ε_{tr} and A_0 (and, below, C_+ , C_* , $\bar{m}$).

Step 1 (base). By [I, Lemma 7.1(b)] at f_j , $G_b^{(j)}(a_j + rb) = \text{Exc}^{(j)}(rb) + \nu_j \Psi(rU^*b/\nu_j)$, and by (OS2') the summands of $\text{Exc}^{(j)}(rb)$ vanish off F_j . On F_j , Lemma 6.7(c) gives the summand $2(r\beta_i - |a_j(i)|)_+$ with $\beta_i := (s_i^{(j)}b(i))_-$. Assume $c_b C_+ \leq \frac{1}{4}$. Then $r\beta_i \leq |a_j(i)|/4 + rC_*U_*(i)$; a positive summand forces $rC_*U_*(i) > \frac{3}{4}|a_j(i)|$, so $|a_j(i)| < 2C_*rU_*(i)$, and then $2(r\beta_i - |a_j(i)|)_+ \leq 2r\beta_i \leq \frac{1}{2}|a_j(i)| + 2rC_*U_*(i) \leq 3rC_*U_*(i)$. Hence

$$\text{Exc}^{(j)}(rb) \leq \text{Fl}_*(r) := 3C_*r\bar{m}(2C_*r),$$

and $\text{Fl}_*(r)/r^2 \rightarrow 0$ as $r \rightarrow 0$, independently of j , t and (b, ω) . Fix $r_1 > 0$ with $\text{Fl}_*(r) \leq \min(1, \varepsilon_{\text{tr}}q_0/20)r^2$ for $0 < r \leq r_1$, and require $t_1 \leq r_1$ (then $r \leq c_b t \leq r_1$). The Hilbert part is estimated as in Lemma 2.18, and

$$G_b^{(j)}(a_j + rb) \leq \hat{G}_b := \frac{r^2}{2}h^{(j)}(b)(1 + K_A c_b) + \text{Fl}_*(r).$$

Step 2 (blocks) is unchanged. *Step 3 (rebalancing).* Since $\text{Fl}_*(r) \leq r^2$, the bound $|\varepsilon_m| \leq K_3 r^2$ holds after enlarging K_3 by an additive constant depending only on f ; the error term $|q^*(A) - 1| + q^*(A - a_j) \leq 2q^*(rb) \leq 2(1 + \|U\|)c_b A_0$ is unchanged. *Step 4 (constants).* The only new contribution to $\hat{\Gamma}$ is $q_0^{(j)}\text{Fl}_*(r) \leq \frac{r^2}{2} \cdot \frac{\varepsilon_{\text{tr}}}{10}$ (as $q_0^{(j)} \leq 1$); absorb it by replacing $\varepsilon_{\text{tr}}/5$ by $\varepsilon_{\text{tr}}/10$ in the other conditions of that step. The new condition $c_b \leq 1/(4C_+)$ is of the form $c_b \leq \kappa_1$, and the new condition $t_1 \leq r_1$ does not involve A_2 or γ_B . The side— case is the same with $-r, -b, -\omega$. $\square$

With $C_* = 0$ (or $U_* = 0$) the hypothesis on $\bar{m}$ is void and no flips occur. For the constant sequence $f_j = f$ and U_* with $m_{U_*}(y) = o(y)$ the lemma is the one-sided expansion with sparse flips used in Subsection 6.5.

Lemma 6.11 (First-order pin and bounded free switching at any F). *Let T be admissible, $g \in \mathcal{C}(f)$, $\eta \leq \eta_*$ with $\eta_{\Gamma}(\eta) \leq 1$, $0 < t \leq \min(t_{\eta}, 1)$, and let $(B_{\pm}, \Theta_{\pm})$ be a two-sided decomposition of g at scale t .*

- (a) $|\sum_m \sum_k \Delta\theta_{k,m} u_{k,m}(\hat{z})| = |\Delta B(\hat{z})| \leq t/q_0$.
- (b) *For every finite set $\mathcal{U}$ of pairs (k, m) , $m \in I$, there is $C_{\mathcal{U}} < \infty$, depending only on f and $\mathcal{U}$, with*

$$\max_{(k,m) \in \mathcal{U}} |\Delta\theta_{k,m}| \leq C_{\mathcal{U}} \left(1 + \left\| \sum_{(k,m) \notin \mathcal{U}} \Delta\theta_{k,m} u_{k,m} \right\|_1 \right).$$

Proof. (a) is [I, (15)] together with $|q_0 B_{\pm}(\hat{z})| \leq t/2$ [I, Lemma 8.7(a)]. (b) Let $P^{\perp}$ be the projection onto $e^{\perp}$ and $\Phi(c) := (P^{\perp}U^* \sum_{\mathcal{U}} c_{k,m} u_{k,m}, (\sum_{\mathcal{U}} c_{k,m} u_{k,m})(\hat{z}))$ for $c \in \mathbb{R}^{\mathcal{U}}$. If $\Phi(c) = 0$, then $U^* \sum_{\mathcal{U}} c_{k,m} u_{k,m} = \mu e$ for some μ , hence $\sum_{\mathcal{U}} c_{k,m} u_{k,m} =$

$(\mu/\nu)a$ (U^* is injective), hence $0 = (\mu/\nu)a(\hat{z}) = \mu/\nu$; so $\sum_{\mathcal{U}} c_{k,m} u_{k,m} = 0$ and $c = 0$ by (T-b). Thus $\|c\|_{\infty} \leq C_{\mathcal{U}} \|\Phi(c)\|$. Apply this to $c = \Delta\theta|_{\mathcal{U}}$: by [I, (15)], $\sum_{\mathcal{U}} \Delta\theta_{k,m} u_{k,m} = -\Delta B - R$ with $R := \sum_{(k,m) \notin \mathcal{U}} \Delta\theta_{k,m} u_{k,m}$; $\|P^{\perp} U^* \Delta B\| \leq 2(2\nu/q_0)^{1/2}$ by [I, Lemma 8.7(d)], $|\Delta B(\hat{z})| \leq t/q_0 \leq 1/q_0$ by (a), $\|P^{\perp} U^* R\| \leq \|U\| \|R\|_1$ and $|R(\hat{z})| \leq (1 + \|U\|) \|R\|_1$. $\square$

Part (b) is [I, Lemma 8.29] without the hypothesis that F be finite: the possibility $a \in Y$ is harmless because the a -direction is pinned at first order by (a).

Lemma 6.12 (Un-switching pairs). *Let T be admissible and let $l = (k, m)$ be a carrier with $w_m(k) = 0$ (a d -neutral carrier; then $k \in Q_m$). For $c \in \mathbb{R}$ the pair $P_l(c) := (cu_{k,m}, -(c/\lambda_{k,m})e_k$ in block m) represents the zero functional, $d_m(e_k) = 0$, its base part vanishes at ξ , and*

$$\Gamma_w(P_l(c)) \leq c^2 C_{\text{un}}(l)^2, \quad C_{\text{un}}(l)^2 := \frac{q_0 \|U\|^2}{\nu} + \frac{\sigma_m}{m^2 C_m}.$$

Consequently, if (b, ω) represents g , so does $(b, \omega) + P_l(c)$, and $\sqrt{\Gamma_w((b, \omega) + P_l(c))} \leq \sqrt{\Gamma_w(b, \omega)} + |c| C_{\text{un}}(l)$.

Proof. $M_m > 0$ and $w_m(k) = 0$ give $k \in Q_m$. $R_m^* e_k = \lambda_{k,m} u_{k,m}$ and $d_m(e_k) = \Phi_m(k)^2 w_m(k)/C_m = 0$, so the pair represents $cu_{k,m} - cu_{k,m} = 0$. By [I, Lemma 1.7] off P_m , $\lambda_{k,m} q_0 u_{k,m}(\hat{z}) = \zeta_m(k) = \sigma_m \Phi_m(k)^2 w_m(k)/C_m = 0$, so $u_{k,m}(\xi) = 0$. Further $h(u_{k,m}) \leq \|U\|^2 \|u_{k,m}\|_1^2 / \nu \leq \|U\|^2 / \nu$ and $H_m(e_k / \lambda_{k,m}) \leq \Phi_m(k)^2 / (\lambda_{k,m}^2 C_m) = 1/(m^2 C_m)$. $\sqrt{\Gamma_w}$ is a seminorm on pairs. $\square$

6.3. Engineered recovery with raised support.

Definition 6.13 (Two-piece data at arbitrary base support). Two-piece data for $g \in X^*$ at f are defined by [I, Definition 7.7] read without the hypothesis that F be finite: a pair (b, ω) , $b \in \ell_1$, $\omega_m \in c_{00}$, $\text{supp } \omega_m \subseteq Q_m$, is side-+ (side-) admissible if it represents g , $b_j = 0$ for $j \notin F \cup K$ and $z_j b_j \geq 0$ (≤ 0) for $j \in K$; no condition is imposed on F . We put $b^{\theta} := \frac{1}{2}(b^+ + b^-)$, $v := b^+ - b^-$, and Δd_m , κ_w as in [I, Definition 7.7]. The data satisfy *one-sided cushion sparsity* (CS-side) if the profiles of $(sb^+)_-$ and $(sb^-)_+$ (restricted to F) are sparse:

$$m_{(sb^+)_-}(x) = o(x) \quad \text{and} \quad m_{(sb^-)_+}(x) = o(x) \quad (x \rightarrow 0^+).$$

No condition is imposed on b^{θ} ; for finite F , (CS-side) is void.

Theorem 6.14 (Engineered recovery with raised support). *Let T be admissible, $f \in S_{p^*}$ (F arbitrary), and let $g \in \mathcal{C}(f)$ carry two-piece data $(b^{\pm}, \omega^{\pm})$ satisfying (CS-side). Let $\rho \in (0, 1)$ with $\rho^2 \kappa_w < 1$, and assume that $I_- := \{m : \Delta d_m < 0\}$ satisfies the scrambling condition (SC) of [I, Definition 7.16] (no assumption if $I_- = \emptyset$). Then there are norm-attaining first rows $f'_i \rightarrow f$ with finite base supports and $g'_i \in \mathcal{C}(f'_i)$ with $g'_i \rightarrow \rho g$. In particular $(f, \rho g) \in \text{cl NA}((c_0, p), \ell_2^2)$, and $(f, g) \in \text{cl NA}((c_0, p), \ell_2^2)$ if $\kappa_w \leq 1$.*

Proof. We modify the proof of [I, Theorem 7.18] and its approximants [I, Definition 7.12]. Put $\delta := (1 - \rho^2 \kappa_w)/2$ and $m(x) := m_{(sb^+)_-}(x) + m_{(sb^-)_+}(x)$.

Approximants. For a window N_w , a scale $s_1 \in (0, 1]$ and a cut-off $N'' > N_w$ put $\mathbf{m}_j := 4\rho s_1 |b_j^{\theta}|$ for $j \in K \cap [1, N_w]$, $\mathbf{r}_j := (8\rho s_1 |b_j^{\theta}| - |a_j|)_+$ for $j \in F \cap [1, N_w]$, and

$$a'' := a1_{[1, N_w]} + \sum_{j \in K \cap [1, N_w]} \mathbf{m}_j z_j e_j^* + \sum_{j \in F \cap [1, N_w]} \mathbf{r}_j s_j e_j^*,$$

so that $|a_j''| = \max(|a_j|, 8\rho s_1|b_j^\theta|)$ with sign s_j on $F \cap [1, N_w]$ (the support is *raised* where the average data need it). All other definitions ($a' = a''/q^*(a'')$, e' , $z'_j = z_j$ for $j \leq N''$ and $z'_j = 0$ beyond, $\hat{x}'$, x' , f') are those of [I, Definition 7.12]. The window N_w is chosen *after* s_1 , with

$$(N0) \quad \alpha(N_w) \leq s_1^2, \quad \rho \sum_{j \in F, j > N_w} (s_j v_j)_- \leq \delta s_1/32,$$

and then $N'' > N_w$ with [I, (11), (12)] and $\rho V_{>N''} \leq \delta s_1/32$, where now $V_{>N''} := \sum_{j \in F \cup K, j > N''} |v_j|$. This is possible because $v \in \ell_1$. Then $\text{supp } a' = (F \cap [1, N_w]) \cup \{j \in K \cap [1, N_w] : b_j^\theta \neq 0\}$ is finite and $z' = \text{sign } a'$ there, so f' attains its norm and has finite base support ([I, Lemma 7.13(a)]); support coordinates in (N_w, N'') become contacts of f' with signs s_j , those beyond N'' free coordinates. Moreover $\|a'' - a\|_1 \leq s_1^2 + 12\rho s_1\|b^\theta\|_1$, so (E1) of [I, Lemma 7.13] holds with 4 replaced by 12, an additional s_1^2 , and K_e replaced by $K'_e := 2\|U\|(12\rho\|b^\theta\|_1 + 1)/\nu$; (E2) is unchanged, since $\hat{x}' - \hat{z} = -z1_{(N'', \infty)} + U(e' - e)$. The remaining parts (E3)–(E5) of [I, Lemma 7.13] and [I, Lemmas 7.14, 7.15, 7.17] use the approximants only through (E1), (E2) and the forced data of f' ; they hold with the adjusted constants. At late stages $q^*(a'') \leq 2$, hence $|a'_j| \geq |a''_j|/2 \geq |a_j|/2$ on $F \cap [1, N_w]$.

Step 0. The condition (T_1) “ $\rho T_0(\|b^+\|_\infty + \|b^-\|_\infty) \leq a_{\min}/8$ ” is replaced by

$$2\rho|\tau|m(4\rho|\tau|) \leq \min\left(\frac{\delta}{16}, 1\right)\tau^2 \quad (0 < |\tau| \leq T_0),$$

which holds for small T_0 by (CS-side) and Lemma 6.7(d); the constant $K_\#$ is fixed after this uniform bound, so that the bounds $\hat{G}_b \leq K_\#\tau^2$ below include the flips. The other conditions and the order of choices are unchanged.

Steps 1, 2 are verbatim, with $\beta^\theta := b^\theta 1_{[1, N_w]}$ and $\beta^\pm := b^\pm 1_{[1, N_w]} \pm \frac{1}{2}v1_{(N_w, \infty)}$; $g'' - g = -b^\theta 1_{(N_w, \infty)} + (\text{block terms}) \rightarrow 0$.

Step 3 (base). We claim

$$\text{Exc}'(\tau B^\diamond) \leq \frac{1}{2}\rho|\tau|V_{>N''} + \rho|\tau| \sum_{j \in F, j > N_w} (s_j v_j)_- + \text{Fl}(\tau), \quad \text{Fl}(\tau) \leq 2\rho|\tau|m(4\rho|\tau|),$$

and $\text{Exc}'(\tau B^\theta) = 0$. On $\text{supp } a'$, $a'_j + \tau B_j^\diamond = (1 - \tau\rho c)a'_j + \tau\rho\beta_j^\diamond$ with $|\tau\rho c| \leq \frac{1}{2}$, and the summand of Exc' equals $|x+y| - |x| - \text{sign}(x)y$ with $x := (1 - \tau\rho c)a'_j$, $y := \tau\rho\beta_j^\diamond$. (i) $j \in F \cap [1, N_w]$: $s_j x \geq |a'_j|/4$. For $\diamond = \theta$ ($|\tau| \leq s_1$), $|y| \leq \rho s_1|b_j^\theta| \leq |a'_j|/8$: no flip. For $\diamond = \pm$ ($\text{sign } \tau = \diamond$) the anti-sign part of y is at most $\rho|\tau|\beta_j$ with $\beta = (sb^+)_-$, resp. $(sb^-)_+$; since $|a'_j| \geq |a_j|$, a flip needs $|a_j| < 4\rho|\tau|\beta_j$ and costs at most $2\rho|\tau|\beta_j$ (Lemma 6.7(c)); the sum is at most $2\rho|\tau|m(4\rho|\tau|)$. (ii) Window contacts with mass, and contacts in $[1, N'']$: verbatim as in [I, Theorem 7.18]. (iii) $j \in F \cap (N_w, N'')$: $j \notin \text{supp } a'$, $z'_j = s_j$, $\beta_j^\theta = 0$ and $\beta_j^\pm = \pm \frac{1}{2}v_j$; for $\diamond = \text{sign } \tau$ the summand is $2(s_j\tau\rho\beta_j^\diamond)_- = \rho|\tau|(s_j v_j)_-$. (iv) $j > N''$, $j \in F \cup K$: $z'_j = 0$ and the summand is at most $\frac{1}{2}\rho|\tau||v_j|$ (and 0 for θ). (v) Off $F \cup K$, $b^\pm$ and v vanish. This proves the claim. Hence $\hat{G}_b$ of [I, Theorem 7.18] acquires the additional terms $\rho|\tau| \sum_{j \in F, j > N_w} (s_j v_j)_- + \text{Fl}(\tau)$.

Steps 4, 5. The rebalancing is unchanged ($\text{Fl}(\tau) \leq \tau^2$). In Step 5 the first-order base terms are at most $\frac{1}{2}\rho|\tau|V_{>N''} + \rho|\tau| \sum_{j > N_w} (s_j v_j)_- + |\tau|o(s_1) \leq |\tau|\delta s_1(\frac{1}{64} + \frac{1}{32} + \frac{1}{64}) \leq \delta\tau^2/16$ for $|\tau| > s_1$ at late stages, as in [I, Theorem 7.18], and the flips contribute $c'\text{Fl}(\tau) \leq \frac{\tau^2}{2} \cdot \frac{\delta}{8}$. Hence $p^*(f' + \tau g') \leq 1 + \frac{\tau^2}{2}(\rho^2\kappa_w + \frac{6\delta}{8}) \leq 1 + \frac{\tau^2}{2}(1 - \delta)$ for $0 < |\tau| \leq T_0$. *Step 6* is verbatim. $\square$

The raises do for support coordinates what the window masses do for contacts: they absorb the two-sided use of the average data below scale s_1 , at total mass $O(s_1)$ whatever b^θ is. For data satisfying the two-sided sparsity of $|b^\theta|$, $(sb^+)_-$, $(sb^-)_+$, or bounded ratios $|b^\pm| \leq C|a|$ on F , the theorem was obtained without raising; the point here is that b^θ need not be sparse.

Theorem 6.15 (Flip-extended engineered recovery). *Let $T, f, g, (b^\pm, \omega^\pm), \rho$ and I_- be as in Theorem 6.14, but instead of (CS-side) and $\rho^2 \kappa_w < 1$ assume that for some $t_0 > 0$*

$$\rho^2 \kappa^{\text{fl}} < 1, \quad \kappa^{\text{fl}} := \max_{\diamond=\pm} \left(\Gamma_w(b^\diamond, \omega^\diamond) + \sup_{0 < r \leq t_0} \text{Fl}_r(b^\diamond) \right),$$

where $\text{Fl}_r(b^+) := 2q_0 \text{Fl}_{(sb^+)_-}(r)/r^2$ and $\text{Fl}_r(b^-) := 2q_0 \text{Fl}_{(sb^-)_+}(r)/r^2$. Then $(f, \rho g) \in \text{clNA}((c_0, p), \ell_2^2)$.

Proof. Run the proof of Theorem 6.14 with $\delta := (1 - \rho^2 \kappa^{\text{fl}})/2$ and $T_0 \leq t_0/(2\rho)$, without the flip condition of Step 0. In Step 3(i), at late stages $s_j x \geq |a_j|/(1 + \epsilon)$ with $\epsilon \rightarrow 0$ along the construction ($c \rightarrow 0, q^*(a'') \rightarrow 1$), so $\text{Fl}(\tau) \leq \text{Fl}_\beta(\rho|\tau|(1 + \epsilon)) \leq \frac{\rho^2 \tau^2}{2q_0} (1 + \epsilon)^2 \sup_{0 < r \leq t_0} \text{Fl}_r(b^\diamond)$; this is $O(\tau^2)$, so $K_\#$ exists, and its contribution $c' \text{Fl}(\tau)$ to $\hat{\Gamma}$ is at most $\frac{\rho^2 \tau^2}{2} (\sup_r \text{Fl}_r(b^\diamond) + o(1))$ since $c' \rightarrow q_0$. The θ -data carry no flips, and $\Gamma_w(b^\theta, \omega^\theta) \leq \kappa_w \leq \kappa^{\text{fl}}$ by convexity. Step 5 then gives $p^*(f' + \tau g') \leq 1 + \frac{\tau^2}{2} (\rho^2 \kappa^{\text{fl}} + \frac{5\delta}{8} + o(1)) \leq 1 + \frac{\tau^2}{2} (1 - \delta)$ at late stages. $\square$

Theorem 6.16 (Windowed recovery at arbitrary base support). *Let T be admissible, $f \in S_{p^*}$ (F arbitrary), $g \in \mathcal{C}(f)$, $\rho \in (0, 1)$ and $\eta_0 \in (0, 2]$ with $\rho^2(1 + \eta_0) \leq (1 + \rho^2)/2$. Suppose there are $c_b > 0$, $C_+ \geq 1$, $\beta \in \ell_1$ with $\beta \geq 0$ and $m_\beta(x) = o(x)$, and triples (T_j, n_j, K_j) with $T_j \rightarrow 0$, $n_j \geq 24\rho^2 K_j/(c_b(1 - \rho^2))$ and $K_j T_j/n_j \rightarrow 0$, such that for every j and every $t \in \{T_j 2^{1-s} : 1 \leq s \leq n_j\}$ there are a functional g_t and two-piece data $(b^\pm, \omega^\pm)$ for g_t with:*

- (i) $p^*(f + r g_t) \leq 1 + \frac{r^2}{2} (1 + \eta_0)$ for $|r| \leq c_b t$;
- (ii) $p^*(g - g_t) \leq K_j t$;
- (iii) $\Gamma_w(b^\pm, \omega^\pm) \leq 1 + \eta_0/2$;
- (iv) $(s_j b_j^+)_- \leq C_+ |a_j|/t + \beta_j$ and $(s_j b_j^-)_+ \leq C_+ |a_j|/t + \beta_j$ for $j \in F$;
- (v) either $\Delta d_m \geq 0$ for all these data and all m , or the set of blocks m in which some of these data have $\Delta d_m < 0$ satisfies (SC).

Then $(f, \rho g) \in \text{clNA}((c_0, p), \ell_2^2)$, along norm-attaining first rows with finite base supports.

Proof. By [I, Lemma 8.43] (whose proof uses only (i), (ii), $g \in \mathcal{C}(f)$ and the convexity of p^*), $\rho \bar{g}_j \in \mathcal{C}(f)$ for large j and $\rho \bar{g}_j \rightarrow \rho g$, where $\bar{g}_j$ is the average of the g_t over the n_j scales. The averages $\bar{D}_j$ of the data are two-piece data for $\bar{g}_j$ (the representations and the side conditions are linear), $\Delta \bar{d}_m$ is the average of the Δd_m , so (v) passes to $\bar{D}_j$ (a subset of a set of blocks satisfying (SC) satisfies (SC)), and $\kappa_w(\rho \bar{D}_j) \leq \rho^2(1 + \eta_0/2) < 1$ by convexity of Γ_w . By convexity of $x \mapsto x_\pm$, (iv) holds for $\bar{D}_j$ with t replaced by the least scale $T'_j := T_j 2^{1-n_j}$, and Lemma 6.7(b) gives $m(x) \leq 2m_{C_+|a|/T'_j}(2x) + 2m_\beta(2x) = 2m_\beta(2x) = o(x)$ for $x < T'_j/(2C_+)$: (CS-side). Theorem 6.14 applied to the mate $\rho \bar{g}_j$ with data $\rho \bar{D}_j$ gives $(f, \rho' \rho \bar{g}_j) \in \text{clNA}$ for all $\rho' < 1$, hence $(f, \rho \bar{g}_j) \in \text{clNA}$; let $j \rightarrow \infty$. $\square$

All window theorems of this section are instances of Theorem 6.16 (Lemma 6.10 turns (iv) and the block size conditions into (i)), applied at f or, through Theorem 2.19, at nearby first rows.

We turn to the exact one-sided coefficient. We say that f is *block-tame* if every Q_m is finite, there are no degenerate peaks, and (MS) [I, Definition 5.5] holds at ξ ; nothing is assumed about F or K (for finite F this is (BT) of [I, Definition 7.8]). For g with $g(\xi) = 0$ the one-sided coefficients $\gamma^\pm(g)$ are defined as in [I, Definition 7.7], with side admissibility as in Definition 6.13.

Theorem 6.17 (Exact one-sided coefficient at arbitrary base support). *Let T be admissible, $f \in S_{p^*}$ block-tame (F arbitrary), and $g \in X^*$ with $g(\xi) = 0$. Then:*

- (a) $\liminf_{t \rightarrow 0^\pm} 2(p^*(f + tg) - 1)/t^2 \geq \gamma^\pm(g)$;
- (b) if $\gamma^\pm(g) < \infty$, the infimum defining it is attained;
- (c) for every side-+ admissible pair (b, ω) for g ,

$$\limsup_{t \rightarrow 0^+} \frac{2(p^*(f + tg) - 1)}{t^2} \leq \Gamma_w(b, \omega) + 2q_0 \limsup_{t \rightarrow 0^+} \frac{\text{Fl}_{(sb)_-}(t)}{t^2},$$

and the same for $t \rightarrow 0^-$, side-- pairs and $(sb)_+$;

- (d) if some side- $\pm$ pair attaining $\gamma^\pm(g)$ has a sparse profile $(sb)_\mp$, then $\lim_{t \rightarrow 0^\pm} 2(p^*(f + tg) - 1)/t^2 = \gamma^\pm(g)$;
- (e) every $g \in \mathcal{C}(f)$ has two-piece data with $\kappa_w = \max(\gamma^+(g), \gamma^-(g)) \leq 1$.

Proof. (a), (b): Steps 2–4 of the proof of [I, Theorem 7.10] use F only through the test points of Step 1, and the duality of Step 4 only through the cone $C_+ := \{\chi_c \in c_{00} : \chi_c = 0 \text{ on } F, z_j \chi_{c,j} \leq 0 \text{ on } K\}$, whose polar condition $\sup_{\chi_c \in C_+} 2b_\pi(\chi_c) < \infty$ is exactly side-+ admissibility in the sense of Definition 6.13 (no condition on F). We replace the test points of Step 1, which used a Gram system on F , by the following ones, valid for every F . Let $k \in H$ with $\langle k, e \rangle = 0$ and $\chi_c \in C_+$, put $\kappa := \|k\|^2/(2q_0)$, $k_2 := -\kappa e$, $\chi := Uk + \chi_c$, $\chi_2 := Uk_2$, $\eta_t := \xi + t\chi + t^2\chi_2$, $\mathbf{h}_t := q_0 e + tk + t^2 k_2$ and $Z_t := \eta_t - U\mathbf{h}_t$. Since $\xi = q_0 z + q_0 Ue$, $Z_t = q_0 z + t\chi_c$; as χ_c vanishes on F , moves contacts inward and is finitely supported, $\|Z_t\|_\infty \leq q_0$ for small t . Moreover $\|\mathbf{h}_t\|^2 = (q_0 - t^2\kappa)^2 + t^2\|k\|^2 = q_0^2 + t^4\kappa^2$. As $B_{q^{**}} = B_{\ell_\infty} + U(B_H)$, $q^{**}(\eta_t) \leq \max(\|Z_t\|_\infty, \|\mathbf{h}_t\|) \leq q_0 + O(t^4)$, and $a(\eta_t - \xi) = t\langle U^*a, k \rangle + ta(\chi_c) + t^2\langle U^*a, k_2 \rangle = -t^2\kappa\nu$. Hence the base excess satisfies $E_q(t\chi + t^2\chi_2) \leq \frac{t^2}{2} \frac{\nu}{q_0} \|k\|^2 + O(t^4)$, which is the bound of Step 1. The block test of Step 2 uses χ_2 only through $q(\chi_2)$ and through a term $w(t^2\delta_2)$ that cancels, and accepts $\chi_2 = Uk_2 \in c_0$; the rest is unchanged. (c) In the proof of [I, Proposition 7.9], $G_b(a + tb) = \text{Fl}_{(sb)_-}(t) + \nu\Psi(tU^*b/\nu)$ by [I, Lemma 7.1(b)] and Lemma 6.7(c) (off F the summands vanish by side admissibility). If the $\limsup$ of $\text{Fl}_{(sb)_-}(t)/t^2$ is infinite there is nothing to prove; otherwise $\hat{G}_b := \text{Fl}_{(sb)_-}(t) + \frac{t^2}{2}h(b)(1 + K_1t) = O(t^2)$, the rebalancing of that proof applies ([I, Proposition 7.3] and the transfer data hold at every f), and the additional term in $\hat{\Gamma}$ is $q_0\text{Fl}_{(sb)_-}(t)$. (d) follows from (a) and (c), and (e) from (a), (b) and $p^*(f + tg) \leq s(t) \leq 1 + t^2/2$. $\square$

Corollary 6.18 (Block-tame first rows with infinite support). *Let T be admissible and $f \in S_{p^*}$ block-tame (F arbitrary), and suppose that $V_f := \sum_m (\sum_{k \in Q_m} \lambda_{k,m} |u_{k,m}| + |R_m^* w_m|)$ has a sparse profile. Then $(f, g) \in \text{clNA}((c_0, p), \ell_2^2)$ for every $g \in \mathcal{C}(f)$ such that $|g|1_F$ has a sparse profile.*

Proof. By Theorem 6.17(e) there are two-piece data with $\kappa_w \leq 1$, $b^\pm = g - \sum_m R_m^*(\omega_m^\pm - d_m(\omega_m^\pm)w_m)$ and $\omega_m^\pm \in \mathbb{R}^{Q_m}$; hence $|b^\pm - g| \leq CV_f$ with $C := \max_\pm \max(\|\omega^\pm\|_\infty, \max_m |d_m(\omega_m^\pm)|)$, and $(sb^+)_-, (sb^-)_+ \leq |g| + CV_f$ on F . By Lemma 6.7(b) these profiles are sparse. For block-tame f every set of blocks satisfies (SC) (the remark after [I, Definition 7.16]; its proof concerns the blocks only). Theorem 6.14 applies for every $\rho < 1$. $\square$

6.4. Window-pinned mates and room off the support. In this subsection T is a signature-ladder operator (see the standing conventions), with windows $\mathcal{W}(l)$, $T_{\text{hi}}(l)$, $T_{\text{lo}}(l)$, n_l^w , carriers l , u_l , λ_l , $\Delta\theta_l$ and $v_l := \delta_l h_l / n_l$ as in [I, Section 8]. Definitions [I, 8.3, 8.30, 8.33] are read verbatim for arbitrary F ; in particular $J_\gamma = \{j \notin F : |z_j| \leq 1 - \gamma\}$ excludes F . For f with arbitrary F , a mate $g \in \mathcal{C}(f)$ is *window-pinned* if the condition of [I, Definition 8.30] holds.

Proposition 6.19 (The clamped window certificate). *Let T be a signature-ladder operator, $f \in S_{p^*}$ (F arbitrary), $g \in \mathcal{C}(f)$, $\eta \leq \eta_*$ with $\eta_\Gamma(\eta) \leq 1$, l_* a window index, $t \in \mathcal{W}(l_*)$ with $t \leq \min(t_\eta, 1)$, and let $(B_\pm, \Theta_\pm)$ be a two-sided decomposition of g at scale t with*

$$(\text{PIN}_K) \quad \sum_l |\Delta\theta_l| \leq Kt \quad (K \geq 1).$$

Let b^{cl} be the clamp of [I, Lemma 8.28] ($b^{\text{cl}} := 0$ off F), $M_t := \min\{M : \alpha(M) \leq t^2\}$, $F_t := F \cap [1, M_t]$, $a_t := a|_{F_t}$, $\kappa_t := (b^{\text{cl}}|_{F_t})(\hat{z})/a_t(\hat{z})$, $b_t := b^{\text{cl}}|_{F_t} - \kappa_t a_t$, let ω_m^c be as in [I, Definition 8.14], and $c_t := (b_t, (\omega_m^c)_m)$. Assume $Kt \leq 1$ and $t^2 \leq 1/(2(1 + \|U\|))$, and put $K_\kappa := \frac{1}{2q_0} + (1 + \|U\|)(2K + \frac{3}{2q_0} + 2)$. Then:

- (a) $a_t(\hat{z}) \geq \frac{1}{2}$; c_t is a balanced finite certificate with $\text{supp } b_t \subseteq F_t$ finite and $b_t(\xi) = 0$; on $\text{supp } \omega_m^c$ one has $k \in Q_m$, $\text{gap}_m(k) \geq t^2$ and $|\omega_m^c(k)| \leq 2\text{gap}_m(k)/t$;
- (b) $|\kappa_t| \leq 2K_\kappa t$, and if $2K_\kappa t^2 \leq 1$ then $|b_t(j)| \leq 3|a_j|/t$ for all j ;
- (c) $\|B_+ - b_t\|_1 \leq (2K + \frac{3}{2q_0} + 2 + 2K_\kappa)t$ and $\|g - g_{c_t}\|_1 \leq K'_2 Kt$;
- (d) $\sqrt{\Gamma_w(c_t)} \leq \sqrt{1 + \eta_\Gamma(\eta)} + K'_4 Kt$,

where K'_2, K'_4 depend only on f , N and the operator.

I, Proposition 8.13(b),(c). hold with this K : their proofs use only (PIN_K) , [I, Lemma 8.7(b)] and [I, Lemma 8.8]. (a) $|a_t(\hat{z}) - 1| = |(a - a_t)(\hat{z})| \leq \|\hat{z}\|_\infty \alpha(M_t) \leq (1 + \|U\|)t^2 \leq \frac{1}{2}$, since $a(\hat{z}) = 1$; so $b_t(\hat{z}) = 0$ and $b_t(\xi) = q_0 b_t(\hat{z}) = 0$. The statement on ω_m^c is [I, Proposition 8.15(a)], whose proof does not use F . (b) We have $(b^{\text{cl}}|_{F_t})(\hat{z}) = B_+(\hat{z}) - (B_+|_{F^c})(\hat{z}) - ((B_+ - b^{\text{cl}})|_{F})(\hat{z}) - (b^{\text{cl}}|_{F \setminus F_t})(\hat{z})$, with $|B_+(\hat{z})| \leq t/(2q_0)$ [I, Lemma 8.7(a)], $\|B_+|_{F^c}\|_1 \leq Kt + t/q_0$ [I, Proposition 8.13(b)], $\|(B_+ - b^{\text{cl}})|_{F}\|_1 \leq \|\Delta B|_{F}\|_1 + t/(2q_0) \leq Kt + t/(2q_0)$ [I, Lemma 8.28], and $\|b^{\text{cl}}|_{F \setminus F_t}\|_1 \leq (2/t)\alpha(M_t) \leq 2t$. As $\|\hat{z}\|_\infty \leq 1 + \|U\|$, this gives $|(b^{\text{cl}}|_{F_t})(\hat{z})| \leq K_\kappa t$ and $|\kappa_t| \leq 2K_\kappa t$. Then $|b_t(j)| \leq 2|a_j|/t + 2K_\kappa t|a_j| \leq 3|a_j|/t$. (c) $B_+ - b_t = B_+|_{F^c} + (B_+ - b^{\text{cl}})|_{F} + b^{\text{cl}}|_{F \setminus F_t} + \kappa_t a_t$; use the bounds of (b). The block part of $g - g_{c_t}$ is bounded as in the proof of [I, Proposition 8.15(c)], which uses only [I, Lemmas 8.8, 8.10], (P2) and [I, Proposition 8.13(a),(c)]. (d) As in the proof of [I, Proposition 8.15(d)]: $\sqrt{\Gamma_w}$ is a seminorm, $\Gamma_w(B_+, \Theta_+) \leq 1 + \eta_\Gamma(\eta)$, and the remainder pair has $q_0 h(x) \leq \|U\|^2 \|x\|_1^2 / \nu$ and $\sigma_m H_m(X_m) \leq (\sum_k \lambda_{k,m} |X_m(k)|)^2 / C_m$, both at most $(\text{const} \cdot Kt)^2$ by (c). $\square$

Theorem 6.20 (Window-pinned mates at arbitrary base support). *Let T be a signature-ladder operator and $f \in S_{p_N^*}$ (F arbitrary). If $g \in \mathcal{C}(f)$ is window-pinned at f , then $(f, g) \in \text{clNA}((c_0, p_N), \ell_2^2)$. No condition on the contact set, on the rooms of the signature sets or on the blocks of f is needed.*

Proof. Fix $\rho \in (0, 1)$, choose $\eta_0 \in (0, 2]$ with $\rho^2(1 + \eta_0) \leq (1 + \rho^2)/2$, $\kappa_0 := \frac{1}{2}(\sqrt{1 + \eta_0/2} - 1)$, $\varepsilon_{\text{tr}} := \eta_0/2$, and $\eta \leq \eta_*$ with $\eta_\Gamma(\eta) \leq 1$ and $\sqrt{1 + \eta_\Gamma(\eta)} \leq 1 + \kappa_0$. Let c_b, t_1 be given by Lemma 6.10 for the constant sequence $f_j = f$, $U_* = 0$, $C_+ := 3$, $A_0 := 4$, $A_2 := 1$, $\gamma_B := 1$. Let (l_j, K_j) be the data of window pinning and $K := K_j + 6 \leq 7K_j$: by [I, Lemma 8.10] and (P2) the fine carriers contribute at most $6t^2$, so (PIN_K) holds on $\mathcal{W}(l_j)$. Since $K_j T_{\text{hi}}(l_j) \rightarrow 0$, for large j every dyadic scale t of $\mathcal{W}(l_j)$ satisfies $t \leq \min(t_\eta, t_1, 1)$, $Kt \leq 1$, $t^2 \leq 1/(2(1 + \|U\|))$, $2K_\kappa t^2 \leq 1$ and $K'_4 Kt \leq \kappa_0$. The clamped certificate c_t of Proposition 6.19 then has $\Gamma_w(c_t) \leq (1 + 2\kappa_0)^2 = 1 + \eta_0/2 \leq 2$ and $t\|b_t\|_1 \leq 3\|a\|_1 \leq A_0$; since $\text{supp } b_t \subseteq F$, the pair is admissible on both sides, $(\text{OS2}')$ holds with $C_+ = 3$ on both sides, and every block coordinate is of kind [1]. Lemma 6.10 gives $p^*(f + rg_{c_t}) \leq 1 + \frac{r^2}{2}(1 + \eta_0)$ for $|r| \leq c_b t$, and $p^*(g - g_{c_t}) \leq (1 + \|U\|)K'_2 Kt \leq K''_j t$ with $K''_j := 7(1 + \|U\|)K'_2 K_j$. As $n_{l_j}^w/K''_j \rightarrow \infty$ and $K''_j T_{\text{hi}}(l_j)/n_{l_j}^w \rightarrow 0$, Corollary 6.4 (with $t^{(j)} := T_{\text{hi}}(l_j)$, $n_j := n_{l_j}^w$) gives $(f, \rho g) \in \text{clNA}$. Let $\rho \rightarrow 1$. $\square$

Theorem 6.21 (Infinite base support with signature room). *Let T be a signature-ladder operator and $N \geq 1$. Every $f \in S_{p_N^*}$ satisfying (SR) of [I, Definition 8.3], with F finite or infinite, belongs to $\mathcal{R}$. In particular the claim of [I, Remark 8.46] holds.*

I , Proposition 8.13(a). holds at arbitrary F : its proof uses [I, Lemma 8.7(b)] (where $J_\gamma \subseteq F^c$), the pinning inequality [I, Lemma 8.11] (which uses only (P1) and the sets $S_l \cap J_\gamma$) and [I, Lemma 8.12]. As in the proof of [I, Theorem 8.18], $K_1 \Lambda_f(l) T_{\text{hi}}(l) \rightarrow 0$ and $n_l^w/\Lambda_f(l) \rightarrow \infty$, because $\Lambda_f(l) \leq \varpi_0^{-l^2} \Lambda^\circ(l)$ and (P3). So every $g \in \mathcal{C}(f)$ is window-pinned with $K_l := K_1 \Lambda_f(l)$, and Theorem 6.20 applies. $\square$

Theorem 6.22 (Sign-mixed room at arbitrary base support). *Let T be a signature-ladder operator and $N \geq 1$. Let $f \in S_{p_N^*}$ (F arbitrary) satisfy*

$$\mathfrak{r}_l := \min_{\sigma=\pm 1} \sum_{s \in S_l \setminus F} v_l(s)(1 + \sigma z_s) > 0 \quad (l \in \mathcal{L}_N)$$

and $(W^\pm)$ of [I, Definition 8.33]. Then $f \in \mathcal{R}$.

Proof. The proof of [I, Lemma 8.34] does not use the finiteness of F (the rooms are computed on $S_l \setminus F$, and [I, Lemma 8.24] holds at every F). Hence, as in the proof of [I, Theorem 8.35], every mate is window-pinned with $K_j := (1/q_0 + 22)\Lambda_f^\pm(l_j)$; apply Theorem 6.20. $\square$

At infinite F the support itself can provide room at the scales of a window, through the cushions. For $M \geq 0$ let $F_{\leq M} := F \cap [1, M]$,

$$\mathfrak{r}_l^{[M]} := \min_{\sigma=\pm 1} \sum_{s \in S_l \setminus F_{\leq M}} v_l(s)(1 + \sigma z_s), \quad \Lambda_f^c(l, M) := \prod_{l'' \leq l, l'' \in \mathcal{L}_N} \left(1 + \frac{3}{\mathfrak{r}_{l''}^{[M]}}\right),$$

$(z_s = s_s \text{ on } F)$ and $M(t) := \min\{M : \alpha(M) \leq t^2\}$.

Theorem 6.23 (Cushion room). *Let T be a signature-ladder operator, $N \geq 1$, and let $f \in S_{p_N^*}$ satisfy*

$$(W^c) \quad \liminf_{l \rightarrow \infty} \frac{\Lambda_f^c(l, M(T_{\text{lo}}(l)))}{l^{2^3} \Lambda^\circ(l)} = 0$$

(with $\Lambda_f^c(l, M) := \infty$ if some factor has $\mathfrak{r}_{l''}^{[M]} = 0$). Then $f \in \mathcal{R}$. Since $\mathfrak{r}_l^{[M]} \geq \mathfrak{r}_l$, $(W^\pm)$ implies (W^c) .

Proof. Let $t \in \mathcal{W}(l_*)$, $t \leq \min(t_\eta, 1)$, a decomposition at scale t , and $M \geq M(t)$. Put $E_l^c := \sum_{s \in S_l \setminus F \leq M} \phi_{z_s}(\Delta B(s))$. By [I, Lemma 8.24] off F , and on $F \cap (M, \infty)$ by $\phi_{s_j}(x) = 2(s_j x)_-$ [I, Lemma 8.23(a)] and Lemma 6.9,

$$\sum_l E_l^c \leq \frac{t}{q_0} + 2 \sum_{j \in F, j > M} \left(\frac{2|a_j|}{t} + \mathfrak{f}_j^+ + \mathfrak{f}_j^- \right) \leq \left(\frac{2}{q_0} + 4 \right) t.$$

For $s \in S_l$, $-\Delta B(s) = \Delta \theta_l v_l(s) + r_l(s)$ with $r_l(s) := \sum_{l' > l} \Delta \theta_{l'} y_{l'}(s)/n_{l'}$ ((P1), [I, (15)]), so $|\Delta \theta_l| v_l(s) (1 + z_s \text{sign } \Delta \theta_l) \leq \phi_{z_s}(\Delta B(s)) + 2|r_l(s)|$ [I, Lemma 8.23(c),(d)]; summing over $s \in S_l \setminus F \leq M$ gives $\mathfrak{r}_l^{[M]} |\Delta \theta_l| \leq E_l^c + 2 \sum_{l' > l} \pi_{l', l} |\Delta \theta_{l'}|$. Unrolling as in the proof of [I, Lemma 8.34], $\sum_l |\Delta \theta_l| \leq (2/q_0 + 26) \Lambda_f^c(l_*, M)t$. Since $M(\cdot)$ is nonincreasing, $M := M(T_{\text{lo}}(l_*))$ serves all scales of $\mathcal{W}(l_*)$. By (W^c) and (P3) every mate is window-pinned; Theorem 6.20 applies. $\square$

Remark 6.24 (Scope of cushion room). If z is constant of modulus 1 on $S_l \setminus F \leq M_0$ for some $l \in \mathcal{L}_N$ and some M_0 (the signature set is swallowed, off a finite set, by contacts and support coordinates of one sign), then $\mathfrak{r}_l^{[M]} = 0$ for all $M \geq M_0$, and (W^c) fails whatever the decay of a . So Theorem 6.23 adds to Theorem 6.22 only for signature sets on which a is sign-mixed (or z is sign-mixed off F), and then only for fast decay of a on those sets. Signature sets swallowed by the support are treated in Subsections 6.5 and 6.7.

6.5. Swallowed carriers and cushion sparsity. In this subsection T is a signature-ladder operator. We extend [I, Definition 8.37] to arbitrary base support.

Definition 6.25 (Bad carriers at arbitrary base support). Let $f \in S_{p_N^*}$ (F arbitrary). For $l \in \mathcal{L}_N$ let $\mathfrak{r}_l := \min_{\sigma=\pm 1} \sum_{s \in S_l \setminus F} v_l(s)(1 + \sigma z_s)$ (the room off the support; 0 if $S_l \subseteq F$), and $\mathcal{B} := \{l \in \mathcal{L}_N : \mathfrak{r}_l = 0\}$ (the bad carriers). For $l \in \mathcal{B}$ let ϵ_l be the sign with $z \equiv \epsilon_l$ on $S_l \setminus F$ ($\epsilon_l := 1$ if $S_l \subseteq F$). Put $\mathcal{B}_K := \{l \in \mathcal{B} : S_l \setminus F \text{ infinite}\}$ (contact-swallowed) and $\mathcal{B}_F := \mathcal{B} \setminus \mathcal{B}_K$ (swallowed by the support: $S_l \setminus F$ is finite), $\mathcal{B}_{\text{pk}} := \{l \in \mathcal{B} : k(l) \in P_{m(l)}\}$ and $\mathcal{B}_{\text{np}} := \mathcal{B} \setminus \mathcal{B}_{\text{pk}}$. For $l \in \mathcal{B}$ let $\tau_l := -\epsilon_l \Delta \theta_l$ (for a given two-sided decomposition) and let $q_l = \epsilon_l \Phi_{m(l)}(k(l)) w_{m(l)}(k(l)) / (m(l) C_{m(l)})$ be its d -weight (as in Definition 4.2, with the sign ϵ_l of l). For good l let $S_l^* := S_l \setminus (F \cup \bigcup_{l' \in \mathcal{B}, l' > l} \text{supp } y_{l'})$, $\mathfrak{r}_l^* := \min_{\sigma=\pm 1} \sum_{s \in S_l^*} v_l(s)(1 + \sigma z_s)$ and $\Lambda_f^*(l) := \prod_{l'' \leq l, l'' \in \mathcal{L}_N \setminus \mathcal{B}} (1 + 3/\mathfrak{r}_{l''}^*)$. Finally $U_{\mathcal{B}}(j) := \sum_{l \in \mathcal{B}} |u_l(j)|$ and $m_{\mathcal{B}} := m_{U_{\mathcal{B}}}$ (profile on F , Definition 6.6). Consider:

- (W*) $\mathfrak{r}_l^* > 0$ for every good l , and $\liminf_l \Lambda_f^*(l) / (l^{2^3} \Lambda^\circ(l)) = 0$;
- (H2) every block m has a peak k with $\alpha_m(k) \neq 0$ and $j(k, m) \notin \mathcal{B}$;
- (H3-inf) no $l \in \mathcal{B}_K$ has $k(l)$ a degenerate peak with $\text{sign } w_{m(l)}(k(l)) = \epsilon_l$, and no $l \in \mathcal{B}_F$ has $k(l)$ a degenerate peak;
- ($\mathcal{B}_{\text{fin}}$) $\mathcal{B}$ is finite;

- (CS_B) $m_B(y) = o(y)$ as $y \rightarrow 0^+$ (*cushion sparsity*);
 (H4-inf) $\sup_{l \in \mathcal{B}} \sup_{j \in F} |u_l(j)|/|a_j| < \infty$ (*cushion domination*).

For finite F , $\mathcal{B}_F = \emptyset$, (H3-inf) is (H3), and under $(\mathcal{B}_{\text{fin}})$ the product Λ_f^* differs from that of [I, Definition 8.37] by a constant factor. The product is taken over good carriers only because carriers of $\mathcal{B}_F$ may have no room at all.

Lemma 6.26 (Cushion sparsity). *Assume $(\mathcal{B}_{\text{fin}})$. (a) (H4-inf) implies (CS_B). (b) (CS_B) holds iff the profile of $v_l 1_{S_l \cap F}$ is sparse for every $l \in \mathcal{B}$ with $S_l \cap F$ infinite.*

Proof. (a) If $|u_l(j)| \leq c_B |a_j|$ on F , then $U_B \leq |\mathcal{B}| c_B |a|$ on F , and the set defining $m_B(y)$ is empty for $y < 1/(|\mathcal{B}| c_B)$. (b) $T_B := \bigcup_{l \in \mathcal{B}} \text{supp } y_l$ is finite. By (P1), $U_B = v_l$ on $S_l \setminus T_B$ for $l \in \mathcal{B}$, and U_B vanishes off $T_B \cup \bigcup_{l \in \mathcal{B}} S_l$. Each of the finitely many $j \in F \cap T_B$ has $a_j \neq 0$ and leaves the sets defining the profiles for small y . $\square$

Lemma 6.27 (Exact switching at arbitrary base support). *Let $f \in S_{p_N}^*$ (F arbitrary) satisfy (W*), (H2), (H3-inf) and $(\mathcal{B}_{\text{fin}})$, let $g \in \mathcal{C}(f)$, $\eta \leq \eta_*$ with $\eta_\Gamma(\eta) \leq 1$, and put $K_* := (1/q_0 + 22)\Lambda_f^*(l_*)$. There are C_f and l_f , depending only on f , N and the operator, such that for $l_* \geq l_f$, $t \in \mathcal{W}(l_*)$ with $t \leq \min(t_\eta, 1)$, and every two-sided decomposition of g at scale t :*

- (a) $\sum_{l \notin \mathcal{B}} |\Delta \theta_l| \leq K_* t$ and $\sum_{l > l_*} |\Delta \theta_l| \leq 6t^2$; and [I, Lemma 8.39](a)–(c) hold;
- (b) there is $\tau' \in \mathbb{R}^{\mathcal{B}}$ with $\tau'_l \geq 0$ ($l \in \mathcal{B}_K$), $\tau'_l = 0$ ($l \in \mathcal{B}_{\text{pk}}$), $\sum_{m(l)=m} q_l \tau'_l = 0$ for every m , such that $V' := \sum_{l \in \mathcal{B}} \epsilon_l \tau'_l u_l 1_{F^c}$ is z -signed and supported in K , $\sum_{l \in \mathcal{B}} |\tau_l - \tau'_l| \leq C_f K_* t$ and $\|\Delta B 1_{F^c} - V'\|_1 \leq C_f K_* t$.

Proof. (a) The good inequalities in the proof of [I, Lemma 8.38] use the sets $S_l^* \subseteq F^c$, the budget [I, Lemma 8.24] (valid at every F) and (P1); bad carriers never occur on their right sides, because the targets of all bad carriers are removed from S_l^* and other signatures vanish on S_l . Unrolling over the good carriers gives the bound. The proof of [I, Lemma 8.39] uses only [I, Lemma 8.27], [I, (17)], (a) and (H2). (b) Let $l_f \geq \max \mathcal{B}$ and $l_f \geq \max_m l_m^\circ$ ([I, Lemma 8.39]). We run case $(\mathcal{B}_{\text{fin}})$ of the proof of [I, Lemma 8.40] with two changes: $T_0 := \bigcup_{l \in \mathcal{B}} \text{supp } y_l \cup \bigcup_{l \in \mathcal{B}_F} (S_l \setminus F)$ (finite), and the cone $\mathcal{Z}_f$ has no sign row for $l \in \mathcal{B}_F$. For $j \notin F \cup T_0$ the only bad vector that can be nonzero at j is the signature of the $l \in \mathcal{B}_K$ with $j \in S_l$, where $z_j = \epsilon_l$; hence the cost $\mathbf{c}(\tau) := \sum_{j \notin F} \phi_{z_j}(L_j(\tau))$ equals $\sum_{l \in \mathcal{B}_K} 2(\tau_l)_- \mathbf{m}_l + \sum_{j \in T_0 \setminus F} \phi_{z_j}(L_j(\tau))$ with $\mathbf{m}_l := \|v_l 1_{S_l \setminus (F \cup T_0)}\|_1 > 0$. The bad peaks are controlled by [I, Lemma 8.39](c) and (H3-inf): either $\alpha_{m(l)}(k(l)) \neq 0$, or $l \in \mathcal{B}_K$ is at a degenerate peak with sign $w(k(l)) = -\epsilon_l$, and then $\tau_l \leq \lambda_l K_d t$ and $(\tau_l)_- \leq \mathbf{c}(\tau)/(2\mathbf{m}_l)$. The rest of the proof (Hoffman's bound, [I, (18)]) is unchanged. $\square$

Lemma 6.28 (Common shift and deep assignment). (a) *Let $x, y \in \mathbb{R}$ and $A \geq 0$ with $x - y \geq -2A$. Then $\{\sigma : x - \sigma \geq -A, y - \sigma \leq A\} = [y - A, x + A] \neq \emptyset$, and its point σ closest to 0 satisfies $|\sigma| = (-x - A)_+ + (y - A)_+$.*

(b) *Let $\alpha \geq 0$, $A := 2\alpha$, $\varphi^\pm \geq 0$, and let $x, y_0, \xi, e \in \mathbb{R}$ with $x \geq -\alpha - \varphi^+$, $y_0 \leq \alpha + \varphi^-$ and $x - y_0 = \xi + e$. Put $\beta := x - \sigma$, where σ is given by (a) for $(x, x - \xi, A)$, if $\xi \geq -2A$, and $\beta := -A$ if $\xi < -2A$. Then (i) $|x - \beta| \leq \varphi^+ + \varphi^- + |e|$; (ii) $\beta \geq -A$; (iii) $\beta - \xi \leq A$ if $\xi \geq -2A$, and $\beta - \xi = |\xi| - A \leq |\xi|$ if $\xi < -2A$; (iv) $|\beta| \leq A + |\xi|$.*

Proof. (a) The interval is nonempty because $y - A \leq x + A$; at most one of $y - A > 0$, $x + A < 0$ holds. (b) If $\xi \geq -2A$, then $y := x - \xi = y_0 + e$; (ii), (iii) follow from

(a), and $|\sigma| \leq (\alpha + \varphi^+ - 2\alpha)_+ + (\alpha + \varphi^- + |e| - 2\alpha)_+ \leq \varphi^+ + \varphi^- + |e|$; (iv): $-A \leq \beta \leq x - (y - A) = \xi + A$. If $\xi < -2A$, (ii)–(iv) are clear, $x + A \geq \alpha - \varphi^+ \geq -\varphi^+$, and $x = y_0 + \xi + e \leq \alpha + \varphi^- - 4\alpha + |e|$, so $x + A \leq \varphi^- + |e|$; hence $|x - \beta| = |x + A| \leq \varphi^+ + \varphi^- + |e|$. $\square$

Theorem 6.29 (Cushion-sparse swallowing). *Let T be a signature-ladder operator and $N \geq 1$. Every $f \in S_{p_N^*}$ (F arbitrary) satisfying (W^*) , $(H2)$, $(H3\text{-inf})$, $(\mathcal{B}_{\text{fin}})$ and $(CS_{\mathcal{B}})$ belongs to $\mathcal{R}$. In particular this holds under $(H4\text{-inf})$ in place of $(CS_{\mathcal{B}})$.*

Proof. Fix $g \in \mathcal{C}(f)$ and $\rho \in (0, 1)$; choose $\eta_0, \kappa_0, \varepsilon_{\text{tr}}, \eta$ as in the proof of Theorem 6.20. Let $l_* \geq l_f$, $t \in \mathcal{W}(l_*)$ with $t \leq \min(t_\eta, 1)$, and a two-sided decomposition at scale t . Constants $C_f, C_\tau, K_\kappa, A_0^R, A_2$ below depend only on f, N and the operator; K_2^R, K_4^R are such constants times $\Lambda_f^*(l_*)$.

Step 1 (pinning, exact switching). Lemma 6.27 gives (a) and τ', V' as in (b).

Step 2 (bounded switching). Lemma 6.11(b) with $\mathcal{U} := \{(k(l), m(l)) : l \in \mathcal{B}\}$ gives $|\Delta\theta_l| \leq C_{\mathcal{U}}(1 + K_*t)$ for $l \in \mathcal{B}$. Assume $K_*t \leq 1$ and $C_f K_*t \leq 1$; then $|\tau'_l| \leq 2C_{\mathcal{U}} + 1 =: C_\tau$. Put $X := \sum_{l \in \mathcal{B}} \epsilon_l \tau'_l u_l$. Then $X 1_{F^c} = V'$, $|X_j| \leq C_\tau U_{\mathcal{B}}(j)$, and $\Delta B - X = \sum_{l \in \mathcal{B}} \epsilon_l (\tau_l - \tau'_l) u_l - \sum_{l \notin \mathcal{B}} \Delta\theta_l u_l$, so $\|\Delta B - X\|_1 \leq (C_f + 1)K_*t$.

Step 3 (exact data). Let $\chi, \mathbf{e}_\pm$ be given by [I, Lemma 8.26] for $e := \Delta B 1_{F^c} - V'$: $B_+ 1_{F^c} = \chi V' + \mathbf{e}_+$, $B_- 1_{F^c} = -(1 - \chi)V' + \mathbf{e}_-$, $\|\mathbf{e}_\pm\|_1 \leq C_f K_*t + t/q_0$. For $j \in F$ apply Lemma 6.28(b) with $\alpha := |a_j|/t$, $\varphi^\pm := \mathfrak{f}_j^\pm$, $x := s_j B_+(j)$, $y_0 := s_j B_-(j)$, $\xi := s_j X_j$ and $e := s_j (\Delta B - X)_j$ (the hypotheses are Lemma 6.9), and let $b_1^+(j) := s_j \beta_j$; $b_1^+ := 0$ off F . With $A_j := 2|a_j|/t$ and $D_t := \{j \in F : s_j X_j < -2A_j\}$ (the deep set), the lemma gives

$$\|B_+ 1_F - b_1^+\|_1 \leq \frac{t}{2q_0} + (C_f + 1)K_*t, \quad s_j b_1^+(j) \geq -A_j,$$

$$s_j (b_1^+ - X)_j \leq \begin{cases} A_j & j \notin D_t, \\ |X_j| & j \in D_t, \end{cases}$$

and $|b_1^+(j)| \leq A_j + |X_j|$. Put $\kappa := (b_1^+ + \chi V')(\hat{z})$, $b^+ := b_1^+ + \chi V' - \kappa a$, $b^- := b^+ - X$; in block m let $\omega_m^+ := \omega_m^c$ [I, Definition 8.14] off the coordinates $k(l)$, $l \in \mathcal{B}_{\text{np}}$, $\omega_m^+(k(l)) := \omega_{+,m}(k(l))$ for $l \in \mathcal{B}_{\text{np}}$, $\omega_m^- := \omega_m^+ + \sum_{l \in \mathcal{B}_{\text{np}}, m(l)=m} (\epsilon_l \tau'_l / \lambda_l) e_{k(l)}$, and $g_t := b^+ + \sum_m R_m^*(\omega_m^+ - d_m(\omega_m^+) w_m)$. Then: (a) Off F , $b^+ = \chi V'$ and $b^- = -(1 - \chi)V'$, so the pairs are side-+, resp. side- admissible; $b^+ - b^- = X = \sum_{l \in \mathcal{B}_{\text{np}}} \epsilon_l \tau'_l u_l = \sum_m R_m^*(\omega_m^- - \omega_m^+)$ (as $\tau' = 0$ on $\mathcal{B}_{\text{pk}}$) and $d_m(\omega_m^-) - d_m(\omega_m^+) = \sum_{m(l)=m} q_l \tau'_l = 0$; so both pairs represent g_t and are d -neutral; $b^+(\hat{z}) = 0$ and $b^\pm(\xi) = 0$ [I, Lemma 3.2]. (b) Since $(b_1^+ + \chi V')(\hat{z}) = B_+(\hat{z}) - \mathbf{e}_+(\hat{z}) - (B_+ 1_F - b_1^+)(\hat{z})$, [I, Lemma 8.7(a)] gives $|\kappa| \leq K_\kappa t$. If $K_\kappa t^2 \leq 1$, then on F : $(s_j b_j^+)_- \leq 3|a_j|/t$, and $(s_j b_j^-)_+ \leq 3|a_j|/t$ off D_t and $\leq |X_j| \leq C_\tau U_{\mathcal{B}}(j)$ on D_t . (c) $t\|b^\pm\|_1 \leq A_0^R := 3 + 3C_\tau |\mathcal{B}|$ (use $\sum_F A_j \leq 2/t$, $\|U_{\mathcal{B}}\|_1 \leq |\mathcal{B}|$). (d) $B_+ - b^+ = \mathbf{e}_+ + (B_+ 1_F - b_1^+) + \kappa a$; off F , $B_- - b^- = \mathbf{e}_-$; on F , $B_- - b^- = (B_+ - b^+) - (\Delta B - X)$; all are $O(K_*t)$ in ℓ_1 . The block parts are compared with $\Theta_\pm$ exactly as in the proof of [I, Lemma 8.41(b),(c)]. Hence $\|g - g_t\|_1 \leq K_2^R t$ and $\Gamma_w(b^\pm, \omega^\pm) \leq (\sqrt{1 + \eta_\Gamma(\eta)} + K_4^R t)^2$. (e) As in [I, Lemma 8.41(d)], every block coordinate is of kind [1], or of kind [2] with $\gamma_B := \gamma_{\mathcal{B}} > 0$ and $A_2 := 4 + C_\tau / \lambda_{\mathcal{B}}$, $\lambda_{\mathcal{B}} := \min_{\mathcal{B}} \lambda_l$.

Step 4 (local validity). If moreover $K_4^R t \leq \kappa_0$ and $t \leq t_1$, Lemma 6.10 (constant sequence, $U_* := U_{\mathcal{B}}$, $C_+ := 3$, $C_* := C_\tau$, $A_0 := A_0^R$, A_2 , γ_B ; its hypothesis on $\bar{m}$ is $(CS_{\mathcal{B}})$) applies to the $+$ data for $0 < r \leq c_b t$ and to the $-$ data for

$-c_b t \leq r < 0$; both represent g_t and have $\Gamma_w \leq (1 + 2\kappa_0)^2 = 1 + \eta_0/2 \leq 2$. So $p^*(f + rg_t) \leq 1 + \frac{r^2}{2}(1 + \eta_0)$ for $|r| \leq c_b t$, and $p^*(g - g_t) \leq (1 + \|U\|)K_2^R t$.

Step 5. By (W^*) choose window indices l_j with $\Lambda_f^*(l_j)/(l_j 2^{l_j^3} \Lambda^\circ(l_j)) \rightarrow 0$. By (P3) all the smallness conditions above hold on $\mathcal{W}(l_j)$ for large j , and $K_j := (1 + \|U\|)K_2^R(l_j)$, $n_j := n_{l_j}^w$, $T_j := T_{\text{hi}}(l_j)$ satisfy $n_j \geq 24\rho^2 K_j/(c_b(1 - \rho^2))$ and $K_j T_j/n_j \rightarrow 0$. Theorem 6.16 applies with $C_+ := 3$ and $\beta := C_\tau U_B$, whose profile is sparse by (CS_B) and Lemma 6.7(b); condition (v) holds since the data are d -neutral. Hence $(f, \rho g) \in \text{cl NA}$; let $\rho \rightarrow 1$. The last sentence follows from Lemma 6.26(a). $\square$

Remark 6.30 (Scope of Theorem 6.29; non-sparse swallowing). (a) In the model $v_l(s) \approx 2^{-s}$ on S_l , $|a_s| \approx v_l(s)^{1+\beta_0}$ on $S_l \cap F$, (CS_B) holds iff $\beta_0 < 1$; ratio profiles $|a_s|/v_l(s) = 1/s$ or $2^{-\sqrt{s}}$ are covered, whereas (H4-inf) needs $\inf |a_s|/v_l(s) > 0$. (b) The deep set is where the method of Theorem 6.29 departs from exact cushion bounds: every two-piece data with $b^+ - b^- = X$ on F must have $(s_j b_j^+)_- > A_j$ or $(s_j b_j^-)_+ > A_j$ on D_t , and fixed data whose anti-sign part on $S_l \cap F$ is $|c|v_l$ pay the flip cost $\text{Fl}_{|c|v_l}(r) \geq r|c|m_{v_l}(r|c|/2)$ at scale r (each j with $|a_j| < r|c|v_l(j)/2$ contributes at least $r|c|v_l(j)$), which is not $o(r^2)$ for non-sparse profiles. This is a statement about fixed data, not a non-recovery statement. (c) *Heuristic, not used.* For critical profiles a window decomposition at scale t can share the flips between the cushions of both sides, while fixed data built from it carry them on one side; in the exactly self-similar model ($|a_s| = v_l(s)^2$, gaps 1) the resulting loss is the factor 2 in the flip coefficient, and averaging over a window does not remove it. (d) (*Proved in our sources, not reproduced here.*) Under additional hypotheses (bounded gaps, allowedness rule (c), a of constant sign on $S_l \cap F$, $\text{supp } u_l \subseteq F$, d -neutral bad carriers, quasi-monotone ratios) super-critical swallowing is harmless for signature-ladder operators with these features, by un-switching (Lemma 6.12) and Theorem 6.14; and without profile conditions one obtains recovery of the shrunk fibre $\rho_* \mathcal{C}(f)$ for an explicit $\rho_* < 1$. For T_{final} both statements are contained in Theorems 6.38 and 6.40 below (whose hypotheses (W^*) , (H2), (H3-inf), $(\mathcal{B}_{\text{fin}})$ are among those of the cited results; the rows of the first statement can fail (ND') and are then covered by Theorem 6.40), and we do not reproduce the proofs for general signature-ladder operators. The non-sparse case for operators without a fast-decaying diagonal base is open.

6.6. Box-dominated support. Here T is an arbitrary admissible operator. Fix an enumeration $\mathbf{l} : \mathbb{N} \times I \rightarrow \mathbb{N}$ of the carriers, put $\mathcal{C}_L := \{(k, m) : \mathbf{l}(k, m) \leq L\}$, $L(t) := \min\{L : \sum_{\mathbf{l}(k, m) > L} \lambda_{k, m} \leq t^3\}$, and

$$M_f(L) := \sum_{m \in I} \max \left\{ \frac{1}{\mu_{k, m}} : (k, m) \in \mathcal{C}_L, k \in P_m \right\} \quad (\max \emptyset := 0),$$

the inverse margins of the coarse peaks.

Theorem 6.31 (Box-dominated support). *Let T be admissible, $N \geq 1$, and let $f \in S_{p_N}^*$ satisfy*

- (BD) $\text{Bx}(j) \leq C_a |a_j|$ for every $j \in \mathbb{N}$;
- (SC_I) the set I of all blocks satisfies (SC) of [I, Definition 7.16];
- (W_M) for every $n \in \mathbb{N}$, $\liminf_{T' \rightarrow 0+} T' M_f(L(2^{-n} T')) = 0$.

Then $f \in \mathcal{R}$. Condition (BD) forces $F = \mathbb{N}$; every carrier is swallowed by the support, and no room, pinning, design feature, or condition on the mates or on the strict non-peaks is used.

Proof. $F = \mathbb{N}$: for every j some $u_{k,m}$ has $u_{k,m}(j) \neq 0$, because $\{u_{k,m}\}_k$ is dense in S_q^* ; so $\text{Bx}(j) > 0$ and $a_j \neq 0$. (SC_I) excludes degenerate peaks, so $\mu_{k,m} > 0$ at every peak. Fix $g \in \mathcal{C}(f)$, $\rho \in (0, 1)$, and $\eta_0, \kappa_0, \varepsilon_{\text{tr}}, \eta$ as in the proof of Theorem 6.20. An f -constant depends only on f, g, ρ, N, T .

Step 1 (exact data at one scale). Let $t \leq \min(t_\eta, 1)$, L with $\sum_{l(k,m) > L} \lambda_{k,m} \leq t^3$, and $(B_\pm, \Theta_\pm)$ a two-sided decomposition at scale t , with $d_\pm, \omega_\pm$ as in [I, Lemma 8.8]. Put $\mathcal{Q}_m := \mathcal{Q}_m \cap \{k : (k, m) \in \mathcal{C}_L\}$ (finite), $\omega_m^\pm := \omega_{\pm, m} 1_{\mathcal{Q}_m}$ and $b_0^\pm := g - \sum_m R_m^*(\omega_m^\pm - d_m(\omega_m^\pm)w_m)$. Both pairs represent g . Drop m . (a) At a peak k , $\sigma|\alpha(k)| = \lambda_k \mu_k$ [I, (2)], so by [I, Lemma 8.8(c)] the coarse peaks contribute $\sum \lambda_k |\omega_\pm(k)| \leq (t/2) \max 1/\mu_k$; outside $\mathcal{C}_L$, $|\omega_\pm| \leq 4/t$ gives $\lambda_k |\omega_\pm(k)| \leq 4\lambda_k/t$. Hence $E_\pm := \sum_m \sum_{k \notin \mathcal{Q}_m} \lambda_{k,m} |\omega_{\pm, m}(k)| \leq \frac{t}{2} M_f(L) + 4t^2$. (b) $|d_\pm - d(\omega^\pm)| \leq t/\sigma + E_\pm/m$ ([I, Lemma 8.8(d)] and $|d(\omega)| \leq \|D\omega\|_2 \leq \sum_k \Phi_k |\omega(k)|$). (c) $b_0^\pm - B_\pm = \sum_m R_m^*(\omega_{\pm, m} 1_{\mathcal{Q}_m^c}) - \sum_m (d_{\pm, m} - d_m(\omega_m^\pm)) R_m^* w_m$, so $\|b_0^\pm - B_\pm\|_1 \leq K_1 t$ with $K_1 := C_f(1 + M_f(L))$, and, if $t \leq t_f$ and $t^2 M_f(L) \leq 1$, pointwise $|(b_0^\pm - B_\pm)_j| \leq 5\text{Bx}(j)/t \leq 5C_a |a_j|/t$ (use $|(R_m^* w_m)_j| \leq \sum_k \lambda_{k,m} |u_{k,m}(j)|$ and $|d_\pm - d(\omega^\pm)| \leq 1/t$). (d) $X := b_0^+ - b_0^- = \sum_m R_m^*((\omega_m^- - \omega_m^+) - \Delta d_m^0 w_m)$ with $\Delta d_m^0 := d_m(\omega_m^-) - d_m(\omega_m^+)$; since $|\omega^\pm| \leq 4/t$ and $\|\Phi_m\|_2 \leq \frac{1}{2}$, $|\Delta d_m^0| \leq 4/t$ and $|X_j| \leq 12\text{Bx}(j)/t \leq 12C_a |a_j|/t$.

Step 2 (common shift). Let $A_j := (6C_a + 1)|a_j|/t$, $x_j := s_j b_{0,j}^+$ and $y_j := s_j b_{0,j}^-$. By Lemma 6.9 and (c), $x_j \geq -A_j - f_j^+$ and $y_j \leq A_j + f_j^-$; by (d), $x_j - y_j \geq -2A_j$. Let σ_j be the point of $[y_j - A_j, x_j + A_j]$ closest to 0; by Lemma 6.28(a), $|\sigma_j| \leq f_j^+ + f_j^-$. Put $\tilde{\sigma} := \sum_j s_j \sigma_j e_j^*$ ($\|\tilde{\sigma}\|_1 \leq t/(2q_0)$), $\kappa' := -\tilde{\sigma}(\hat{z})$, $b^\pm := b_0^\pm - \tilde{\sigma} - \kappa' a$ and $g_t := g - \tilde{\sigma} - \kappa' a$. Both pairs $(b^\pm, \omega^\pm)$ represent g_t , $b^\pm(\xi) = 0$, $b^+ - b^- = X$, and for $t^2 \leq q_0/(1 + \|U\|)$ (so that $|\kappa'| \leq 1/t$), $(s_j b_j^+)_- \leq (6C_a + 2)|a_j|/t$ and $(s_j b_j^-)_+ \leq (6C_a + 2)|a_j|/t$. Moreover $p^*(g - g_t) \leq (1 + \|U\|)(\|\tilde{\sigma}\|_1 + |\kappa'|) \leq Kt$ with $K := (1 + \|U\|)^2/q_0$. As $F = \mathbb{N}$ there is no side condition off F : these are two-piece data for g_t , with Δd_m^0 of arbitrary sign.

Step 3 (Γ_w). The pair $(b^+, \omega^+ - d(\omega^+)w)$ differs from (B_+, Θ_+) by a base part of ℓ_1 -norm $\leq (K_1 + K)t$ and the block part $-\omega_+ 1_{\mathcal{Q}^c} + (d_+ - d(\omega^+))w$, for which $\sigma_m H_m \leq \sigma_m (E_+/m)^2/C_m$. As $\sqrt{\Gamma_w}$ is a seminorm and $\Gamma_w(B_+, \Theta_+) \leq 1 + \eta_\Gamma(\eta)$, $\sqrt{\Gamma_w}(b^+, \omega^+) \leq \sqrt{1 + \eta_\Gamma(\eta)} + K_4 t$ with $K_4 := C'_f(1 + M_f(L))$; the same for the $-$ data.

Step 4 (local validity). $\|B_\pm\|_1 \leq \|g\|_1 + \sum_k \lambda_k |\Theta_\pm(k)| \leq \|g\|_1 + 6/t$, so $t\|b^\pm\|_1 \leq A_0 := \|g\|_1 + 8$ once $(K_1 + K)t^2 \leq 1$. For $k \in \mathcal{Q}_m$ and $s_k := \text{sign } w_m(k)$, [I, Lemma 8.8(f)] gives $s_k \omega^+(k) \leq \frac{3}{2} \text{gap}(k)/t$, and $s_k \omega^+(k) \geq -4/t$; if $w_m(k) = 0$ both signs are allowed and $|\omega^+(k)| \leq \frac{3}{2} \text{gap}(k)/t$. So every block coordinate of the $+$ data is of kind [1] or of kind [3] (side $+$) with $A_2 := 4$, and symmetrically for the $-$ data. Lemma 6.10 (constant sequence, $U_* = 0$, $C_+ := 6C_a + 2$, $A_0, A_2 = 4$, $\gamma_B = 1$) gives f -constants c_b, t_1 , independent of L , with $p^*(f + r g_t) \leq 1 + \frac{r^2}{2}(1 + \eta_0)$ for $|r| \leq c_b t$, provided $K_4 t \leq \kappa_0$ and $t \leq t_1$.

Step 5 (windows of fixed length). Let $n_0 \geq 24\rho^2 K/(c_b(1 - \rho^2))$ be an integer. By (W_M) with $n = n_0$ there are $T_i \rightarrow 0$ with $T_i M_f(L_i) \rightarrow 0$, $L_i := L(2^{-n_0} T_i)$. Use the scales $T_i 2^{1-s}$, $1 \leq s \leq n_0$, all with the coarse level L_i . For large i all conditions

of Steps 1–4 hold at these scales. Theorem 6.16 applies with the triples (T_i, n_0, K) , $C_+ := 6C_a + 2$ and $\beta := 0$: (v) holds by (SC_I) . $\square$

Remark 6.32. (a) Under (BD) every switching, even of box size, is absorbed by the cushions, so exact two-piece data exist at every scale without pinning; what two-piece data cannot carry (peak usage, fine carriers, the w -direction of non- d -neutral data) is moved into the base at second-order cost controlled by the margins, and the closeness constant K is an f -constant: windows of fixed length suffice. Critical mates are covered. (b) For signature-ladder operators $L(t) \leq l$ for $t \in \mathcal{W}(l)$, so (W_M) follows from $M_f(l)T_{\text{hi}}(l) \rightarrow 0$ along a subsequence. (c) In this method (BD) cannot be weakened to a sparse profile of Bx on F : the box-size switching then exceeds the cushions by $\approx \text{Bx}(j)/t$ on a set whose flip cost at scales $r < t$ is of order r^2 times a factor growing as $t \rightarrow 0$. The example “ $F = \mathbb{N}$, $a > 0$ ” is covered when a box-dominates; for the designed operator, thin a and infinitely many swallowed carriers are treated in Subsection 6.7 (Theorem 6.42).

6.7. Raise transfer at the designed operator. From now on $T = T_{\text{final}}$ and $U = U_{\text{final}}$, with diagonal base $U^*e_s^* = \mu_s^*k_s$ (Lemma 3.2). For each level we use one sub-window w_l of level l as the l -th window; all results of the previous subsections hold for this signature-ladder operator. We use raise companions (Definition 2.1(b)), the raise-transfer Lemma 2.15 with its extension to small corrections of arbitrary sign on a finite set (Remark 2.16), Corollary 2.17 and Theorem 2.19. For a carrier k we write $\gamma_k := \langle U^*u_k, U^*a \rangle$. With the diagonal base, the value change of a carrier under a move of a on F depends only on $u_k|_F$: $u_k(\hat{z}') - u_k(\hat{z}) = \langle U^*u_k, e' - e \rangle$ when z is fixed, and $\langle U^*u, U^*x \rangle = \sum_s (\mu_s^*)^2 u(s)x_s$.

Lemma 6.33 (Value-preserving raises). *Let $f \in S_{p^*}$ (F arbitrary), let L_0 be a finite set of carriers, and assume*

$$(\text{ND}'_{L_0}) \quad a|_F \notin \text{span}\{u_k|_F : k \in L_0\}.$$

There are a finite set $F_0 \subseteq F$ and $c_0, C_0 > 0$ such that for every raise Δa with $\text{supp } \Delta a \cap F_0 = \emptyset$ and $\|U^\Delta a\|_H \leq c_0$ there is $\Delta a'' \in \ell_1(F_0)$ with $\|\Delta a''\|_1 \leq C_0\|U^*\Delta a\|_H$ and $|\Delta a''_s| \leq |a_s|/2$, such that the first row $f^\# := f[(a + D)/q^*(a + D), z]$, $D := \Delta a + \Delta a''$, satisfies $u_k(\hat{z}^\#) = u_k(\hat{z})$ for every $k \in L_0$. One may take $C_0 \geq 1$, and the radius can be chosen as $c_0 \geq c_f \min\{1, \min_{s \in F_0} |a_s|\}/(1 + C_0)^2$ with $c_f > 0$ depending only on ν (not on L_0); if $F_0 = \emptyset$, no correction is needed.*

Proof. Let $V_0 := \{c \in \mathbb{R}^{L_0} : u_c|_F = 0\}$, $u_c := \sum_k c_k u_k$. (i) If $x \in \ell_1$ is supported in F and $c \in V_0$, then $\langle U^*u_c, U^*x \rangle = \sum_s (\mu_s^*)^2 u_c(s)x_s = 0$. Hence for every $A \in \ell_1(F) \setminus \{0\}$ the vector $(\langle U^*u_k, U^*A/\|U^*A\| - e \rangle)_{k \in L_0}$ lies in $V_0^\perp$. (ii) Define $\Lambda : \ell_1(F) \rightarrow \mathbb{R}^{L_0}$ by $\Lambda(x)_k := \sum_{s \in F} (\mu_s^*)^2 x_s (u_k(s) - a_s \gamma_k / \nu^2)$. A vector c annihilates $\Lambda(\ell_1(F))$ iff $u_c(s) = a_s (\sum_k c_k \gamma_k) / \nu^2$ for all $s \in F$, i.e. iff $u_c|_F \in \mathbb{R}a|_F$; by (ND'_{L_0}) this means $u_c|_F = 0$, i.e. $c \in V_0$. So $\Lambda(\ell_1(F)) = V_0^\perp$, and there is a finite $F_0 \subseteq F$ with $\Lambda(\mathbb{R}^{F_0}) = V_0^\perp$; choose $F'_0 \subseteq F_0$ with $|F'_0| = \dim V_0^\perp$ such that $\Lambda|_{\mathbb{R}^{F'_0}}$ is a bijection onto $V_0^\perp$. (iii) Let Y_F be the closure of $U^*(\ell_1(F))$ in H ; by (i), $\langle U^*u_c, h \rangle = 0$ for $c \in V_0$ and $h \in Y_F$. For $y \in Y_F$ and $x \in \mathbb{R}^{F'_0}$ near 0 put $e(y, x) := (U^*a + y + U^*x) / \|U^*a + y + U^*x\|$ and $G(y, x)_k := \langle U^*u_k, e(y, x) - e \rangle$. Since $e(y, x) - e \in Y_F$, G is a smooth map from a neighbourhood of $(0, 0)$ in $Y_F \times \mathbb{R}^{F'_0}$ to $V_0^\perp$, $G(0, 0) = 0$, and $D_x G(0, 0)\xi = \langle U^*u_k, P^\perp U^*\xi \rangle / \nu = \Lambda(\xi)_k / \nu$, an isomorphism $\mathbb{R}^{F'_0} \rightarrow V_0^\perp$. The implicit function theorem gives a neighbourhood of 0 in Y_F and

a smooth map $y \mapsto x(y)$ with $G(y, x(y)) = 0$, $x(0) = 0$ and $\|x(y)\|_1 \leq C_0\|y\|$. For $\|U^*\Delta a\| \leq c_0$ small, $|x_s| \leq |a_s|/2$ on the finite set F'_0 . Put $\Delta a'' := x(U^*\Delta a)$. (iv) *Quantitative radius*. On $\{\|v - U^*a\| \leq \nu/2\}$ the map $v \mapsto v/\|v\|$ has first derivatives bounded by C/ν and second derivatives bounded by C/ν^2 ; hence $\|D_y G\| \leq C/\nu$, and $D_x G$ is C/ν^2 -Lipschitz in (y, x) for $\|y\| + \|U^*x\| \leq \nu/4$. Let B be the inverse of $D_x G(0, 0)$ on $V_0^\perp$ and enlarge C_0 so that $C_0 \geq \max\{1, 2\|B\|C/\nu\}$. Then $x \mapsto x - BG(y, x)$ is a contraction of the ball $\{\|x\|_1 \leq C_0\|y\|\}$ into itself as soon as $\|y\|(1 + C_0) \leq c\nu^2/(1 + \|B\|)$, and its fixed point $x(y)$ satisfies $|x_s| \leq |a_s|/2$ on F'_0 as soon as $C_0\|y\| \leq \min_{F_0} |a_s|/2$. Since $\|B\| \leq C_0\nu/C$, both conditions hold for $\|y\| \leq c_f \min\{1, \min_{F_0} |a_s|\}/(1 + C_0)^2$ with c_f depending only on ν . Then $a + D$ has support F and sign z there, $e^\# = e(U^*\Delta a, \Delta a'')$ and $u_k(\hat{z}^\#) - u_k(\hat{z}) = G(U^*\Delta a, \Delta a'')_k = 0$. $\square$

If (ND'_{L_0}) fails, some nontrivial combination u_c has $u_c|_F = \kappa a|_F$ with $\kappa \neq 0$, and every move of a on F changes $\sum_k c_k u_k(\hat{z})$ by exactly $-\kappa\nu\|e' - e\|^2/2$ (a second-order drift of one sign); this case is treated in Lemma 6.39 below.

For $W \in \ell_1$, $W \geq 0$, the *raise room* of W is

$$M^W(y) := \sum \{\mu_s^* W_s : s \in F, |a_s| < yW_s\},$$

$$(\text{RR}_W) \quad M^W(y) = O(y^{1+\varepsilon}) \text{ for some } \varepsilon > 0.$$

(RR_W) holds, for instance, if $|a_s| \geq c(\mu_s^*)^{1/(1+\varepsilon)}$ on F : then $|a_s| < yW_s$ forces $\mu_s^* < (y\|W\|_\infty/c)^{1+\varepsilon}$. Since $\log_2(1/\mu_s^*) = 2^{2^s}$, (RR_W) holds for every power profile $|a_s| \approx v_l(s)^{1+\beta_0}$; it fails only for μ -thin supports. (RR_W) is not a cushion condition: it holds for critical and super-critical profiles.

Theorem 6.34 (Support swallowing with raise room). *Let $T = T_{\text{final}}$ and $N \geq 1$. Let $f \in S_{p_N}^*$ (F arbitrary) satisfy (W^*) , $(H2)$, $(H3\text{-inf})$, $(\mathcal{B}_{\text{fin}})$, and, with $U_{\text{np}} := \sum_{l \in \mathcal{B}_{\text{np}}} |u_l|$, $(\text{ND}'_{\mathcal{B}_{\text{np}}})$ and $(\text{RR}_{U_{\text{np}}})$. Then $f \in \mathcal{R}$. No condition on the profile of a on the swallowed signature sets (sparse, critical, super-critical or mixed), on the d -neutrality of the bad carriers, or on their targets is imposed.*

Proof. Fix $g \in \mathcal{C}(f)$, $\rho \in (0, 1)$ and $\eta_0, \kappa_0, \varepsilon_{\text{tr}}, \eta$ as in the proof of Theorem 6.20, and let $\theta_* := (1 - \rho^2)/(12\rho^2)$. The idea is to run the proof of Theorem 6.29 not at f but at a raised companion f_j , where the support dominates the bounded switching, so that the deep set is empty; Theorem 2.19 transfers the recovery back to f . Let $C_\tau, K_\star(l), K_2^R(l), c_b, t_1$ be the constants of the proof of Theorem 6.29 (with $U_* = 0$ in Step 4), computed at f with a factor 2 of room; we show below that they serve at the companions.

Step 1 (windows). By (W^*) choose levels $l_j \rightarrow \infty$ with $\Lambda_f^*(l_j)/(l_j 2^{l_j^3} \Lambda^\circ(l_j)) \rightarrow 0$, and let w_j be the sub-window used at level l_j . Put $K_j := 2(1 + \|U\|)K_2^R(l_j)$ (a constant times $\Lambda_f^*(l_j)$) and $n_j := \lceil 48\rho^2 K_j/(c_b(1 - \rho^2)) \rceil$ and $\theta_j := \theta_*/l_j \leq \theta_*$. Choose $m_j \in \mathbb{N}$ minimal with

$$C_R T_j^\varepsilon \log(e/T_j) \leq \theta_j c_b^2 4^{-n_j}, \quad T_j := T_{\text{hi}}(w_j) 2^{-m_j},$$

where ε is the exponent of $(\text{RR}_{U_{\text{np}}})$ and C_R the constant of Step 2 below. Then $m_j = O((n_j + \log(l_j/(\theta_* c_b^2)))/\varepsilon)$, and $\log l_j = o(n(w_j))$; also $n_j = O(\Lambda_f^*(l_j)) = o(n(w_j))$ by (W^*) and (GW1) ; so for large j the scales $\mathcal{S}_j := \{T_j 2^{1-i} : 1 \leq i \leq n_j\}$ lie in $\mathcal{W}(w_j)$, and $K_j T_j \rightarrow 0$.

Step 2 (the companion). Put $y_j := 4C_\tau T_j$,

$$\Delta a_s := s_s(y_j U_{\text{np}}(s) - |a_s|)_+ \quad (s \in F),$$

let $\Delta a''$ be the correction of Lemma 6.33 for $L_0 := \mathcal{B}_{\text{np}}$, $D_j := \Delta a + \Delta a''$, $\lambda_j := q^*(a + D_j)$ and $f_j := f[(a + D_j)/\lambda_j, z]$. The raised set $\{s \in F : |a_s| < y_j U_{\text{np}}(s)\}$ leaves every finite subset of F as $j \rightarrow \infty$, so it misses F_0 for large j , and $\max_{s \in \text{supp } \Delta a} \mu_s^* \rightarrow 0$. Hence: $\|\Delta a\|_1 \leq y_j |\mathcal{B}| \rightarrow 0$; $\|U^* \Delta a\| \leq y_j M^{U_{\text{np}}}(y_j) \leq C y_j^{2+\varepsilon}$ and $\|U^* \Delta a\| = o(\|\Delta a\|_1)$; $\lambda_j \geq 1 + \|\Delta a\|_1 - \|\Delta a''\|_1 - \|U^* D_j\| \geq 1$ for large j ; $\lambda_j \rightarrow 1$ and $f_j \rightarrow f$ (Lemma 2.14(d); D_j is not a raise, but the proofs of Lemma 2.14(b), (c), (e) use only that z is fixed, and (d) follows from $\|D_j\|_1 \rightarrow 0$ and $\lambda_j \rightarrow 1$); and with $\varepsilon_e := \|U\| \|e_j - e\| \leq 2\|U\| \|U^* D_j\|/\nu \leq C y_j^{2+\varepsilon}$, Corollary 2.17 (with the correction term of Remark 2.16) and Lemma 2.12 give

$$p^*(f_j + r\rho g) \leq 1 + \frac{s(\lambda_j \rho r) - 1}{\lambda_j} + \varepsilon'_j,$$

$$\varepsilon'_j := 2\|U^* D_j\| + 2\|\Delta a''\|_1 + p^*(L^*(w_j - w)) \leq C_R T_j^{2+\varepsilon} \log(e/T_j),$$

so $\varepsilon'_j \leq \theta_j c_b^2 (T_j 2^{-n_j})^2 \leq \theta_* c_b^2 (T_j 2^{-n_j})^2$ by the choice of m_j . Moreover: (i) $|a_s^{(j)}| \geq y_j U_{\text{np}}(s)/\lambda_j \geq 2C_\tau T_j U_{\text{np}}(s)$ for $s \in F \setminus F_0$ (raised or not); (ii) z, F, K, J , the rooms, $\mathcal{B}$, the signs ϵ_l and T_0 are those of f ; the values $u_l(\hat{z}_j) = u_l(\hat{z})$, $l \in \mathcal{B}_{\text{np}}$, are preserved (Lemma 6.33), and all block scalars move by $O(\varepsilon_e)$ (Lemma 2.12(i)). At a strict non-peak, [I, Lemma 1.7] gives $q_l = \epsilon_l q_0 u_l(\hat{z})/\sigma_m(l)$; hence the d -rows of the exact-switching cone at f_j are positive multiples (one factor per block) of those at f , and since the remaining rows depend only on $z, F, T_0, \mathcal{B}$ and the peak statuses, the cone of Lemma 6.27 is the same set at f_j and at f , with Hoffman constants within a factor tending to 1 (the d -rows are rescaled by block factors $q_0^{(j)} \sigma_m / (q_0 \sigma_m^{(j)}) \rightarrow 1$). The gaps of the bad strict non-peaks and the margins of the non-degenerate bad peaks and of the peaks of (H2) are f -constants, so these statuses persist for large j .

Step 3 (decompositions at f_j). Let $g_j := g - (g(\xi_j)/q_0^{(j)})a^{(j)}$; then $g_j(\xi_j) = 0$ and $p^*(g - g_j) = O(\varepsilon_e)$, since $\|\xi_j - \xi\|_\infty = O(\varepsilon_e)$. By the display of Step 2 with $\rho = 1$ and Corollary 2.17, $p^*(f_j + t g_j) \leq 1 + \frac{t^2}{2}(1 + \eta'_j)$ for $t \in [T_j 2^{-n_j}, T_j]$, with $\eta'_j \leq (\lambda_j - 1)\lambda_j^2 + 2\theta_j c_b^2 + O(\varepsilon_e/t) \rightarrow 0$: here $\varepsilon'_j \leq \theta_j c_b^2 t^2$ on these scales by the choice of θ_j in Step 1, $\theta_j \rightarrow 0$, $\lambda_j \rightarrow 1$, and $\varepsilon_e/t \leq C y_j^{2+\varepsilon} 2^{n_j}/T_j \rightarrow 0$. The lemmas of [I, Section 8] used in the proof of Theorem 6.29 ([I, Lemmas 8.5–8.8, 8.10, 8.24, 8.26–8.28, 8.38, 8.39]) use the mate only through the budget $p^*(f' + t g') - 1 \leq t^2/2$ and $g'(\xi') = 0$; they hold at f_j for g_j on these scales with constants changed by factors at most 2. The uniform smallness [I, Lemma 8.6] holds along (f_j) : its proof by contradiction applies to sequences of pairs (j_n, t_n) with $t_n \rightarrow 0$, which force $j_n \rightarrow \infty$, and $f_{j_n} \rightarrow f$.

Step 4 (exact data at f_j without deep set). Run Steps 1–3 of the proof of Theorem 6.29 at f_j for g_j at a scale $t \in \mathcal{S}_j$. The bounded-switching constant is uniform: the map Φ of Lemma 6.11(b) depends continuously on $(e_j, \hat{z}_j)$, so $|\tau'_l| \leq C_\tau$ for large j . Since $\tau' = 0$ on $\mathcal{B}_{\text{pk}}$, $|X_s| \leq C_\tau U_{\text{np}}(s) \leq |a_s^{(j)}|/(2t)$ for $s \in F \setminus F_0$ by (i) and $t \leq T_j$; on the finite set F_0 , $|a_s^{(j)}| \geq |a_s|/4$ and $|X_s| \leq C_\tau |\mathcal{B}| \leq |a_s|/(2T_j)$ for large j . Hence the deep set is empty at every scale of $\mathcal{S}_j$. The bad carriers of $\mathcal{B}_K$ at degenerate peaks with sign $w(k(l)) = -\epsilon_l$ are handled as follows: their row $\tau_l = 0$ of the cone is kept; at f_j , $k(l)$ is either a peak, and then $\tau_l \leq \lambda_l K_{dt}$ by [I, Lemma 8.39(c)],

or a strict non-peak of gap $\mathbf{g}_j = O(\varepsilon_e)$, and then [I, Lemma 8.8(f)] and [I, (16)] give $\tau_l \leq \lambda_l(K_d t + 3\mathbf{g}_j/t)$ and $\lambda_l(|\omega_+(k)| + |\omega_-(k)|) \leq |\tau_l| + \lambda_l(K_d t + 6\mathbf{g}_j/t)$; in both cases $(\tau_l)_- \leq \mathbf{c}(\tau)/(2\mathbf{m}_l)$. As $\mathbf{g}_j = o(t^2)$ on $\mathcal{S}_j$, the violation of the row is $O(K_* t)$, and the window certificate vanishes at $k(l)$ (it vanishes at peaks and at coordinates of gap $< t^2$). Steps 1–3 of the proof of Theorem 6.29 then give d -neutral two-piece data at f_j for a functional $g_{j,t}$ with $(s^{(j)}b^+)_- \leq 3|a^{(j)}|/t$ and $(s^{(j)}b^-)_+ \leq 3|a^{(j)}|/t$ on F , $p^*(g_j - g_{j,t}) \leq \frac{1}{2}K_j t$, $\Gamma_w \leq 1 + \eta_0/2$ (the budget factor $1 + \eta'_j$ of Step 3 enters the bound of [I, Lemma 8.7(d)] multiplicatively: apply it to $g_j/(1 + \eta'_j)^{1/2}$; this gives $\Gamma_w \leq (1 + \eta'_j)(1 + \eta_\Gamma(\eta)) + o(1)$, which is $\leq 1 + \eta_0/2$ for large j because $\eta'_j \rightarrow 0$), and the block size conditions of kinds [1], [2] with uniform A_2, γ_B .

Step 5 (local validity). Lemma 6.10 for the sequence $f_j \rightarrow f$ with $U_* = 0$, $C_+ = 3$ gives, with c_b, t_1, j_0 independent of j , $p^*(f_j + r g_{j,t}) \leq 1 + \frac{r^2}{2}(1 + \eta_0)$ for $|r| \leq c_b t$, $t \in \mathcal{S}_j$, $j \geq j_0$.

Step 6. We verify Theorem 2.19 with the rows $f_j, \lambda_j, \varepsilon'_j, c_j := c_b$, the windows (T_j, n_j) and K_j : (a) is Step 2; (b) holds with $p^*(g - g_{j,t}) \leq O(\varepsilon_e) + \frac{1}{2}K_j t \leq K_j t$ and Step 5; (c), (d) hold by Steps 1–2; for (e), the averaged data over $\mathcal{S}_j$ are d -neutral two-piece data at f_j for $\bar{g}_j$ with $\kappa_w \leq 1 + \eta_0/2$ and side parts at most $3|a^{(j)}|/(T_j 2^{1-n_j})$ on F (convexity), so they satisfy (CS-side) at f_j , and if $\rho \bar{g}_j \in \mathcal{C}(f_j)$, Theorem 6.14 at f_j (with $L_- = \emptyset$) applied to $\rho \bar{g}_j$, whose data $\rho \bar{D}_j$ have $\kappa_w < 1$, gives $(f_j, \rho \bar{g}_j) \in \text{clNA}$. Hence $(f, \rho g) \in \text{clNA}$; let $\rho \rightarrow 1$. $\square$

Remark 6.35. (a) The ℓ_1 -mass of the raise is about $y_j m_{U_{\text{np}}}(y_j)$, of order y_j^2 for critical and much larger for super-critical profiles; for these profiles it is not $o(T_j^2)$, so Theorem 2.20 (which needs $p^*(f_j - f) = o(T_{\text{lo}}^2)$) does not apply. Theorem 2.19 needs only the transfer error ε'_j , which involves the Hilbert footprint of the raise and the normer change, but not its ℓ_1 -mass. (b) At the raised row the deep set is empty. (*Heuristic:* the factor-2 loss of Remark 6.30(c), which we attribute to deep assignment, therefore does not arise in this proof.) (c) The theorem concerns p_N . It does not by itself say anything about individual first rows of Martín's norm p : [I, Lemma 1.5] transfers density for p_N (for infinitely many N) to p , not membership of a row in $\mathcal{R}$.

The same raise recovers *fixed* d -neutral data without any sparsity condition.

Theorem 6.36 (Fixed data at raised rows). *Let $T = T_{\text{final}}$, $f \in S_{p^*}$ (F arbitrary), and let $g \in \mathcal{C}(f)$ carry d -neutral two-piece data $(b^\pm, \omega^\pm)$ with $\rho^2 \kappa_w < 1$. Put $W := (sb^+)_- + (sb^-)_+$ on F and let L_0 be the set of carriers (k, m) with $k \in \text{supp } \omega_m^+ \cup \text{supp } \omega_m^-$. If (RR_W) and (ND'_{L_0}) hold, then $(f, \rho g) \in \text{clNA}((c_0, p), \ell_2^2)$.*

Proof. For $y \rightarrow 0$ let f_y be the raise of f along W at level y ($\Delta a_s := s_s(yW_s - |a_s|)_+$) followed by the correction of Lemma 6.33 for L_0 ; as in Step 2 above, $\lambda_y \geq 1$, $\lambda_y \rightarrow 1$, $f_y \rightarrow f$, $\varepsilon'_y = O(y^{2+\varepsilon} \log(1/y))$, and $|a_s^{(y)}| \geq yW_s/\lambda_y$ off the fixed finite set F_0 . (1) For ω supported in strict non-peaks of block m , $d_m(\omega) = (mq_0/\sigma_m) \sum_k \Phi_m(k) u_{k,m}(\hat{z}) \omega(k)$ by [I, Lemma 1.7]; the values of L_0 are preserved and their gaps are f -constants, so at f_y (for small y) $d_m^{(y)} = (q_0^{(y)} \sigma_m / (q_0 \sigma_m^{(y)})) d_m$ on these vectors. Hence the data remain d -neutral, and since $b^+ - b^- = \sum_m R_m^*(\omega_m^- - \omega_m^+)$ does not involve the row, both pairs represent at f_y one functional G_y ; replacing $b^\pm$ by $b^\pm - \kappa_y a^{(y)}$, $\kappa_y := G_y(\xi_y)/q_0^{(y)}$, we get two-piece data at f_y (same z, F, K) for a functional g_y with $g_y(\xi_y) = 0$ and $p^*(g - g_y) = O(\varepsilon_e \log(e/\varepsilon_e))$

(the normer change), and $\Gamma_w^{(y)} \rightarrow \Gamma_w$. (2) Fix $\rho_1 \in (\rho, 1)$ with $\rho_1^2(\kappa_w + 2\varepsilon_{\text{tr}}) < 1$. $(s^{(y)}b^+)_-, (s^{(y)}b^-)_+ \leq W \leq \lambda_y|a^{(y)}|/y \leq 3|a^{(y)}|/t$ off F_0 for $t \leq T := y/4$ (and on F_0 for small y , up to the shift $\kappa_y a^{(y)}$). Apply Lemma 6.10 at f_y ($U_* = 0$, $C_+ = 4$, block coordinates of kind [2] with $A_2 := 1$, γ_B half the least gap) to the scaled data $\rho_1(b^\pm - \kappa_y a^{(y)}, \omega^\pm)$ of $\rho_1 g_y$, whose coefficient is $\rho_1^2 \Gamma_w^{(y)} \leq \rho_1^2(\kappa_w + o(1)) < 1 \leq 2$, so that hypothesis (OS1) of that lemma holds (the unscaled data may have $\Gamma_w > 2$ when $\rho < 1/\sqrt{2}$). It gives $p^*(f_y + r\rho_1 g_y) \leq 1 + \frac{r^2}{2}(\rho_1^2 \Gamma_w^{(y)} + \varepsilon_{\text{tr}})$ on the respective sides for $|r| \leq c_b T$, uniformly in small y . (3) For $|r| \leq c_b T$, (2) gives $p^*(f_y + r\rho_1 g_y) \leq 1 + \frac{r^2}{2}(1 - \delta) \leq s(r)$ for some $\delta > 0$ and small T ; for $|r| > c_b T$, Corollary 2.17, $\varepsilon'_y = o(T^2)$, $p^*(g - g_y) = o(T)$ and [I, Lemma 3.6(a)] give $p^*(f_y + r\rho_1 g_y) \leq s(r)$. So $\rho_1 g_y \in \mathcal{C}(f_y)$ for small y . (4) Theorem 6.14 at f_y for the mate $\rho_1 g_y$ with data $\rho_1(b^\pm - \kappa_y a^{(y)}, \omega^\pm)$ (side parts $\leq 4|a^{(y)}|/T$, so (CS-side) holds), with the factor ρ/ρ_1 , gives $(f_y, \rho g_y) \in \text{clNA}$; and $(f_y, \rho g_y) \rightarrow (f, \rho g)$. $\square$

Since every norm-attaining first row has finite base support [I, Proposition 1.10(c)], the approximants produced in Theorems 6.14, 6.34 and 6.36 have finite base supports; so for the pairs covered by these theorems the infinite-support half of Lemma Z holds (compare Proposition 6.48 below).

The raise room (RR) is removed by a pinning lemma: an unraised support coordinate that would be deep for the data pins the switching carrier through the mate's own flip budget. For a window index l_* let $T(l_*) := \bigcup_{l'' \leq l_*} \text{supp } y_{l''}$ and $T_B := \bigcup_{l \in B} \text{supp } y_l$.

Lemma 6.37 (Pinning by an unraised deep coordinate). *Let T be a signature-ladder operator, $f' \in S_{p^*}$ a first row with forced data $(a', z', \dots)$, $F' := \text{supp } a'$, and $g' \in X^*$ with $g'(\xi') = 0$ and $p^*(f' + r g') \leq 1 + (1 + \eta')r^2/2$ for $r = \pm t$, where $\eta' \leq 1$. Let $(B_\pm, \Theta_\pm)$ be a two-sided decomposition of g' at f' at scale $t \in \mathcal{W}(l_*)$, $l \leq l_*$ a carrier, ϵ_l a sign, $\tau_l := -\epsilon_l \Delta \theta_l$, $\sigma := \text{sign } \tau_l$, and let $s \in S_l \cap F'$ with $s \notin T(l_*)$ and $\text{sign}(a'_s)\epsilon_l = -\sigma$. Then*

$$|\tau_l|v_l(s) \leq \frac{2|a'_s|}{t} + \varphi_s + |r_s|, \quad \varphi_s := \mathfrak{f}_s^+ + \mathfrak{f}_s^-,$$

where $\sum_{s \in F'} \varphi_s \leq (1 + \eta')t/(2q'_0)$ and $r_s := \sum_{l'' > l_*, s \in \text{supp } y_{l''}} \Delta \theta_{l''} y_{l''}(s)/n_{l''}$ satisfies $\sum_s |r_s| \leq 8t^2$. Consequently, if $|\tau_l|v_l(s) \geq \frac{16}{5}|a'_s|/t$, then $|\tau_l| \leq \frac{8}{3}(\varphi_s + |r_s|)/v_l(s) \leq C_q t/v_l(s)$ with $C_q := \frac{8}{3}(2/q'_0 + 8)$. If $s \in (S_l \cap T(l_*) \cap F') \setminus T_B$, the same holds with r_s replaced by $r_s + r'_s$, $r'_s := \sum \Delta \theta_{l''} y_{l''}(s)/n_{l''}$ over the good $l'' \leq l_*$ with $s \in \text{supp } y_{l''}$; then $|r'_s| \leq \frac{4}{3} \sum_{\text{good}} |\Delta \theta_{l''}|$.

Proof. By (P1), for $s \in S_l$: $u_l(s) = v_l(s)$, $u_{l'}(s) = 0$ for $l' < l$, and $u_{l''}(s) = y_{l''}(s)/n_{l''}$ for $l'' > l$; for $l < l'' \leq l_*$, $s \in \text{supp } y_{l''}$ would put s into $T(l_*)$. So by [I, (15)], $\Delta B(s) = \epsilon_l \tau_l v_l(s) - r_s$. The proof of [I, Lemma 8.28] uses only the base budget; with the budget $(1 + \eta')t^2/2$ it gives $s_s B_+(s) \geq -|a'_s|/t - \mathfrak{f}_s^+$, $s_s B_-(s) \leq |a'_s|/t + \mathfrak{f}_s^-$ ($s_s := \text{sign } a'_s$) and the stated bound for $\sum \varphi_s$; hence $s_s \Delta B(s) \geq -2|a'_s|/t - \varphi_s$. As $s_s \epsilon_l = -\sigma$, $s_s \Delta B(s) = -|\tau_l|v_l(s) - s_s r_s$, which gives the first inequality. By [I, Lemma 8.10] and (P2), $\sum_s |r_s| \leq \frac{4}{3} \sum_{l'' > l_*} 6\lambda_{l''}/t \leq 8T_{l_0}(l_*)^3/t \leq 8t^2$. Under the deep hypothesis $2|a'_s|/t \leq \frac{5}{8}|\tau_l|v_l(s)$, which gives the second inequality. For $s \in T(l_*) \setminus T_B$ the additional terms come from coarse good targets containing s ($\|y_{l''}\|_1/n_{l''} \leq \frac{4}{3}$). $\square$

Theorem 6.38 (Support swallowing of any profile). *Let $T = T_{\text{final}}$ and $N \geq 1$. Every $f \in S_{p_N}^*$ (F arbitrary) satisfying (W^*) , $(H2)$, $(H3\text{-inf})$, $(\mathcal{B}_{\text{fin}})$ and $(ND'_{\mathcal{B}_{\text{np}}})$ belongs to $\mathcal{R}$. In particular μ -thin supports are allowed: no raise-room condition is needed.*

Proof. We describe the changes to the proof of Theorem 6.34. Write $\mathcal{B}' := \mathcal{B}_{\text{np}}$, $U_{\text{np}} = \sum_{l \in \mathcal{B}'} |u_l|$, $\delta_{\mathcal{B}} := \min_{l \in \mathcal{B}'} \delta_l$, and let $H \geq 1$ be the maximum, over $L' \subseteq \mathcal{B}'$, of the Hoffman constants (for the violation measure of [I, (18)] augmented by $\sum_{l \in L'} |\tau_l|$) of the cones $\mathcal{Z}_f(L') := \mathcal{Z}_f \cap \{\tau_l = 0 : l \in L'\}$, where $\mathcal{Z}_f$ is the cone of Lemma 6.27; there are finitely many, so H is an f -constant. Let $C'_f K_\star t$ bound the violation of $\mathcal{Z}_f$ by the actual amplitudes (proof of Lemma 6.27), $C_q := \frac{8}{3}(4/q_0 + 8)$, and $A := 96(1 + \|U\|)\rho^2/(c_b(1 - \rho^2))$.

Step 1' (thresholds). For a level l and an integer J put

$$K_1(l, J) := \frac{5}{4}C_q[(1 + 2K_\star(l))2^{s_{\max}(l)} + 2^J]/\delta_{\mathcal{B}},$$

$$K_{i+1} := 4H(C'_f K_\star + |\mathcal{B}'|K_i) + 2K_i \quad (1 \leq i \leq |\mathcal{B}'| + 1),$$

and $K'(l, J) := C_B(K_\star + K_{|\mathcal{B}'|+2})$, with an f -constant C_B such that $K'(l, J)t$ bounds the closeness constants of Step 3 of the proof of Theorem 6.29 when $C_f K_\star t$ is replaced there by $K_{|\mathcal{B}'|+2}t$. By induction, $K'(l, J) \leq C'_B[(1 + K_\star(l))2^{s_{\max}(l)} + 2^J]/\delta_{\mathcal{B}}$ with an f -constant C'_B : the dependence on 2^J is additive.

Step 1'' (windows and depth). Take levels l_j as in Theorem 6.34, the sub-window w_j of level l_j , $T_j := T_{\text{hi}}(w_j)$ and $L_j := \log_2(1/T_j)$. For $n \in \mathbb{N}$ put $\theta_j := \theta_\star/l_j$, $x_j(n) := \theta_j c_b^2 T_j 4^{-n}/C_e$ (C_e an f -constant fixed in Step 2') and $J_j(n) := \min\{J : \mu_\star^j \leq x_j(n)^2\}$. By Lemma 3.2(d), $2^{J_j(n)} \leq 2\log_2(2\log_2(1/x_j(n))) \leq 2\log_2(2(L_j + 2n + \log_2 l_j + C))$. Let n_j be the least $n \geq 1$ with $n \geq AK'(l_j, J_j(n))$; it exists because the right side grows at most logarithmically in n ; put $J_j := J_j(n_j)$. By Proposition 3.24(a), $n(w_j) \geq l_j 2^{L_j} \Lambda^\circ(l_j) \Omega(w_j)$ and $T_j \leq 2^{-L_j^3}/(l_j \Omega(w_j) \Lambda^\circ(l_j))$, where $\Omega(w_j) \geq \Omega_K(l_j) \geq 2^{s_{\max}(l_j)}/\delta_{\mathcal{B}}$ for large j ; moreover $L_j \leq l_j^3 + l_j + Q(w_j) \leq n(w_j)$ (Proposition 3.14(f)), $T_j n(w_j) \leq 2$, and $\log_2 n(w_j)/n(w_j) \rightarrow 0$ (Proposition 3.24(b)). With (W^*) this gives $n_j = o(n(w_j))$, $K'(l_j, J_j)T_j \rightarrow 0$ and $J_j \rightarrow \infty$. So $\mathcal{S}_j := \{T_j 2^{1-i} : 1 \leq i \leq n_j\} \subseteq \mathcal{W}(w_j)$.

Step 2' (raise only the deep part). Let $y_j := 4C_\tau T_j$ and

$$R_j := \left\{ s \in F \cap \bigcup_{l \in \mathcal{B}'} S_l : s \notin T_{\mathcal{B}}, s \geq J_j, |a_s| < y_j U_{\text{np}}(s) \right\},$$

$$\Delta a_s := s_s(y_j U_{\text{np}}(s) - |a_s|)_+ 1_{R_j}(s),$$

followed by the correction $\Delta a''$ of Lemma 6.33 for $L_0 := \mathcal{B}'$ ($J_j > \max F_0$ for large j), and let f_j be the resulting row. Then $\|U^* \Delta a\| \leq \mu_{J_j}^* \|\Delta a\|_1 \leq \mu_{J_j}^* y_j |\mathcal{B}|$, so $\lambda_j \geq 1$ for large j , and with $\mu := \mu_{J_j}^* \leq x^2$, $x := x_j(n_j)$, Lemma 2.12 and Lemma 6.33 give $\varepsilon'_j \leq C\mu y_j \log(e/(\mu y_j))$; since $\mu \log(e/\mu) \leq \sqrt{\mu} \leq x$ and $x^2 \log(1/y_j) \leq x$ for large j , $\varepsilon'_j \leq 2Cxy_j = 8CC_\tau \theta_j c_b^2 T_j^2 4^{-n_j}/C_e \leq \theta_j c_b^2 (T_j 2^{-n_j})^2$ once $C_e \geq 8CC_\tau$; as in Step 3 of the proof of Theorem 6.34, the budget factor $1 + \eta'_j$ then tends to 1, and the bound $\Gamma_w \leq 1 + \eta_0/2$ of the window data holds for large j . Further $\|\Delta a\|_1 \leq y_j |\mathcal{B}| \rightarrow 0$, $\lambda_j \rightarrow 1$, $f_j \rightarrow f$, and Step 2(ii) of the proof of Theorem 6.34 holds verbatim.

Step 4' (exact data without deep set). Fix $t \in \mathcal{S}_j$ and a two-sided decomposition of g_j at f_j at scale t , with actual amplitudes τ_l , $l \in \mathcal{B}'$. The $|\mathcal{B}'|$ numbers $|\tau_l|$ miss

one of the $|\mathcal{B}'| + 1$ intervals $(K_i t, K_{i+1} t]$; fix such an i_0 , let $L' := \{l : |\tau_l| \leq K_{i_0} t\}$ (pinned) and call the other $l \in \mathcal{B}'$ kept ($|\tau_l| > K_{i_0+1} t$). Let τ' be a nearest point of $\mathcal{Z}_f(L')$ to τ (the amplitudes of $\mathcal{B}_{\text{pk}}$ are treated as in Theorem 6.34). Then $\sum |\tau_l - \tau'_l| \leq H(C'_f K_* + |\mathcal{B}'| K_{i_0}) t \leq K_{i_0+1} t/4$, so $\tau'_l = 0$ on L' , and for kept l : $\text{sign } \tau'_l = \text{sign } \tau_l$ and $\frac{3}{4}|\tau_l| \leq |\tau'_l| \leq \frac{5}{4}|\tau_l|$; also $|\tau'_l| \leq C_\tau$. *Claim:* $X' := \sum_{l \in \mathcal{B}'} \epsilon_l \tau'_l u_l$ satisfies $s_j X'_j \geq -4|a_j^{(j)}|/t$ for all $j \in F$ (the deep set is empty). (i) $j \in T_{\mathcal{B}} \cap F$ (finite): $|a_j^{(j)}| \geq |a_j|/4$, so $4|a_j^{(j)}|/t \geq |a_j|/T_j \geq C_\tau |\mathcal{B}| \geq |X'_j|$ for large j . (ii) $j \notin T_{\mathcal{B}} \cup \bigcup_{l \in \mathcal{B}'} S_l$: $X'_j = 0$ by (P1). (iii) $j \in S_l \setminus T_{\mathcal{B}}$ with $\tau'_l = 0$: $X'_j = 0$. (iv) $j \in S_l \setminus T_{\mathcal{B}}$, l kept: $X'_j = \epsilon_l \tau'_l v_l(j)$. If $s_j \epsilon_l = \text{sign } \tau'_l$ there is nothing to prove; otherwise suppose $|\tau'_l| v_l(j) > 4|a_j^{(j)}|/t$. If $j \geq J_j$, then $|a_j^{(j)}| \geq y_j v_l(j)/2$ (whether or not $j \in R_j$), so $4|a_j^{(j)}|/t \geq 2y_j v_l(j)/T_j = 8C_\tau v_l(j) > |\tau'_l| v_l(j)$, a contradiction. If $j < J_j$, then $|\tau_l| v_l(j) \geq \frac{4}{5} |\tau'_l| v_l(j) > \frac{16}{5} |a_j^{(j)}|/t$, and Lemma 6.37 at $f' = f_j$ gives $|\tau_l| \leq C_q t / v_l(j) \leq \frac{5}{4} C_q 2^{J_j} t / \delta_{\mathcal{B}}$ if $j \notin T(l_j)$, and (with $|\tau'_j| \leq \frac{4}{3} K_* t$ by Lemma 6.27(a) at f_j) $|\tau_l| \leq C_q (1 + 2K_*) t / v_l(j) \leq \frac{5}{4} C_q (1 + 2K_*) 2^{s_{\max}(l_j)} t / \delta_{\mathcal{B}}$ if $j \in T(l_j) \setminus T_{\mathcal{B}}$ (we used $v_l(j) \geq \frac{4}{5} \delta_{\mathcal{B}} 2^{-j}$). In both cases $|\tau_l| \leq K_1 t < K_{i_0+1} t$, contradicting “kept”. Hence Steps 1–3 of the proof of Theorem 6.29, run at f_j with τ' (closeness $K_{|\mathcal{B}'|+2} t$ instead of $C_f K_* t$), give d -neutral two-piece data at f_j for a functional $g_{j,t}$ with $(s^{(j)} b^+)_-, (s^{(j)} b^-)_+ \leq 3|a_j^{(j)}|/t$ on F , $p^*(g_j - g_{j,t}) \leq K'(l_j, J_j)t$ and $\Gamma_w \leq (\sqrt{1 + \eta_\Gamma(\eta)} + K'_4 t)^2$ with $K'_4 = O(K'(l_j, J_j))$.

Steps 5, 6 are as in Theorem 6.34, with $K_j := 2(1 + \|U\|)K'(l_j, J_j)$, so that $n_j \geq AK'(l_j, J_j) = 48\rho^2 K_j / (c_b(1 - \rho^2))$ by Step 1". $\square$

The pinning constant of Lemma 6.37 at the depth J is of order 2^J , and a raise beyond J has footprint at most μ_j^* . The doubly exponential decay of μ^* makes $2^J = O(\log \log(1/x))$ when $\mu_j^* \leq x^2$, which is absorbed by every window; for $\mu_s = 2^{-s^2-1}$ or $\mu_s = 2^{-2^s}$ the corresponding fixed point of Step 1" need not exist (Remark 3.3).

Lemma 6.39 (Structure of the failure of (ND')). *Let T be a signature-ladder operator, $f \in S_{p^*}$ (F arbitrary), L_0 a finite set of carriers, $T_0 := \bigcup_{k \in L_0} \text{supp } y_k$, and suppose that $\sum_{k \in L_0} c_k u_k|_F = \kappa a|_F$ with $\kappa \neq 0$. Then $F \setminus T_0 \subseteq \bigcup \{S_k : k \in L_0, c_k \neq 0\}$ and $a_s = (c_k/\kappa)v_k(s)$ for $s \in S_k \cap F \setminus T_0$.*

Proof. Let $s \in F \setminus T_0$. If s lies in no S_k , $k \in L_0$, then $u_k(s) = 0$ for all $k \in L_0$ (signatures of L_0 and targets of L_0 miss s), so $\kappa a_s = 0$, contradicting $s \in F$. If $s \in S_k$, then $u_{k'}(s) = 0$ for $k' \in L_0 \setminus \{k\}$ (disjoint signature sets, targets in T_0) and $u_k(s) = v_k(s)$ by (P1); so $\kappa a_s = c_k v_k(s)$, and $a_s \neq 0$ forces $c_k \neq 0$. $\square$

Theorem 6.40 (Removal of (ND')). *Let $T = T_{\text{final}}$ and $N \geq 1$. Every $f \in S_{p_N^*}$ (F arbitrary) satisfying (W*), (H2), (H3-inf) and $(\mathcal{B}_{\text{fin}})$ belongs to $\mathcal{R}$.*

Proof. If $(\text{ND}'_{\mathcal{B}_{\text{np}}})$ holds, this is Theorem 6.38. Otherwise Lemma 6.39 with $L_0 := \mathcal{B}_{\text{np}}$ gives $F \setminus T_{\mathcal{B}} \subseteq \bigcup_{c_k \neq 0} S_k$ with $|a_s| = |c_k/\kappa| v_k(s)$ for $s \in S_k \cap F \setminus T_{\mathcal{B}}$. For $s \in F \setminus T_{\mathcal{B}}$ and $l \in \mathcal{B}$, $u_l(s) = 0$ unless $s \in S_l$ (bad targets lie in $T_{\mathcal{B}}$); and s lies in exactly one S_k , so $|u_l(s)|/|a_s| \leq \max_{c_k \neq 0} |\kappa/c_k|$. On the finite set $F \cap T_{\mathcal{B}}$ the ratios are finite. So (H4-inf) holds, and Theorem 6.29 gives $f \in \mathcal{R}$, with no raise. $\square$

Thus (ND') fails only when the support of a is carried, up to finitely many coordinates, by the signature sets of the kernel carriers with profiles proportional

to the signatures: then every bad-carrier switching on F is cushion-dominated and no raise is needed. Together, Theorems 6.38 and 6.40 settle infinite base support with finitely many bad carriers for T_{final} , under the same hypotheses (W^*) , $(H2)$, $(H3\text{-inf})$ as at finite F .

We turn to infinitely many bad carriers. Then bounded switching (Lemma 6.11) is not available, and the switching on F is of box size. For a level l_* put $\mathcal{B}(l_*) := \mathcal{B} \cap [1, l_*]$, $\mathcal{B}_{\text{np}}(l_*)$, $\mathcal{B}_{\text{pk}}(l_*)$ likewise, $T_{\mathcal{B}}(l_*) := \bigcup_{l \in \mathcal{B}(l_*)} \text{supp } y_l \subseteq [1, s_{\max}(l_*)]$, and $\text{Bx}_*(s) := \sum_{l \in \mathcal{B}_{\text{np}}(l_*)} \lambda_l |u_l(s)|$. The rooms $\mathfrak{r}_l^*$ of good $l \leq l_*$ are computed on $S_l \setminus (F \cup \bigcup_{l' \in \mathcal{B}(l_*), l' > l} \text{supp } y_{l'})$, and $K_*(l_*) := (1/q_0 + 22)\Lambda_f^*(l_*)$ with these rooms. Further

$$\gamma_{\mathcal{B}}(l_*) := \min \left(\min_m M_m, \min \{ \text{gap}_{m(l)}(k(l)) : l \in \mathcal{B}_{\text{np}}(l_*) \} \right),$$

$$\mu_{\mathcal{B}}(l_*) := \min \{ \mu_{k(l), m(l)} : l \in \mathcal{B}_{\text{pk}}(l_*), \alpha_{m(l)}(k(l)) \neq 0 \}$$

($\mu_{\mathcal{B}}(l_*) := 1$ if there is no such l). Let $\mathcal{Z}(l_*) \subseteq \mathbb{R}^{\mathcal{B}(l_*)}$ be the cone of Lemma 6.27 built from $\mathcal{B}(l_*)$, with $T_0 := T_{\mathcal{B}}(l_*) \cup \bigcup_{l \in \mathcal{B}_F(l_*)} (S_l \setminus F)$, and for $P \subseteq \mathcal{B}_{\text{np}}(l_*)$ and $\Sigma \subseteq T_{\mathcal{B}}(l_*) \cap F$ let

$$\mathcal{Z}(P, \Sigma) := \mathcal{Z}(l_*) \cap \{ \tau_l = 0 : l \in P \} \cap \left\{ s_j \sum_{l \in \mathcal{B}(l_*)} \epsilon_l \tau_l u_l(j) \geq 0 : j \in \Sigma \right\}.$$

Let $H(l_*) \geq 1$ be a constant such that for all P, Σ and $\tau \in \mathbb{R}^{\mathcal{B}(l_*)}$, $\text{dist}_1(\tau, \mathcal{Z}(P, \Sigma))$ is at most $H(l_*)$ times the sum of: the cost $\mathfrak{c}(\tau)$ of Lemma 6.27, $\sum_{l \in P \cup \mathcal{B}_{\text{pk}}(l_*)} |\tau_l|$, the moduli of the d -sums, and the negative parts of the Σ -rows; it exists by Hoffman's theorem (finitely many systems; the conversion of $\mathfrak{c}$ into violations involves the factors $1/(2m_l)$ and $1/(1-|z_j|)$, $j \in T_0 \setminus (F \cup K)$). Let $\psi(J) := 4 \sum_{l \in \mathcal{B}} \lambda_l \|u_l 1_{[J, \infty)}\|_1$, so that $\psi(J) \rightarrow 0$ as $J \rightarrow \infty$.

Lemma 6.41 (Window data at a deep-raised row). *Let $T = T_{\text{final}}$, $f \in S_{p_N}^*$ (F arbitrary, $\mathcal{B}$ possibly infinite) satisfy $(H2)$ and $(H3\text{-inf})$, let $g \in \mathcal{C}(f)$, and let $\eta_0, \kappa_0, \varepsilon_{\text{tr}}, \eta$ be as in the proof of Theorem 6.20. There are f -constants C'_f, C''_f, K_4 and a level l_f such that the following holds for $l_* \geq l_f$ with $\mathfrak{r}_l^* > 0$ for all good $l \leq l_*$ and $(\text{ND}'_{\mathcal{B}_{\text{np}}(l_*)})$. Let J be an integer with $J > \max F_0$ (F_0 of Lemma 6.33 for $L_0 := \mathcal{B}_{\text{np}}(l_*)$), let*

$$R := \{s \in F : s \geq J, |a_s| < 4\text{Bx}_*(s)\}, \quad \Delta a := \sum_{s \in R} s_s (4\text{Bx}_*(s) - |a_s|) e_s^*$$

(a deep raise at box level, of mass $\leq \psi(J)$ and footprint $\leq \mu_f^* \psi(J)$), let $\Delta a''$ be the correction of Lemma 6.33, and $f' := f[(a+D)/q^*(a+D), z]$, $D := \Delta a + \Delta a''$, with $\lambda' := q^*(a+D) \leq 2$. Put

$$K^{\text{pk}} := C'_f (1 + K_*(l_*)) (1 + l_*/\mu_{\mathcal{B}}(l_*)),$$

$$K_1 := \frac{5}{4} C_q [(1 + 2K_*(l_*)) 2^{s_{\max}(l_*)} + 2^J] / \delta_{\min}(l_*) + 4 + 4K^{\text{pk}},$$

$$K_{i+1} := 4H(l_*) (K^{\text{pk}} + l_* K_i + s_{\max}(l_*) K_1) + 2K_i \quad (1 \leq i \leq l_* + 1),$$

$$K_{\text{top}} := K_{l_*+2}.$$

Let $t \in \mathcal{W}(w_{l_*})$ be a dyadic scale with $t \leq \min(t_\eta, 1)$, and let g' be a functional with $g'(\xi') = 0$ and $p^*(f' + rg') \leq 1 + (1 + \eta')r^2/2$ at $r = \pm t$, $\eta' \leq 1$. Assume:

- (S1) every $j \in T_{\mathcal{B}}(l_*) \cap F$ with $j < J$ has $|a_j| \leq t^2$ or $|a_j| \geq K_{\text{top}} t^2$;
- (S2) $K_{\text{top}} t \leq \min(1, \kappa_0/K_4)$;

(S3) at f' , the bad strict non-peaks $l \leq l_*$ have gaps $\geq \gamma_{\mathcal{B}}(l_*)/2$, and the non-degenerate bad peaks $l \leq l_*$ are peaks with margins $\geq \mu_{\mathcal{B}}(l_*)/2$.

Then every two-sided decomposition of g' at f' at scale t yields d -neutral two-piece data $(b^\pm, \omega^\pm)$ at f' for a functional g'_t with $(s'b^+)_- \leq 3|a'|/t$ and $(s'b^-)_+ \leq 3|a'|/t$ on F , $p^*(g' - g'_t) \leq C_f'' K_{\text{top}} t$, $\Gamma_w(b^\pm, \omega^\pm) \leq (\sqrt{(1+\eta')(1+\eta_\Gamma(\eta))} + C_f'' K_{\text{top}} t)^2$, $t\|b^\pm\|_1 \leq 9$, and block coordinates of kind [1] or of kind [2] with $A_2 := 12$, $\gamma_B := \gamma_{\mathcal{B}}(l_*)/2$.

Proof. (1) *Pinning.* Lemma 6.27(a) at f' at level l_* with $\mathcal{B}(l_*)$ in place of $\mathcal{B}$: the fine bad carriers $l > l_*$ enter the good inequalities only through their targets, with total contribution $\leq 8t^2$ ([I, Lemma 8.10], (P2)); so $\sum_{\text{good}} |\Delta\theta_l| \leq K_* t$ (up to a factor 2 for the budget at f'). For a non-degenerate bad peak $k = k(l)$, $l \leq l_*$, [I, (16)] at f' with $\Delta\Theta(k) = -\epsilon_l \tau_l / \lambda_l$ gives $\text{sign}(w'(k)) \epsilon_l \tau_l / \lambda_l = |\omega_+(k)| + |\omega_-(k)| + \Delta d M'$, where $0 \leq |\omega_+(k)| + |\omega_-(k)| \leq t/(\sigma'|\alpha'(k)|) = t/(\lambda_l \mu_k)$ by [I, Lemma 8.8(c)] and [I, (2)], and $|\Delta d M'| \leq K_{dt}$ ([I, Lemma 8.39(a)] at f'); by (S3), $|\tau_l| \leq 2t/\mu_{\mathcal{B}}(l_*) + \lambda_l K_{dt}$. Degenerate bad peaks are handled by (H3-inf) as in Lemma 6.27. Hence the violation of $\mathcal{Z}(l_*)$ by the actual amplitudes $\tau = (\tau_l)_{l \in \mathcal{B}(l_*)}$ is at most $K^{\text{pk}} t$ (the fine bad d -terms are $O(t^2)$ and the fine bad targets on T_0 contribute $\leq 8t^2$). (2) *Pigeonhole and projection.* The at most l_* numbers $|\tau_l|$, $l \in \mathcal{B}_{\text{np}}(l_*)$, miss one of the $l_* + 1$ intervals $[K_i t, K_{i+1} t]$; let i_0 be such an index, $P := \{l : |\tau_l| \leq K_{i_0} t\}$ (pinned), and call the others kept. Let $\Sigma := \{j \in T_{\mathcal{B}}(l_*) \cap F : j < J, |a_j| \leq t^2\}$. For $j \in \Sigma$, $j \notin R$ and $|a'_j| \leq \frac{3}{2}|a_j|$, so [I, Lemma 8.28] at f' and [I, (15)] give $s_j \sum_{\mathcal{B}(l_*)} \epsilon_l \tau_l u_l(j) \geq -3|a_j|/t - \varphi_j - |r_j| \geq -(3 + K^{\text{pk}})t$, where r_j collects the good coarse, fine and bad-peak carriers at j . So the Σ -rows are violated by at most $s_{\max}(l_*) K_1 t$ in total. Let τ' be a nearest point of $\mathcal{Z}(P, \Sigma)$ to τ : $\sum |\tau - \tau'| \leq H(l_*)(K^{\text{pk}} + l_* K_{i_0} + s_{\max}(l_*) K_1) t \leq K_{i_0+1} t/4$. Hence $\tau' = 0$ on P , and for kept l , $\text{sign } \tau'_l = \text{sign } \tau_l$ and $\frac{3}{4}|\tau_l| \leq |\tau'_l| \leq \frac{5}{4}|\tau_l| \leq \frac{15}{2}\lambda_l/t$ ([I, Lemma 8.10]). (3) *No deep set.* Let $X' := \sum_{l \in \mathcal{B}_{\text{np}}(l_*)} \epsilon_l \tau'_l u_l$; then $|X'_j| \leq \frac{15}{2} \text{Bx}_*(j)/t$. We claim $s_j X'_j \geq -4|a'_j|/t$ for all $j \in F$. If $j \geq J$: $|a'_j| \geq 2\text{Bx}_*(j)$ (whether or not $j \in R$, as $\lambda' \leq 2$), so $4|a'_j|/t > |X'_j|$. If $j < J$ and $j \in S_l \setminus T_{\mathcal{B}}(l_*)$ for some $l \in \mathcal{B}_{\text{np}}(l_*)$: $X'_j = \epsilon_l \tau'_l v_l(j)$, which vanishes if l is pinned; if l is kept and $s_j X'_j < -4|a'_j|/t$, then Lemma 6.37 at f' applies to the actual τ_l and gives $|\tau_l| \leq K_1 t$, contradicting “kept”. If $j < J$ and $j \in T_{\mathcal{B}}(l_*) \cap F$: for $j \in \Sigma$ the Σ -row gives $s_j X'_j \geq 0$; otherwise $|a_j| \geq K_{\text{top}} t^2$ by (S1), and $s_j X_j(\tau) \geq -2|a'_j|/t - K^{\text{pk}} t$, $|X'_j - X_j(\tau)| \leq K_{\text{top}} t/4$ and $2|a'_j|/t \geq K_{\text{top}} t/2 \geq (K^{\text{pk}} + K_{\text{top}}/4)t$ (as $|a'_j| \geq |a_j|/4$ and $K_{\text{top}} \geq K_1 \geq 4K^{\text{pk}}$), so $s_j X'_j \geq -4|a'_j|/t$. In all other cases $X'_j = 0$. (4) *Data.* Steps 2–3 of the proof of Theorem 6.29 at f' with τ' (empty deep set, closeness $K_{\text{top}} t$ instead of $C_f K_* t$; bounded switching is not used, only $|\tau'_l| \leq \frac{15}{2}\lambda_l/t$) give the data; the bound for Γ_w is [I, Lemma 8.7(d)] applied to $g'/(1+\eta')^{1/2}$, whose budget at $r = \pm t$ is $r^2/2$, plus the closeness term. Since $\sum_F A_j \leq 2/t$ and $\|X'\|_1 \leq \frac{15}{2} \sum_l \lambda_l/t \leq 3/t$, $t\|b^\pm\|_1 \leq 9$; at $k = k(l)$, $|\omega^-(k)| \leq |\omega^+(k)| + |\tau'_l|/\lambda_l \leq (4 + \frac{15}{2})/t$; and (S3) gives the gap bound for kind [2]. $\square$

For a level l let $C_{\text{VP}}(l) := C_0 2^{\max F_0} / \min_{s \in F_0} |a_s|$, with $F_0, C_0 \geq 1$ the data of Lemma 6.33 for $L_0 := \mathcal{B}_{\text{np}}(l)$, and $C_{\text{VP}}(l) := 1$ if $F_0 = \emptyset$; so $C_{\text{VP}}(l) \geq 2^{\max F_0} \geq 1$ (as $|a_s| \leq 1$). We may and do assume $\gamma_{\mathcal{B}}(l), \mu_{\mathcal{B}}(l) \leq 1$ (replace them by their

minimum with 1). Put

$$\Xi_f(l) := \frac{(4H(l)l + 3)^{l+2} (1 + K_*(l)) (1 + C_{VP}(l))}{\gamma_B(l) \mu_B(l)}.$$

Theorem 6.42 (Infinitely many bad carriers). *Let $T = T_{\text{final}}$ and $N \geq 1$. Let $f \in S_{p_N^*}$ (F arbitrary, $\mathcal{B}$ possibly infinite) satisfy (H2), (H3-inf), $\mathfrak{r}_l^* > 0$ for every good l , $(\text{ND}'_{\mathcal{B}_{\text{np}}(l)})$ for every l , and*

$$(\text{W}_{\text{inf}}) \quad \liminf_{l \rightarrow \infty} \frac{\Xi_f(l)}{l^{l^3} \Lambda^\circ(l)} = 0.$$

Then $f \in \mathcal{R}$. For finite $\mathcal{B}$ the rates H , C_{VP} , γ_B , μ_B are f -constants for large l , and (W_{inf}) implies (W^) (because $\Xi_f(l) \geq 1 + K_*(l) \geq \Lambda_f^*(l)$); finite $\mathcal{B}$ is covered under (W^*) alone by Theorems 6.38 and 6.40.*

Proof. We follow the proof of Theorem 6.38, with Lemma 6.41 in place of Step 4'. (i) *Levels.* By (W_{inf}) choose l_j with $\Xi_f(l_j)/(l_j 2^{l_j^3} \Lambda^\circ(l_j)) \rightarrow 0$, and let w_j be the sub-window of level l_j , $L_j := \log_2(1/T_{\text{hi}}(w_j))$. (ii) *Local validity.* Lemma 6.10 with $U_* = 0$, $C_+ = 3$, $A_0 = 9$ gives c_0, t_1 depending only on f ; with $A_2 = 12$ and $\gamma_B = \gamma_B(l_j)/2$, $c_b(l_j) = c_0 \gamma_B(l_j)/24$. (iii) *Sub-window position.* For $n, p \in \mathbb{N}$ consider the scales $\mathcal{S}(n, p) := \{T_{\text{hi}}(w_j) 2^{-p} 2^{1-i} : 1 \leq i \leq n\}$. For an integer $J > \max F_0$ write $K_{\text{top}}(J)$ for the constant of Lemma 6.41 with $l_* = l_j$ and this J . Condition (S1) with $K_{\text{top}}(J)$ at all scales of $\mathcal{S}(n, p)$ excludes, for each of the at most $s_{\text{max}}(l_j)$ coordinates $j' \in T_B(l_j) \cap F$, at most $n + \log_2 K_{\text{top}}(J) + 2$ values of p ; so some $p \leq P(n, J) := (s_{\text{max}}(l_j) + 2)(n + \log_2 K_{\text{top}}(J) + 2)$ satisfies it at all scales of $\mathcal{S}(n, p)$. (iv) *Depth.* Put $\theta_j := \theta_*/l_j$ and, for (n, J) , in this order: $K_{\text{top}}(J)$, $P(n, J)$, $x(n, J) := \theta_j c_b(l_j)^2 \gamma_B(l_j) \mu_B(l_j) 2^{-2(L_j + P(n, J) + n)} / (C_e(1 + C_{VP}(l_j)))$. For each n let $J(n)$ be the least $J \geq 1 + \max F_0$ (or $J \geq 1$ if $F_0 = \emptyset$) with $\mu_J^* \leq x(n, J)^2$. It exists: $\log_2(1/\mu_J^*) = 2^{2^J}$, while $\log_2(1/x(n, J))$ grows at most linearly in J (since K_1 , hence $K_{\text{top}}(J)$, depends on J through the summand 2^J , $\log_2 K_{\text{top}}(J) \leq J + C_j$, and so $P(n, J)$ is at most linear in J). Moreover $2^{J(n)} \leq \max\{2^{1+\max F_0}, C \log_2(L_j + n + \log_2 l_j + C_j)\}$ with C_j independent of n , and $2^{1+\max F_0} \leq 2C_{VP}(l_j)$ (Lemma 3.2(d)). Put $P := P(n, J(n))$. The raise of Lemma 6.41 at depth $J(n)$ has footprint $\leq C\mu_{J(n)}^* \leq Cx(n, J(n))^2$, which is at most the radius $c_f/(1+C_{VP}(l_j))^2$ of Lemma 6.33 for C_e large; with Lemma 6.33 and Lemma 2.12 its transfer error is $\varepsilon' \leq C(1+C_{VP})\sqrt{\mu_{J(n)}^*} \leq C(1+C_{VP})x(n, J(n)) \leq \theta_j c_b^2(T'_{\text{lo}})^2$, where $T'_{\text{lo}} \geq T_{\text{hi}}(w_j) 2^{-P-n}$ is the least scale, once C_e is a large f -constant; and the values, block scalars, gaps and margins move by $\leq C\sqrt{\mu_{J(n)}^*} \leq \frac{1}{2} \min(\gamma_B, \mu_B)$, which gives (S3) (the values of $\mathcal{B}_{\text{np}}(l_j)$ are preserved, and the peak margins are Lipschitz in the values and the block scalars by Lemma 2.7). Let n_j be the least n with $n \geq AK_{\text{top}}(J(n))/c_b(l_j)$, $A := 96(1 + \|U\|)C_f''\rho^2/(1 - \rho^2)$; it exists because the right side depends on n only through $2^{J(n)}$, which grows at most logarithmically in n (this is the fixed point $J \rightarrow K_{\text{top}} \rightarrow n \rightarrow P \rightarrow x \rightarrow J$). Unrolling $K_{i+1} \leq (4H(l_j)l_j + 2)K_i + 4H(l_j)(K^{\text{pk}} + s_{\text{max}}(l_j)K_1)$ gives

$$K_{\text{top}} \leq (1 + s_{\text{max}}(l_j))(4H(l_j)l_j + 3)^{l_j+1}(K_1 + K^{\text{pk}})$$

(the cruder bound without the factor $1 + s_{\text{max}}(l_j)$ is false in general, because of the term $s_{\text{max}}K_1$ of the recursion). Using this, $(1 + s_{\text{max}})2^{s_{\text{max}}}/\delta_{\text{min}} \leq 2\Omega_K(l_j)$,

Proposition 3.24 and Proposition 3.14(f), one obtains

$$n_j + P_j \leq C \Xi_f(l_j) \Omega_K(l_j) s_{\max}(l_j) \log_2 n(w_j), \quad K_{\text{top}} T_{\text{hi}}(w_j) \rightarrow 0,$$

and, since $l2^{t^3} \Lambda^\circ(l) \Omega_K(l) s_{\max}(l) \log_2 n(w) \leq \text{Des}(l)^{1/2} \cdot 2 \log_2 Q(w) \leq Q(w)$ and $n(w) \geq l2^{t^3} Q(w)$, $(n_j + P_j)/n(w_j) \rightarrow 0$ by (W_{inf}) . So the chosen scales lie in $\mathcal{W}(w_j)$, and (S1), (S2), (S3) hold for large j . (v) *Conclusion*. The raise mass is $\leq \psi(J(n_j)) \rightarrow 0$, so $\lambda_j \rightarrow 1$ and $f_j \rightarrow f$; the budget hypothesis of Lemma 6.41 holds by Corollary 2.17, as in Step 3 of Theorem 6.34, with $\eta'_j \rightarrow 0$ by the choice of θ_j . Lemma 6.41 at f_j for g_j gives exact d -neutral data at every scale with cushion-compatible side parts, and since $\eta'_j \rightarrow 0$ and $K_{\text{top}} T_{\text{hi}}(w_j) \rightarrow 0$ these data satisfy $\Gamma_w \leq 1 + \eta_0/2$ on the chosen scales for large j ; Lemma 6.10 gives the pieces, and Theorem 2.19 together with Theorem 6.14 at f_j concludes as in Theorem 6.34. $\square$

Remark 6.43 (The rates in Ξ_f). $K_\star(l)$ contains the rooms of the good carriers, as in $(W^\star)$; $H(l)$ the Hoffman constants of the growing exact cones (through the d -rows $q_l = \epsilon_l q_0 u_l(\hat{z})/\sigma_m$ and the contact pattern on $T_{\mathcal{B}}(l) \setminus F$, and the factors $1/\mathfrak{m}_l$); $C_{\text{VP}}(l)$ the conditioning of the value-preserving correction; $\gamma_{\mathcal{B}}(l)$ the gaps of the bad strict non-peaks; and $\mu_{\mathcal{B}}(l)$ the margins of the bad peaks, which are needed both to control the rows $\tau_l = 0$ at bad peaks (through $\sigma|\alpha(k)| = \lambda_k \mu_k$) and to keep the bad peaks peaks under the raise. For finite $\mathcal{B}$ they are f -constants. For infinite $\mathcal{B}$ they are f -dependent rates of the kind that the clean sub-windows of Section 3 convert into design constants at finite F ; no box domination, no raise room and no bounded switching is used.

If (ND') fails at some level, it fails at all larger levels, and the value-preserving correction is not needed.

Proposition 6.44 (Infinitely many bad carriers without (ND')). *Let $T = T_{\text{final}}$, $N \geq 1$, and let $f \in S_{p_N}^*$ satisfy (H2), (H3-inf) and $\mathfrak{r}_l^\star > 0$ for every good l . Suppose (ND') $_{\mathcal{B}_{\text{np}}(l_0)}$ fails for some l_0 , so that by Lemma 6.39 there are finitely many kernel carriers k , numbers $c_k \neq 0$ and $\kappa \neq 0$ with $a = (c_k/\kappa)v_k$ on $S_k \cap F \setminus T_{\mathcal{B}}(l_0)$ and $F \setminus T_{\mathcal{B}}(l_0) \subseteq \bigcup_k S_k$. Put $m_k^F(l) := \|v_k 1_{S_k \cap F \setminus T_{\mathcal{B}}(l)}\|_1$. If (W_{inf}) holds with $C_{\text{VP}}(l)$ replaced by $1 + \max_k 1/m_k^F(l)$, then $f \in \mathcal{R}$.*

Proof. We run the proof of Lemma 6.41 with $f' = f$ (no raise and no correction: $R := \emptyset$, $D := 0$; its hypothesis (ND') is used only for the correction $\Delta a''$, which is not needed here) at levels $l_* \geq l_0$, with $K_1 \geq 3C_q(1 + 2K_\star)/\min_k m_k^F(l_*)$ added to its definition, and show that the deep set is empty. Write $\hat{S}_k := S_k \cap F \setminus T_{\mathcal{B}}(l_*)$. On $\hat{S}_k$ the sign of a is constant and $|a_s| = |c_k/\kappa|v_k(s)$. If the switching of a kernel carrier k is anti-signed there and $|\tau_k| \geq 3|c_k/\kappa|/t$, summing the first inequality of Lemma 6.37 over $s \in \hat{S}_k$ gives $(|\tau_k| - 2|c_k/\kappa|/t)m_k^F \leq t/q_0 + 8t^2 + \frac{4}{3}K_\star t$, hence $|\tau_k| \leq 3C_q(1 + 2K_\star)t/m_k^F \leq K_1 t$, so k is pinned. Thus a kept kernel carrier has $|\tau'_k| \leq \frac{5}{4}|\tau_k| < \frac{15}{4}|c_k/\kappa|/t$, and on $\hat{S}_k$, $s_s X'_s \geq -\frac{15}{4}|c_k/\kappa|v_k(s)/t > -4|a_s|/t$. At $j \in S_k \cap F \cap T_{\mathcal{B}}(l_*) \setminus T_{\mathcal{B}}(l_0)$ the non-kernel part of X'_j is at most $\frac{15}{2} \sum_{l > k, j \in \text{supp } y_l} \lambda_l |y_l(j)|/(n_l t) \leq 5 \cdot 2^{-2j} \delta_k/t$ by Proposition 3.15(d), which is below $\frac{1}{4}|c_k/\kappa|v_k(j)/t$ once $2^j \geq 20n_k|\kappa/c_k|$. The finitely many remaining coordinates of F (including those of $S_l \cap F$ for non-kernel bad l , which lie in $T_{\mathcal{B}}(l_0)$) have fixed $|a_j| > 0$ and are thick ($|a_j| \geq K_{\text{top}} t^2$) at the scales of large levels, where the argument of step (3) of the proof of Lemma 6.41 applies. The data of Lemma 6.41

then live at f itself, and Theorem 6.16 concludes, with the window arithmetic of Theorem 6.42. $\square$

When $F \cap S_k$ has bounded gaps beyond $s_{\max}(l)$, $1/m_k^F(l) \leq C_k 2^{s_{\max}(l)}/\delta_k$ is absorbed by $\Omega_K(l)$, and no rate is added. For very sparse $F \cap S_k$ the rate $1/m_k^F(l)$ is genuine.

6.8. Deep raises at clean sub-windows, and the residual. The finite-support master theorems (Sections 4 and 5) work at clean sub-windows with companions that close rooms, raise thresholds and tune values. To transport them to infinite base support one raises *first*, deep and at box level, and then exactifies at the raised row. The next two lemmas are the ingredients that are proved; the transport itself is only sketched (Remark 6.50). Recall the rate objects (O1)–(O12) of Definition 3.7, clean sub-windows (Definition 3.18), and $s_t(l)$ of Definition 3.5(5).

Lemma 6.45 (A deep raise preserves the classification). *Let $T = T_{\text{final}}$, $f \in S_{p_N}^*$ (F arbitrary), l a level, $J > s_t(l)$, and let $f^r := f[(a + \Delta a)/\lambda_r, z]$ for a raise Δa supported in $F \cap [J, \infty)$, $\lambda_r := q^*(a + \Delta a) \in [1, 1 + 2\|\Delta a\|_1]$. Then:*

- (a) *the values at f^r of the rate objects (O1), (O2), (O6) and (O11) of level l equal those at f , and the values of (O10) are those at f divided by λ_r ;*
- (b) *every value $u_k(\hat{z})$ moves by at most $2\|U^*\Delta a\|/\nu$, and there is $M_f(l) = C_f^l \text{Des}(l)$ (C_f an f -constant) such that the value of every object of (O3)–(O5), (O7), (O8), (O12) of level l moves by at most $M_f(l)\|U^*\Delta a\|$, provided $M_f(l)\|U^*\Delta a\| \leq 1$.*

Consequently, if w is a clean sub-window of level l for f , $M_f(l)\|U^\Delta a\| \leq b(w)^2$ and $\|\Delta a\|_1 \leq \frac{1}{2}$, then every object of (O1)–(O8), (O10)–(O12) that is tiny at f has value $\leq 2b(w)$ at f^r , and every robust one has value $\geq u(w)/2$.*

Proof. (a) z and $F = \text{supp}(a + \Delta a)$ are unchanged, so the rooms (O1), the target rooms (O2) and the shift costs (O6) (which depend only on F , on $z|_{F^c}$ and on design data) are unchanged. Since $J > s_t(l)$, the coordinates entering (O10), (O11) are not raised, so $a_s^r = a_s/\lambda_r$ there, and the ratios in (O11) are unchanged. (b) $e^r = U^*(a + \Delta a)/\|U^*(a + \Delta a)\|$ is invariant under normalization, so $\|e^r - e\| \leq 2\|U^*\Delta a\|/\nu$ and $|u_k(\hat{z}^r) - u_k(\hat{z})| \leq \|U^*u_k\|\|e^r - e\|$ with $\|U^*u_k\| \leq q^*(u_k) = 1$; the block scalars move by $C_f\|U^*\Delta a\|$ (Lemma 2.12(i)). (O3), (O4) are Lipschitz in the values and the block scalars with constants $\leq C_f/\Phi_{\min}(l)^2$, $\Phi_{\min}(l) := \min_{l'' \leq l} \Phi_{l''}$, and $1/\Phi_{l''} \leq D_{\text{room}}(l) \leq \text{Des}(l)^{1/6}$; (O5) is linear in the values divided by $\Phi_{\max} \geq \Phi_{\min}$; the singular values in (O7) are 1-Lipschitz in the operator norm (Weyl); the minors (O8) have Lipschitz constant $\text{Lip}(l) \leq \text{Des}(l)$. For (O12), $\gamma_k^r a_s^r/(\nu^r)^2 = (\gamma_k + \langle U^*u_k, U^*\Delta a \rangle) a_s/(\nu^2 + 2\langle U^*a, U^*\Delta a \rangle + \|U^*\Delta a\|^2)$ for $s \leq s_t(l)$: the factor λ_r cancels, so the entries of Γ move by $C_f\|U^*\Delta a\|$, and the determinants of order at most l (entries bounded by C_f) move by at most $l!C_f^l$ times that. The last statement follows since $b(w) \leq 1$, $u(w)/b(w) \geq 2^{64}$, and, for (O10), since $\lambda_r \leq 1 + 2\|\Delta a\|_1 \leq 2$ by the hypothesis $\|\Delta a\|_1 \leq \frac{1}{2}$ (without it a robust (O10) object would only have value $\geq u(w)/\lambda_r$). $\square$

Since $\|U^*\Delta a\| \leq \mu_J^*\|\Delta a\|_1$, the requirement $M_f(l)\|U^*\Delta a\| \leq b(w)^2$ holds for a box-level deep raise as in Lemma 6.41 as soon as $C_f^l \text{Des}(l) \mu_J^* \psi(J) \leq b(w)^2$, and there $\|\Delta a\|_1 \leq \psi(J) \leq \frac{1}{2}$ for large J ; by Lemma 3.2(d) the corresponding 2^J is of order $l \log_2(C_f) + \log_2 \log_2(\text{Des}(l)/b(w))$. No value-preserving correction is needed:

the classification at w survives with factors 2, and all exactifications are done at f^r with its own values. The shift-pinning rates (O9) are not covered by the lemma: their definition is an infimum over unbounded τ with value-dependent coefficients, and we do not prove their stability.

At infinite base support the exactifying moves of Section 4 (banks and pulls) may fall on support coordinates. The following first-order calculus of moves on the support replaces them there.

Lemma 6.46 (Moves on the support). *Let $T = T_{\text{final}}$ and let $f^\circ = f[a^\circ, z^\circ]$ be a first row with base support F° and $\nu^\circ := \|U^*a^\circ\|$, $e^\circ, \gamma_k^\circ := \langle U^*u_k, U^*a^\circ \rangle$. For a finite set $Y \subseteq F^\circ$ and $\Delta \in \mathbb{R}^Y$ with $|\Delta_x| \leq |a_x^\circ|/2$ let $A := a^\circ + \sum_{x \in Y} \Delta_x e_x^*$ and $f^\Delta := f[A/q^*(A), z^\circ]$. Put $\alpha_x := (\mu_x^*)^2 a_x^\circ / (\nu^\circ)^2$.*

(a) *For every carrier k ,*

$$u_k(\hat{z}^\Delta) - u_k(\hat{z}^\circ) = \frac{1}{\nu^\circ} \sum_{x \in Y} (\mu_x^*)^2 u_k(x) \Delta_x - \frac{\gamma_k^\circ}{\nu^\circ} \sum_{x \in Y} \alpha_x \Delta_x + R_k,$$

$$|R_k| \leq \frac{C}{(\nu^\circ)^2} \left(\sum_{x \in Y} \mu_x^* |\Delta_x| \right)^2,$$

with an absolute constant C , provided $\sum_x \mu_x^ |\Delta_x| \leq \nu^\circ/2$.*

(b) (Pairs.) *Let l'' be a carrier, $L \geq l''$ a level and $s < s'$ in $S_{l''} \cap F^\circ$ beyond $s_{\max}(L)$, and put $\mathbf{a}_x := a_x^\circ / v_{l''}(x)$. For $\Delta_{s'} \in \mathbb{R}$ let $\Delta_s := -(\alpha_{s'}/\alpha_s) \Delta_{s'}$. Then $|\Delta_s| \leq (\mu_{s'}^* / \mu_s^*)^2 (|a_{s'}^\circ| / |a_s^\circ|) |\Delta_{s'}|$, every carrier $k \leq L$ with $k \neq l''$ satisfies $|u_k(\hat{z}^\Delta) - u_k(\hat{z}^\circ)| \leq |R_k|$, and*

$$u_{l''}(\hat{z}^\Delta) - u_{l''}(\hat{z}^\circ) = \frac{(\mu_{s'}^*)^2 v_{l''}(s')}{\nu^\circ} \left(1 - \frac{\mathbf{a}_{s'}}{\mathbf{a}_s} \right) \Delta_{s'} + R_{l''}.$$

(c) (Single moves are not private.) *A single move Δ_x at a coordinate x with $u_k(x) = 0$ changes $u_k(\hat{z})$ at first order by $-\gamma_k^\circ \alpha_x \Delta_x / \nu^\circ$.*

Proof. (a) $U^*A = U^*a^\circ + \sum_x \Delta_x \mu_x^* k_x$, and $e^\Delta = U^*A / \|U^*A\|$ does not depend on the normalization of A ; z° is fixed, so $u_k(\hat{z}^\Delta) - u_k(\hat{z}^\circ) = \langle U^*u_k, e^\Delta - e^\circ \rangle$. The map $h \mapsto (U^*a^\circ + h) / \|U^*a^\circ + h\|$ has derivative $h \mapsto (h - \langle h, e^\circ \rangle e^\circ) / \nu^\circ$ at 0 and second derivative bounded by $C/(\nu^\circ)^2$ on $\|h\| \leq \nu^\circ/2$. With $h = \sum_x \Delta_x \mu_x^* k_x$, $\langle h, e^\circ \rangle = \sum_x \Delta_x (\mu_x^*)^2 a_x^\circ / \nu^\circ$ and $\langle U^*u_k, k_x \rangle = \mu_x^* u_k(x)$, $\langle U^*u_k, e^\circ \rangle = \gamma_k^\circ / \nu^\circ$, which gives the formula; $\|U^*u_k\| \leq 1$. (b) $\alpha_s \Delta_s + \alpha_{s'} \Delta_{s'} = 0$ removes the second term of (a). For $k \leq L$, $k \neq l''$, u_k vanishes at s, s' (Theorem 3.12 and (P1): beyond $s_{\max}(L)$ only l'' is nonzero on $S_{l''}$ among the carriers $\leq L$). For l'' : $(\mu_s^*)^2 v_{l''}(s) \Delta_s = -(\mu_{s'}^*)^2 a_{s'}^\circ v_{l''}(s) \Delta_{s'} / a_s^\circ = -(\mu_{s'}^*)^2 v_{l''}(s') (\mathbf{a}_{s'} / \mathbf{a}_s) \Delta_{s'}$. (c) is (a) with $Y = \{x\}$. $\square$

Since μ^* decreases, the deeper coordinate s' must do the work: $|\Delta_s| \leq u^{-1} |\Delta_{s'}|$ when $|a_s^\circ| \geq u |a_{s'}^\circ|$, whereas pairing in the other order would require astronomically large moves. If $|a_s^\circ|, |a_{s'}^\circ| \geq u$, the spread $|1 - \mathbf{a}_{s'} / \mathbf{a}_s| \geq u$ and $s, s' \leq s_t(L)$, the private effect in (b) is at least $\text{Des}(L)^{-1} u |\Delta_{s'}|$ (Proposition 3.16(c)), so robust pairs are two-sided private tuning resources with design constants; a single anchor coordinate s_0 with $u_k(s_0) = 0$ for the tuned carriers can replace the partner by choosing Δ_{s_0} so that $\sum_x \alpha_x \Delta_x = 0$. By (c), a single robust support move is not an admissible tuning resource: its common term moves every value at first order. Where the tuning coordinates are not in F , Lemma 4.15 applies verbatim (its proof uses the base support only through hypothesis (T2), which keeps the bank

and pull coordinates off the support, and through Lemma 4.14, which holds for every $A^0 \in \ell_1$).

Proposition 6.47 (Reduction through raised companions). *Let T be admissible, $f \in S_{p^*}$, $g \in \mathcal{C}(f)$, $\rho \in (0, 1)$ and η_0 as in Theorem 2.19. Suppose there are first rows f_j^{ex} , raises Δa_j of f_j^{ex} with $\lambda_j := q^*(a_j^{\text{ex}} + \Delta a_j) \geq 1$, the raise companions f_j , and windows (T_j, n_j) , numbers $c_j \in (0, 1]$, $K_j \geq 0$ and functionals $g_{j,t}$ satisfying (b)–(e) of Theorem 2.19 at f_j , and*

$$\varepsilon'_j := p^*(f_j^{\text{ex}} - f) + 2\|U^* \Delta a_j\| + p^*(L^*(w_j - w_j^{\text{ex}})) \leq \theta_* c_j^2 (T_j 2^{-n_j})^2.$$

Then $(f, \rho g) \in \text{cl NA}((c_0, p), \ell_2^2)$.

Proof. By Lemma 2.15 at f_j^{ex} and $p^*(f_j^{\text{ex}} + \lambda_j r \rho g) \leq s(\lambda_j \rho r) + p^*(f_j^{\text{ex}} - f)$, hypothesis (a) of Theorem 2.19 holds with this ε'_j (as $\lambda_j \geq 1$); apply that theorem. $\square$

The finite-support companions of Sections 4 and 5 have $p^*(f_j^{\text{ex}} - f) = o(T_{\text{lo}}^2)$; the proposition reduces the transport of the finite-support master theorems to the production of exact window data, cushion-compatible on F , at raised exactifying companions.

Proposition 6.48 (The infinite-support half of Lemma Z). *Let T be a signature-ladder operator and $N \geq 1$. The following are equivalent:*

- (i) $\text{NA}((c_0, p_N), \ell_2^2)$ is dense;
- (ii) Lemma Z [I, Problem 8.21] holds at every f with finite base support, and (LSC-trunc) holds: for every $f \in S_{p_N^*}$, $g \in \mathcal{C}(f)$, $\rho < 1$ and $\varepsilon > 0$ there is a first row f' with finite base support, $p_N^*(f' - f) < \varepsilon$ and $\text{dist}(\rho g, \mathcal{C}(f')) < \varepsilon$.

(LSC-trunc) holds for every pair (f, g) covered by Theorems 6.3, 6.14, 6.16, 6.20–6.23, 6.29, 6.31, Corollary 6.18, and, for $T = T_{\text{final}}$, Theorems 6.34–6.42 and Proposition 6.44.

Proof. For both operators [I, Theorem 8.20] holds (its proof uses [I, Theorem 8.18] and [I, Lemma 8.4(a)], and Lemma 3.22(b)). (i) $\Rightarrow$ (ii): [I, Theorem 8.20] gives Lemma Z everywhere, and [I, Lemma 1.16] gives norm-attaining approximants, which have finite base support [I, Proposition 1.10(c)], with $\rho g \in \text{LiC}(f_n)$. (ii) $\Rightarrow$ (i): given $(f, g, \rho, \varepsilon)$, (LSC-trunc) gives f' with finite F' and $g' \in \mathcal{C}(f')$ with $p^*(g' - \rho g) < \varepsilon$; Lemma Z at f' for (g', ρ', ε) gives $f'' \in \mathcal{R}_0$ and $g'' \in \mathcal{C}(f'')$ with $p^*(f'' - f') < \varepsilon$ and $p^*(g'' - \rho' g') < \varepsilon$. Then $p^*(f'' - f) < 2\varepsilon$ and $p^*(g'' - \rho' \rho g) < 2\varepsilon$; as $\rho \rho'$ ranges over $(0, 1)$, Lemma Z holds at f , hence everywhere, and [I, Theorem 8.20] gives (i). For the last statement: each of the listed results produces norm-attaining approximants $f'_i \rightarrow f$ (finite base support) and $g'_i \in \mathcal{C}(f'_i)$ with $g'_i \rightarrow \rho g$ (Theorem 2.19 through rows $f_j \rightarrow f$, each of which is recovered along norm-attaining rows by Theorem 6.14). $\square$

Remark 6.49 (Truncation is not the obstruction). In Theorem 6.14 the support beyond the window N_w becomes contacts of the norm-attaining approximant, at a first-order cost controlled by the tail of the *fixed* vector $v = b^+ - b^-$, and the near support is raised where the average data need it. So, once exact data with one-sided cushion sparsity exist (fixed, or window data at f , or at a companion with small transfer error), truncating the support costs nothing; what remains at infinite F is the production of such data. This is an interpretation of the proofs, not a theorem about pairs without exact data.

Remark 6.50 (Transport of Master Theorem III' to infinite support; sketch). *Claim (sketch, not proved here)*. For T_{final} and every N , a first row f with infinite F belongs to $\mathcal{R}$ unless, at all but finitely many levels, every clean sub-window w exhibits (C_w^*) (Definition 5.16, read at the deep-raised row), or a tiny *tuning regularity modulus* (below), or a failure of (H2), or a bad degenerate peak with the swallowing sign none of whose block's donor peaks has a tuning resource. *Intended route*: at a clean w , (1) a deep box-level raise $f \rightarrow f^r$ beyond $J > s_t(l)$ (Lemma 6.45; no value-preserving correction); (2) at f^r , the moves (C1)–(C4) of Section 4 and the exactification of tiny minors of Section 5, with banks and pulls replaced by the support moves of Lemma 6.46 where the coordinates lie in F ; (3) exact d -neutral window data at the companion, cushion-compatible on F by the argument of Lemma 6.41 (Lemma 6.37 and the pigeonholes); (4) Proposition 6.47 and Theorem 6.14. *Items not written out*: the transplant of exact data at infinite F (the finite-support transplant copies B_+1_F and uses [I, Lemma 8.9]; it must be rerun with the clamped data of Theorem 6.29 and Lemma 6.37 at the decomposition row); the stability of the shift-pinning rates (O9) under the deep raise; the exactification of tiny minors with the additional pinned faces and contact rows of Lemma 6.41 (these are subsystems, whose minors are rate objects); the joint fixed point of the support tuning with the threshold buffer; and the raise level $8Bx_*$ needed because the box rows allow $|\tau'_l| \leq 12\lambda_l/t$. *Tuning regularity modulus*: the first-order resource map (private two-sided columns of contacts, convertible thin support coordinates, robust pairs and anchors of Lemma 6.46; one-sided pull columns; and single-support columns $((\mu_s^*)^2/\nu)(u_k(s) - \gamma_k a_s/\nu^2)_{k \in L_0}$ at robust support coordinates) must map the cone of admissible moves onto $\mathbb{R}^{L_0}$ with a right inverse of norm at most $(\text{Des}(l)/u(w))^C$. Its modulus is a finite minimum of determinants in $(a_s)_{s \leq s_t(l)}$ and values, i.e. built from rate objects; when only single-support columns exist on proportional profiles $a = c_k v_k$, it reduces to the Sherman–Morrison denominator $|1 - \sum_k c_k \gamma_k/\nu^2|$ of Remark 3.9, which vanishes in particular when (ND') fails (then $a|_F = \sum_k c_k u_k|_F$); the converse is not claimed. (A description of this residual as the conjunction of “every usable coordinate is thick support” and “nearly proportional profile” alone is too narrow: thick bank coordinates with off-support pull coordinates of the wrong orientation and tiny spreads also lack resources.)

Remark 6.51 (The residual at infinite base support). (a) *Recovered* (proved above). *Rows* (every mate is recovered, i.e. the row is in $\mathcal{R}$): first rows with signature room (SR), sign-mixed room or cushion room at any F (Theorems 6.21–6.23); finitely many cushion-sparse swallowed carriers for every signature-ladder operator, under (W^*) , (H2), (H3-inf) (Theorem 6.29); box-dominated support $F = \mathbb{N}$ for every admissible operator under (SC_I) and (W_M) (Theorem 6.31); for T_{final} : finitely many bad carriers with any support profile, under (W^*) , (H2), (H3-inf) (Theorems 6.38, 6.40), and infinitely many bad carriers under (H2), (H3-inf) and (W_{inf}) (Theorem 6.42, Proposition 6.44). *Pairs* (a given mate is recovered): window-pinned mates at any F (Theorem 6.20); cushion-sparse mates of block-tame rows, with V_f of sparse profile (Corollary 6.18); and, for T_{final} , mates ρg with fixed d -neutral data, $\rho^2 \kappa_w < 1$, (RR_W) and (ND'_{L_0}) (Theorem 6.36). (b) *Open for T_{final} at infinite F* , for pairs (f, g) with g not window-pinned: (b1) the rates: rooms of good carriers decaying too fast for (W^*) (the support-free part of approximate swallowing), and, for infinitely many bad carriers, failure of (W_{inf}) , i.e. of the rates $H(l)$, $C_{\text{VP}}(l)$, $\gamma_B(l)$, $\mu_B(l)$ (and $1/m_k^F(l)$ in Proposition 6.44); these would become design

constants through the transport of Remark 6.50; (b2) failure of (H2), and bad degenerate peaks with the swallowing sign, (H3-inf), whose treatment needs a donor with a tuning resource; (b3) the finite-support residual (C^*) of Definition 5.16, read at deep-raised rows; (b4) a tiny tuning regularity modulus (Remark 6.50); (b5) the scrambling condition (SC) for non- d -neutral data at raised rows below the scale of the normer change, which enters only inside (C^*). (c) For signature-ladder operators without a fast-decaying diagonal base, non-sparse support swallowing (critical and super-critical profiles beyond the special cases of Remark 6.30(d)) and infinitely many swallowed carriers without box domination remain open. (d) No counterexample is known in any of these classes. (*Interpretation, not a statement proved here:*) every infinite-support mechanism examined is either harmless after a deep raise (thin, critical, super-critical, μ -thin supports, and box-size switching under (W_{inf})), pinned by the mate's own flip budget (thin shallow coordinates), cushion-dominated (failure of (ND')), or a rate phenomenon of the same kind as at finite base support.

7. STATUS AND OPEN PROBLEMS

Throughout this section $T = T_{\text{final}}$ with base U_{final} (Section 3), except where another operator is named, and $N \geq 1$.

7.1. What is proved.

- (a) T_{final} is admissible, satisfies the conclusion of Martín's Lemma B, does not depend on N , and satisfies every hypothesis of [I, Section 8] on every choice of sub-windows (Theorem 3.12, Lemma 3.22). Hence density of $\text{NA}((c_0, p_N), \ell_2^2)$ is equivalent to Lemma Z for T_{final} and N [I, Theorem 8.20], and Lemma Z for infinitely many N gives density for Martín's norm p built with T_{final} (Remark 3.13).
- (b) *Finite base support.* Every $f \in S_{p_N^*}$ with finite base support that is not a (C^*) row belongs to $\mathcal{R}$ (Corollary 5.17). Among (C^*) rows, the block-tame ones are in $\mathcal{R}$ [I, Corollary 7.22] (for instance the self-aligned rows, Corollary 5.30); so are the rows in $\mathcal{R}_0 \cup \mathcal{R}_0^\pm \cup \mathcal{R}_S$ [I, Section 8]. For the norm built with the modified design $D^{U1'}$, the rows with finite base support covered by Master Theorem IV' are in $\mathcal{R}$ (Theorem 5.68, Corollary 5.69). Constant-sign maximal contact never produces a (C^*) row (Proposition 5.21).
- (c) *Infinite base support.* The rows listed in Remark 6.51(a) are in $\mathcal{R}$ and the pairs listed there are in clNA ; for each of them (LSC-trunc) holds (Proposition 6.48).

7.2. The main problems.

Problem 7.1 (Lemma Z for the designed operator). Let $N \geq 1$ and $T = T_{\text{final}}$. Is it true that for every $f \in S_{p_N^*}$, $g \in \mathcal{C}(f)$, $\rho < 1$ and $\varepsilon > 0$ there is $f' \in \mathcal{R}_0$ with $p_N^*(f' - f) < \varepsilon$ and $\text{dist}(\rho g, \mathcal{C}(f')) < \varepsilon$? Equivalently [I, Theorem 8.20], is $\text{NA}((c_0, p_N), \ell_2^2)$ dense?

Problem 7.1 is open for every N , including $N = 1$. By Proposition 6.48, it has a positive answer if and only if Lemma Z holds at every first row with finite base support and (LSC-trunc) holds. By Corollary 5.17, [I, Lemma 1.16] and [I, Lemma 8.4(a)], the first condition follows from a positive answer to Problem 5.80; the second is the following problem.

Problem 7.2 (Lower semicontinuity through truncations). Let $N \geq 1$ and $T = T_{\text{final}}$. Does (LSC-trunc) of Proposition 6.48 hold, i.e. for every $f \in S_{p_N}^*$, $g \in \mathcal{C}(f)$, $\rho < 1$ and $\varepsilon > 0$, is there a first row f' with finite base support, $p_N^*(f' - f) < \varepsilon$ and $\text{dist}(\rho g, \mathcal{C}(f')) < \varepsilon$?

Problem 7.3 (Martín's norm). Is $\text{NA}((c_0, p), \ell_2^2)$ dense, for Martín's norm p built with T_{final} , or with any admissible operator?

A positive answer to Problem 7.1 for infinitely many N answers Problem 7.3 positively for T_{final} [I, Lemma 1.5]. The row-wise results of Sections 4 to 6 do not transfer to p , since their hypotheses depend on N (Remark 5.84(d), Remark 5.70(b)).

7.3. Finite base support: the residual (C*). Problem 5.80 asks whether every (C*) row is in $\mathcal{R}$. (C*) rows exist (Proposition 5.72); the nested-tuning rows of Proposition 5.73 are (C*) rows that are not block-tame, but their existence is only sketched (Remark 5.74), and only their mates with exact data are known to be recovered (Corollary 5.76). Two routes are available; each leaves precise gaps.

Route through absorbers and levers (Subsection 5.6). Master Theorem IV' needs a $\varkappa$ -neutral lever in every activity class with active shift, and in mixed classes the absence of frustrated positive carriers. The first requirement is not a technicality: a donor raise moves the block scalar $\varkappa$ by the same order as the raise, and the exact shifted system depends on a block only through ratios (Corollary 2.11, Proposition 5.46), so this route needs a second, $\varkappa$ -neutral lever in every such block (Remark 5.47; no claim is made that a one-lever argument cannot succeed). An earlier argument in which a donor raise alone exactifies the block scalars (which would have put every row with finite base support in $\mathcal{R}$ for $N = 1$) fails for this reason; that conclusion is not proved and is not used.

Problem 7.4 (Status coherence without a lever). For the design $D^{U1'}$ and $N \geq 1$: let f have finite base support and let w be a clean sub-window with an active block, an active near-threshold switching carrier d , and no inactive Ω -carrier of robust relative position, and suppose that the exact zero set of the tiny minors (Definition 5.52), in the ratio variables and near the ratios of f , lies in $\{\varrho_d \geq 1\}$. Is $f \in \mathcal{R}$? In particular, can the window data be completed with d a peak whose excess $|\omega_+(d)| + |\omega_-(d)|$ is $O(Kt/\lambda_d)$, with K of the form $C_f(\text{Des}/u)^C$, or can a $\varkappa$ -neutral lever be supplied by other means? (By Proposition 5.46, ϱ_d at an exact completion is a function of the ratio variables up to fine terms, so under this hypothesis every exact completion near the ratios of f has $\varrho_d \geq 1$ up to fine terms, and d cannot be kept strictly below its threshold by such a completion. Remark 5.81 records a sketch for carriers with inactive excess and a heuristic genericity route.)

Problem 7.5 (Mixed activity classes). For the design $D^{U1'}$ and $N \geq 2$: are the rows with finite base support whose clean sub-windows carry mixed activity classes in $\mathcal{R}$? One route is an exact completion (Lemma 5.63) at which no carrier of a negative block is bad in the sense of Lemma 5.66 (Remark 5.82); finite models indicate that mixing can force the bad state at every exact completion, which is numerical evidence only and concerns this route, not the recoverability of the rows.

Route through exact coupling and uniform composition (Remark 5.83).

Problem 7.6 (Exact coupling). With two or more shifted blocks, or without a free-ray carrier with $q < 0$, can the coupling rows $\Delta d_m = d_m^\#(\omega^-) - d_m^\#(\omega^+)$ be

satisfied exactly inside the exact combinatorial cone, with constants of the form $C_f(\text{Des}/u)^C$? (One shifted block with a free-ray carrier: Proposition 5.79.) Together with this, coarse exactness at the companion modulo the coupling rows and the contributions of carriers beyond the level (item (RT*c) of Remark 5.83) is only sketched.

Problem 7.7 (Uniform composition). Are there d -consistent engineered approximants (the normalized values of the finitely many coarse switching carriers re-tuned exactly to their values at the row) with per-piece bounds $T_0 \geq c_p t$ uniform in the scale, such that the averaging of Theorem 2.20 can be carried out at one engineered approximant for all pieces of a window, and the violation mass stays below the late threshold of the companion? (Items (S2a'), (S2b) are sketches and (S2c) is a heuristic, Remark 5.83. The route uses d -consistency because without it window masses change the coarse block data at first order, which gives a mismatch of order up to s_1/t^2 ; the upper bound is proved, finite models indicate that it is attained up to a constant, which is numerical evidence only, and that the mismatch cannot then be rebalanced is heuristic.)

Remark 7.8 (Further open points at finite base support). (a) For the nested-tuning rows of Proposition 5.73, Lemma Z is not proved for general mates; a route through exact window data with a shift column and transplant to block-tame companions is only a sketch (Remark 5.84(a)). (b) The existence of these rows is a sketch (Remark 5.74). (c) Earlier sketches of a stable shifted resonance theorem required, besides Problems 7.4 and 7.5: exact absorption of fine-peak residues on finitely many coarse free coordinates (now proved by the zero-value absorber pairs of Lemma 5.51 and Proposition 5.59, for $D^{U1'}$); the scrambling condition at companions (proved in configuration (i), Lemma 5.62); a shift direction constant along a window (proved through the one-ray normalization, Proposition 5.54); and a joint completion (Lemma 5.63). The exactification of the block scalars (ϑ_m, A_m) is the part that remains, in the form of Problem 7.4.

7.4. Infinite base support. For first rows with infinite base support, Problem 7.2 is open for T_{final} outside the classes of Remark 6.51(a). The obstructions are the following.

Problem 7.9 (Transport of Master Theorem III'). Let $N \geq 1$ and $T = T_{\text{final}}$. Is every $f \in S_{p_N}^*$ with infinite base support in $\mathcal{R}$ unless, at all but finitely many levels, every clean sub-window exhibits (C*) at the deep-raised row, a tiny tuning regularity modulus (Problem 7.11), a failure of (H2), or a degenerate bad peak with the swallowing sign whose block has no donor with a tuning resource? The intended route (Remark 6.50) combines a deep raise (Lemma 6.45), the moves of Sections 4 and 5 with support moves (Lemma 6.46), exact window data at the companion (Lemma 6.41) and Proposition 6.47; it is a sketch. The items not written out are: the transplant of exact data at infinite base support, with clamped data at the decomposition row; the stability of the shift-pinning rates under the deep raise; the exactification of tiny minors with the additional pinned faces and contact rows; and the joint fixed point of the support tuning with the threshold buffer.

Problem 7.10 (Infinitely many bad carriers). Let $N \geq 1$ and $T = T_{\text{final}}$. Is every $f \in S_{p_N}^*$ satisfying the hypotheses of Theorem 6.42 other than (W_{inf}) in $\mathcal{R}$? The rate condition involves the quantities $H(l)$, $C_{\text{VP}}(l)$, $\gamma_{\mathcal{B}}(l)$, the bad-peak margin rate

$1/\mu_{\mathcal{B}}(l)$, and, when (ND') fails, the kernel rate $1/m_k^F(l)$ of Proposition 6.44; the same question arises for rooms of good carriers that decay too fast for (W*). These rates would become design constants through Problem 7.9.

Problem 7.11 (Tuning regularity). At a clean sub-window of a first row with infinite base support, does the first-order resource map of Remark 6.50 (private two-sided columns of contacts, convertible thin support coordinates, robust pairs and anchors of Lemma 6.46, one-sided pull columns, and single-support columns at robust support coordinates) map the cone of admissible moves onto $\mathbb{R}^{L_0}$ with a right inverse of norm at most $(\text{Des}(l)/u(w))^C$, or can the first rows at which it does not be recovered otherwise? On proportional profiles the modulus reduces to the Sherman–Morrison denominator of Remark 3.9, which vanishes in particular when (ND') fails (then $a|_F = \sum_k c_k u_k|_F$); the converse is not claimed.

Problem 7.12 (Condition (H2) and degenerate bad peaks). Are first rows with infinite base support at which (H2) fails, or which have a degenerate bad peak with the swallowing sign (failure of (H3-inf)) whose block has no donor peak with a tuning resource, in $\mathcal{R}$?

Problem 7.13 (Scrambling along deep raises). For non- d -neutral data at raised rows, does the scrambling condition (SC) [I, Definition 7.16] transfer along a deep raise below the scale of the change of the normer? This enters only through shifted (non- d -neutral) data, i.e. inside (C*) at infinite base support (Problem 7.9), and in the extension of Theorem 6.36 to non- d -neutral fixed data, which is open (Theorem 6.36 itself assumes d -neutral data and uses no (SC)).

Remark 7.14 (Other operators). For signature-ladder operators without a fast-decaying diagonal base, non-sparse support swallowing (critical and super-critical flip profiles beyond the special cases of Remark 6.30) and infinitely many swallowed carriers without box domination remain open (Remark 6.51(c)). For a general admissible operator, Problem 7.3 and the density for p_N are open for every N ; the resonance obstruction [I, Theorem 6.6] shows that the intrinsic mechanisms of Part I do not suffice in general, and Theorem 2.28 shows that *block-tame* approximating rows for a mate whose profile takes different values at two coordinates s_1, s_2 outside the data supports must put s_1 or s_2 into a bank, the support or the support of a strict non-peak, or be aligned with maximal contact; nothing is proved about approximating rows that are not block-tame. (Such mates exist at self-aligned rows of T_{final} for $N = 1$, Proposition 5.32.)

Remark 7.15 (Evidence). No counterexample is known in any of the classes above. Every mechanism examined at finite base support is either removed at a clean sub-window or reduced to (C*); every mechanism examined at infinite base support is harmless after a deep raise, pinned by the flip budget of the mate, cushion-dominated, or a rate phenomenon of the same kind as at finite base support (Remark 6.51(d)). This is a description of the state of the proofs, not evidence for either answer.

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